



Program Structure & Syllabus

Bachelor of Technology (Computer Science Engineering)

Duration: 04 Years

Batch: 2024-2028

School of Computer Science Engineering and Technology

Vision of the University:

To be a global leader in education, research, and innovation, empowering higher learning ecosystem

Mission of the University:

- Empower all the members of the Bennett ecosystem and provide thought leadership, focus on nation building and prepare our graduates to move with the Times.
- Cultivate international partnerships and collaborations with academic institutions, industry, and government organizations and provide a rigorous and innovative education that equips students with the knowledge, skills, and ethical values required to excel in their careers.
- Foster a culture of lifelong learning, adaptability, and critical thinking, ensuring graduates are prepared to tackle challenges in all academic fields.
- Drive interdisciplinary research and innovation, pushing the boundaries of human knowledge, addressing pressing global issues and solving real world problems.
- Enhance a collaborative environment that encourages faculty and students to engage in research, innovation, and entrepreneurship, creating a lasting impact on society.
- Promote diversity, equity, and inclusion, ensuring that all individuals, regardless of background, feel welcomed, respected, and empowered.
- Prepare students to become global citizens, capable of addressing global challenges and contributing to the well-being of communities worldwide.
- Provide a globally connected career services networking with graduate employers and alumni.
- Foster a strong sense of ethical responsibility in our graduates, emphasizing the importance of ethical conduct, sustainability, and social impact in professional practice.
- Commit to ongoing assessment and improvement of our programs and invest in modern infrastructure and advanced technology to support teaching, research, and innovation adapting to the evolving needs of students, industries, and society.

Vision of the Department:

International Excellence and National Relevance in teaching, research, and technology development.

Mission of the Department:

1. Strive to continuously adapt and improve teaching-learning pedagogy and encourage the use of technology and online learning.
2. Innovate through Entrepreneurial framework, Research Projects, Quality publications and Consultancy.
3. Transparency, Teamwork and Timely execution will be three mantras of the department.

Program Educational Objectives (PEO):

PEO1: Provide graduates with a solid foundation in engineering principles to solve complex computer science problems effectively.

PEO2: Equip graduates with skills in software development, project management, and communication to design and implement innovative solutions using emerging technologies.

PEO3: Foster a mindset of continuous learning, enabling graduates to adapt to technological advancements and contribute to the field through research and development activities.

PEO to Mission Statement Mapping:

Mission Statements			
PEO Statements	M1	M2	M3
PEO1	1	2	1
PEO2	2	3	3
PEO3	3	1	2

1: Low

2: Medium

3: High

Program Outcomes (POs) and Program Specific Outcomes (PSOs):

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PSO1: Analyse a complex problem and propose its solution using various computer system concepts.

PSO2: Apply software development and project management methodologies on various platforms using emerging technologies to design, develop and implement solutions of problems.

PSO3: Foster the field of computing by taking a deep dive into its latest trends and going through new research & development activities.

Mapping of POs/PSOs to PEOs:

PEO Statements			
Program Outcomes	PEO1	PEO2	PEO3
PO1	3	2	
PO2	3	1	
PO3	2	3	
PO4	3		1
PO5	1	3	2
PO6		2	
PO7	2	2	
PO8		2	
PO9		3	2
PO10		2	1
PO11		2	
PO12		2	3
PSO1	3	2	1
PSO2	2	3	1
PSO3	1	2	3

1: Low

2: Medium

3: High

List of Courses:

Semester: I

S. No.	Course Name	Course Code	Credit	L	T	P
1	Computational Thinking and Programming	CSET101	5	2	1	4
2	Engineering Calculus	EMAT101L	4	3	1	0
3	Electromagnetics and Mechanics	EPHY111L	5	3	1	2
4	Environment and Sustainability	CSET108	3	3	0	0
5	Foundations of Innovation and Entrepreneurship	CSET107	2	2	0	0
6	New Age Life Skills	CSET103	2	2	0	0
Total Credits			20	14	3	6
*CSET102, CSET103, CSET107 will be run in both Sem-I and Sem-II. Half of CSE students will get CSET102 and other half (CSET103+CSET107) in Semester - I and vice versa in Semester - II.						

Semester: II

S. No.	Course Name	Course Code	Credit	L	T	P
1	Object Oriented Programming using Java	CSET109	6	3	1	4
2	Digital Design	CSET105	4	3	0	2
3	Discrete Mathematical Structures	CSET106	4	3	1	0
4	Linear Algebra and Ordinary Differential Equations	EMAT102L	4	3	1	0
5	New Age Life Skills	CSET103	2	2	0	0
6	Introduction to Electrical and Electronics Engineering	CSET102	4	2	1	2
Total Credits			22	14	4	8

Semester: III

S. No.	Course Name	Course Code	Credit	L	T	P
1	Information Management Systems	CSET201	4	3	0	2
2	Data Structures using C++	CSET243	7	3	1	6
4	Probability and Statistics	CSET240	5	3	1	2
5	Software Engineering	CSET205	4	3	0	2
6	Specialization Core - I		4	3	0	2
Total Credits			24	15	2	14

Semester: IV

S. No.	Course Name	Course Code	Credit	L	T	P
1	Microprocessors and Computer Architecture	CSET203	4	3	0	2
2	Design and Analysis of Algorithms	CSET244	7	3	1	6
3	Computer Networks	CSET207	4	3	0	2
4	Ethics for Engineers, Patents, Copyrights and IPR	CSET208	1	1	0	0
5	Operating Systems	CSET209	4	3	0	2
6	Design Thinking & Innovation	CSET210	2	0	0	4
7	Specialization Core - II		4	3	0	2
Total Credits			26	16	1	18
*For AI Specialization students CSET301 will run in sem IV and (Specialization Core-II+CSET208) will run in Semester V						
1	Refresher Course for Industry Career	CSET398				

Semester: V

S. No.	Course Name	Course Code	Credit	L	T	P
1	Artificial Intelligence and Machine Learning	CSET301	5	3	0	4
2	Automata Theory and Computability	CSET302	4	3	1	0
3	High Performance Computing	CSET305	4	3	0	2
4	Seminar on Special Topics in Emerging Areas	CSET303	1	0	1	0
5	Competitive Programming	CSET304	2	0	0	4
6	Specialization Elective - I		3	2	0	2
7	Elective - II		3	2	0	2
8	Open Elective - I		3	3	0	0
Total Credits			25	16	2	14

Semester: VI

S. No.	Course Name	Course Code	Credit	L	T	P
1	Undergraduate Research in Computer Science Engineering	CSET306	1	0	1	0
2	HSS - Elective		3	3	0	0
3	Specialization Elective – III		3	2	0	2
4	Elective - IV		3	2	0	2
5	Open Elective - II		3	3	0	0
6	Elective - V		3	2	0	2
7	Open Elective-III		3	3	0	0
	Total Credits		19	15	1	6
1	Summer Internship	CSET399				

Semester: VII

S. No.	Course Name	Course Code	Credit	L	T	P
1	Capstone Project	CSET402	6	0	0	12
2	Specialization Elective – VI		3	2	0	2
3	Open Elective - IV		3	3	0	0
	Total Credits		12	5	0	14
	OR					
1	Professional Practical Internship	CSET497	12	0	0	24

Semester: VIII

S. No.	Course Name	Course Code	Credit	L	T	P
1	Industrial Project/R&D Project/Start-up Project	CSET499	12	0	0	24
	Total Credits		12	0	0	24

Specialization Core I and II

S. No.	Course Name	Specialization	Course Code	Credit	L	T	P
1	Statistical Machine Learning	AI-1	CSET211	4	3	0	2
2	Blockchain Foundations	Blockchain-1	CSET212	4	3	0	2
3	Linux and Shell Programming	Cyber Security-1	CSET213	4	3	0	2
4	Data Analysis using Python	Data Science-1	CSET214	4	3	0	2
5	Graphics and Visual Computing	Gaming-1	CSET215	4	3	0	2
6	UI/UX Design for Human Computer Interface	CSE-1	CSET216	4	3	0	2
7	Software Development with DevOps	DevOps-1	CSET217	4	3	0	2
8	Full Stack Development	Full Stack-1	CSET218	4	3	0	2
9	Quantum Computing Foundations	Quantum Computing-1	CSET219	4	3	0	2
10	Unmanned Aerial Vehicles	Drones-1	CSET220	4	3	0	2
11	Robotic Process Automation Essentials	RPA-1	CSET221	4	3	0	2
12	Microcontrollers, Robotics & Embedded Systems	IOT-1	CSET222	4	3	0	2
13	Augmented Reality Foundations	AR & VR -1	CSET223	4	3	0	2
14	Product Design Principles and Practices	PDT-1	CSET238	4	3	0	2
15	Cloud Computing	Cloud Computing-1	CSET224	4	3	0	2
16	Fundamentals in Swift Programming	Mobile Technology - 1	CSET241	4	2	0	4
17	UX Design Foundations	UI/UX -1	CSET245	4	3	0	2
18	Cloud Computing	CSE-2	CSET224	4	3	0	2
19	Intelligent Model Design using AI	AI-2	CSET225	4	3	0	2
20	Blockchain Engineering	Blockchain-2	CSET226	4	3	0	2
21	System and Network Security	Cyber Security-2	CSET227	4	3	0	2
22	Data Mining and Predictive Modelling	Data Science-2	CSET228	4	3	0	2
23	Game Engine and Architecture	Gaming-2	CSET229	4	3	0	2
24	DevOps Practices and Principles	DevOps-2	CSET230	4	3	0	2
25	Programming Methodologies for Backend Development	Full Stack-2	CSET231	4	3	0	2
26	Design of Cloud Architectural Solutions	Cloud Computing-2	CSET232	4	3	0	2
27	Machine Learning with Quantum Computing	Quantum Computing-2	CSET233	4	3	0	2

28	Drone Remote Sensing	Drones-2	CSET234	4	3	0	2
29	Digital Bots Development	RPA-2	CSET235	4	3	0	2
30	IoT Networks and Protocols	IOT-2	CSET236	4	3	0	2
31	VR and 360 Video Production	AR & VR -2	CSET237	4	3	0	2
32	Product Design Architecture and Delivery	PDT-2	CSET239	4	3	0	2
33	Advance Swift Programming	Mobile Technology - 2	CSET242	4	2	0	4
34	UI/UX Design for Human Computer Interface	UI/UX -2	CSET246	4	3	0	2

Elective-I, III, VI

Specialization Electives: Artificial Intelligence

S. No.	Course Name	Course Code	Credit	L	T	P
1	Advanced Computer Vision and Video Analytics	CSET340	3	2	0	2
2	Cognitive Modelling	CSET341	3	2	0	2
3	Probability and Random Processes	CSET342	3	2	0	2
4	AI in Healthcare	CSET343	3	2	0	2
5	Image and Video Processing	CSET344	3	2	0	2
6	Information Retrieval and Search Engine	CSET345	3	2	0	2
7	Natural Language Processing	CSET346	3	2	0	2
8	Social Network Analysis	CSET347	3	2	0	2
9	Reinforcement Learning	CSET348	3	2	0	2
10	Special Topics in Artificial Intelligence	CSET349	3	2	0	2
11	Emerging Topics in Artificial Intelligence	CSET417	3	2	0	2
12	Introduction to Generative Artificial Intelligence	CSET419	3	2	0	2
13	Advanced Topics in Generative Artificial Intelligence	CSET424	3	2	0	2
14	Generative AI and Large Language Models on AWS	SCSE3042	3	2	0	2
15	Prompt Engineering for Generative AI	SCSE3043	3	2	0	2

Specialization Electives: Blockchain

S. No.	Course Name	Course Code	Credit	L	T	P
1	Blockchain Technologies: Platforms & Applications	CSET350	3	2	0	2
2	Smart Contracts and Solidity Programming	CSET351	3	2	0	2
3	Digital Currencies and Blockchain	CSET352	3	2	0	2
4	Blockchain Policy: Legal, Social and Economic Impact	CSET353	3	2	0	2
5	Cyber Security with Blockchain	CSET354	3	2	0	2
6	Web Development for Blockchain Applications	CSET355	3	2	0	2
7	Security and Privacy for Big Data Analytics	CSET356	3	2	0	2
8	Modern Cryptography	CSET357	3	2	0	2
9	Cloud Infrastructure and Services	CSET358	3	2	0	2
10	Special Topics in Blockchain	CSET359	3	2	0	2
11	Emerging Topics in Blockchain	CSET416	3	2	0	2

Specialization Electives: Cyber Security

S. No.	Course Name	Course Code	Credit	L	T	P
1	Cyber Security with Blockchain	CSET354	3	2	0	2
2	Malware Analysis for Mobile Devices	CSET360	3	2	0	2
3	Modern Cryptography	CSET357	3	2	0	2
4	Security and Privacy for Big Data Analytics	CSET356	3	2	0	2
5	Device Level IoT Security	CSET361	3	2	0	2
6	Vulnerability Analysis in Network Protocols	CSET362	3	2	0	2
7	Penetration Testing, Auditing and Ethical Hacking	CSET363	3	2	0	2
8	Forensics and Cyber Law	CSET364	3	2	0	2
9	Web Security	CSET365	3	2	0	2
10	Special Topics in Information Security	CSET366	3	2	0	2
11	Cloud Security and Compliances	CSET462	3	2	0	2
12	Emerging Topics in Cyber Security	CSET415	3	2	0	2

Specialization Electives: Data Science

S. No.	Course Name	Course Code	Credit	L	T	P
1	Digital Marketing and Trend Analysis	CSET367	3	2	0	2
2	Structural Equation Modelling	CSET368	3	2	0	2
3	Time Series Analysis	CSET369	3	2	0	2
4	Data Visualization and Dashboards	CSET370	3	2	0	2
5	Security and Privacy for Big Data Analytics	CSET356	3	2	0	2
6	Social Network Analysis	CSET347	3	2	0	2
7	Big Data Analytics and Business Intelligence	CSET371	3	2	0	2
8	Advanced Database Management System	CSET372	3	2	0	2
9	Satellite Data Analysis	CSET373	3	2	0	2
10	Special Topics in Data Science	CSET374	3	2	0	2
11	Emerging Topics in Data Science	CSET414	3	2	0	2

Specialization Electives: Gaming

S. No.	Course Name	Course Code	Credit	L	T	P
1	VR Gaming	CSET375	3	2	0	2
2	Augmented Reality	CSET337	3	2	0	2
3	Game Mechanics and Game Physics	CSET377	3	2	0	2
4	Game Programming with HTML5	CSET378	3	2	0	2
5	AI for Games	CSET379	3	2	0	2
6	Animation and Rendering Techniques	CSET380	3	2	0	2
7	Pixel and Poly Arts for Games	CSET440	3	2	0	2
8	Game Design, Development and Programming	CSET441	3	2	0	2
9	Special Topics in Gaming	CSET442	3	2	0	2
10	Emerging Topics in Gaming	CSET413	3	2	0	2

Specialization Electives: Robotic Process Automation

S. No.	Course Name	Course Code	Credit	L	T	P
1	Digital Process Automation Architecture	CSET443	3	1	0	4
2	Enterprise Process Automation Frameworks	CSET444	3	2	0	2
3	Cloud Infrastructure and Services	CSET358	3	2	0	2
4	Continuous Integration and Deployment in DevOps	CSET445	3	2	0	2
5	Server-Side Development Frameworks: Express and Spring	CSET446	3	2	0	2
6	Special Topics in RPA	CSET447	3	2	0	2
7	Emerging Topics in RPA	CSET412	3	2	0	2

Specialization Electives: DevOps

S. No.	Course Name	Course Code	Credit	L	T	P
1	Build and Release Management in DevOps	CSET448	3	2	0	2
2	Cloud Services Development and Operations	CSET449	3	2	0	2
3	Source and Version Control in DevOps	CSET450	3	2	0	2
4	Continuous Integration and Deployment in DevOps	CSET445	3	2	0	2
5	Software Craftsmanship in DevOps	CSET451	3	2	0	2
6	Software Containerization in DevOps	CSET452	3	2	0	2
7	System Provisioning and Configuration Management in DevOps	CSET453	3	2	0	2
8	Test Automation in DevOps	CSET454	3	2	0	2
9	Cloud Infrastructure and Services	CSET358	3	2	0	2
10	Special Topics in DevOps	CSET456	3	2	0	2
11	Emerging Topics in DevOps	CSET411	3	2	0	2

Specialization Electives: Full Stack

S. No.	Course Name	Course Code	Credit	L	T	P
1	Front-End Web UI Frameworks and Tools: Bootstrap	CSET457	3	2	0	2
2	Front-End Web UI Frameworks and Tools: React Native	CSET458	3	1	0	4
3	Server-Side Development Frameworks: Express and Spring	CSET446	3	2	0	2
4	Web Development for Blockchain Applications	CSET355	3	2	0	2
5	Cloud Infrastructure and Services	CSET358	3	2	0	2
6	Advanced Database Management System	CSET372	3	2	0	2
7	Front-End UI Frameworks and Tools: Flutter and Django	CSET459	3	1	0	4
8	Special Topics in Full Stack	CSET460	3	2	0	2
9	Emerging Topics in Full Stack	CSET410	3	2	0	2

Specialization Electives: Cloud Computing

S. No.	Course Name	Course Code	Credit	L	T	P
1	Cloud Services Development and Operations	CSET449	3	2	0	2
2	Cloud System Administration and Operations	CSET461	3	2	0	2
3	Cloud Security and Compliances	CSET462	3	2	0	2
4	AWS Cloud Support Associate	CSET463	3	2	0	2
5	Developing Solutions for Microsoft Azure	CSET464	3	2	0	2
6	Google Associate Cloud Engineer	CSET465	3	2	0	2
7	Software Containerization in DevOps	CSET452	3	2	0	2
8	Cloud Infrastructure and Services	CSET358	3	2	0	2
9	Special Topics in Cloud Computing	CSET466	3	2	0	2
10	Emerging Topics in Cloud Computing	CSET409	3	2	0	2

Specialization Electives: Drones

S. No.	Course Name	Course Code	Credit	L	T	P
1	Drone Communication	CSET467	3	2	0	2
2	Drone Applications, Components and Assembly	CSET468	3	2	0	2
3	UAV Simulation	CSET469	3	2	0	2
4	Drone Swarming	CSET470	3	2	0	2
5	UAV Mission Planning and Deployments	CSET471	3	2	0	2
6	Auto Pilot and Flight Control	CSET472	3	2	0	2
7	Special Topics in Drone Technology	CSET473	3	2	0	2
8	Emerging Topics in Drone Technology	CSET408	3	2	0	2

Specialization Electives: Quantum Computing

S. No.	Course Name	Course Code	Credit	L	T	P
1	Quantum Computing for Data Analysis	CSET474	3	2	0	2
2	Quantum Information Theory	CSET475	3	2	0	2
3	Quantum Computer Architecture	CSET476	3	2	0	2
4	Quantum Cryptography	CSET477	3	2	0	2
5	Special Topics in Quantum Computing	CSET478	3	2	0	2
6	Cloud Infrastructure and Services	CSET358	3	2	0	2
7	Security and Privacy for Big Data Analytics	CSET356	3	2	0	2
8	Emerging Topics in Quantum Computing	CSET407	3	2	0	2

Specialization Electives: IoT and Robotics

S. No.	Course Name	Course Code	Credit	L	T	P
1	Robotics: Dynamics and Controls	CSET479	3	2	0	2
2	IoT Analytics	CSET480	3	2	0	2
3	IoT: Security and Attacks	CSET420	3	2	0	2
4	Device Level IoT Security	CSET361	3	2	0	2
5	Humanoids	CSET421	3	2	0	2
6	Mobile IoT Networks	CSET422	3	2	0	2
7	Special Topics in IoT	CSET423	3	2	0	2
8	Emerging Topics in IoT and Robotics	CSET406	3	2	0	2

Specialization Electives: Augmented & Virtual Reality

S. No.	Course Name	Course Code	Credit	L	T	P
1	Augmented Reality and ARCore	CSET425	3	2	0	2
2	Animation and Rendering Techniques	CSET380	3	2	0	2
3	3D Models for Augmented and Virtual Reality	CSET376	3	2	0	2
4	Building Social VR with 3D Characters	CSET426	3	2	0	2
5	Virtual Reality: Interface, Application and Design	CSET427	3	2	0	2
6	Special Topics in AR and VR	CSET428	3	2	0	2
7	Emerging Topics in Augmented & Virtual Reality	CSET405	3	2	0	2

Specialization Electives: Product Design Technology

S. No.	Course Name	Course Code	Credit	L	T	P
1	Art, Craft and Design	CSET434	3	2	0	2
2	3D Models for Augmented and Virtual Reality	CSET376	3	2	0	2
3	Agile Software Development	CSET455	3	2	0	2
4	Product Design Technologies and Tools	CSET437	3	0	0	6
5	Animation and Rendering Techniques	CSET380	3	2	0	2
6	Special Topics in Product Design Technology	CSET438	3	2	0	2
7	Design and Manufacturing for Digital Products	CSET433	3	2	0	2
8	Fashion Design Technology	CSET435	3	2	0	2
9	Furniture Design	CSET436	3	2	0	2
10	Emerging Topics in Product Design Technology	CSET404	3	2	0	2

Specialization Electives: Mobile Technologies

S. No.	Course Name	Course Code	Credit	L	T	P
1	iOS Application Development	CSET396	3	2	0	2
2	Mobile Application Development	CSET397	3	2	0	2
3	Design Patterns and UI Testing	CSET439	3	2	0	2
4	iOS Machine Learning	CSET494	3	0	0	6

5	Backend in iOS	CSET495	3	2	0	2
6	Front-End Web UI Frameworks and Tools: Bootstrap	CSET457	3	2	0	2
7	Special Topics in Mobile Technologies	CSET496	3	2	0	2
8	Emerging Topics in Mobile Technologies	CSET403	3	2	0	2

Specialization Electives: UI/UX

S. No.	Course Name	Course Code	Credit	L	T	P
1	User Centered Design	CSET313	3	2	0	2
2	Strategic UI/UX Design: From Concept to Market Success	CSET314	3	2	0	2
3	UX Research and Data Analytics	CSET315	3	2	0	2
4	UI/UX Design Testing and Evaluation	CSET316	3	0	0	6
5	Product Management for UX Designers	CSET317	3	2	0	2
6	Special Topics in Full Stack	CSET460	3	2	0	2
7	Build and Release Management in DevOps	CSET448	3	2	0	2
8	Source and Version Control in DevOps	CSET450	3	2	0	2
9	Continuous Integration and Deployment in DevOps	CSET445	3	2	0	2

General Electives (Elective II, IV, V)

S. No.	Course Name	Course Code	Credit	L	T	P
1	Current Technology Project	CSET319	3	2	0	2
2	Information Retrieval	CSET320	3	2	0	2
3	User Centered Design	CSET321	3	2	0	2
4	Secure Coding	CSET322	3	2	0	2
5	Compiler Construction	CSET323	3	2	0	2
6	Software Project Management	CSET324	3	2	0	2
7	Digital Communication	CSET325	3	2	0	2
8	Soft Computing	CSET326	3	2	0	2
9	Embedded Systems	CSET327	3	2	0	2
10	Distributed Computing	CSET328	3	2	0	2
11	Engineering Optimization	CSET329	3	2	0	2
12	Wireless Networks	CSET330	3	2	0	2
13	Combinatorics	CSET331	3	2	0	2
14	Mobile and Networked Embedded Systems	CSET332	3	2	0	2
15	Problem Solving using C	CSET333	3	2	0	2
16	Programming using C++	CSET334	3	2	0	2

17	Deep Learning	CSET335	3	2	0	2
18	Bioinformatics and Computational Genomics	CSET336	3	2	0	2
19	Computational Geometry	CSET338	3	2	0	2
20	Advances in AI	CSET339	3	2	0	2
21	Image and Video Processing	CSET344	3	2	0	2
22	Natural Language Processing	CSET346	3	2	0	2
23	Reinforcement Learning	CSET348	3	2	0	2
24	Cloud Infrastructure and Services	CSET358	3	2	0	2
25	Forensics and Cyber Law	CSET364	3	2	0	2
26	Satellite Data Analysis	CSET373	3	2	0	2
27	Game Mechanics and Game Physics	CSET377	3	2	0	2
28	Game Programming with HTML5	CSET378	3	2	0	2
29	AI for Games	CSET379	3	2	0	2
30	Android Application using Kotlin	CSET393	3	2	0	2
31	Modern and Contemporary Application in CS	CSET394	3	2	0	2
32	Latest Advances in Engineering and Technology	CSET395	3	2	0	2
33	Emerging Topics in Mobile Technologies	CSET403	3	2	0	2
34	Emerging Topics in Product Design Technology	CSET404	3	2	0	2
35	Emerging Topics in Augmented & Virtual Reality	CSET405	3	2	0	2
36	Emerging Topics in IoT and Robotics	CSET406	3	2	0	2
37	Emerging Topics in Quantum Computing	CSET407	3	2	0	2
38	Emerging Topics in Drone Technology	CSET408	3	2	0	2
39	Emerging Topics in Cloud Computing	CSET409	3	2	0	2
40	Emerging Topics in Full Stack	CSET410	3	2	0	2
41	Emerging Topics in DevOps	CSET411	3	2	0	2
42	Emerging Topics in RPA	CSET412	3	2	0	2
43	Emerging Topics in Gaming	CSET413	3	2	0	2
44	Emerging Topics in Data Science	CSET414	3	2	0	2
45	Emerging Topics in Cyber Security	CSET415	3	2	0	2
46	Emerging Topics in Blockchain	CSET416	3	2	0	2
47	Emerging Topics in Artificial Intelligence	CSET417	3	2	0	2
48	IoT: Security and Attacks	CSET420	3	2	0	2
49	Virtual Reality: Interface, Application and Design	CSET427	3	2	0	2
50	Network Drivers and Protocols	CSET429	3	2	0	2
51	Digital Signal Processing	CSET430	3	2	0	2

52	Special Topics in Computer Science	CSET431	3	2	0	2
53	Advanced Microprocessor	CSET432	3	2	0	2
54	Software Craftsmanship in DevOps	CSET451	3	2	0	2
55	Agile Software Development	CSET455	3	2	0	2
56	Cloud System Administration and Operations	CSET461	3	2	0	2
57	Drone Applications, Components and Assembly	CSET468	3	2	0	2
58	Computational Topo-geometrical Data Analysis	CSET498	3	2	0	2
59	Ethics and Social Implications of AI	SCSE4040	3	3	0	0
60	Prompt Engineering	SCSE4041	3	3	0	0

Honours-I, II, III, IV, V, VI

S. No.	Course Name	Course Code	Credit	L	T	P
	Semester IV					
1	Hons-I		3	2	0	2
	Semester V					
2	Hons-II		3	2	0	2
3	Hons-III		3	2	0	2
	Semester VI					
4	Hons-IV		3	2	0	2
5	Hons-V		3	2	0	2
	Semester VII					
6	Hons-VI		3	2	0	2
Total Credits			18	12	0	12

Minor in AI (I, II, III, IV)

S. No.	Course Name	Course Code	Credit	L	T	P
	Semester III					
1	Data Structure Fundamentals	CSET291	4	4	0	0
	Semester IV					
2	Introduction to Algorithm Design and Development	CSET292	4	4	0	0
	Semester V					
3	Artificial Intelligence	CSET391	4	4	0	0
	Semester VI					
4	Applied AI	CSET392	4	4	0	0
Total Credits			16	16	0	0

CS Open Electives-I, II, III, IV

S. No.	Course Name	Course Code	Credit	L	T	P
1	Applications of AI	CSET381	3	3	0	0
2	Web Technologies	CSET382	3	3	0	0
3	Semantic Technology	CSET383	3	3	0	0
4	IT Support Technologies	CSET384	3	3	0	0
5	Software Engineering	CSET385	3	3	0	0
6	Programming Languages	CSET386	3	3	0	0
7	Artificial Intelligence for Creative Expression	CSET387	3	3	0	0
8	AI and Public Policy	CSET388	3	3	0	0
9	Linux Apache MySQL PHP (LAMP)	CSET389	3	3	0	0
10	Data Structures and Algorithms	CSET390	3	3	0	0
11	Software Testing	CSET481	3	3	0	0
12	Computing Start-ups	CSET482	3	3	0	0
13	Career Skills for IT Companies	CSET483	3	3	0	0
14	Cyber security: Impact on Govts, Policies and Economics	CSET484	3	3	0	0
15	AI and Society	CSET485	3	3	0	0
16	Special Topics in Computer Engineering	CSET486	3	3	0	0
17	Search Engine Optimization	CSET489	3	3	0	0
18	Growth Hacking	CSET488	3	3	0	0
19	Digital Marketing	CSET487	3	3	0	0
20	Advanced Skill Enhancement	CSET492	3	3	0	0
21	Advanced Industry Certification	CSET491	6	6	0	0
22	Global Experience and Practicum	CSET418	3	0	0	6
23	International Acquaintance and Externship	CSET493	3	0	0	6

*The course CSET491 will be considered equivalent to two open elective courses

HSS Electives

S. No.	Course Name	Course Code	Credit	L	T	P
1	Financial Accounting and Management	CSET307	3	3	0	0
2	Communication Skills for Global Professionals	CSET308	3	3	0	0
3	Organizational Behaviour	CSET309	3	3	0	0
4	Marketing Strategies and Planning	CSET310	3	3	0	0
5	Micro and Macro Economics	CSET311	3	3	0	0

*In Special Circumstances Open Electives can be run as General Electives and Vice Versa. Also, Specialization Electives can be run as General Electives and Vice Versa

Semester I

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET101	Computational Thinking and Programming	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	1	4	5
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Solve given problem in Python by using standard programming constructs.

CO2: Build programs using the features of object-oriented programming languages.

CO3: Make use of Python inbuilt APIs to write functions.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	2			3				2			2	3		
CO2	2		2	3		2				3			2	1	2
CO3	3	3	3	2	3				2						2

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

08 lecture hours

Why Python, Applications of Python, Versions of Python, Number system, Binary, Octal, Literals, Variables, Data types, Operators, Control Structures if/else/elif, Nested if/else, Lists/Nested Lists, Tuple.

Module II:

10 lecture hours

Iterative Control, while loop, For loop, Range, Break, Continue, Pass, Nested Loops, Loop inside if, List Operations, List comprehension, Membership operator, Functions, Default argument, required arguments, keyword argument, variable arguments, Function invoking, Nested functions, Lambda functions, Map Reduce, Recursive Functions, Factorial, GCD using Euclid Algorithm, Dictionary Operations, Sets, Enumerators.

Module III:

10 lectures hours

String Operations, Slicing, File Handling, Exception Handling, Modules, Namespaces, Local, Global, and Built-in Namespaces, Object Oriented Programming, Classes and Objects, Encapsulation, Data abstraction, Inheritance, Polymorphism, Interactive Graphics, Displaying Images, Generating Colors, Graphics Objects, Entry Objects, Turtle, Tkinter, Working with Widgets, Controlling Layout with Geometry Managers, Events and Event Handlers, Developing Interactive applications.

Studio Work/Laboratory Experiments:

The lab component of this course is designed to introduce online-coding tools such as Microsoft Azure, Colab to the students and provide hands-on experience with the concepts taught in the lectures.

Text Books :

1. Dierbach, Charles. *Introduction to computer science using python: A computational problem-solving focus*. Wiley Publishing, 2012. ISBN 9789332584686.

Reference Books :

1. Martinez, D. and Jesús, S. D, *Applied Computational Thinking with Python: Design Algorithmic Solutions for Complex and Challenging Real-world Problems*. Packt Publishing, 2021. ISBN 89351507314.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
EMAT101L	Engineering Calculus	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	1	0	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Convergence and divergence of sequence and series.

CO2: Continuity, differentiability and integrability for functions of one variable and multi variables. Approximating the smooth function with a polynomial.

CO3: Computing the area and volume of functions up to three variables.

CO4: Solving the problems using computational software packages.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	2			3				2			2	3		
CO2	2	3		3						3			2		
CO3	3	3	3	2	3				2						2

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

12 lecture hours

The Real Number System, Archimedean Property, Convergence of a Sequence, Monotone Sequences, Cauchy Criterion, Bolzano-Weierstrass Theorem, Limit inferior and Limit Superior. Infinite Series, Convergence Tests and Alternating series. Limit, Continuity, Existence of Maxima, Intermediate Value Property.

Module II:

12 lecture hours

Differentiability, Rolle's Theorem, Mean Value Theorem. Convexity, Concavity, L'Hospital Rule, Fixed Point Iteration Method, Taylor's Theorem, Taylor Series, Power Series. Riemann Integration, Fundamental Theorems of Calculus, Riemann Sum. Improper Integrals, Beta-Gamma Functions, Differentiation under integration.

Module III:

08 lectures hours

Area between two curves; Polar Coordinates, Volumes by slicing, Washer and Shell Methods, Length of a plane curve, Areas of Surfaces of Revolution. Review of vectors, Calculus of Vector Valued Functions, Functions of Several Variables: limit, Continuity and Differentiability, Chain Rule, Directional Derivative, Gradient.

Module IV:

10 lectures hours

Mixed Derivative Theorem, MVT, Extended MVT, Taylor's Theorem in Multiple Variables, Hessian, Maxima, Minima, Second Derivative Test, Lagrange Multiplier Method. Double Integrals, Change of

Variable in a Double and Triple Integrals, Area of a Parametric Surface and surface integral. Surface Area, Surface Integrals, Line Integrals Green's Theorem and Applications.

Text Books :

1. Weir, Maurice D., Joel Hass, and Frank R. Giordano. *Thomas' calculus*. 14th ed. Pearson Education India, 2018. ISBN 9789353060411.
2. Ross, Kenneth A. *Elementary Analysis the Theory of calculus*. 2nd ed. Springer publication, 2013. ISBN 978-1461462705.

Reference Books :

1. Stewart, James. *Calculus*. 7th ed. Cengage Learning, 2012. ISBN 0538497815.
2. Ghorpade, Sudhir R., and Balmohan Vishnu Limaye. *A course in calculus and real analysis*. 1st ed. New York: Springer, 2006. ISBN 978-0387305300.
3. Bartle, Robert G., and Donald R. Sherbert. *Introduction to real analysis*. 4th ed. New York: Wiley, 2014. ISBN 812655181X.
4. Kreyszig, Herbert, and Erwin Kreyszig. *Advanced Engineering Mathematics, Student Solutions Manual and Study Guide*. 10th ed. John Wiley & Sons, 2010. ISBN 978-8126554232.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET108	Environment and Sustainability	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	3	0	0	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1 : Demonstrate analytical thinking skills concerning environmental topics.

CO2 : Demonstrate an ability to combine the many disciplines and fields that intersect with environmental concerns.

CO3 : Illustrate an integrative approach to environmental issues with a focus on sustainability.

CO-PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2			3				2			2	3		2
CO2	2	2		2						3			2	2	1
CO3	3	3	3	2	3				2					2	2

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

06 lecture hours

Environmental Sustainability, Ecosystems, and Climate Change: Overview of Environmental Sustainability, Sustainability and Population Growth, Exploration of Growth Curves, Way forward – Ecosystems and climate change introduction – Ecosystem dynamics –Tragedy of commons - Tragedy of commons solutions – Ecosystems and extinctions – Weather vs. Climate – Climate changes in the past – Climate change in the present – Climate processes.

Module II:

05 lecture hours

Biodiversity and its conservation: Biodiversity – Value of biodiversity – Threats to biodiversity - Conservation of biodiversity – Case studies.

Module III:

08 lectures hours

Natural resources: Natural Resources and Their Challenges, Renewable and Non-renewable Energy Sources, Water's Role in Agriculture, Agriculture's Connection to Food Production, Addressing Issues and Solutions in Agriculture and Food Production.

Module IV:

08 lectures hours

Environmental pollution: Causes, effects and control measures of various types of pollutions - Air pollution - Water pollution - Wastewater treatment - Soil pollution - Noise pollution - Thermal pollution - - Solid waste management - E-waste - Case studies on pollution.

Module V:

07 lectures hours

Ethical Perspectives on the Environment, Global Environmental Policies, Environmental Legislation in India, Environmental Impact Assessment, Challenges Encountered in Enforcing Environmental Laws.

Module VI:**08 lectures hours**

Life cycle analysis and sustainability thinking: Introduction of LCA – Methodology of LCA – ISO 14044 Significance of LCA – Case studies of LCA .

Text Books :

1. Bharucha, Erach. *Textbook of environmental studies for undergraduate courses*. 2nd ed. Universities Press, 2013. ISBN 978-8173718625.

Reference Books :

1. Lodish, Harvey, Arnold Berk, S. Lawrence Zipursky, Paul Matsudaira, David Baltimore, and James Darnell. *Molecular Cell Biology*. 8th ed. WH Freeman, 2016. ISBN 978-1319067748.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET107	Foundations of Innovation and Entrepreneurship	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	0	2	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To discuss the basic understanding of the relevant concepts and practices of entrepreneurship.

CO2: To recognize the essential criteria in the decision to become an entrepreneur or opt for entrepreneurship.

CO3: To practice with the basic tools and tactics required to manage and grow an entrepreneurial firm.

CO-PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2			3				2			2	3		
CO2	2	2		3					1	3			2		3
CO3	3	3	3	2	3				2						2

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

DISCOVERING THE ENTREPRENEUR & ENTREPRENEURSHIP: The psychology of an entrepreneur, what is an entrepreneur? The entrepreneurial mindset: looking inside the black box, entrepreneurial expertise – orientation, do: Interact with a start-up founder, do: discover a problem to solve.

Module II:

09 lecture hours

OVERVIEW OF ENTREPRENEURSHIP: Evolving concept of entrepreneurship & entrepreneurial environment, entrepreneurial opportunity –the customer, market, and competition, how to avoid the pitfalls of entrepreneurship – The dark side, myths & realities about entrepreneurship, do: know your customer, do: perform market study –competition analysis.

Module III:

09 lectures hours

WHERE DO GOOD IDEAS COME FROM? Creativity in entrepreneurship, the creative process–stages/steps, creative thinking – techniques, sources for generation of Ideas, and role of Imagination, effectuation, Improvisation, analogous thinking, etc, problem finding & solving – divergent /convergent, associative approaches and do: generate viable Ideas, do: prototype solution.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET103	New Age Life Skills	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	0	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: By the end of this course, students should be able to express their role in improving their quality of life.

CO2: By the end of this course, students should be able to understand human psychology and its applications to improve human achievements and happiness to substantial extent.

CO3: By the end of this course, students should be able to reflect on the aspects of leadership, motivation, and stress management.

CO4 : By the end of this course, students should be able to appreciate the nature of positive and proactive thinking and assertiveness.

CO5: By the end of this course, students should be able to apply concepts, tools and frameworks of emotional intelligence and stress management

CO-PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1						3	2	3	3	3	3			3	1
CO2						3	2	3	3	3	3			3	1
CO3						3	2	3	3	3	3			3	1
CO4						3	2	3	3	3	3			3	1
CO5						3	2	3	3	3	3			3	1

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

4 lecture hours

Introduction to Life Skills, the critical role of management skills, Developing Self-awareness: Key dimensions, enigma of self-awareness, Understanding and Appreciating Individual differences, Self-esteem, Case studies and Practice Sessions.

Module II:

4 lecture hours

Introduction to communicating and thinking, Communication, Thinking and reasoning, Proactive thinking, Positive thinking, Assertiveness, Emotional Intelligence, Values and Ethics, Case studies and Practice Sessions.

Module III:

4 lectures hours

Setting goals, Goal Setting: Setting SMART Goals, what are the main components, Managing time and prioritizing, Time Management matrix, Balancing Time and Goal, case studies, Practice Sessions.

Module IV:**4 lectures hours**

Diagnostic Surveys for Creative Problem Solving, Problem Solving, Creativity and Innovation, Steps in Analytical Problem Solving, Multiple Approaches to creativity, Conceptual Blocks, Case studies and Practice Sessions.

Module V:**4 lectures hours**

Attitudes, what are the main components? Behavior and attitude, Major job attitude, Job satisfaction, Satisfied and dissatisfied employees on the workplace, Personality, The Myers-Briggs Type Indicator, The Big Five Personality Model, other personality traits relevant to Life skills, Case studies and Practice Sessions.

Module VI:**5 lectures hours**

Diagnostic Surveys for Building Effective Teams, Team Development Behaviour, Developing Credibility, Leading Teams, Exercises in Building Effective Teams, Team Development Behaviour, Developing Teams and Work, Leading Teams, Team Membership, Developing Credibility, Exercises in Building Effective Teams, Case studies and Practice Sessions.

Module VII:**4 lectures hours**

Making Oral and Written Presentations, Introduction to Technical Communication, Process of communication, Essential Elements of Effective Presentations, speaking as a Leader, presenting yourself professionally, developing your Interpersonal Skills, Argument Building, develop your facilitation skills, Making Formal Presentations, Creative Writing for social media, developing self-brand, Case studies and Practice Sessions.

Module VIII:**4 lectures hours**

Diagnostic Survey for Gaining Power and Influence, Building a Strong Power Base and Using Influence wisely, Strategies for gaining organizational power, Transforming power into influence, activities for gaining power and influence, Diagnostic Survey for Motivating Others, Increasing Motivation and Performance, Diagnosing Work Performance Problems, Enhancing Individual abilities, Fostering a Motivating Work Environment, Elements of an Effective Motivation Program, Activities for Motivating Others, Case Studies and Practice Sessions.

Module IX:**4 lectures hours**

Diagnostic Survey for Managing Conflict, Interpersonal Conflict Management, Diagnosing the Type of Interpersonal Conflict, Selecting the Appropriate Conflict Management Approach, Resolving Interpersonal Confrontations Using the Collaborative Approach, Conflict Management Strategies, activities for Improving Managing Conflict Skills, Case Studies and Practice Sessions.

Module X:**5 lectures hours**

Diagnostic Survey for Empowering and Delegating, Management Dilemma Involving Empowerment, The Meaning of Empowerment, Dimensions of Empowerment, Self-Efficacy, Self-Determination, Personal Consequence, How to Develop Empowerment, providing Support, providing Information, Providing Resources, Inhibitors to Empowerment, Delegating Work, Advantages of Empowered Delegation, Review of Delegation Principles, Case Studies and Practice Sessions.

Studio Work/Laboratory Experiments:

1. Group Presentations
2. Group Activities
3. Role Plays
4. Group Decision Making activity
5. Group Treasure Hunt
6. Creating Team profiles
7. Creating Self Profiles

Text Books :

1. Whetten David A. and Kim S. Cameron, *Developing Management Skills (8th ed.)*, Pearson, 2017. ISBN 9789332584686, ISBN 9332584680.
2. Wadkar Alka, *Life Skills for Success (1st ed.)*, Sage Publications, 2016. ISBN 9789351507314, ISBN 9351507319.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
Course	EPHY111L: Mechanics and Electromagnetism	L	T	P	C
Owning School/Department	Physics	3	1	2	5
Pre-requisites/Exposure	NIL				

Course Outcomes (COs)

On completion of this course, the students will be able to:

- CO1: Calculate electrostatic fields and potentials, produced by regularly shaped charged bodies.
- CO2: To understand magnetic fields produced by variously shaped current-carrying bodies.
- CO3: To appreciate the interconnectedness of electric and magnetic phenomena and to realize the significance of Maxwell's equations,
- CO4: Learn how to apply Newton's Laws and understand the concept of energy
- CO5: Solve problems involving rotational motion
- CO6: Get introduced to central forces and conversion of two body problems into one body problem
- CO7: Perform basic physics experiments, tabulate observations, analyze data and draw conclusion

CO-PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	2	1								3		2
CO2	3	3	2	2									3		2
CO3	3	3	2		2		2						3		2
CO4	3	3	2	2	1								3		2
CO5	3	3	2	2									3		2
CO6	3	3	2	2									3		2
CO7				3	3								2		3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module 1

10

lecture hours

Vector operators and coordinate systems, Gauss' law and its applications; Electrostatic potential; Electric fields in matter; Electric polarization, Bound charges, Displacement vector; Electric Permittivity and dielectric constant

Module 2

11

lecture hours

Biot-Savart law; Ampere's law and applications; Magnetic fields in matter, Magnetization, Bound currents; Faraday's law of electromagnetic induction; Displacement current and the generalized Ampere's law; Maxwell's equations; Electromagnetic waves

Module 3

13

lecture hours

Mathematical prerequisites, Kinematics, Newton's Laws, Work-Energy Theorem, Conservation of Momentum and Energy, Conservative Forces and Potential Energy, Non-conservative forces, Rotational Motion, Conservation of Angular Momentum, Moment of Inertia, Rigid Body Motion, Euler's Equations, Non-Inertial Frames of Reference, Pseudo Forces, Coriolis, and Centrifugal Forces, Central Forces, Kepler's Laws, Planetary Motion.

Harmonic Oscillator: Free, Damped, and Forced Oscillations, Resonance, Special Theory of Relativity: Lorentz Transformation, Length contraction, Time Dilation, Energy-Mass-Momentum Relations.

List of Experiments

1. Estimation of Error in Vernier Calipers & Screw Gauge
2. Study of Hall Effect
3. Determination of Moment of Inertia of a Flywheel
4. Determination of Specific Charge (e/m) of Electron
5. Determination of Coefficient of Viscosity
6. Verification of Biot-Savart Law
7. Determination Young's Modulus
8. Determination of Equipotential Lines
9. Determination Coefficient of Static Friction
10. Verification of Principle of Superposition using Helmholtz Coils
11. Additional Experiments

Text Books :

1. D. J. Griffiths, Introduction to Electrodynamics (4th Edition), Pearson Education India Learning Private Limited, 2015, ISBN 978-9332550445.
2. D. Kleppner & R. Kolenkow, An Introduction to Mechanics (1st Edition), McGraw-Hill Education, 2017, ISBN 978-0070647787.

Reference Books :

1. D. Halliday, R. Resnick, J. Walker, Fundamentals of Physics, John Wiley & Sons, 2021, ISBN 978-1119773512,
2. H. D. Young, R. A. Freedman, University Physics with Modern Physics (14th Edition) , Pearson, 201, ISBN 978-0194577199.

SEMESTER

II

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET109	Object Oriented Programming using Java	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	1	4	6
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To examine different programming structures in a platform independent language such as wrapper classes, collections, exceptions, and multithreading.

CO2: To explain the concepts of object-oriented programming like encapsulation, abstraction, inheritance and polymorphism.

CO3: Make use of GUI and database-based programming to develop Applications for real life problems.

CO-PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3		3	3								3		
CO2	3	3		3	3								3	3	3
CO3			3		3								3	3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

Handling user input/output using Scanner class, Platform independence of Java (JVM), Operators (unary, arithmetic, logical, shift left, shift right, ternary, assignment) Exploring bitwise operators, compound assignment, if/else, Using switch for menu-driven applications while, For, do-while, Introduction to Abstraction and Polymorphism Objects Lifecycle: creation, dereference, garbage collection, Wrapper classes (Boolean, Integer, Double, Character), Autoboxing, unboxing, String (concat, indexOf, split, length, toLowerCase, toUpperCase, replace, trim), Array declaration, instantiation, multidimensional arrays, Designing methods for modularity and reusability Static variables, Static methods, Static blocks, Constructors and destructors, Instance Initialization Blocks, Security aspects in class design Encapsulation, Inheritance, single, multilevel, Hierarchical.

Module II:

11 lecture hours

Polymorphism, Function Overloading method overriding, Concrete Class Abstraction abstract methods, non- availability of multiple inheritance, interfaces, Interfaces vs abstract class, Embedded Interface Anonymous class, Inbuilt Packages, User defined packages, Array List collection, LinkedList Collection Vector collection, Exception handling, Checked and unchecked exception, try, catch, finally, Propel, Propagate, Data Stream Handling.

Module III:

11 lectures hours

Thread and process, Parallel Computing Concurrent Programming synchronization, Swing features, JavaFX Features MySQL database, NoSQL Databases getConnection, createStatement, executeQuery, JEE (client-server architecture for web based applications), Microservices Architecture Running Servlet, Generic Servlet, HTTP Servlet, Servlet Config, Web Filters Servlet to handle Get and Post Methods, Startups on programming, Session Management.

Module IV:**10 lectures hours**

JSPs, Struts framework, Spring MVC framework Regular expressions (Lambda expressions), Collection framework Concurrency in Java, HashMap, Linked Hash Map, Tree Map, Kotlin for Android WebSockets JPA (Java Persistence API), Microservices with Spring Boot.

Studio Work / Laboratory Experiments:

Students will gain hands-on experience on core JAVA. Students will be able to solve simple to medium level computation problems in Java. Emphasis will be that the student code themselves as much as they can. They learn to debug the programs resulting in error free code. Instructor will be giving assignments based on the content covered in the lecture classes in the corresponding week.

Text Books :

1. Schildt, Herbert. *Java: the complete reference*. 10th ed. McGraw-Hill Education Group, 2014. ISBN 978-93392120.
2. Bloch, Joshua. *Effective java (the java series)*. 1st ed. O'Reilly Media, Inc., 2017. ISBN 978-0134686097.

Reference Books :

1. Anuradha A. Puntambekar, *ObjectOriented Programming*. 1st ed. UNICORN Publishing Group, 2020. ISBN 9789333223819.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET105	Digital Design	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	2	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Identify appropriate truth table and examine gate level implementation from combinational logic function.

CO2: Build sequential circuits.

CO3: Experiment with the circuit diagram for combinational and sequential logic using Verilog.

CO-PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1				2							1	3		
CO2	1		2	1		1								3	
CO3			3	2											2

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

Digital Logic, Analog v/s Digital Systems, Data Representation, Binary Number System, Decimal Number System, Octal Number System, Hexa-Decimal Number System, Binary Arithmetic, Unsigned and Signed Numbers, Signed Magnitude, 1's Complement, 2's Complement, Fixed- and Floating-Point Numbers, Weighted Codes, BCD Code, Basic Logic Gates, Universal Logic Gates, Truth Table

Module II:

10 lecture hours

Laws of Boolean Algebra, Reduction of Boolean Expression using Boolean Laws, Conversion of Boolean Expression to Logic Diagram, Conversion of Logic Diagram to Boolean Expression, Boolean Function Representation using SoP and PoS form, Standard Sum of Products, Standard Product of Sum, Two Variable K-Map, Implicants, Prime Implicants and Essential Prime Implicants, Three Variable K-Map, Four Variable K-Map, Don't Care Condition, Design and Analysis of Combinational Circuits, Half Adder, Full Adder and Carry Propagation, Subtractor, Four Bit Binary Adder-Subtractor, Excess-3, Gray Code, Parity, Hamming Code, Binary Multiplier, Magnitude Comparator, Multiplexer, Implement Boolean Function using Mux, DeMultiplexer, Implement Full Subtractor using De-Mux.

Module III:

10 lectures hours

4 * 2 Encoder, Octal to Binary Encoder, Priority Encoder, 2*4 Decoder, Implement Full Adder using Decoder, BCD to 7-segment Display Decoder, Sequential Circuits, Sequential v/s Combinational Circuits, Clock, SR Latch, D Latch, Edge Trigger and Level Triggered, SR Flip Flop Circuit Diagram, SR Flip Flop Truth Table, SR Flip Flop Characteristic Table, SR Flip Flop Excitation Table, JK Flip Flop Circuit Diagram, JK Flip Flop Truth Table, JK Flip Flop Characteristic Table, JK Flip Flop Excitation Table, Race Around Condition, Master Slave JK Flip Flop, D Flip Flop, T Flip Flop, State Diagram, State Equation and State Table, Mealy State Machine, Moore State Machine, State Reduction, State Assignment, ASM Chart, Analysis using JK Flip Flop.

Module IV:**12 lecture hours**

Register, Parallel Register, Shift Registers, Bi-Directional Shift Register, Universal Shift Register, Counter, Asynchronous v/s Synchronous Counter, Ripple Counter (UP, DOWN, UP/DOWN), Synchronous Counter (UP, DOWN, UP/DOWN), Ring Counter, Johnson's Counter, Programmable Logic, Types of PLDs, Architecture of PROM, PLA Architecture, PAL Architecture, System on Chip Design, Logic Synthesis in Quantum Computing.

Studio Work / Laboratory Experiments:

In this course students will start with basic digital components such as Arithmetic and logical operation, Memory etc. Then finally design soft IP. The Lab will use Altera Quartus prime Lite tool for design and FPGA Altera DEII utilize for physical implementation.

Text Books :

1. Mano, M. Morris. *Digital Design: with an Introduction to the Verilog HDL, VHDL and System Verilog*. 6th ed. Pearson, 2018. ISBN 978- 0134549897.

Reference Books :

1. Wakerly, John F. *Digital Design: Principles and Practices*. 5th ed. Prentice Hall, 2008. ISBN 9780134460093.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET106	Discrete Mathematical Structures	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	1	0	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

- CO1 : To explain logical notation to illustrate sets, relations, functions, and integers.
CO2 : To examine induction hypotheses and prove elementary properties of modular arithmetic.
CO3 : Experiment and solve critical examples of algebraic structures and graph theory.

CO-PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1											2	3		
CO2		3	1	2										3	1
CO3		1	1	2										2	3

1=weakly related 2= moderately related 3=strongly related

Course Contents:

Module I:

11 lecture hours

Introduction and applications of Discrete mathematical structures, Proposition, Logical operators, Converse, Inverse, Contrapositive, Compound propositions, Precedence of logical operators, Tautology, Contradiction, Logical Equivalence, Derived implications, Well-formed formula, Tautological Implication, Logical equivalence laws, Rules of inference, Predicates and Quantifiers, Nested quantifiers, English to logic conversion, Direct proof, Proof by contradiction, Proof by induction, Russell's paradox.

Module II:

11 lecture hours

Representation of Sets, Types of Sets, Power Set, Venn Diagrams, Operations on Sets, Partition of Sets, Fuzzy Sets. Functions, Types of Functions, Sum and Product of Functions, Relation, Relation vs Function, Different Types of Relations, Graphical Representation of Relations, Matrix Representation of Relations, Closure of relations.

Module III:

10 lectures hours

Representation of integers, Binary operations, Divisibility, Euclidean Theorem for GCD, Residue classes, Linear congruence, Chinese remainder theorem, inclusion-exclusion principle, Binomial coefficients, Permutation, and combination, Pigeonhole principle, Recurrence relations and generating functions, semi-group, Monoid, and group, Abelian group, Cyclic group, Addition modulo m, Multiplication modulo m, Ring, Field, and integral domain.

Module IV:**09 lectures hours**

Partially ordered set, Elements of Posets, properties of lattices, Bounded, distributive, Complemented Lattice, Graphs, Homomorphism and Isomorphism, Euler graph, Hamiltonian circuit, Handshaking lemma (Havel Hakimi theorem), Graph Score Theorem, Bipartite graphs, Kionig's Theorem, Graph coloring, Chromatic number, Clique, Matching, Trees, forest, rooted trees, Binary trees, height of trees.

Text Books :

1. Bisht, Raj Kishor, and H. S. Dhami. *Discrete mathematics*. 1st ed. Oxford University Press. 2015. ISBN 97- 80199452798.
2. Rosen, Kenneth H. *Discrete mathematics and its applications*. 8th ed. The McGraw Hill Companies,, 2007. ISBN 978- 9390727353.

Reference Books :

1. Fortney, Jon Pierre. *Discrete Mathematics for Computer Science: An Example-based Introduction*. 1st ed. CRC Press, 2020. ISBN 978-1000296806.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
EMAT102L	Linear Algebra and Ordinary Differential Equations	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	1	0	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Solving system of linear equations by using Gaussian elimination to reduce the augmented matrix to row echelon form or to reduced row echelon form.

CO2: To be familiar with the concepts of dimension of a subspace and the rank and nullity of a matrix.

CO3: To calculate the eigenvalues and eigenvectors of a square matrix using the characteristic polynomial.

CO4: Fundamental concepts of ordinary differential equations

CO5: Solving linear differential equations of both first and higher order

CO-PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3												2		
CO2	2													2	
CO3	2											2			3
CO4	1		2										1		
CO5			2									2	1		

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

12 lecture hours

Review of Matrices, System of linear equations, Gauss elimination method, Elementary matrices, Invertible matrices, Gauss-Jordan method for finding inverse of a matrix, Review of Basic properties of determinant, Cofactor expansion, Determinant method for finding inverse of a matrix, Cramer's Rule, Vector space, Subspace and Examples, Linear span, Linear independence and dependence and Examples, Basis, Dimension, Extension of a basis of a subspace, Intersection and sum of two subspace, Examples. Linear transformation, Kernel and Range of a linear map, Rank-Nullity Theorem, Rank of a matrix, Row and column spaces, Solvability of system of linear equations, some applications, Inner product, Cauchy-Schwartz inequality.

Module II:

09 lecture hours

Orthogonal basis, Gram-Schmidt orthogonalization process, Orthogonal projection, Orthogonal complement, Projection theorem, Fundamental subspaces. Fundamental subspaces and their relations, An application (Least square solutions and least square fittings). Eigen-values, Eigen- vectors, Characterization of a diagonalizable matrix, Diagonalization: Example, An application. Diagonalization of a real symmetric matrix, Representation of a real linear map by matrices.

Module III:**12 lectures hours**

Introduction to DE, Order of DE, First Order ODE $F(x, y, y') = 0$. Concept of solution (general solution, singular solution, implicit solution etc.) Geometrical interpretations (direction fields), Separable form, Reduction to separable form, Exact equations, Integrating factors, Linear equations, Bernoulli equation, orthogonal trajectories, Picard's existence and uniqueness theorem (without proof), Picard's iteration method, Second order linear ODE: fundamental system and general solutions of homogeneous equations, Wronskian, Reduction of order, Characteristic equations: real distinct roots, complex roots, repeated roots, Non-homogeneous equations: Undetermined coefficients and Variation of parameters.

Module IV:**09 lectures hours**

Extension to higher order differential equations, Euler-Cauchy equation, Real analytic solutions of Linear second order equations, Linear system of Differential equations, Fundamental Set, Linearly independent solutions, Laplace transform: Laplace and inverse Laplace transforms, First shifting theorem, Transforms of derivative and integral, Differentiation and Integration of transforms, unit step function, Second shifting theorem, Convolution and applications, Initial value problems.

Text Books :

1. Strang, Gilbert. *Introduction to linear algebra*. 4th ed. Wellesley-Cambridge Press, 2006. ISBN 978- 0030105678.
2. Kreyszig, Erwin. *Advanced Engineering Mathematics 10th Edition with Wiley Plus Set*. John Wiley & Sons, 2010. ISBN 978-0470458365.

Reference Books :

1. Hoffmann, Kenneth, and Ray Alden Kunze. *Linear algebra*. 2nd ed. Prentice-Hall, 2004. ISBN 9789332550070.
2. Simmons, George F. *Differential equations with applications and historical notes*. 2nd ed. McGraw-Hill Education (India) Pvt Limited, 2002. ISBN 978-0070530713.
3. Coddington, Earl A. *An introduction to ordinary differential equations*. 1st ed. DoverPublication, 1989. ISBN 9780486659428.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET102	Introduction of Electrical and Electronics Engineering	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	1	2	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1 : To articulate the fundamental parameters governing an electrical circuit such as current, voltage.

CO2 : To explain basic electronic components such as resistors, capacitors, and inductors.

CO3 : To make use of concepts, working, and application of various circuits using the components on a breadboard.

CO-PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	1				2							3		
CO2	2	1				1								2	
CO3		1		1		1						1			2

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

07 lecture hours

Basic Circuit Theory: Charge, current, voltage, electric field, conductance, resistance, Ohm's law; current source, voltage source, dc, ac, periodic signal concept, examples of different periodic signals, their average, RMS value, resistors in series and parallel, voltage and current division. Kirchhoff's current law (KCL) and Kirchhoff's voltage law (KVL), Nodal analysis and Mesh analysis.

Module II:

07 lecture hours

Network theorems: Maximum power transfer theorem, Source Transformation, Superposition theorem, Thevenin's theorem, Norton's theorem. Introduction to capacitor (C), introduction to phasor diagram, series and parallel combination of capacitors, Impedance and frequency dependency, Introduction to inductor (L), introduction to phasor diagram, series and parallel combination of inductors, Impedance and frequency dependency, units of L, C measurements, lowpass, highpass and bandpass filter, transfer function, Single phase motors, dc motors, stepper motors, their applications.

Module III:

08 lectures hours

Diodes and its applications, Semiconductor Materials: Electrons and Holes, Intrinsic and Extrinsic Semiconductors, doping of n and p-type semiconductors, Diode circuits: PN Junction diode and its applications: Half wave, center tapped full wave rectifier circuit, bridge rectifier circuit, the efficiency of rectifier circuits, ripple factor, rectifiers with filter circuits.

Module IV:**06 lectures hours**

Op-Amp: Characteristics of an Op-amp. Inverting and Non-inverting amplifier, Integrator, Differentiator.

Text Books :

1. Charles, K. Alexander, and N. O. Matthew. *Fundamentals of electric circuits*. 1st ed. McGraw-hill Education, 2017. ISBN 9789353165505.
2. Boylestad, Robert L., and Louis Nashelsky. *Electronic devices and circuit theory*. 1st ed. Pearson Education India, 2009. ISBN 9788131703144.
3. Tsividis, Yannis. *A First Lab in Circuits and Electronics*. 1st ed. Wiley, 2002. ISBN 9780471386957.

Reference Books :

1. Neamen, Donald A. *Microelectronics: circuit analysis and design*. 1st ed. New York: McGraw-Hill, 2009. ISBN 978-0073380643.
2. Bell, David A. *Electronic instrumentation and measurements*. 3rd ed. Oxford University Press India, 2013. ISBN 978-0195696141.

Semester III

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET201	Information Management Systems	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	2	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Articulate the competent understanding of database systems design and ER Modelling.

CO2: Build database systems and understand new developments and trends in databases.

CO3: Construct databases and make use of efficient SQL queries to retrieve and manipulate data as required.

CO-PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	1	3	3	3				3	2	2	1	2		
CO2	3	2	3	3	3				1	2	2	2	3		
CO3	3	2	3	3	3				3	2	2	1	3		

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

Purpose of IMS, Real-life applications of Data-Intensive systems, Typical system challenges, Data independence, Database system architecture levels, Role of several databases, ER diagram: Entity-set, Attributes, Relationships, Cardinality ratio, EER diagram: Specialization, Generalization, Constraints of EER, Aggregation, ER to Relational model, Relational model, Structure of relational databases, Constraints of relational model, Relational algebra: Basic and derived operator, Tuple relational calculus.

Module II:

10 lecture hours

Functional dependency – definition, trivial and non-trivial FD, Armstrong's axioms, closure of FD set, Closure of attributes, Irreducible set of FDS, Normalization, 1NF, 2NF, 3NF, BCNF, Decomposition using FD, Dependency preservation, Multivalued dependency, 4NF, join dependency, 5NF, Query optimization, Measures of query cost: selection operation, sorting, join, Evaluation of expressions, Transformation of relational expressions, Estimating statistics of expression results.

Module III:

10 lectures hours

Properties of transactions, Serializability of transactions, testing for serializability, System recovery, Two- Phase Commit protocol, Recovery and Atomicity, Log-based recovery, concurrent executions of transactions, Locking mechanism, Solution to concurrency related problems, Deadlocks, Two-phase locking protocol, Isolation, Intent locking, Discretionary Access Control, Mandatory Access Control, Authentication, Authorization and access control, DAC, MAC and RBAC models.

Module IV:**12 lectures hours**

Pipelining, Streaming algorithms framework, Turnstile model, Cash register models, Sliding window model, Data warehouse, Operational data store, Star schema, Snowflake schema, Data cube concept, OLAP, Cube and Roll-up, NoSQL database systems framework, Column stores, RDF stores, HBase, Big Data, Hadoop MapReduce architecture, Distributed Database Systems framework (DDS), Need for Data Privacy, Privacy law, Anonymity models, Privacy in Cloud and Big Data.

Studio Work / Laboratory Experiments:

Entity-Relationship model: Design process, constraints, Keys, Design issues, E-R diagrams, weak entity sets, extended E-R features – generalization, specialization, aggregation, reduction to E-R database schema. Basics of SQL, DDL, DML, DCL, structure – creation, alteration, defining constraints – Primary key, foreign key, unique, not null, check, IN operator Functions - aggregate functions, Built-in functions –numeric, date, string functions, set operations, subqueries, correlated sub-queries, use of group by, having, order by, join and its types, Exist, Any, All, view and its types. Transaction control commands – Commit, Rollback, save point.PL/SQL Concepts: - Cursors, Stored Procedures, Stored Function, Database Triggers, No SQL queries using Mongo DB and Firebase.

Text Books :

1. Silberschatz, Abraham, Henry F. Korth, and Shashank Sudarshan. *Database system concepts*. 7th ed. McGraw-Hill, 2011. ISBN 9780078022159.
2. Ramez Elmasri and Sham Navathe, *Fundamentals of Database Systems*. 1 ed. Pearson/Addison Wesley, 2016. ISBN 9780133970779.

Reference Books :

1. Negi, Mukesh. *Fundamental of Database Management System: Learn essential concepts of database systems*. 1st ed. BPB Publications, 2019. ISBN 9789388176626.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET243	Data Structures using C++	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	3	1	6	7	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Articulate the design, use and associated algorithms of fundamental and abstract data structures.

CO2: Examine various searching and sorting techniques based on complexity analysis for applicative solutions.

CO3: Demonstrate hands-on experience on implementing different data structures.

CO4: Build optimized solutions for real-world programming problems using efficient data structures.

CO-PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	3	2								2	2		
CO2	3	3	3	3								1	3		
CO3	2	3	3	3								2	3		
CO4	3	2	2										3		

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

Essentials of Data Structures, Initiating C++: The First Program, Lifecycle of C++ Program Execution, Fundamentals of Object-Oriented Programming (OOP), C++ Inheritance Principles, Multiple inheritance, Friend Function, Run-time Polymorphism, Time Complexity: Asymptotic Analysis, Big-Omega, Big-Theta, Big-Oh Notation, Handling Arrays, Insertion, Deletion, Traversal, Linear Search, Recursion, Binary Search, Tower of Hanoi, Sorting, Bubble Sort, Selection Sort, Insertion Sort, Merge Sort, Shell Sort, Counting Sort.

Module II:

10 lecture hours

Navigating C++: Pointers Essentials, Structures and Unions in C++, Exploring Linked Lists in C++, Implementing Singly Linked Lists, Traversal Techniques for Singly Linked Lists in C++, Searching, Insertion, Deletion, Polynomial Handling, Circular Linked List, Traversal, Insertion, Deletion, Stacks, Traversal, Insertion, Deletion, Infix to Postfix Conversion, Post-fix Expression Evaluation, Deques, Queues, Simple Queue Insertion, Deletion, Traversal, Circular Queue Insertion, Deletion, Traversal.

Module III:

11 lectures hours

Tree Data Structures, Height, Complete, Full, Perfect Trees, Binary Search Trees, Pre-Order, InOrder, Post-Order, BST Searching, BST Insertion, BST Deletion, Heaps, Min-Max Heaps, HeapSort, Hashing, Hash Functions, Hash Tables, Hashing Collision Resolution Strategies: Separate Chaining, Open Addressing, Double Hashing, Graphs, Different Types of Graphs, Graphs Representations, Incidence Matrix, Adjacency Matrix, Graphs Traversals: BFS, DFS, Topological Sort.

Module IV:**11 lectures hours**

Maintaining Height Equilibrium: AVL Trees, Balanced Factor and Rotations in AVL Trees, Operations in AVL Trees: Insertion and Deletion, Exploring Red-Black Trees: Insertion and Deletion, B-Trees: Efficient Insertion and Deletion Operations, Enhanced Structure: B+ Trees and their Insertion and Deletion Operations, Deletion, Disjoint Sets, Path Compression, Union Finding Algorithm, van Emde Boas Tree.

Studio Work / Laboratory Experiments:

The laboratory of Data structures is designed to provide a practical exposure to the students about the concepts and topics taught in the classroom sessions. Implementing the learnt concepts using C++ will help the students to have a better understanding of the subject.

Text Books :

1. Carey, John, Shreyans Doshi, and Payas Rajan. *C++ Data Structures and Algorithm Design Principles: Leverage the power of modern C++ to build robust and scalable applications*. 1st ed. Packt Publishing Ltd, 2019. ISBN 9781838827919.
2. Kanetkar, Yashavant. *Data Structures Through C: Learn the fundamentals of Data Structures through C*. 1st ed. Cengage Learning, 2019. ISBN 9789388511360.

Reference Books :

1. Hu, Yang. *Algorithms C++*. 1st ed. Cengage Learning, 2020. ISBN 9798676695750.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET240	Probability and Statistics	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	1	2	5
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain different standard distributions, and how these can be used to model different real life situations.

CO2: To make students examine different statistical tests to datasets, and make the correct data driven decision.

CO3: To implement real life problems of Probability and Statistics using programming languages.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3		3	3								3		
CO2	3	3		3	3								3	3	3
CO3			3		3								3	3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

Probability, Conditional Probability, Independence, Law of total probability, Bayes' theorem, Monty hall problem, Random Variables: discrete and continuous, Expected value, Variance, Properties of Variance, PMF, CDF, Joint Probability, Joint Conditional Probability, Convolution, Numerical samples, Correlation, Covariance.

Module II:

14 lecture hours

Discrete Distributions, Bernoulli, Binomial, Poisson, Geometric, Applications of Discrete Distribution, Continuous Random Variables, Continuous domain and correlation with discrete domain, Continuous Distribution: Probability density function, Cumulative distribution function, Uniform Distribution, Normal Distribution, Standard normal distributions, Z Scores, Z tables, Exponential, applications of distributions, Sampling distribution, Central Limit Theorem, Confidence Interval Estimation, Known sigma and unknown sigma based estimation, Markov Inequality, Chebyshev inequality, WLLN, Outlier detection, Zero shot learning.

Module III:

18 lectures hours

Descriptive statistics, Visualization of central tendency and variability, Hypothesis Testing, Z Test, T Test, F test, ANOVA, AB Testing, Importance of parameter estimation, Least square, Linear regression and multiple regression, Parameter estimation using LSE, Maximum likelihood, Parameter estimation using MLE, Markov chains, Statistical Inference.

Studio Work / Laboratory Experiments:

The students learn to perform numerical analysis, compute odds of any event, check distribution and both linear and non-linear regression. In addition to this, students build search engines (like google), implement ideas of dynamical system (via Markov models), play with the concepts of central limit theorem and law of large numbers. Lastly, everyone is encouraged to try and test scientific hypothesis via the paradigm of statistical hypothesis testing and confidence interval estimation. The students are also given hands on training in the language R.

Text Books :

1. Ross, Sheldon M. *Introduction to probability and statistics for engineers and scientists*. 6th ed. Academic press, 2020. ISBN 9780128177471.
2. Baron, Michael. *Probability and statistics for computer scientists*. 3rd ed. Chapman and Hall book. 2019. ISBN 978-1584886419.

Reference Books :

1. Paul L. Meyer, *Introductory Probability and Statistical Applications*. 2nd ed. Addison-Wesley, 2017. ISBN 978-0201047141.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET205	Software Engineering	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	2	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

- CO1 : To articulate understanding of Software Engineering as an iterative and systematic process.
CO2 : To examine the software development process to complement the technical understanding of software products.
CO3 : To implement development life cycle through the IDE, UML, and Git.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	1		3		2				2				3	2	
CO2	1		2		3				2				2	3	
CO3	3		2		3				2						2

1=weakly related 2= moderately related 3=strongly related

Course Contents:

Module I:

09 lecture hours

Importance of Software Engineering, Phases of software development lifecycle, SDLC case study, Software process model, Waterfall model, V Model, Prototyping model, Incremental model, RAD Model, Spiral model, Choosing a model, Lifecycle documents, Agile model, Need of agile, Version Control System, Distributed VC, Git repository, Git online and Desktop, Check-in, and check-out code in repository, Create branch and merging branch, Git + Eclipse/IDE.

Module II:

09 lecture hours

Agile manifesto, Agile principles, Agile development methods, Extreme programming (XP), XP principles, Test first development, Refactoring, Pair programming, Scrum, Product backlog, Sprint cycle, Continuous integration, Requirements engineering, Issues in capturing requirements, Requirement elicitation, Requirement analysis, Functional and Non-functional requirements, Requirement specification, Requirement prioritization, User stories, Acceptance criteria, Requirement validation and verification, UML, Use case, Use case Diagram, Include and extend relationship, Generalization in use Case, Top down and bottom-up approach in use case diagram, Guidelines for creating use case diagrams.

Module III:

11 lectures hours

UML behavioural diagrams, Activity diagram, Activity diagram with swimlane, UML structural diagrams, Class diagram, Relationships in class diagram, Sequence diagram, Rules of creating sequence diagram, Description of relationship between use-case, activity, and sequence diagram, Architectural design, Cohesion, Coupling, Early locking of architecture, Architectural pattern, MVC pattern, Layered

architecture, Repository architecture, Client server architecture, Software architect, roles, and responsibilities, Pipe and filter architecture.

Module IV:

13 lectures hours

Reverse engineering, Horseshoe model, Software cost estimation, Cost estimation factors, COCOMO model, Software quality assurance and testing, Software testing for competitive advantage, Testing strategies, Designing test cases, Black box testing, Equivalence partitioning, Boundary value analysis, White box testing, Black box vs. white box testing, Control flow testing technique, Control flow graph, Cyclometric complexity, Levels of coverage, Integration testing, Top down and bottom-up integration, Continuous Integration testing, Regression testing, Acceptance testing, System testing, Performance testing, Software engineering standards, ISO standards, CMM.

Studio Work / Laboratory Experiments:

The course will cover labs in the following areas: Getting used to eclipse and GitHub, hands-on session on software design via Visual Paradigm, Designing and inspecting SRS, software testing using Junit approach.

Text Books :

1. Pressman, Roger S. *Software engineering: a practitioner's approach*. 8th ed. McGraw Hill International, 2019. ISBN 978-1259253157.
2. Sommerville, Ian. *Software Engineering*, 10th ed. Pearson Education India, 2017. ISBN 978-9332582699.

Reference Books :

1. Summers, Boyd. *Effective Methods for Software Engineering*. 1 ed. CRC Press, 2020. ISBN 9781000052710.

Semester IV

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET203	Microprocessors and Computer Architecture	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	2	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Explain about modern architecture and microprocessors and the design techniques.

CO2: Examine various design alternative of computer architecture based on CPU performance, memory, I/O.

CO3: Understand instruction sets and implement with assembly program and design application specific processor using HDL.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	2	3		3				1				2	2	
CO2	2	3	3	3					1				2	3	
CO3	2	2	2	3					3	3			3	3	

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

Using Microprocessors and Computer Architecture, Microprocessor Architecture, CPU Registers, Segment Registers, General Purpose Registers, Bus Interface Unit, Address bus, data bus and control bus, Control Unit, ALU, Memory unit, Overview of architecture and contrast between 32- and 64-bit processors, Microprocessor Pin Configuration, Microcontroller, Comparison with microprocessor, AI Processor, Comparison of AI Processor with regular Microprocessor, ARM processor Architecture, Instruction Set Architecture, Instruction interpretation and execution, MIPS Instructions format, ARM – Instruction Set Architecture.

Module II:

11 lecture hours

Addressing Modes, Immediate, Direct, Indirect, PC-relative, Indexed, Subroutine Call, Small-scale operations and their RTL description, Arithmetic and logic unit, Signed Magnitude Addition-Subtraction Algorithm's register setup, Register configuration of 2's complement Addition Subtraction Algorithm, Signed Magnitude Multiplication Algorithm, Booth's multiplication algorithm, CPU – Memory interconnections, Organization of memory modules, Associative memory, Cache Memory, Cache Memory Mapping Techniques, Associative, Direct, Set Associative.

Module III:

10 lectures hours

Interfacing of memory chips, Methods for allocating addresses and decoding, Interfacing of I/O devices, LEDs toggle-switches as examples, Memory mapped and isolated I/O structure, Input/Output techniques, Logic-driven, managed input/output, Interrupt controlled I/O transfer, Different types of

Interrupts, DMA controller, Buses and connecting I/O devices to CPU and memory, Secondary Storage – solid-state drive (SSD), Comparison of SSD with HDD.

Module IV:

11 lectures hours

Pipelining, pipeline hazards, Problems with pipeline architectural design, Instruction level parallelism and advanced issues, RISC and CISC paradigm, Design issues of a RISC processor, Multiprocessor system and its characteristics, Topics of Advancement in Microprocessor Research, Parallel processing concepts, Parallelism algorithm for multiprocessor systems, Overview of GPU Architecture, Comparison between GPU and CPU Architecture, Interconnection network, Cache in multiprocessor systems and related problems, Cache coherence protocols.

Studio Work / Laboratory Experiments:

In this course students will start with basic components of CPU such as ALU, Memory etc. Then finally combine all components and develop a processor. The Lab will use Altera Quartus prime Lite for design. MIPS for assembly programming.

Text Books :

1. Hennessy, John L., and David A. Patterson. *Computer architecture: a quantitative approach*. 5th ed. Morgan Kaufmann, 2012. ISBN 978-9381269220.
2. Hall, Douglas V. *MICROPROCESSORS AND INTERFACING*. 3 ed. McGraw Hill Education, 2017. ISBN 978-1259006159.

Reference Books :

1. Stallings, William. *Computer organization and architecture: designing for performance*. 10th ed. Pearson Education India, 2016. ISBN 978- 0132936330.
- .

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET244	Design and Analysis of Algorithms	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	3	1	6	7	
Pre-requisites/Exposure	CSET243					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Examine and Analyze the asymptotic performance of algorithms.

CO2: Explain various algorithmic techniques for solving problems.

CO3: Experiment to find and develop optimal solutions by applying different algorithmic strategies for polynomial and non-polynomial problems.

CO-PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	2	1				2	2	2	2	2		
CO2	3	3	3	2	1				2	2	2	2	3		
CO3	1	2		3	1				2	2	2	2	3		

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

12 lecture hours

Introduction to algorithm, What is Time Complexity and Space Complexity, Order of Growth; Approximation; Asymptotic Notations : Big Oh, Theta, Omega, Amortized analysis, Analysing control statement, Loop Invariant, Recurrence Relations Introduction, Back Substitution Method, Recursion Tree Method, Master's Theorem, Divide and Conquer Algorithm, Multiplying large Integers Problem, Median of two sorted arrays, Binary search, Quick Sort, Merge Sort, Max-Min problem, Strassen's Matrix Multiplication, Radix Sort, Bucket Sort.

Module II:

14 lecture hours

Greedy Algorithm: General Characteristics, Knapsack Problem, Huffman code, Activity selection problem, Minimum Spanning trees, Prim's algorithm, Kruskal's algorithm with Disjoint sets, Shortest paths: Dijkstra's Algorithm, Graphs Algorithms:- Applications of DFS- bi-connectivity, Topology Sort, Articulation point, Connected components, Max-Flow, Min-Cut, Ford-fuelkerson, Dynamic Programming:- Introduction, Principle of Optimality, Calculating Binomial Coefficient, 0-1 Knapsack, Matrix chain multiplication, Longest Common Subsequence, All Points Shortest path Floyd Warshall, Largest Divisible Subset.

Module III:

08 lectures hours

Backtracking and Branch and Bound: - State-Space Search Tree, eight queen's problem, Graph Colouring, Hamiltonian Cycle, Travelling Salesman Problem using Branch and Bound Approach, String Matching Algorithms, Naive string-matching algorithm, Knuth Morris-Pratt algorithm.

Module IV:**08 lectures hours**

Introduction to NP-Completeness: - P and NP, NP Complete and NP-Hard, Approximation algorithms, Travelling Salesman problem, Randomized Algorithms: Randomized Quick Sort, Computational Geometry: Convex hull, Online Algorithms: K Server Problem.

Studio Work / Laboratory Experiments:

The students will be implementing the fundamental design paradigms like dynamic paradigm, greedy algorithms, graphs traversing and several others.

Text Books :

1. Cormen, Thomas H., Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein. *Introduction to algorithms*. 4th ed. MIT press, 2022. ISBN 9780262367505.
2. Horowitz, Ellis, Sartaj Sahni, and Sanguthevar Rajasekaran. *Computer algorithms C++: C++ and pseudocode versions*. 2nd ed. The Orient Blackswan, 2019. ISBN 9386235145.

Reference Books :

1. Karumanchi, Narasimha. *Algorithm Design Techniques: Recursion, Backtracking, Greedy, Divide and Conquer, and Dynamic Programming*. 1st ed. CareerMonk Publications, 2018. ISBN 978-8193245255.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET207	Computer Networks	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	2	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Examine the functionality of the different layers within network architecture.

CO2: Illustrate TCP/IP model suite protocols.

CO3: Design the networks for organization and select the appropriate networking architecture and technologies, subnetting and routing mechanism.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	1		2	2	1				2	1	2	2		
CO2	3	3	3	3	3		2		2	2	2	2	3		
CO3			2		2	2	1			2	1	2	3		

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

09 lecture hours

Why Computer Networks: Applications of Networks, Transmission Media, Connecting Devices, Local Area Networks: LAN topologies: Bus topology, Ring topology, Star topologies, Mesh topology, Hybrid topology, OSI reference model, TCP/IP Protocol suite, Physical Layer: Services, Line coding scheme, Modulation, Multiplexing, Switching methods, Ethernet, Bluetooth, Wi-Fi, Wi-Fi Direct, WPA/WPA2/WPA3, Data Link layer: Services, Framing, Switches.

Module II:

08 lecture hours

Reliable Data Delivery: Error detection, Error Correction, Flow control: Stop and wait, Go Back-N, Flow control: S-R Protocol, Error control (Retransmission techniques, timers), Medium Access sublayer - Channel Allocations, LAN protocols /ALOHA protocols, CSMA, CSMA/CD, Network Layer Protocols: Services (IP, ICMP), IP addressing, sub netting, Super netting (CIDR), IPV4, IPV6.

Module III:

09 lectures hours

Routing and Forwarding, Static and dynamic routing, Unicast and Multicast Routing, Distance-Vector Routing, Link-State Routing, Shortest path computation-Dijkstra's algorithm, Address mapping-ARP, RARP, BOOTP, DHCP, Transport Layer: Services, UDP and TCP segment formats, connection establishment and termination, Expert Lecture from Industry, Congestion control, Congestion control: Open Loop and closed loop, Quality of service, Flow characteristics, Techniques to improve QoS.

Module IV:**07 lectures hours**

Session Layer: Services, Protocols, Presentation layer: Services, Protocols, Application layer: Services, DNS, SIP, RTP, Telnet/SSH, HTTP, HTTPS, Remote login, Electronic mail, SMTP, FTP Commands and Replies, WWW, SNMP, Addressing Schemes, Uniform Resource Identifiers.

Module V:**09 lectures hours**

Principles of Cryptography, Symmetric key, Public key, Authentication protocols, Digital signatures, Firewalls, Security in different layers: Secure E-mail SSL, IP security, Advanced Topics in CN: Dark Net, CASS: Content-Aware Search System, Service-centric networking, Software-defined networking, Cloud Systems: Services, Data centre, 4G and 5G Networks, Body area sensor Networks, Satellite networks, SWARM networks.

Studio Work / Laboratory Experiments:

Study of different types of networks cables and practically implement the cross-wired cable and straight through cable using clamping tool. Configure a network topology, connect different networks, static routing and dynamic routing, virtual LAN, RIP and OSPF using packet tracer. Also, Wireshark will be used for network troubleshooting, analysis, software, and communications protocol development.

Text Books :

1. Forouzan, Behrouz A. *Data communications and networking*. 5th ed. McGraw Hill, 2021. ISBN 1260597822.
2. Wetherall, David J., and Andrew S. Tanenbaum. *Computer networks*. 6th ed. Pearson Education, 2021. ISBN 9780137523214.

Reference Books :

1. Kurose, James F. *Computer networking: A top-down approach featuring the internet*. 6th ed. Pearson Education India, 2017. ISBN 9781485832535.

Name of Program	Bachelor of Technology (Computer Science Engineering)						
CSET208	Ethics for Engineers, Patents, Copyrights and IPR	L	T	P	C		
Owning School/Department	School of Computer Science Engineering and Technology	1	0	0	1		
Pre-requisites/Exposure	-						

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Explain and Practice the professional ethics for engineers.

CO2: Examine and understand the patent law, and how patents are prosecuted and enforced.

CO3: Articulate the importance of intellectual property laws in modern engineering.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	1		1			3	1	3				2			2
CO2	1		1			3	1	3			2	2			2
CO3	1		1			3	1	3				2			3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

08 lecture hours

Why Ethics, Patents, Copyrights, and IPR, Moral issues, Types of inquiry, Moral dilemmas, Moral autonomy, Theories about right action, Kohlberg's theory, Gilligan's theory, Models of Professional Roles, Self-interest, customs, and religion, uses of ethical theories, Patents, Patentable Subject Matter, Novelty, Non-Obviousness, Patenting Process, Infringement and Searching, Patent Applications, Claim Drafting, Patent Prosecution, Design Patents, Business Method Patents, Foreign Patent Protection, Computer-Related Inventions, Patent Enforcement, Technical Design-Around.

Module II:

06 lecture hours

Copyrights Subject matter of Copyright, Rights of the owners of the copyright, Authorship, ownership, licensing, assignment of Copyright, Registration of Copyright & Authorities, Copyrights for Technology Protection, Intellectual Property Rights, IP Law Overview, Mask Works, Trade Secrets, Trademarks, Engineers as Expert Witnesses.

Text Books :

1. Rockman, Howard B. *Intellectual property law for engineers and scientists*. 2nd ed. Wiley-IEEE Press, 2020. ISBN 978-1119381976.

Reference Books :

1. Racherla, Uday S. *Intellectual Assets for Engineers and Scientists: Creation and Management*. 1st ed. CRC Press, 2018. ISBN 978-1498788472.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET209	Operating Systems	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	2	4
Pre-requisites/Exposure	CSET203				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain the structure and the services provided by the Operating System.

CO2: To define, restate, discuss, and explain the policies for scheduling, deadlocks, memory management, synchronization, system calls, and file systems.

CO3: To examine distributed Operating System, cloud computing, and Virtualization.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3	2		2								3	2	
CO2	3	3	3	3									3		
CO3	2		3		3								2		3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

09 lecture hours

Operating system: Organization, abstraction provided by OS, features and roles, OS evolution, Operating system architecture, OS examples; Process management: Process control block, system calls and interrupts, context switching, scheduler and dispatcher, process states and life cycle, multithreading, kernel vs. user level threads, process vs. thread; CPU scheduling algorithms: FCFS, SJF, STRF, priority, round robin, multilevel queue and feedback scheduling, highest response ration next, lottery scheduling.

Module II:

12 lecture hours

Memory management, techniques, Contiguous: Fixed and variable length partitioning, Non-contiguous: Paging, translation lookaside buffer (TLB), multilevel paging, segmentation, segmented paging; Virtual memory, dynamic loading, demand paging, page fault, thrashing, Page replacement algorithms: First in First Out (FIFO), Least Recently Used (LRU), Optimal Page Replacement; File organization, access mechanism, file allocation methods - Contiguous allocation, linked allocation, indexed allocation, Indexing techniques: Single indirect block, double indirect blocks; Storage management, storage devices – HDD, SSD; disk architecture, disk performance metrics; Disk scheduling algorithms: FCFS, SSTF, SCAN, C-SCAN, LOOK, C-Look, NOOP and deadline scheduler.

Module III:

12 lecture hours

Distributed systems, network vs. distributed OS, robustness analysis, design issues; Remote procedure call (RPC), structure, parameter passing, handling partial failures, SunRPC and XDR; Distributed file systems, Servers: stateless and stateful, REST; Distributed shared memory, architecture, design principles, consistency model; Security systems calls, authentication & authorization, reliability, availability & privacy, common attacks, crypto systems, Kerberos, access control lists; OS design:

MAC, and iOS; Virtualization: Types, models; Cloud computing, architecture, service and deployment models, cloud challenges.

Module IV:

09 lecture hours

Inter process communication, shared memory method, message passing method and its types; Process synchronization, critical section problem (CSP), synchronization constructs, hardware solutions to CSP- Lock variables, software solutions to CSP-Peterson solutions, strict alteration; Classical synchronization problem: Producer consumer problem, dining philosopher problem, reader writer problem, monitor; Deadlock, necessary condition for deadlock, deadlock handling mechanism: prevention, avoidance, detection; resource allocation graph, deadlock recovery.

Studio Work / Laboratory Experiments:

Students will gain practical experience with the implementation and use of operating system functions such as process management, inter-process communication, process synchronization, memory management and file systems etc. Moreover, students will have exposure to the tools for measuring and monitoring of operating systems related parameters and services.

Text Books :

1. Silberschatz, Abraham, Peter B. Galvin, and Greg Gagne. *Operating System Concepts*. 10th ed. John Wiley & Sons, 2018. ISBN 978-1-119-32091-3.
2. Stallings, William. *Operating systems: internals and design principles*. 9th ed. Prentice Hall Press, 2021. ISBN 978-0134670959.

Reference Books :

1. Andrew, S. Tanenbaum, and Bos Herbert. *Modern operating systems*. 1st ed. Pearson Education, 2021. ISBN 9789332575776.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET210	Design Thinking & Innovation	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	0	0	4	2
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Explain the concepts of design thinking.

CO2: Conceptualize and ideate solution for an existing problem using Computing resources and computing devices.

CO3: Implement the concept to create and prototype the idea into near workable solution.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2		3	2	2	3				1			3		2
CO2	3		2										3		
CO3				2	3								2	3	

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

56 lecture hours

Students are set to embark on a group project designed to foster a holistic understanding of the design thinking approach applied in project development. Continuous assessments will gauge their performance against various project-related criteria. The project entails the creation of a poster, video, blog, report, and presentation, enabling students to explore different deliverables and associated documentation. The focus is on tackling real-world problems and actionable issues, utilizing technology and improved design for effective solutions.

Studio Work / Laboratory Experiments:

The course will be taught using a combination of the best practices of teaching-learning. Multiple environments will be used to enhance the outcomes such as seminar, self-learning, MOOCs, group discussions and ICT based tools for class participation along with the classroom sessions. The teaching pedagogy being followed includes more exposure to hands-on experiment and practical implementations done in the lab sessions. To make the students aware of the industry trends, one session of expert lecture will be organized to provide a platform to the students for understanding the relevant industry needs.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET398	Refresher Course for Industry Career	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	0	0	0	0
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1 : Understand the workplace practices and competencies necessary for professional and academicsuccess.

CO2 : Explore career preferences, professional goals and develop necessary skills.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3		3	3	3	3	3	3	3	3	2	3		
CO2	3	3		3	3					2	2		3	3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

00 lecture hours

This course will familiarize the students with the process of placements through training and assessments. Fundamentals of courses, communication & behavioural skills, workshops on relevant topics and seminars would be conducted. This will help the students to gain practical experience about the workplace before receiving their undergraduate degrees. They would be able to establish proficiency in a range of business or industry skills appropriate for internships, placements, including professional and inter-cultural communication through written, verbal, and non-verbal means.

Semester V

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET301	Artificial Intelligence and Machine Learning	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	4	5
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1 : Articulate various supervised and unsupervised machine learning approaches.

CO2 : Build and evaluate models generated from data.

CO3 : Implement systems to solve real life problem using AI and Machine Learning approaches.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	3		2	3								3	3	
CO2	2	3	3	3	3								2	3	
CO3	3	3	3	3	3							3	3	3	

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

08 lecture hours

Designing a learning system, Types of machine learning: Problem based learning, Supervised learning, Unsupervised learning, Reinforcement learning, Linear Regression: Weights and Features, Applications, Cost Functions, Finding best fit line, Gradient Descent Algorithm: Learning Algorithm, First order derivatives, Linear regression using gradient descent, Learning rate, Logistic Regression, Sigmoid Function, Cost Function for Logistic Regression, Multi-class classification, Probability Distribution, SoftMax Function, Polynomial Regression, Performance Metrics: Classification (Confusion Matrix, Accuracy, Precision, Recall, F1-score, ROC-AUC), Regression (MSE, MAE, RMSE, R2 Score).

Module II:

07 lecture hours

Decision Tree, Selecting Best Splitting Attribute, CART (Gini Index). ID3 (Entropy, Information Gain), Hyperparameters in Decision tree, Issues in Decision tree learning. Overfitting and Underfitting, Bias and Variance, Cross Validation. Ensemble Learning (Concord's Theorem), Bagging, Bootstrap and Aggregation, Random Forest. Boosting, AdaBoost, Gradient Boost. Feature Engineering, Feature Selection, Feature Extraction.

Module III:

09 lecture hours

Artificial Neural Network, Neural network representation, Perceptron model, Stepwise v/s Sigmoid function, Multilayer perceptron model, Matrix Calculus (Jacobian, Hessian Matrix), Computation Graph, Backpropagation Algorithm, Activation Functions, Stochastic Gradient Descent, Batch Gradient Descent, Mini-Batch Gradient Descent, Vanishing and Exploding Gradients, Overfitting Problem, Regularization (Ridge, Lasso, Elastic), Dropout and Early Stopping, Bayesian Learning: Bayes theorem and concept learning, Naïve Bayes classifier, Gibbs Algorithm, Bayesian belief networks, The EM

algorithm, Support Vector Machines, Hyperplane, Support Vectors, Kernels, Non-Parametric Regression, Locally weighted regression, K-nearest neighbour.

Module IV:

07 lecture hours

Unsupervised learning (clustering, Association rule learning, Dimensionality reduction), Common distance Measures, k-means clustering, Elbow method, Hierarchical Clustering – agglomerative and divisive, Dendrogram, Similarity measures for hierarchical clustering, DBSCAN, Cluster Quality (R index, Silhouette Coefficient), Dimensionality Reduction, Principal Component Analysis, Singular Vector Decomposition, T-distributed Stochastic Neighbour Embedding.

Module V:

11 lecture hours

Population Based Algorithms: Genetic Algorithm, Fitness Function, Selection, Crossover, Mutation, Swarm Optimization, Particle Swarm Optimization, Ant-Colony Optimization, Reinforcement Learning, Actors, State, Reward Policy, Actions, Computer Vision, Convolutional Neural Networks, Deep Learning for Sequential Data, Recurrent Neural Network, LSTM, GRU, Natural Language Processing, Word Embeddings, Transformers (BERT and GPT), Building and Deploying ML models (MLFlow), MLOps, Need of MLOps, ML Production Infrastructure.

Studio Work / Laboratory Experiments:

It is concerned with the design, analysis, implementation, and applications of programs that learn from experience. Learning algorithms can also be used to model aspects of human and animal learning.

Text Books :

1. Mitchell, Tom M. *Machine Learning*. 1st ed. McGraw Hill, 2017. ISBN 978-1259096952.
2. Alpaydin, Ethem. *Introduction to machine learning*. 4th ed. Phi, 2020. ISBN 978-8120350786.

Reference Books :

1. Campesato, Oswald. *Artificial intelligence, machine learning, and deep learning*. 1st ed. Mercury Learning and Information, 2020. ISBN 9781683924665.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET302	Automata Theory and Computability	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	3	1	0	4	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate Chomsky hierarchy, and different languages and grammars.

CO2: To design machines and develop efficient mathematical models for real-life problems.

CO3: To examine computational cost and complexity for computer-based problems.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	2											3		
CO2	3	3	3	2	3								3	3	
CO3	2	3	3	2									3		3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

12 lecture hours

Introduction, Sets, Relation, and Functions, Mathematical Logics, Formal Languages, Language Classification, Finite Automata: Finite State Machine, DFA, NFA, epsilon-NFA, Equivalence between NFA to DFA, Equivalence between epsilon-NFA to DFA, Minimization of DFA, Regular Language, Regular Expression, Regular Expression to Finite Automata, Finite Automata to Regular Expression, Arden's Theorem, State Elimination Method, Pumping Lemma, Closure Properties of Regular Languages, Decision Properties of Regular Languages, Finite Automata with Output, Mealy Machine, Moore Machine, Equivalence between Mealy and Moore Machine, Properties and Limitation of FSM.

Module II:

09 lecture hours

Context Free Grammar, Context Free Language, Regular Grammar, Left and Right Associative Grammar, Left Derivation and Right Derivation, Removal of Ambiguity in Grammar, Normalization of Grammar, CNF, GNF, Deterministic Push Down Automata, Non-Deterministic Push Down Automata, Comparison between DPDA and NPDA, Pumping Lemma for CFG, Closure Properties of CFG, Context Sensitive Grammar.

Module III:

11 lecture hours

Turing Machine: Definition and Design, Church-Turing Thesis, State Transition Diagram for Turing Machine, Un-restricted Grammar, Universal Turing Machine, Combining Turing Machines, Variations of Turing Machine, Multi stack PDA, Multi-tape TMs, Single sided infinite tape TMs, Non-deterministic TMs, Multi-head TMs, Post Correspondence Problem, Halting Problem, Enumerability of TMs, Acceptability of TMs, Decidability of TMs, Rice Theorem, Chomsky Hierarchy, Diagonalization.

Module IV:**10 lecture hours**

Computational Complexity: The concept of a reduction, P, NP, and NP-completeness, NP-Hardness, The Cook-Levin Theorem, Space Complexity Classes (PSPACE and NPSPACE), Probabilistic State Automaton, Markov Chain Model, Probabilistic Turing machine, Cellular Automata, Quantum Computing.

Text Books :

1. Peter, Linz. *An Introduction to Formal Languages and Automata*. 6th ed. Jones & Bartlett Learning, 2016. ISBN 9781284077247.
2. Sarkar, Bikash Kanti and Ambuj Kumar. *INTRODUCTION TO THEORY OF COMPUTATION*. 1st ed. The Orient Blackswan, 2019. ISBN 9789386235757.

Reference Books :

1. Wigderson, Avi. *Mathematics and computation: A theory revolutionizing technology and science*. 1st ed. Princeton University Press, 2019. ISBN 978-0691189130.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET303	Seminar on Special Topics in Emerging Areas	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	0	1	0	1
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate speech preparation and presentation techniques, audience awareness and self-awareness along with the cultivation of self-confidence.

CO2: To examine the ability to present scientific material in visual, written and oral form including the formulation of an effective presentation on a topic in the Computer Science domain.

CO3: To explain an ability to listen to a scientific presentation with the discussion of the strengths and weaknesses of a speaker's presentation.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	2											2		
CO2	3	3	3	2									2	3	2
CO3	2	2	2	3									2		2

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

14 lecture hours

This course focuses on developing students' proficiency in researching, selecting, and presenting topics within computer science through seminars. The curriculum encompasses an exploration of emerging engineering research areas and industry requirements. Students will undergo training in effective presentation skills, including verbal communication, visual aid design, and written content structuring. Practical application is emphasized through individual and group exercises, peer reviews, and feedback sessions. Guest lectures from industry professionals provide insights into current and future trends. The course culminates in students delivering final seminars, incorporating learned skills. Assessment methods include topic proposal and research, mid-term presentation skills assessment, seminar preparation, peer-reviewed practice presentations, and a final seminar presentation. Through this holistic approach, students enhance both their presentation capabilities and their understanding of the dynamic landscape of computer science.

Text Books :

1. Geraci, Angelo. *Special Topics in Information Technology*. 1st ed. Springer International Publishing, 2021. ISBN 9783030624765.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET304	Competitive Programming	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	0	0	4	2	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To be able to solve a complex problem in given deadline in a competitive setup.

CO2: Find optimal solution by implementing most suitable algorithm.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1		3	3	3	3				2	2			3	3	2
CO2		3	3	3	3					1			3	3	2

1=weakly related

2= moderately related

3=strongly related

Studio Work / Laboratory Experiments:

This course having multiple problems statements from the following topics in each lab. Coding Exercise and Mathematics; Basic Data Structures (Array); LinkedList, Searching and Sorting, Stack and Queue; Bit manipulation, Divide and Conquer, Greedy and Dynamic Programming (DP); Backtracking, Branch and bound, Tree algorithms; Combinatorial Games; Basic Graph Algorithms; Shortest Path Algorithms and Network, Flow Problems; String Algorithms, Randomized Algorithms, Approximation Algorithms, Geometric Algorithms, Optimization Algorithms, Game Theory, Online algorithms.

Text Books :

1. Laaksonen, Antti. *Competitive programmer's handbook*. 1st ed. Springer, 2018. ISBN 978-3319725468.
2. Laaksonen, Antti. *Guide to competitive programming*. 1st ed. Springer International Publishing, 2020. ISBN 9783030393577.

Reference Books :

1. Dürr, Christoph, and Jill-Jênn Vie. *Competitive programming in Python: 128 algorithms to develop your coding skills*. 1st ed. Cambridge University Press, 2020. ISBN 9781108716826.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET305	High Performance Computing	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	2	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1 : Compare and Analyze data parallel and task parallel algorithms and their serial versions for searching and sorting tasks on matrix, tree, and graph data structures.

CO2 : Explain distributed memory, message passing modelling and mapping of parallel programs to physical processors.

CO3 : Implement serial and parallel algorithms for different applications using HPC libraries.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	2	2	2									1		1
CO2	2	3	2	3									1	2	
CO3	3		3	3	2								3	2	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

06 lecture hours

Why High Performance Computing, Applications of High Performance Computing, Parallel programming Software platforms, Cloud computing, Grid computing and Cluster computing, Multi-core CPUs, Graphical Processing Units, SISD, SIMD, MISD, MIMD, Communication architecture, Shared memory, Cache memory, Virtual memory, Data parallelism, Task parallelism, Bit-level parallelism, Instruction-level parallelism, Concurrency, Decomposition, Mapping.

Module II:

13 lecture hours

PRAM, NUMA, Multithreading vs Multiprocessing, Shared memory model, OpenMP, Distributed memory model, Message passing interface, Computational graph, Multithreaded DAG model, Brent's theorem, Work optimality, Weak scaling, Dependencies in code, Data races, Race conditions, Sync and Spawn, Parallel for loops, Parallel merge sort, Parallel bubble sort, Parallel shear sort, Parallel quick sort, Comparator networks, Bitonic sequences, Bitonic splits, Bitonic merge, Bitonic sort.

Module III:

11 lecture hours

PRAM, NUMA, Multithreading vs Multiprocessing, Shared memory model, OpenMP, Distributed memory model, Message passing interface, Computational graph, Multithreaded DAG model, Brent's theorem, Work optimality, Weak scaling, Dependencies in code, Data races, Race conditions, Sync and Spawn, Parallel for loops, Parallel merge sort, Parallel bubble sort, Parallel shear sort, Parallel quick sort, Comparator networks, Bitonic sequences, Bitonic splits, Bitonic merge, Bitonic sort.

Module IV:**12 lecture hours**

Distributed BFS, Graphs and adjacency matrix, Matrix based BFS, CUDA programming, Parallel matrix operations, Sparse vs Dense matrices, BLAS, LAPAC, Cluster Computing, Job Scheduling, Load Balancing, Resource Allocation, Code optimization, Memory management for Parallel computing, Distributed memory sorting, Graph Partitioning, Parallel graph operations, Advanced research topics and tools in HPC, SLURM, LSF, InfiniBand, libraries for Cloud HPC, OpenACC, Docker, containers and Kubernetes, Shell, Power Shell.

Studio Work / Laboratory Experiments:

Implementing the concepts of parallel programming using OpenMP and MPI in C.

Text Books :

1. Robey, Robert, and Yuliana Zamora. *Parallel and high performance computing*. 1 ed. Manning, 2021. ISBN 9781638350385.
2. Sterling, Thomas, Maciej Brodowicz, and Matthew Anderson. *High performance computing: modern systems and practices*. 1 ed. Elsevier Science, 2017. ISBN 9780124202153.

Reference Books :

1. Dimov, Ivan, and Stefka Fidanova, eds. *Advances in High Performance Computing: Results of the International Conference on "High Performance Computing" Borovets, Bulgaria, 2019*. Vol. 902. Springer Nature, 2020. ISBN 9783030553477.

Semester VI

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET306	Undergraduate Research in Computer Science Engineering	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	0	1	0	1	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Build sufficient background and contextual knowledge by conducting a thorough literature search.

CO2: Make use of knowledge gained from literature to identify the gaps in the present study to construct the research goal(s)/question(s) and execute the study.

CO3: Write a coherent scientific report/research paper including context, methods, discussion.

CO-PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	2	3											3	
CO2	1	2	3											3	
CO3	1	2	3											3	

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

14 lecture hours

Importance of know-how of research in undergraduate studies The research process begins with formulating a clear and concise research problem, preceded by a concise description of past studies within the field. Establishing the connection between current research and prior works is crucial for context. Subsequently, stating research goals and outlining specific questions under each goal shapes the study's direction. Describing the methodology and detailing the data collection plan is pivotal, followed by performing statistical tests and coding procedures on the collected data for analysis. Presenting the study's results with visual aids provides clarity. Acknowledging study limitations and proposing future work are imperative for refining research. Finally, concluding with implications of the results elucidates their potential impact on the world at large, showcasing the significance of the study within its field. Other responsibilities include time management (monitoring speakers' times) and moderating the question/answer period. Professionalism: All students enrolled in this course are graduate students (apprentices) in a chosen field of study. As such, you are considered to be professionals. You are expected to conduct yourselves in a professional manner by meeting all course expectations (deadlines) as required and without the need for constant reminder.

Text Books :

1. Robson, Colin. *How to Do a Research Project: a guide to undergraduate students*. 1 ed. Wiley, 2016. ISBN 9781118691326.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET399	Summer Internship	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	0	0	0	0
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To gain a better understanding of the workplace and developed and demonstrated workplace competencies necessary for professional and academic success.

CO2: To clarify career preferences and professional goals.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2		3			2	1	1	3	2			3	1	2
CO2	2		3					1		2		3	3		

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

00 lecture hours

Summer internship program allows students to gain practical experience in the workplace before receiving their undergraduate degrees. The student identifies companies willing to hire him/her on a full-time basis for an 4-8 week period. A basic premise for this course is that career-related work experiences during graduation benefit students by allowing them to gain comfort and confidence in job performance and job search skills, explore compatibility with specific careers and companies, and become more mature professionally. Students will gain an understanding of workplace dynamics, professional expectations, and the influence of culture on both. Build proficiency in a range of business or industry skills appropriate to the field of the internship placement, including professional and inter-cultural communication through written, verbal, and non-verbal means. Refine and clarify professional and career goals through critical analysis of the internship experience or research project.

Semester VII

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET402	Capstone Project	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	0	0	12	6
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Finalize the Capstone Project problem which is of the level of representative of the Undergraduate work.

CO2: Evaluate the proposal keeping in view the different stakeholders including potential customers and market analysis.

CO3: Assess the idea from feasibility, scope, timeline and delivery point of view.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3		3	3								3		
CO2	3	3		3	3								3	3	3
CO3			3		3								3	3	3

1=weakly related

2= moderately related

3=strongly related

Studio Work / Laboratory Experiments:

Initial Milestones of the Capstone Project should be completed in this course. In this semester students should have a full understanding of the problem he is working upon and should be able to do all the preliminary analysis and documentation required for Capstone. Student should be able to discuss with potential stakeholders about the viability, risks and opportunities associated with the project. He should also assess it from the point of view of scope and limitations, so that it can be completed in time with desired functionalities. Students will be discussing the proposed project with the instructors and iterate to make it more optimistic and workable for the issuing facing the society.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET497	Professional Practical Internship	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	0	0	24	12
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Identification of the existing problem in the workflow for performing a specific task.

CO2: Implementation of the solutions developed for the assigned/identified real world problems.

CO3: Develop technical writing skills for presenting the solution designed for the problem statement following ethical and legal standards.

CO-PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3		3	3								3		
CO2	3	3		3	3								3	3	3
CO3			3		3								3	3	3

1=weakly related

2= moderately related

3=strongly related

Studio Work / Laboratory Experiments:

The purpose of the semester long project is to enable the students to further explore and learn about the application of the theoretical concepts studied in the previous semesters and develop the understanding related to the implementation, design and theoretical aspects of the computer science and its application to the practical problems. Student will extend and deepen the knowledge of the computer science while working. Student will be learning what it's like to be part of a workforce: how to deal with the fellow workers, including your bosses; how to take suggestions, (including criticism); how to contribute in the team, and so on. Immersed in this absorbing and exciting world, it's easy to lose sight of the academic world, but remember that the project semester is really part of your university education. The developed projects would be assessed based on the following points: The relevance and novelty of the problem statement. The design of the proposed solution. Use of latest tools/technology for implementing the designed solution. Ability to collaborate, coordinate and work in a team. Presentation and technical reporting of the solution designed for the real-world problem. Ethical and legal standards followed for developing the solution.

Semester VIII

Name of Program	Bachelor of Technology (Computer Science Engineering)						
CSET499	Industrial Project	R&D Project	Start-up Project	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology			0	0	24	12
Pre-requisites/Exposure	-						

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Identification, formulation and analysis of the existing problem in the (non-automated) workflow for performing a specific task. Design and implementation of the automated solutions for the assigned/identified real world problems.

CO2: Technical report writing. Practice and further development of skills in time management and reporting within an industrial or research laboratory setting.

CO3: Contribute to an ethical and professional work culture and to learn to work in diverse teams.: Integration and application of knowledge and skills acquired in earlier course work.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	2	2	1	1				3		3	2	2	3	
CO2	2	2						1		2		2	3		
CO3								3	3			2			3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

336 lecture hours

The purpose of the project semester is to further develop the understanding related to the implementation, design and theoretical aspects of the computer science and its application to the practical problems. Many of the subjects that a student had studied in the university have a direct impact on what the student will be doing in the software house / industry. Student will extend and deepen the knowledge of the computer science while working. Student will be learning what it's like to be part of a workforce: how to deal with the fellow workers, including your bosses; how to take suggestions, (including criticism); how to contribute in the team, and so on. Immersed in this absorbing and exciting world, it's easy to lose sight of the academic world, but remember that the project semester is really part of your university education. Computer Science Engineering department has developed several learning outcomes for it, and department will evaluate a student based on these tasks and goals to see how well a student has achieved the learning outcomes. There are few points to worth about the of the semester project, which are as following: It is the largest single piece of work you will do during your degree course. It is the part of the curriculum that allows you to specialize in a topic you are good at or enjoy. It allows you to show off a wide range of the skills and knowledge learned during your course. It encourages integration of material learned in a number of course units. It gives you an opportunity to learn higher level of project related skills as given in the learning outcomes for project semester. A company involved in developing Websites, Mobile Apps, Business applications, Societal applications in different domains of Computer Science Engineering. Company / Academic Institution/ research institute must deal in the software related projects. Work assigned having a profile of system analyst, Tester, Quality Analyst, Data Scientist, Data Analytics, Programmer, Software Engineer or similar is

acceptable. Student must not opt for the profile related to sales or marketing. Student may opt for startups. Project must be related to the phases of software development life cycle. Projects having some research insights will be appreciated. No Self-Certification or online training courses will be considered as a substitute of project semester. However, these can be good value additions to the real industry projects and may add value to your assessment.

Specialization

Core I and II

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET211	Statistical Machine Learning	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	2	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate key features and methods of Statistical Machine Learning (SML).

CO2: To formulate and design the given application as a statistical machine learning problem.

CO3: To implement and evaluate common statistical machine learning techniques.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	POS3
CO1	3	2	2	2						2			2		2
CO2	2	3	2	2						2			2	2	2
CO3	2	2	3	3						2			2		3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I

7 lecture hours

Statistical Theory, Supervised Learning, Unsupervised Learning, Data and Types, Feature variable, Machine Learning, Statistics terms, Supervised learning, Concentration inequalities, Generalization bounds, Plugin classifiers, Least-squares methods, Bias vs Variance, Theory of generalization, Understand Underfitting, Overfitting, Parametric methods, Maximum likelihood, Bayes algorithm, Minimax algorithm, Expectation-Maximization, Advantages and Disadvantages, Applications of EM Algorithm, Use case of EM Algorithm.

Module II

10 lecture hours

Bayesian versus Non-Bayesian approaches, Density estimation, Gaussian Distributions, Gaussian Mixture Models, Gaussian Discriminant Analysis, Independent Component Analysis, Convexity and Optimization: Convexity, Conjugate functions, Nonparametric classifications methods, Unconstrained optimization, Constrained optimization, Nonparametric methods, KKT conditions, Lagrangian minimization, Primal feasibility, Dual feasibility, Complementary slackness.

Module III

13 lecture hours

Basis pursuit, Polynomial Expansion, Feature maps, The “kernel trick”, Vapnik-Chervonenkis (VC) dimension, VC generalization bounds, Sparsity: High dimensional data, The role of sparsity, Sparsistency, Consistency, Persistency, Sparsity in nonparametric regression, Sparsity in graphical models, Greedy algorithms, Sparse linear regression, Compressed sensing, Nonparametric Methods: Nonparametric regression, Density estimation, Factor Analysis, Matrix Factorization, The bootstrap, Subsampling, Nonparametric Bayes.

Module IV

12 lecture hours

Probability Distributions for modelling, Markov Networks, Hidden Markov Model, Advanced Theory: Concentration of measure, Covering numbers, Learning theory, Exact learning (Dana Angluin), Probably approximately correct learning (PAC learning), VC theory (Vladimir Vapnik and Alexey Chervonenkis), Risk minimization and its approaches, Bundle Methods, Graph Analytics, Graph-based machine learning algorithms, Simulation methods, Variational methods, Tsybakov noise conditions, Surrogate loss functions, Minimax rates for classification, Minimax rates for regression, Manifold methods, Spectral methods.

Studio Work / Laboratory Experiments:

Students will gain practical experience with the implementation of different statistical methods by using different statistical machine learning tools. Eventually, the lab works formulate the problem as a statistical machine learning problem followed by its implementation.

Text Books:

1. Masashi Sugiyama, *Introduction to Statistical Machine Learning* on (1st ed.), Morgan Kaufmann, 2017. ISBN 978-0128021217.
2. T. M. Mitchell, *Machine Learning* (1st ed.), McGraw Hill, 2017. ISBN 978-1259096952.

Reference Books

1. Richard Golden, *Statistical Machine Learning A Unified Framework* (1st ed.), CRC Press 2020. ISBN 9781351051490.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET212	Blockchain Foundations	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	2	4
Pre-requisites/Exposure	-				

Course Outcomes (COs):

On completion of this course, the students will be able to:

CO1: To articulate the fundamentals of blockchain and able to explain cryptographic concepts underlying blockchain technology.

CO2: To examine various types of Blockchain networks and consensus algorithms.

CO3: To make use of wallet transactions, crypto tokens, analyse the block details and Blockchain

CO – PO /PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	2	2	2	2	2	2	2	2	2	2	3	2	
CO2	2	3	2	2	2	2	2	2	2	2	2	2	2	3	-
CO3	2	2	2	2	2	2	2			2			2	3	

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I

11 lecture hours

Why Blockchain Technology, Blockchain Bitcoin blockchain, Blockchain Architecture, Conceptualization, Blockchain components, Cryptocurrencies, Characteristics of cryptocurrencies, Alt coins, Crypto wallets, Creation of Blocks, Wallet Transactions, Transaction details in a Block, Merkle Tree, Hash functions, pseudo random numbers, Puzzle friendly and collision resistant hash, public key cryptosystem, Generation of keys, Digital signatures, Zeroknowledge systems.

Module II

12 lecture hours

Blockchain types-Public Blockchain, Private Blockchain, Federated Blockchain, Permissionless, Permissioned Blockchain Networks, Ethereum blockchain, Go Ethereum, Gas, Gas price, Gas Limit, ETH, MetaMask, Public Test Networks, set up a Ethereum node using Geth, Mining in Blockchain, Steps in Mining, Double spending, Consensus protocols, PoW, Hashcash, Attacks on Bitcoin, Sybil Attacks, 51% Attack, eclipse attacks, DDoS Attacks, Replay Attacks, Byzantine fault, node failure.

Module III

10 lecture hours

Proof of Stake, Difference between PoW vs PoS, Byzantine General Problem, BFT (Byzantine fault tolerance), PBFT (Practical Byzantine fault tolerance), Delegated Proof of Stack, Paxos Consensus algorithm, Raft Algorithm, Solo Miner, Pool Miners, Smart contracts in Blockchain, Solidity, Data types in solidity, Operators, State variables, Global Variables, Local variables.

Module IV

9 lecture hours

Remix, Compilation of smart contracts, Deployment environments, JavaScript Environment, Injected Web3, Web3 Provider, Solidity arrays, Solidity functions, Structs in solidity, Inheritance, Special variables, Solidity mapping, Function overloading, Personal Blockchain network, Ganache, Contract deployment to Ganache network, Modifiers in solidity, Events.

Studio Work / Laboratory Experiments:

In studio work, students will learn to implement various cryptographic primitives such as symmetric/asymmetric cryptosystems, digital signatures and hash functions. Further, students will setup a MetaMask Ethereum wallet in the web browser to create wallets (User Accounts) and use it to send and receive Ethers. Use EtherScan to view the transaction details, explore Blockchain test networks to perform transactions, execute smart contracts, and launch Dapps. This lab provides a platform to create a genesis block to set up the private blockchain network using Go Ethereum (Geth) and Mist, start mining with `miner.start()` command. In addition, students will learn creating and deploying the simple smart contracts like “Hello World”, incrementing/decrementing the counter variable on the Blockchain network.

Text Books:

1. Bettina Warburg, Bill Wanger and Tom Serres, *Basics of Blockchain* on (1st ed.), Independently published, 2019. ISBN 978-1089919445.
2. Holbrook and Joseph, *Architecting enterprise blockchain solutions* (1st ed.), John Wiley & Sons, 2020. ISBN 978-1119557692.

Reference Books:

1. Bashir and Imran, *Mastering blockchain: “Distributed ledger on technology, decentralization, and smart contracts explained* (1st ed.), Packt Publishing Ltd, 2018. ISBN 978-1788839044.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET213	Linux and Shell Programming	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	4	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate Linux commands that are used to manipulate system operations at an admin level.

CO2: To write Shell Programming using Linux commands.

CO3: To design and write applications to manipulate internal kernel-level Linux File systems.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	2	2	2	2	2	2	2	2	2	2	3		
CO2	2	3	2	2	2	2	2	2	2	2	2	2	2		
CO3	2	2	3	3	2	2	2	2	2	2	2	2		3	

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I

8 lecture hours

Linux and Linux utilities, Architecture of Linux, features of Linux, Introduction to vi editor, Linux commands, File handling utilities, security by file permissions, process utilities, disk utilities, Networking commands, Text processing utilities, backup utilities, User management.

Module II

8 lecture hours

Shells need and types, Derived Operators, Linux session, Standard streams, Redirection, Pipes, Tee command, command execution, command-line editing, Quotes, command substitution, job control, aliases, variables, shell/environment customization, Filters, and pipes, File operations.

Module III

12 lecture hours

Grep Operation, Grep Commands, Grep Address, Grep Application, Seed Scripts, operation, Unix file structure, File descriptors, System calls and device drivers, File management, File structures, System calls for file management, Directory API, Process and Process Structure, Process table, Viewing processes, System processes, Process scheduling, zombie processes, orphan process, Fork and its operation, Signals functions, unreliable signals, interrupted system calls, Signal sets, File locking, Threats and Vulnerabilities analysis of Linux- direct, indirect, veiled, conditional, Security Measures in Linux-SSH key pair, Scan Log files , Close Hidden ports, Linux Malwares- Botnets, Ransomware, Rootkits, Socket, Socket communications, UDP, TCP, AWK, Shell Scripting and Security- Password Tester, Permissions and Access Control Lists, Shell Scripting for DevOps- Using environment variables, Bash Script.

Studio Work / Laboratory Experiments:

Students will use LINUX / UBUNTU to gain hands-on experience on LINUX and Shell programming, Linux commands, their uses and practice, editors: vi, nano etc, Introduction to Shell, Shell basic commands, variables Shell programming environments- filters and pipe, Shell programming File handling, Grep its use and commands. Using of Grep with pipe and filters, Unix file structuring, inodes and related system calls. File handling commands and API, Network Penetration testing tools, Wireshark, Nmap, Hash cat, Process management, creation, termination and other useful commands, Process scheduling. Parent, zombie and orphan process, Process system calls. Fork, exec, wait and signal, various commands. Basics of Socket Programming via UDP socket.

Text Books:

1. Mallett, Andrew. *Mastering Linux Shell Scripting*. Packt Publishing Ltd, 2018. ISBN 978-17889905542.
2. Blum, Richard. *Linux command line and shell scripting bible*. Vol. 481. John Wiley & Sons, 2008 ISBN 9781119700930.

Reference Books:

1. Stevens, W. Richard. *UNIX network programming. Volume 2: Interprocess communications*. Prentice hall, 1999. ISBN 9788120307490.
2. IMRAN, BASHIR. "MASTERING BLOCKCHAIN: distributed ledger technology, decentralization, and smart contracts explained, ; distributed ledger.". ISBN 978-1788839044.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET214	Data Analysis using Python	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	4	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate the structured and unstructured data for extracting useful information.

CO2: To work with data through visualization and distributions.

CO3: To make use of real-world datasets utilizing various numerical libraries.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	POS3
CO1	2	2	2	2	2				2	2		2	3		
CO2	2	2	2	2	2				2	2		2	2	3	
CO3	2	2	2	2	2				2	2		2	2		3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I

9 lecture hours

Purpose of data analysis, Structured and Unstructured data, Steps of data analysis, Python Packages for Data Analysis: Numpy, Scipy, Matplotlib, Plotly, NLTK. Data Frames, Usage of frames analytical roles, File handling and reading data for processing, Pre-processing data using multiple python frameworks, Data Formatting, Data Manipulation, Data normalization, Data Merging, Data reshaping, Data Wrangling, Missing value handling, Aggregation function, Data reporting, Data Scrapping using beautifulsoup.

Module II

11 lecture hours

String Manipulations, Demonstrating string functions, A regular expression for data, manipulation, Data Visualization, Using Histograms, Using Boxplots, Plotting data, Venn Diagram, Bar Chart, Pie Chart, Line Chart, Scatter Plots and R2, Grouped charts, Area Charts, Descriptive Statistics, Central tendencies, Analyzing variability, Data Distributions, Random Variables, Bernoulli Distribution, Binomial Distribution, Normal Distribution, Z score, Statistical Properties, Standard Normal Distribution, Correlation: Pearson correlation method.

Module III

8 lecture hours

Exponential distribution, Statistical test, Hypothesis testing, Z-test, Right-tailed test, Two-tailed test, T-Test, Significance of p-value in t-test, Two-sample Z-test, Paired t-test, Introduction to machine learning system, overview of prediction methods, models for classification problems, Sample implementation of machine learning methods on standard datasets, recent start ups based on data analysis, social impact of data analysis.

Studio Work / Laboratory Experiments:

Studio work focuses on Different Data Analysis Methods, Techniques, Algorithms using PythonData manipulation using numpy and scipy. Make use of numpy arrays, matrices, indexing and slicing options with the demonstration of numerical packages for data analysis.

Text Books:

1. Motwani, Bharti. "Data Analytics using Python." (2020). ISBN 8126502959.
2. Klosterman, Stephen. *Data Science Projects with Python: A case study approach to successful data science projects using Python, pandas, and scikit-learn*. Packt Publishing Ltd, 2019. ISBN 978-1838551025.

Reference Books:

1. Mukhiya, Suresh Kumar, and Usman Ahmed. *Hands-On Exploratory Data Analysis with Python: Perform EDA techniques to understand, summarize, and investigate your data*. Packt Publishing Ltd, 2020. ISBN 9781789537253.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET215	Graphics and Visual Computing	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	2	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To demonstrate various graphics primitives and 2-D, 3-D geometric transformations and clipping techniques.

CO2: To comprehend the concepts related three-dimensional object representations.

CO3: To demonstrate the use of OpenGL to create interactive computer graphics applications.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	POS3
CO1	2	3	2	2						2	2	2	3	2	
CO2	2	3	2	2						2	2	2	2	3	
CO3	2	2	3	2						2	2	2		3	2

1=weakly related

2= moderately related

3=strongly related

Module I

10 lecture hours

Importance of Computer Graphics, Graphics Primitives and Attributes, Character Generation, 2D Geometric Transformations, Scaling in Detail, Translation in Detail, 3D Geometric Transformations, Rotations in Detail, Reflection in Detail, Clipping Algorithms, Line Clipping, Polygon Clipping, Viewing Transformations, Transformations in Homogeneous Coordinates, Lighting and Shading Models, Hidden Surface Removal Algorithms, Introduction to Computer Animation.

Module II

13 lecture hours

Cohen Sutherland clipping, Liang Barsky clipping algorithm, Sutherland Hodgeman polygon clipping, Weiler Atherton Polygon clipping algorithm, Three-Dimensional Object Representations, 3D Modelling transformations, Parallel and Perspective projection, Sutherland Hodgeman 3D clipping, Curved lines & Surfaces, Spline representations, Spline specifications, Bezier Curves & surfaces, B-spline curves & surfaces, Rational splines, Displaying Spline curves & surfaces, Rendering, Polygonal representation.

Module III

12 lecture hours

Affine and coordinate system transformations, Visibility and occlusion, Depth buffering, Painter's algorithm, Ray Tracing, Forward rendering, Backward rendering, Phong Shading, Per pixel per vertex Shading, Visualization of 2D/3D scalar fields, Color mapping, ISO surfaces, Direct volume data rendering, Raycasting, Transfer functions, segmentation, Visualization of: Vector fields, flow data.

Module IV

7 lecture hours

Interpolation, Time-varying data, High-dimensional data, Dimension reduction, Parallel coordinates, Non-spatial data, Multi-variate, Texture mapping, Animation Techniques, Keyframing, Transformations and Projections, 2D and 3D Scaling, Translation, Rotation, Shearing, Reflection, Composite transformation, Window to Viewport Transformation, Orthographic projection, Perspective Projections, Boundary Fill Algorithm, Flood Fill algorithm.

Studio Work / Laboratory Experiments:

Lab work would be conducted in OpenGL. All the basic drawing, filling, transformation and clipping algorithms will be covered.

Text Books:

1. Gunawardhana, Pulasthi. *Computer Graphics: Theory and Practical*. Scientific Research Publishing, Inc. USA, 2020. ISBN 9781618969021.
2. Majumder, Aditi, and Meenakshisundaram Gopi. *Introduction to Visual Computing: Core Concepts in Computer Vision, Graphics, and Image Processing*. CRC press, 2018. ISBN 9781482244915.

Reference Books:

1. Thompson, William, Roland Fleming, Sarah Creem-Regehr, and Jeanine Kelly Stefanucci. *Visual perception from a computer graphics perspective*. CRC press, 2011. ISBN 9780367659288.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET216	UI/UX Design for Human Computer Interface	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	4	4	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain UI design with components and user experience designing.

CO2: To implement UI/UX components and libraries.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	2	2						2		2	3		
CO2	2	2	2	2						2		2		3	

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I

11 lecture hours

Historical evolution GUI, Interactive system design: Concept of usability, HCI and software engineering, GUI design and aesthetics, Prototyping techniques, Heuristic Evaluation, Experimental Design, Importance and benefits of good design, Screen design. Scenarios, Design Process. Characteristics of user interface, Web user – Interface popularity, Psychology and Human factors, Conceptual Models, Mistakes and Error, Cognitive models, Socio-Organizational and stakeholder requirements, Social Computing, Experiments designing concepts and methods, Communication and collaboration models, Mobile Ecosystem: Platforms.

Module II

10 lecture hours

Application frameworks, Widgets, Applications, Mobile Design: Elements, Tools, Mobile design case studies, Human interaction with computers, design rules, Universal Design Models and Theories, Interface implementation & evaluation, Communication and collaboration models, Power of the Crowd, crowdsourcing, Internet of things in HCI, Experimental Design. Experiments designing concepts, Exploring design for HCI, GOMS for HCI, Identify Color Guidelines, Stages of action in interaction, Menu types and design, Construction of Prototype, Design Principles for HCI.

Module III

7 lecture hours

HCI for navigation design, Forms and behavior, Design and data for HCI, One factor test with human subjects, Ubiquitous computing with design analysis, Interface implementation, A/B testing, T-tests,

Data assumptions and distributions, Interpretation of non-numeric response, Generalized linear models
Case studies.

Studio Work / Laboratory Experiments:

Students will learn to understand the trouble of interacting with machines and design a system based on user-centered approach.

Text Books:

1. Bhattacharya, Samit. *Human-Computer Interaction: User-Centric Computing for Design*. McGraw-Hill Education, 2019. ISBN 9789353168056.
2. Thakur, Nirmalya, and B. D. Parameshachari, eds. *Human-Computer Interaction and Beyond: Advances Towards Smart and Interconnected Environments (Part II)*. Bentham Science Publishers, 2022. ISBN 9789814998826.

Reference Books:

1. Preece, Jenny, Yvonne Rogers, Helen Sharp, David Benyon, Simon Holland, and Tom Carey. *Human-1. computer interaction*. Addison-Wesley Longman Ltd., 1994. ISBN 9781119547358

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET217	Software Development with DevOps	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	2	4
Pre-requisites/Exposure	-				

Course Outcomes (COs):

On completion of this course, the students will be able to:

CO1: To articulate DevOps engineering and DevOps terminologies to meet the business requirements.

CO2: To construct different applications of DevOps that can be used on different platforms.

CO3: To design and execute projects using different tools, modules, and templates.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	2	2	3					2		2	3		
CO2	2	2	2	2						2		2		3	
CO3	2	2	2	2	1	1				2		2		3	

1=weakly related

2= moderately related

3=strongly related

Course Contents

Module I

12 lecture hours

Why DevOps? , Terminologies, DevOps Stakeholders and roles, DevOps Perspective: People, Process and Product, Building Teams, Boston Consultancy Group (BCG) Matrix, Three Horizons Model, Autonomous Team and its Pros and Cons, Autonomy Criteria, Decoupling Point for Autonomous Team, Case Study on Autonomous Team, DevOps and ISTM (IT Service and Management), Traditional Vs Agile DevOps, DevOps Phases with diagram representation, Introduction of Source and Version Control (SVC), Algorithms for SVC: Lock-Modify-Unlock and Copy-Modify -Merge, Continuous Integration, and Deployment (CICD), Software Craftsmanship, Software Containerization, System Provisioning, and Configuration Management, Testing and Test Automation.

Module II

11 lecture hours

Service-Oriented Architecture and Microservices, Build and Release Management, Virtual Systems, Hypervisor Cloud Computing, Need of Cloud Computing with Application, Delivery Models of Cloud Computing, Deployment Model of Cloud Computing, Git Desktop Usage, and Benefits, Connecting Git Desktop and Git Online, Git Version Control, and Types, Distributed Version Control Systems with Example, Centralized Version Control Systems with Example, Configuration Management, Chef Configuration, Workstation Setup, Configuration of the knife, Test Connection between Knife and Workstation, Organization Setup, Creation and Connection of Node to Organization, Operations on Node, Object and Search, Creation of Environments and Connecting them with Servers.

Module III

6 lecture hours

Puppet, Puppet Working and Architecture, Master and Agents in Puppets, Installing Puppet, Configuring Puppet Master and Agent, Connecting Agents, Puppet Language, Declarative Language, Resources, Files, Exec, Packages, Service in Puppets, Virtual Resources and Exported Resources.

Module IV

13 lecture hours

DevOps Security, Issues in DevOps Security, DevOps Security Needs and Challenges, DevOps Risks Management, Strategies, Policies, Practices for DevOps Security, DevOps Security Tools, DevOps security over Clouds, Security Models, and their Use, Modules and Templates for DevOps, DevOps Manifests, Class Templates Static and Dynamic Content, ERB Module, NTP Module, SSH Module, Sudo Module, Install LAMP with pre-existing Modules.

Studio Work / Laboratory Experiments:

The studio work of this course introduces to setup a home lab for DevOps project for DevOps tools Jenkins, Docker, Ansible, Kubernetes with hands-on experience of concepts taught in the lecture. The lab component also includes installation of Git Bash, Git Hub, Jenkins, Docker, Ansible, Kubernetes.

Text Books:

1. Mark Reed, *DevOps: The Ultimate Beginners Guide to Learn DevOps Step-by-Step* (1st ed.), Publishing Factory, 2020. ISBN 9781647710941.
2. Gene Kim, Kevin Behr, and George Spafford, *The Phoenix Project* (5th ed.), IT Revolution Press, 2019. ISBN 978-1942788294.

Reference Books:

1. Gene Kim and John Willis, *Beyond the Phoenix Project: The Origins and Evolution of DevOps* (1st ed.), IT Revolution Press, 2018. ISBN 9781942788256.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET218	Full Stack Development	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	4	4
Pre-requisites/Exposure	-				

Course- Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To understand the Client-Server architecture and the basics of client-side programming languages and frameworks.

CO2: To examine of client-side languages: HTML, CSS, and JavaScript.

CO3: To implement client-side application development.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	2	2	1	1	2	1	1	2	1	2	3		
CO2	2	2	2	2	1	1	1	1	1	2	2	2		3	
CO3	2	2	2	2	1	1	1	2	1	2	2	2		3	

1=weakly related

2= moderately related

3=strongly related

Course Contents

Module1

8 lecture hours

Why Full Stack Development? Web development vs FullStack Development , Client-Server architecture , Rules of three-tier architecture, MEAN, MERN, Rails, Django Stack and LAMP, MEAN vs MERN stack, Front End Frameworks and Libraries, Web, Web Browser, Web Server, Anatomy of a Website, Developer tools, inspector, Wireframing , Web hosting steps, HTML, HTML Document Object Model, W3C standards for HTML, HTML Validation, Structural markup , Semantic markup, HTML Lists, Links , Absolute versus relative path names , URL: Anatomy, Types, HTML Formatting , HTML Tables, Meta tags, Structural tags, Character entities,escape codes, Image maps, Font awesome Icons , Forms Input Types.

Module II

10 lecture hours

Why CSS? W3C CSS Validator, Syntax, Types, CSS Selectors, Cascading, Inheritance, Specificity, Units of Measure, Width and Height of element, Box Model Layout, Border Box Versus Content Box, Responsive website Design Bootstrap Grid System, CSS pre-processor:Less, Sass and features.

Module III

10 lecture hours

JavaScript, JavaScript Events, JavaScript APIs, Expert Lecture from Industry, Ajax Web App Flow, Ajax Process Refreshing ES6 Specifications and Features ECMA Script, ES6 let and const, The arrow functions, ES6 Classes Inheritance, Spread Operator, Iterators and Generators, React, React in HTML, React in CSS, Built in libraries Leveraging Virtual DOM, Setting up React, Configure git/client,

Creating Repositories, push local repositories to GitHub, Git Branching and Merging , Web Publishing ethics, Hosting a website.

Studio Work / Laboratory Experiments:

In studio work, developers develop both the front-end and the server-side of the application, deploy, debug and maintain their databases and servers. Being a Full Stack Web Developer, you will be at an edge as you make better technical decisions and see the big picture.

Text Books:

1. Flanagan, D., *JavaScript: The Definitive Guide* (7th ed.), Shroffand O'Reilly Media, 2020. ISBN 978935213996.
2. Robbins, J., *Learning Web Design: A Beginner's Guide to HTML, CSS, JavaScript, and Web Graphics* (5th ed.), O'Reilly Media, 2020. ISBN 978-1491960158.

Reference Books:

1. Meloni, J.C. and Kyrnin J, *HTML, CSS, and JavaScript All in One, SamsTeach Yourself* (5th ed.), Pearson, 2018. ISBN 9789389552416.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET219	Quantum Computing Foundations	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	2	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate the differences between quantum and classical computation.

CO2: To understand the mathematical description of quantum states and basic quantum operations.

CO3: To become proficient with engineering requirements for quantum vs classical algorithm implementation.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	1	2	2						2		2	2		
CO2	2	2	2	2						2	2	2		3	
CO3	1	1	2	2						2	2	2		3	

1=weakly related

2= moderately related

3=strongly related

Course Contents

Module I

12 lecture hours

Quantum computing, Evolution of Quantum System on a Classical System, Deterministic Systems, Probabilistic Systems, Quantum System: Physical phenomena, Quantum Superposition principle, Normalization, Orthonormalization, Coherence, Decoherence, Postulates of Quantum Computing-I: Hermitian, Unitary, Pauli matrices, Complex Vector Space, State Space: Hilbert Space. Postulates of Quantum Computing-II: Dirac's Bra-ket Notation, Eigenspace: eigenkets and eigenstate, Dense Matrix, Direct Sum, Tensor, Expectation Values and Variances. Quantum states and Qubit, the atoms of computation, Representing Qubit States, Single Qubit Gates, The Case for Quantum, Multiple Qubits and Entanglement, Qutrit and Qudit, Proving Universality, Classical Computation on a Quantum Computer.

Module II

13 lecture hours

Defining Quantum Circuits, Quantum Teleportation, Variational Quantum eigen solvers, Quantum Algorithms, Deutsch-Jozsa Algorithm: Classical and quantum solution, Constant and Balanced Oracle, Bernstein-Vazirani Algorithm: Classical and quantum solution, Simon's Algorithm: Classical and quantum solution, Upper bound and Lower bound, Quantum Fourier Transform, Quantum Phase Estimation, Shor's Algorithm: Factoring, Reduction from factoring to period-finding, Shor's period-finding algorithm, Continued fractions, Grover's Algorithm: Unstructured search problems, 2-Qubits and 3-Qubits using oracle, Amplitude amplification, Optimality of Grover's Algorithm, Reduction to Three Inequalities, Proofs of the Three Inequalities, Quantum Counting, Controlled-Grover Iteration, Inverse QFT, Classical random walks, Classical Markov

chains, Quantum walks, Quantum Walk search Algorithm, On hypercube, QWSA Collision problem, Finding a triangle in a graph.

Module III

8 lecture hours

The HHL Algorithm, solving linear systems of equations using HHL, Improving the efficiency of the HHL algorithm, Combinatorial Optimization problems, Quantum Approximate Optimization Algorithm, Satisfiability Problems, Solving Satisfiability problems using Grover's Algorithm, Variational Quantum linear solver, Solving the Travelling salesman problem using, phase estimation, Quantum Query Algorithm, The polynomial method, The quantum adversary method, Quantum Complexity Theory, Classical and quantum complexity classes, classically simulating quantum computers in polynomial space.

Module IV

9 lecture hours

Quantum Merlin-Arthur Problem, The local Hamiltonian problem, Local Hamiltonian is QMA-complete, Quantum Encodings, Mixed states and general measurements, Quantum encodings and their limits, Classical error-correction, Quantum error-correction, Quantum error-correcting codes, Fault-tolerant quantum computation, Concatenated codes and the threshold theorem, Quantum error-correction using repetition codes, Measurement error mitigation, Density Matrix and Mixed States, Quantum Image Processing: FRQI and NEQR image representation, Quantum Edge Detection.

Studio Work / Laboratory Experiments:

In-studio work, students will be setting up the environment for programming using Qiskit, they will be designing and analyzing quantum algorithms.

Text Books:

1. Jack D. Hidary, *Quantum Computing: An Applied Approach* (1st ed.), Springer, 2019. ISBN 978-3-030-23921-3.
2. Ronald de Wolf, *Quantum Computing: Lecture Notes* (1st ed.), arXiv, 2021. ISBN arXiv:1907.09415v2.

Reference Books:

1. Venkateswaran Kasirajan, *Fundamentals of Quantum Computing* (1st ed.), Springer, 2021. ISBN 9783030636895.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET220	Unmanned Aerial Vehicles	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	4	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate UAVs, UAV application, needs, deployments, and advancements.

CO2: To understand design concepts, aerodynamic considerations, electronic attachments, and mission- specific deployment of aerial nodes.

CO3: To construct multirotor UAV from scratch using various flight controllers.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	1	2	2						2		2	3		
CO2	2	2		1						2	2	2		3	
CO3	2	1	2	2						2	2	3		3	

1=weakly related

2= moderately related

3=strongly related

Course Contents

Module I

6 lecture hours

Why Unmanned Aerial Vehicle (UAV)? Introduction to UAVs/Drones; Unmanned Aerial Systems (UAS); History and Modern Trends; Technological Advancements; UAV Concept of Operations; UAV Chemical, Biological, Radiological and Nuclear (CBRN); UAV Classification: Wing Geometry Based, Payload Based, Size Based, Mission Based; UAV Anatomy; Mission Planning.

Module II

7 lecture hours

Basic UAV Aerodynamics: Wing Configuration and Thrust; Airframe Configuration; Lift and Drag; Aerodynamic Configuration; Vibration and Noise; Endurance; Gliding Flight; UAV Stability; UAV Propulsion; Propeller Modelling; Motor Modelling.

Module III

8 lecture hours

Coordinate Frame; Earth-Fixed Coordinate Frame; UAV Communication: Radio Wave Communication, Microwave Communication, Line of Sight Communication, Beyond Line-of-Sight Communication; UAV Navigation; Coordinate Frame; Waypoint Tracking Navigation; Sensor Calibration and Measurements; Tracking using GPS; Tracking using Data; State Estimation; Attitude Estimation; Position Estimation.

Module IV

7 lecture hours

UAV Controllability; Flight Control; Position Control: Euler Angles as Outputs, Rotation Matrix as Outputs; Autonomous Control; Semi-Autonomous Control; UAS Safety; Failsafe.

Studio Work / Laboratory Experiments:

In this course students will start with basic Multicopter design principles, components, and assembly precautions. Then finally design and develop a fully functional Multicopter UAV/Drone using off the shelf Flight Controllers.

Text Books:

1. Quan Quan, *Introduction to Multicopter Design and Control* (1st ed.), Springer, 2017. ISBN 9789811033810.

Reference Books:

1. Mohammad Sadraey, *Unmanned Aircraft Design: A Review of Fundamentals* (1st ed.), Morgan & Claypool, 2017. ISBN 1681731681.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET221	Robotic Process Automation Essentials	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To understand the underlying logic/structure related to robotic process automation.

CO2: To examine RPA bots to automate the processes.

CO3: To make use of transactions in business processes using robotic process automation.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	1	1	1		1	2		2		1	1			2
CO2	2		2		3	1			2		1	3		2	1
CO3	2		2		2	1		1		2	3	1	1		2

Course Content:

Module I

10 lecture hours

Process Automation, Application and Evaluation of RPA, Flowchart, Sequence, State Machine, Activities in RPA, Activities to print the log, How to get the input from user, How to send the data to output device i.e. Monitor, Variables & Data type in RPA tool, Managing packages, Publishing of Bot, If Statement, For loop, For each row, Switch, Parallel, Do-while, While, Conversion of Pseudocode in RPA tool, Factorial, Fibonacci code in RPA tool, Selectors, UI Element.

Module II

11 lecture hours

Debugging, Use of Try-Catch for exception handling, .Net Framework editor, .Net Variable, Loops and conditional statement, Extensible Markup Language, Tools for writing XML, Rules for writing XML, Elements attributes and values, Creating the root and child elements, Attended bot, Unattended bot, Floating bot.

Module III

7 lecture hours

Orchestrator, Orchestrator Entities, Tenants, and folder, Robot Provisioning and License Distribution, Expert Lecture from Industry, Unattended Automation with Folders, Orchestrator Resources, Libraries and Templates in Orchestrator, Setting up an Orchestration Process Activities, Queues for Long-running Activities, Tasks Long-running Activities, Jobs Long-running Activities, Formation of RPA Team, Process Design Document/Solution, Design Document. Storage Buckets, Queues, Intermission-Transaction Processing Models, Robots Hosting, Processing Hosting, Triggers and SLAs, Monitoring and Alerts, Advanced Topics in RPA, RPA Frameworks; Transaction, Queue, Machine, Variable declaration.

Studio Work / Laboratory Experiments:

The studio work covers working with the RPA tool Community edition, developing the automation applications as the projects with its detailed demonstration.

Text Books:

1. Murdoch, Richard. *Robotic Process Automation: Guide to Building Software Robots, Automate Repetitive Tasks & Become an RPA Consultant*. Richard Murdoch & RPA Ultra, 2018. ISBN 978-1983036835.
2. Taulli, Tom. "The robotic process automation handbook." *The Robotic Process Automation Handbook*. <https://doi.org/10.1007/978-1-4842-5729-6> (2020). ISBN 9781484257289.

Reference Books:

1. Mullakara, Nandan, and Arun Kumar Asokan. *Robotic process automation projects: build real-world RPA solutions using UiPath and automation anywhere*. Packt Publishing Ltd, 2020. ISBN 9781839210958.

Name of Program	Bachelor of Technology (Computer Science Engineering)						
CSET222	Microcontrollers, Robotics & Embedded Systems			L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology			2	0	4	4
Pre-requisites/Exposure	-						

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain the micro-controllers, robotics, embedded system, and their applicability.

CO2: To build the various robots using Robot Operating System.

CO3: To implement the various Embedded system using microcontrollers and ROS.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2		2		2	3			2	2		3	
CO2	3		3	3	2		2	3	3		2		2	1	2
CO3	3		3		2	2	2		3		3	2		3	

Course Content:

Module I

6 lecture hours

Application of Microcontrollers, Why Embedded system, Application of Robotics, Arduino: Architecture, Serial Port, Serial Communication, Device control using Serial Communication, Arduino Sensors (Humidity, Temperature, Water Detector, Ultrasonic, LDR), Arduino secondary Integration (Relay, DC, Servo motors, RF modules), Pulse width modulation (PWM), I2C communication protocol, BH1750: Digital light sensor, Parallel Communication, Arduino UART, GSM, GPRS Module.

Module II

6 lecture hours

Raspberry Pi: Architecture, Raspberry Pi Port Identification, Raspberry Pi GPIO, Transistorized Switching, Accepting Digital Input on Raspberry Pi, Enabling I2C to Raspberry Pi, Analog, and Digital sensors, BMP180 with Pi, Sensors Interface with Pi LDR, Sensors Interface with Pi DHTT11, Sensors Interface with Pi using Sense HAT.

Module III

9 lecture hours

Fundamentals of Robotics, Robot Operating System (ROS), ROS Essentials: ROS Topics, ROS: Services, Actions, Nodes, Build Robot Environment, Unified Robot Description Format (URDF), ROS parameter server, ROS Services, and parameters, Recording and playing back, reading messages from a bag file, using roscat to edit files in ROS, ROS msg and srv.

Module IV

7 lecture hours

Simple Publisher and Subscriber, Examining the Simple Publisher and Subscribe, Simple Service and Client, Examining the Simple Service and Client, Motion in ROS (ROS Noetic), Working with Pluginlib, Nodelets, and Gazebo Plugins, Robot Navigation (moveit), Grasping, Grasping using MoveIt, creating a pick and place task, Grasping in the Real Robot, ROS Controllers and Visualization Plugins.

Module V

9 lecture hours

Fundamentals of Robotics, Robot Operating System (ROS), ROS Essentials: ROS Topics, ROS: Services, Actions, Nodes, Build Robot Environment, Unified Robot Description Format (URDF), ROS parameter server, ROS Services, and parameters, Recording and playing back, reading messages from a bag file, using roscat to edit files in ROS, ROS msg and srv.

Studio Work / Laboratory Experiments:

Studio work focuses to develop different kind of robots starting from object detection robot, line follower robot. It also consists of introduction to well exist robots such NAO, SOFIA, Pepper.

Text Books:

1. Newman, Wyatt. *A systematic approach to learning robot programming with ROS*. CRC Press, 2017. ISBN 9781498777827.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET223	Augmented Reality Foundations	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	2	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To build a solid background in alternative 3D compositing techniques using computer vision with applications in interactive interfaces – most notably augmented reality interfaces on mobile devices.

CO2: To develop skills in the design and development of interactive augmented reality games and understand the practical issues regarding multi-platform reconstruction.

CO3: To develop interactive augmented reality applications for PC-based mobile devices using a variety of novel input devices.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	POS3
CO1	3	2	3		3		2	2	3			2	2		3
CO2	3		3	2	2			3	3	2		3	2		2
CO3	3		3	2	2		2	3	3	2	3	3	2		

Course Content:

Module 1

10 lecture hours

Industrial applicability of AR, AR Systems Overview, Input and Output Devices for AR, Optical vs. Video, See-Through AR, AR system architecture, Senses, Simple sensing/perception model, Human visual system, 3D Vision, Tracking system, Tracking for Augmented Reality, Importance of Accurate Head Tracking, Tracking Problem, Tracking Technologies.

Module II

10 lecture hours

Head motion prediction, Registration, Calibration – static and dynamic, Real Time Performance, Characteristics – spatial, temporal, system robustness, Scheduling and Fusing Sensor Information, mixed reality UI design, Design of immersive user experience, Alternative Interface Paradigms, Usability guidelines, immersive environments, Space, Scale, Ergonomics.

Module III

10 lecture hours

Physical locomotion techniques, Target based techniques, Steering, Comfort and distress, Gaze direction, comfort range test, Motion Sickness, Simulator Sickness, Cybersickness, AR Interface Design, Properties of AR Environments, Collaborative AR Interfaces, Heterogeneous AR User Interfaces, Tangible and Graspable Interaction.

Module IV

12 lecture hours

Tracking for Augmented Reality, Augmented Reality Interaction, Augmented Reality, Information Browsers, AR Widgets, Graphical Interface Elements, Evaluating AR Interfaces, video mixing, optical blending, projection devices, spatially augmented reality, Immersive virtual reality, Desktop metaphor, mobile/wearable computing, Ubiquitous computing, Tangible user interfaces, Distributed graphics.

Studio Work / Laboratory Experiments:

The laboratory of Augmented reality foundation is designed to provide a practical exposure to the students about the concepts and topics taught in the classroom sessions. Case Studies: - Face-to- Face Collaboration – Shared Space - Remote Collaboration – AR Conferencing, Wearable AR Conferencing - Seamless Collaboration – The Magic Book.

Text Books:

1. Glover, Jesse, and Jonathan Linowes. *Complete Virtual Reality and Augmented Reality Development with Unity: Leverage the power of Unity and become a pro at creating mixed reality applications*. Packt Publishing Ltd, 2019. ISBN 9781838644865.
2. Bowman, Doug, Ernst Kruijff, Joseph J. LaViola Jr, and Ivan P. Poupyrev. *3D User interfaces: theory and practice, CourseSmart eTextbook*. Addison-Wesley, 2004. ISBN 978-0134034324.

Reference Books:

1. Pangilinan, Erin, Steve Lukas, and Vasanth Mohan. *Creating augmented and virtual realities: theory and practice for next-generation spatial computing*. " O'Reilly Media, Inc.", 2019. ISBN 978- 1492044192.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET238	Product Design Principles and Practices	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	2	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate the concepts of product design and development process.

CO2: To apply of product design and development in industrial applications.

CO3: To implement and apply design or redesign of the product.

CO – PO /PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	POS3
CO1	1	2	3												3
CO2	1	2	3												3
CO3	1	2	3												3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I

10 lecture hours

Characteristics of Successful Product, Product Design and Development, Duration and Cost of Product Development, Challenges of Product Development, Structured Methods, Industrial Examples, Organizational Realities, Product Development Process and Organization, Product Development Process, Concept Development: The Front-End Process, Adapting the Generic Product Development Process, Technology-Push Products, Platform Products, Process-Intensive Products, Customized Products, High-Risk Products, Quick-Build Products Digital Products, Product-Service Systems, Complex Systems, Product Development Process Flows, Tyco Product Development Process.

Module II

10 lecture hours

Product Development Organizations, Choosing an Organizational Structure, Distributed Product Development Teams, Tyco Product Development, Opportunity Identification, What Is an Opportunity? Types of Opportunities Tournament Structure of Opportunity Identification, Effective Opportunity, Tournaments, Opportunity, Identification Process, Establish a Charter, Generate and Sense Many Opportunities, Techniques for Generating, Opportunities, Screen Opportunities, Develop Promising Opportunities, Select Exceptional Opportunities, Reflect on the Results and the Product Planning, The Product Planning Process, Identify Opportunities, Evaluate and Prioritize Projects, Competitive Strategy, Market Segmentation, Technological Trajectories, Product Platform Planning, Technology Road mapping, Evaluating Fundamentally New Product Opportunities, Balancing the Portfolio, Allocate Resources and Plan Timing, Resource Allocation, Project Timing, The Product Plan, Complete Pre-Project Planning, Mission Statements, Assumptions and Constraints, Staffing and Other Pre-Project Planning, Reflect on the Results and the Process.

Module III

10 lecture hours

Identifying Customer Needs, Importance of Latent Needs, Process of Identifying Customer Needs, Gather Raw Data from Customers, Choosing Customers, The Art of Eliciting Customer Needs Data, Documenting Interactions with Customers, Interpret Raw Data in Terms of Customer Needs, Organize the Needs into a Hierarchy, Establish the Relative Importance of the Needs, Reflect on the Results and the Process, Product Specifications, What Are Specifications? When Are Specifications Established?, Establishing Target Specifications, Prepare the List of Metrics, Collect Competitive Benchmarking, Set Ideal and Marginally Acceptable, Target Values, Reflect on the, Results and the Process, Setting the Final Specifications, Developing Technical Models of the Product, Developing a Cost Model of the Product, Refine the Specifications, Making Trade- Offs Where Necessary, Flow Down the Specifications as Appropriate, Reflect on the Results and the Process, Target Costing, Concept Generation, The Activity of Concept Generation, Structured Approaches Reduce the Likelihood of Costly Problems.

Module IV

12 lecture hours

Clarify the Problem, Decomposing a Complex Problem into Simpler Subproblems, Focus Initial Efforts on the Critical Subproblems, Search Externally, Interview Lead Users, Consult Experts, Search Patents, Search, Published Literature, Benchmark-Related Products, Generating Solution Concepts, Explore Systematically, Concept Classification Tree, Concept Combination Table, Managing the Exploration Process, Reflecting, Solutions and the Process, Concept Selection, Development Process, Use Method for Choosing a Concept, Structured Method Offers Several Benefits, Overview of Methodology, Concept Screening, Prepare the Selection Matrix, Rate the Concepts, Rank the Concepts, Concept-Screening Matrix Example, Reflect on the Results and the Process, Concept Scoring, Combine and Improve the Concepts, Reflect on the Results and the Process Caveats, Concept-Scoring Matrix Example, Concept Testing, Define the Purpose of the Concept Test, Choose a Survey Population, Choose a Survey Format, Communicate the Concept, Matching the Survey Format with the Means of Communicating the Concept, Issues in Communicating the Concept, Measure Customer Response, Interpret the Results, Reflect on the Results and the Process, Estimating Market Size.

Studio Work / Laboratory Experiments:

In the studio, students will learn the application of structured methods to product development and get a detailed insight into the study and improvement of the development process. The students will be guided to create their development methods, uniquely suited to their personalities, talents, and company environments.

Text Books:

1. Karl T. Ulrich, Steven D. Eppinger and Maria C. Yang., Product Design and Development (7th ed.), McGraw Hill, 2020. ISBN 9390113237

Reference Books:

1. Ali Jamnia, Introduction to Product Design and Development for Engineers (1st ed.), Taylor and Francis, 2018. ISBN 9781138554214

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET224	Cloud Computing	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	2	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate cloud computing principles and their business need.

CO2: To identify the design principles of virtualization techniques in cloud resource management.

CO3: To design and development of cloud architectural solution with its detailed monitoring.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	2	1	1	1	2	2	2	2	2	2	2	2	2	1
CO2	2	2	1	1	1	2	2	2	2	2	2	2	2	2	1
CO3	2	2	3	2	2	2	2	2	2	2	2	2	2	2	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

11 lecture hours

Cloud Computing, Adoption of cloud-based IT resources, Service Models: Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS), Software-as-a-Service (SaaS), Deployment models: Public Cloud, Private Cloud, Hybrid Cloud, Community Cloud, Cloud Computing Characteristics, Challenges of cloud computing, Virtualization concept, Types of virtualizations, Demo of virtualization, Virtualization Merits, Role of virtualization in cloud computing, Virtualization Demerits, VM Placement, VM Migration, VM Migration Demo, VM clustering, Design Issues in VM Clustering, Need of Dockers and Containers, Docker Eco-System, Hypervisor vs Docker.

Modulo II:

12 lecture hours

Microservices, Service-Oriented Architecture, REST API, IP Addressing, Subnetting, Supernetting, Designing of Virtual Private Cloud, Demo of VPC, VPC Peering, VPC Case Study, Cloud Storage, Serverless Computing, Cloud API Gateway, Cloud Databases, Resource Provisioning, Time shared and space shared, Efficient VM Consolidation on cloud server, Task/DAG Scheduling Algorithms, Min-Min, Max-Min, MET, B-level Demo, T-level Demo, Task-VM Mapping, Auto Scaling, Load Balancing.

Module III:

10 lecture hours

Case Study: Cloud Market analysis, Security and Compliances, Shared security model in IAAS/PAAS/SAAS, Shared technology issues, Data loss or leakage, Account or service hijacking, Implementation of cloud security, Security Groups, Network Access Control Lists, Cloud databases, Parallel Query Execution with NoSQL Database, Big Data, Handling Big Data on Cloud Platform,

Map- Reduce framework for large clusters using Hadoop, Design of data applications based on Map Reduce in Apache Hadoop.

Module IV:

9 lecture hours

Comparative study/analysis of public clouds, Edge Computing, Fog Computing, Data Offloading, Cloud-Based DevOps Tools, Task Partitioning, Data Partitioning, Data Synchronization, Distributed File System, Data center, Ongoing Research Topics.

Studio Work / Laboratory Experiments:

Practical experience on global cloud infrastructure by performing experiments on Amazon Web Services (AWS), Google Cloud Platform (GCP) and Microsoft Azure platform. The essential services and their hands-on is compulsory on Core IaaS, PaaS and SaaS.

Text Books :

1. Wang L., Rajiv Ranjan, Jinjun Chen and Boualem Benatallah, *Cloud Computing* (1st ed.), CRC Press, 2017. ISBN 978-1351833097.
2. Hurwitz J. S. and Daniel Kirsch, *Cloud Computing For Dummies* (2nd ed.), Hoboken: John Wiley & Sons, 2020. ISBN 978-1119546658.

Reference Books :

1. Sharma Prerna, Moolchan Sharma and Mohamed Elhoseny, *Applications of Cloud Computing* (1st ed.), CRC Press, 2020. ISBN 9780367904128.

Name of Program	Bachelor of Technology (Computer Science Engineering)						
CSET241	Fundamentals in Swift Programming			L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology			2	0	4	4
Pre-requisites/Exposure	-						

Course-Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate key features of Swift Programming.

CO2: To implement basic concepts and features of Swift using Playgrounds.

CO3: To implement Swift for Mobile Application Development using Xcode.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	3											1	1	2
CO2	1	2	2	3		1							3	2	1
CO3			3	2	2			1	2	1			2	3	

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I

6 lecture hours

Installation and Overview of Swift: Installation of Swift on macOS and Linux, REPL, Package manager, creating a package, Building an Executable, Working with multiple Source File. Introduction to Xcode and Swift Playgrounds: Installation of Xcode, Working with Xcode, create a simple program and execute it using Xcode, working with swift playgrounds, create a simple program and execute it using swift playgrounds.

Module II

6 lecture hours

Introduction of Swift, features of Swift, Data types, constant and variables, operators. Type Annotations, Naming Constants and Variables, Printing Constants and Variables, Semicolons, Integers: Integer Bounds, Int, UInt. Floating-Point Numbers: Double, Float. Type Safety and Type Inference. Numeric Literals, Numeric Type Conversion, Integer Conversion, Integer and Floating-Point Conversion, Boolean.

Module III

6 lecture hours

Strings and Characters: String Literals, Multiline String Literals, Special Characters in String Literals, Initializing an Empty String, String Mutability, Working with Characters, Concatenating Strings and Characters, String Interpolation, Counting Characters, Substrings, Comparing Strings, Prefix and Suffix Equality.

Module IV

6 lecture hours

Control Flow: For-In Loops, While Loops: While, Repeat-While. Conditional Statements: If-else, Switch, Control Transfer Statements: continue, break, fall through, return, throw statements. Functions: Defining and Calling Functions, Function Parameters. Return Values: Functions Without Return Values, Functions with Multiple Return Values, Optional Tuple Return Types Function Argument Labels and Parameter Names: Specifying Argument Labels, Omitting Argument Labels, Default Parameter Values, Variadic Parameters, Function Types, Function Types as Parameter Types.

Studio Work/ Laboratory Experiments:

Students will gain practical experience with the implementation of different swift concepts. The lab work consists of swift programming using Playground and Xcode.

Text Books:

1. Swift Matthew Mathias, John Gallagher, Swift Programming: The Big Nerd Ranch 2nd edition, 2015.
2. Matt Neuberg, iOS 12 Programming Fundamentals with Swift, O'Reilly; 5th edition.
3. IBook Apple Introduction to Swift.

Reference Books:

1. Paris Buttfield-Addison, Jonathon Manning, Tim Nugent Learning Swift: Building App macOS, iOS, and Beyond, O'Reilly Media, Inc., 3rd ed, 2018.
2. Jon Hoffman, Mastering Swift 4, Packt Publishing Limited, 4th edition, 2017.

Name of Program	Bachelor of Technology (Computer Science Engineering)						
CSET245	UX Design Foundations			L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology			3	0	2	4
Pre-requisites/Exposure	-						

Course-Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To learn User Experience Design Methods: Conceptual Models, User Models.

CO2: To learn Systems Thinking and Innovation Frameworks.

CO3: To make use of Wireframing, Prototyping and Information Architecture

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1															
CO2															
CO3															

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module 1

(11 hours)

Gestalt's Principles of Design, User Interface Design & Design Systems, Branding through UI Design, Design Mockups & Developer handoff, UI Design Workshop using Figma, Introduction to Figma, Figma training - Color, typography, proximity, contrast issues, layout design, imagery and graphics in UI. Case Studies.

Module 2

(11 hours)

Understanding hierarchy issue, scale issues, balance/alignment issues, emphasis dominance issue, Principles & Heuristics of HCI: Design Principles & Heuristics, Design Standards & Guidelines, Usability & Accessibility Studies, Conceptual Models, User Models, User Experience Design for Business Applications, Design Thinking & Human Centered Design Process

Module 3

(10 hours)

Responsive design principles, mobile-first design, web vs. mobile, UI considerations. HTML, CSS, Javascript, Storyboarding, Decision Mapping, Decision Flow Diagram, Mental Models, Feedback Loops, Fundamentals of User Experience Design.

Module 4

(10 hours)

Establishing a business case for UI/UX Design: User Research- 1: UX Research methodologies- Qualitative & Quantitative, UX Research Methodologies- Attitudinal & Behavioural User Research- 2: User Personas, Empathy Maps & User Journey Mapping, User Stories & Storyboards.

STUDIO WORK / LABORATORY EXPERIMENTS:

Students will gain hands-on experience on Visual Prototyping and Wireframing, Adobe XD, Figma, InVision, Google Slides, Building simple websites and stylesheets; interactive elements using JavaScript . UX design process, prototyping basics, user flows. Wireframing and prototyping using tools like Axure RP, Sketch, Marvel and Balsamiq. Designing visual elements using Photoshop and Illustrator.

EXTBOOKS/LEARNING RESOURCES:

1. "A Project Guide to UX Design" by Russ Unger and Carolyn Chandler
2. "Lean UX: Designing Great Products with Agile Teams" by Jeff Gothelf and Josh Seiden
3. "The Elements of User Experience" by Jesse James Garrett

REFERENCE BOOKS/LEARNING RESOURCES:

1. The Design of Everyday Things" by Don Norman
2. "About Face: The Essentials of Interaction Design" by Alan Cooper, Robert Reimann, and David Cronin

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET225	Intelligent Model Design using AI	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	3	0	2	4	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate key features of various Machine learning and Deep Learning tools for Intelligent Modelling.

CO2: To examine and implement intelligent applications using Machine Learning and Deep Learning tools.

CO3: To implement Machine Learning and Deep Learning models for design thinking and evaluate them using relevant metrics.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3	3		3		2	1		3	1	2	2	2	3
CO2	3		2		3		3	3	2		2			3	
CO3	3		3	2	3		2	3	2	2	3			3	2

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

8 lecture hours

Why Intelligent Model? Intelligent Citizens of the World, Thinking More Clearly, Empathize, Define, Ideate, Prototype, Test, Using and Understanding Data, Using Models to Decide, Strategize, and Design, Automation/Augmentation Opportunities, AI in Sensitive Applications, AI in Healthcare and Security, Pervasive, AI Systems, AI in IOT Devices, Business Value of Data: Case Study on Data Dependent Company, Data Valuation, Data Quality, Data Driven Marketing and Commerce, Biases in AI Models, Handling Gender, Race, Religion, Cultural Biases, Unintended Biases.

Modulo II:

10 lecture hours

User privacy, Data Privacy aware AI, Data Sharing Regulations, AI Model Security, Attacks on AI models, Adversarial Attack, Relevance Feedback, Adaptive Learning, Online learning, Model fine-tuning, Transfer Learning, Domain Adaption, Explainable AI, ML Accountability and Fairness, Model Interpretation, Class imbalance in Modelling, Handling Data Drift, Human- Machine Co-learning.

Module III:

12 lecture hours

Aligning Mental and Machine Learning Models, Coding Automation, Rapid Prototyping in AI, Market Demand Analysis, Automated Requirement Analysis, Customer opinion analysis, Attribute based opinion mining, Geospatial Analytics, Spatial analysis, Spatial-temporal analysis, Geo-clustering, Market Segmentation, Demographic, Cultural, Behavioural and Psychographic Segmentation, Dynamic Pricing, Price Optimization, Personalization, Recommender Systems, Feature Attribution, SHAP (SHapley Additive exPlanations), Lime, Shapash, Dalex.

Module IV:

12 lecture hours

Speed vs Accuracy Trade-off, optimizing models for resource constrained devices, Deep Learning Model compression, Pruning, Low-rank factorization, Automatic Machine Learning, Neural Architecture Search, Deploying Machine Learning Models, Staging Automation, Cognitive Modelling, Conversational Systems, Data driven persona, User personas, Reasoning Under Uncertainty, Multi-agent System, Reinforcement Learning, General Intelligence, Super Intelligence, Decision Making Systems.

Studio Work / Laboratory Experiments:

Students will gain practical experience with the implementation of different intelligent methods. The lab work consists of exploratory data analysis using tools, libraries and python programming language by using Machine Learning and Deep Learning models.

Text Books :

1. Aggarwal C. C., *Neural Networks and Deep Learning* on (1st ed.), Springer International, 2018. ISBN 978-3030068561.

Reference Books :

1. Kasparov Garry, *Mig Greengard, Deep Thinking: Where on Machine Intelligence Ends and Human Creativity Begins* (1st ed.), John Murray, 2018. ISBN 978-1473653511.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET226	Blockchain Engineering	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	3	0	2	4	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate blockchain concepts and software applications that are used to develop the blockchain use cases.

CO2: To perform compilation, migrating, testing, and deploying the blockchain applications on the decentralized network.

CO3: To make use of building distributed ledger technologies for Blockchain applications and able to solve double spending problems in the digital payments using cryptocurrencies.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	2	2		3	2	1	2	2	2	2	2	2	2	2
CO2	2	2	2		3	2	2	2	2	2	2	2	2	2	2
CO3	2	2	3	2	2	2	1	2	2	2	2	2	2	2	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

Blockchain platforms, distributed ledger technologies, distributed consensus algorithms, endless chains, Proof-of-elapsed time, Proof of Burn, Hyperledger, Frameworks, Fabric network, Transaction flow, Fabric– Membership, Identity management, Fabric components, Fabric deployment, Hyperledger Composer, Application Development, Composer – Network Administration.

Modulo II:

11 lecture hours

IPFS, Decentralized web, Privacy and encryption on IPFS, nodes, IPFS commands, IPFS vs HTTP, Location-based addressing, Content-based addressing, Distributed Hash Table (DHT), Merkle DAGs, IPNS, IPFS Gateway, IPFS Daemon, DNS Link, On-chain Transactions, Off-chain Transactions, Forking, Hard forks, Softworks, Double spending issues, Improving throughputs, Algorand, DAG-Chains, Hyper ledger Indy, Hyperledger Sawtooth.

Module III:

11 lecture hours

Decentralized Autonomous Organizations, DAO Membership, Problems with automated immutable systems, Challenges with the decentralized web, Defi (decentralized finance), Defi on Ethereum, Non-Fungible Tokens (NFTs), NFT use cases, Gas pricing, Ethereum development using Whisper, Swarm, and Raiden Network, State Channels, Case study, Costless Verification: Blockchain Technology, last mile problem, Verification cost reduction in blockchain, Bootstrapping network effects.

Studio Work / Laboratory Experiments:

The laboratory work enables students to get practical experience in various Blockchain platforms like Ethereum and Hyper ledger to design, deploy Decentralised Applications (DApps) on Blockchain network using chain code, which include: 1) Simple Land Registry DApp based on the Ethereum blockchain. 2) A blockchain-based Tollbooth Management System. 3) Creating a decentralized platform for the distribution of scholarship. 4) DApp for Organ Donor list and an Organ Recipient list which will be stored in a distributed data system. 5) A simple decentralized application, where officials can register for birth/death, and generate a certificate which can then be reverified at any point of time in the future (DocVerify), etc.

Text Books :

1. Gaur, Nitin and et al, *Hands-On Blockchain with Hyperledger: Building decentralized applications with Hyperledger Fabric and Composer* (1st ed.), Packt Publishing Ltd, 2018. ISBN 9781788994521.
2. Pathak, Nishith and Anurag Bhandari, *IoT, AI, and Blockchain for. NET: Building a Next Generation Application from the Ground Up* (1st ed.), Apress, 2018. ISBN 9781484237083.

Reference Books :

1. Bashir and Imran, *Mastering blockchain: “Distributed ledger technology, decentralization, and smart contracts explained”* (1st ed.), Packt Publishing Ltd, 2018. ISBN 9781787125445.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET227	System and Network Security	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	2	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate the system vulnerabilities, exploitation, and defense mechanisms.

CO2: To examine the spyware, security issues and different types of attacks.

CO3: To do programming for system and mobile security.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3		3		2	2		3	3		1	2	3	2	
CO2	3	3	2		3	2	3		3	3	2			2	
CO3	3		3	3		2			2	2		3			3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

11 lecture hours

Threats and Vulnerabilities of Windows, Threats and Vulnerabilities of Linux, Controls, Risk Management, Basics of Confidentiality, Integrity and Availability, Generic Security Policies, Security Mechanisms, Assurance, Prevention, and Detection, Security Issues in Windows Operating System (Windows XP, Windows Vista, Windows 7 and Windows 10), Linux Kernel TCP/IP Vulnerabilities, TCP SACKs (25), SACK Panic, Mitigation Schemes and Patches Developed for Each Security Issue, Boot Loader Security Issues, Grub Security Flaw.

Modulo II:

9 lecture hours

Security in Broadcast Networks, Security in Centralised Networks, DDOS Attacks, Volume-based DDoS attacks, Protocol attacks, Application-layer DDoS attacks, Introduction to Malicious Softwares- Botnet, Logical Bombs, Grayware, Spyware, Adware, File Binders, Antivirus, Honeypot, Trapdoor, Latest Malwares Definitions and Their Defence Mechanisms. Android-based Malwares and their Security Patches.

Module III:

9 lecture hours

Introduction to Network Security, Network-based Attacks, Active and Passive Attacks, Phishing and Its Types, Tailgating, Impersonation, Dumpster Diving, Shoulder Surfing, Hoax, Waterhole Attacks, Denial of Service Attacks and its Types, Man in the Middle Attack, Buffer Overflow Attack, Cross-Site Scripting, SQL Injection Attack, Privilege Escalation, Man in the Browser, Zero-Day Attack, SQL Injection Attack, Privilege Escalation, Man in the Browser.

Module IV:

13 lecture hours

Generic Security Software's, Windows Firewalls, Linux Firewalls, Access Control Lists of Firewalls, Types and Examples of Firewalls, Network-based Honeypots and Trapdoors, Virtual Private Network and its Implementation, Network Intrusion Detection System, Network Intrusion Prevention System, Router Security, Switch Security, Proxy Server and its Configurations, Load Balancers, IPv6 and IPv6 Security, Secure Forwarding in Overlay Networks.

Studio Work / Laboratory Experiments:

The lab component is meant to analyse various network security-related concepts and deals with launching different types of attacks and creating a network blueprint of an organization

Text Books :

1. R. Bragg et al, *Network Security: The Complete Reference* (1st ed.), TMH Publications, 2017. ISBN 978- 0070586710.
2. W. Stallings, *Network Security Essentials: Applications and Standards* (3rd ed.), Pearson, 2018. ISBN 978-0132380331.

Reference Books :

1. Joseph Migga Kizza, *Guide to Computer Network Security*, (5th ed.), Springer. 2020. ISBN 978303038140.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET228	Data Mining and Predictive Modelling	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	2	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate data preparation for data mining and analyzing based on pre-processing techniques.

CO2: To examine predictive analysis in various use cases.

CO3: To make use of exploratory data analysis to gain insights and prepare data for predictive modelling.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	2	2		2	2	2		2	2	2	2	2	2	2
CO2	2	3	2		2	2	2	2		2	2	2	2	2	2
CO3	2		2		2			2		2					

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

11 lecture hours

Purpose of Data mining, Procedures of Data Mining, Functionality of Data Mining, Knowledge data discovery process, Data and attribute type, Properties of data, Discrete and continuous attribute, Dataset types, Data quality measurement, Noise Analysis and its importance, Techniques of Data pre-processing, Aggregation, Sampling, Curse of dimensionality, Dimensionality reduction, Feature selection and generation, Discretization and vectorization, Binarization, Attribute transformation correlation, Association rule mining, Apriori algorithm, Rule generation, Pattern Mining in: Multilevel, Multidimensional Space Pattern Mining .

Modulo II:

7 lecture hours

Rule-based reasoning, Memory-based reasoning, measuring data similarity, Similarity Metrics: Distance based measure, Information based measures, Set similarity measure, Jaccard Index, Sorenson Dice Coefficient, Model Selection Problem, Error Analysis, Case study, Startups in Data Analysis.

Module III:

10 lecture hours

Outlier analysis in classification and clustering, Probabilistic models for clustering, Clustering high dimensional data: Subspace clustering, Projection Based clustering, Exploratory data analysis, Data summarization and visualization, Dataset exploration, Data Exploration Tools, Interactive Data Exploration, Predictive models, Design Principles, Parametric Models, Non-Parametric Models, ANOVA, Regression Analysis, Frequent Pattern Mining, Mining Closed and Max Patterns.

Module IV:**14 lecture hours**

Linear discriminant analysis, Fisher discriminant analysis, Time series Model: ARMA, ARIMA, ARFIMA, Factor Analysis, Uncertainty quantification, Forward uncertainty propagation, Inverse uncertainty quantification, Non-Negative Matrix Factorization, Sequential Matrix Factorization. Exact Matrix Factorization, Expert Lecture from Industry, Recommendation System and Collaborative Filtering, Multidimensional Scaling, Mining Textual Data, Temporal mining, Spatial mining, Visual and audio data mining, Ubiquitous and invisible data mining- Privacy, Security, Social Impacts of data mining.

Studio Work / Laboratory Experiments:

Data pre-processing and vectorization. Quality analysis of data. Feature selection and Ranking. Association rule mining and implementation of the Apriori algorithm. Data Similarity and set similarity. Error analysis and model selection. Frequent pattern mining and regression. Discriminant Analysis. Factor Analysis. Matrix Factorization. Recommendation System.

Text Books :

1. Bruce Ratner, *Statistical and Machine-Learning Data Mining: Techniques for Better Predictive Modeling and Analysis* (3rd ed.), Chapman and Hall/CRC, 2017. ISBN 978-1498797603.
2. Dursun Delen, *Predictive Analytics* (1st ed.), Knime, 2020. ISBN 9780136738516 .

Reference Books :

1. Mohammed J. Zaki and Wagner Meira, Jr, *Data Mining and Machine Learning* (1st ed.), Cambridge University Press, 2020. ISBN 9781108473989 .

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET229	Game Engine and Architecture	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	2	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain concepts of underlying game engines. Solve practical 3D game problems.

CO2: To Implement Game control systems for multiplayer games using Human Interface Devices and analyse the basic features of Physics for Games; Illustrate the collision detection systems used in games.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	2	3	2	2	2	2	2	2		2	2	3	2	3
CO2	2	3	2	2	2	2	2		2	2	2	2	2	3	2

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

Game Engine, Engine Differences Across Genres, Runtime engine architecture: Base layer, Memory, layer, Utility, Module, Management, Gameplay, Build layers, Asset Pipeline, Pre-rendering pipeline, Full 3D pipeline, Game profiler, Tools for optimization of games, Types of profilers, Version control in game development, Best Practices in version control, Issues and solutions, Unity engine and features, Unreal engine and features, Amazon Lumberyard and features, cryEngine for VR and features

Modulo II:

10 lecture hours

Engine Levels, Game Engine VR Modes actors, Landscape, Water rendering, Lightening and Environment, Capsule shadow, light mobility, and types of lights, Volumetric clouds, Fog effects, Level streaming, Hierarchal level of details, Blueprints, Variables, functions, and flow control, Timelines and blueprint classes, Render To Texture Blueprint Toolset, Occlusion Culling, Render targets, Real time raytracing, Frame Rendering tool, Programming and scripting, Actor Communication, Blueprint visual scripting, Gameplay Architecture, Gameplay classes and functions, Interfaces, Graphics Programming, Shader development, Threaded rendering, Mesh drawing pipeline, Slate UI Framework.

Module III:

10 lecture hours

Human Interface Devices, Types of HID for games, Interfacing with HID, Devices in practice, features of popular devices, Gameplay Framework, Pawn, Controller, Camera, AI Entities in gaming, Behavior trees, navigation system, Environment query system, Physics in gaming, UMG UI Designer, Networking and multiplayer, Animation Characters and Objects, Animation rigging toolset, Skeletal

Mesh Animation System, Sound in gaming, Ambient Zones, Sound classes, Audio Engine and mixer, Sound Attenuation, Sound cue.

Module IV:

12 lecture hours

Importing and integration media objects, Asset management, Asset Build tool, Automation tool, Testing and Optimization, Performance and profiling, Animation System Architecture, Animation Blending, Animation Pipeline, Skinning and Matrix Palette Generation, Skeletons 496x, Animation controller Collision and Rigid Body Dynamics, Collision/Physics Middleware, The Collision Detection System, Rigid Body Dynamics, and Advance physics features.

Studio Work / Laboratory Experiments:

The lab covers the practice of game engine software development methodologies for Iterative and agile development of a vertical slice of a game from end-to-end e.g. SCRUM. XP, Kanban, TDD etc. and to the design and implementation of subsystems for rendering, collision, physics animation.

Text Books :

1. Jason Gregory, Jeff Lander and Matt Whiting, *Game Engine_Architecture* (3rd ed.), A K Peters/CRC Press, 2018. ISBN 978-1138035454.

Reference Books :

1. Heather Maxwell Chandler, *Game Production Handbook* (3rd ed.), Jones & Bartlett, 2013. ISBN 978- 1449688097.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET230	DevOps Practices and Principles	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	2	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate DevOps engineering and DevOps terminologies to meet the business requirements.

CO2: To construct different applications of DevOps that can be used on different platforms.

CO3: To design and execute projects using different tools, modules, and templates.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	2	2	2	2		2	2	2	2	2	2	3	2	2
CO2	2	2	2	2	3	2	2		2	2	2	2	2	2	2
CO3		2	3	2	3	2	2	2	3	2	3	3	2	2	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

12 lecture hours

Why DevOps? , Terminologies, DevOps Stakeholders and roles, DevOps Perspective: People, Process and Product, Building Teams, Boston Consultancy Group (BCG) Matrix, Three Horizons Model, Autonomous Team and its Pros and Cons, Autonomy Criteria, Decoupling Point for Autonomous Team, Case Study on Autonomous Team, DevOps and ISTM (IT Service and Management), Traditional Vs Agile DevOps, DevOps Phases with diagram representation, Introduction of Source and Version Control(SVC), Algorithms for SVC: Lock-Modify-Unlock and Copy-Modify -Merge, Continuous Integration, and Deployment (CICD), Software Craftmanship, Software Containerization, System Provisioning, and Configuration Management, Testing and Test Automation.

Modulo II:

11 lecture hours

Service-Oriented Architecture and Microservices, Build and Release Management, Virtual Systems, Hypervisor Cloud Computing, Need of Cloud Computing with Application, Delivery Models of Cloud Computing, Deployment Model of Cloud Computing, Git Desktop Usage, and Benefits, Connecting Git Desktop and Git Online, Git Version Control, and Types, Distributed Version Control Systems with Example, Centralized Version Control Systems with Example, Configuration Management, Chef Configuration, Workstation Setup, Configuration of the knife, Test Connection between Knife and Workstation, Organization Setup, Creation and Connection of Node to Organization, Operations on Node, Object and Search, Creation of Environments and Connecting them with Servers.

Module III:

6 lecture hours

Puppet, Puppet Working and Architecture, Master and Agents in Puppets, Installing Puppet, Configuring Puppet Master and Agent, Connecting Agents, Puppet Language, Declarative Language, Resources, Files, Exec, Packages, Service in Puppets, Virtual Resources and Exported Resources.

Module IV:

13 lecture hours

DevOps Security, Issues in DevOps Security, DevOps Security Needs and Challenges, DevOps Risks Management, Strategies, Policies, Practices for DevOps Security, DevOps Security Tools, DevOps security over Clouds, Security Models, and their Use, Modules and Templates for DevOps, DevOps Manifests, Class Templates Static and Dynamic Content, ERB Module, NTP Module, SSH Module, Sudo Module, Install LAMP with pre-existing Modules.

Studio Work / Laboratory Experiments:

The Studio work includes the practical understanding of multiple tools and languages that are used in DevOps Industry. DevOps practices aims at merging development, quality assurance, and operations (deployment and integration) into a single, continuous set of processes.

Text Books :

1. Reed Mark, *DevOps: The Ultimate Beginners Guide to Learn DevOps Step-by-Step* (1st ed.), Publishing Factory, 2020. ISBN 978-1647710941.
2. Gene Kim, Kevin Behr, and George Spafford, *The Phoenix Project* (5th ed.), IT Revolution Press, 2019. ISBN 978- 1942788294.

Reference Books :

1. Gene Kim and John Willis, *Beyond the Phoenix Project: The Origins and Evolution of DevOps* (1st ed.), IT Revolution Press, 2018. ISBN 978-1942788256.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET231	Programming Methodologies for Backend Development	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	4	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate the client-Server architecture and the basics of back-end technologies and tools.

CO2: To examine server-side programming languages like PHP and Node.js.

CO3: To connect with database with the help of server-side scripting-PHP and MySQL, Node.js and MongoDB.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2		3									1	3		
CO2	2		3									1	3		
CO3	2		3									1	3		

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

12 lecture hours

Back-end technologies: servers, applications, and databases, Object-oriented based backend programming languages: PHP, Java, Node.js. MERN Stack: Mongo DB, Express.js, React.js, Node.js. functional-based backend programming languages (Mongo DB), Databases, MySQL, XHTML, PHP- a scripting language. PHP: server-side scripting and its syntax, working with variable and constant, PHP code blocks: arrays, strings, function. PHP error handling and Controlling program flow, looping, and branching, file handling, directories, PHP sessions and security, processing forms on server-side, cookies, Web services, exploring Java-based web technologies: JDBC, servlet, JSP. Web architecture model: client-server model, three-tier model, and service-oriented architecture (SOR), MVC architecture: model, view, controller.

Modulo II:

7 lecture hours

Database programming: Operations and working on metadata, database configuration, connection to MySQL server, execution of MySQL queries, Node.js: Web Applications with Node.js, Core Node.js and Packages, Events Streams, Modules: export, object, class. Loading module from a separate folder, File Systems. MongoDB, Features of MongoDB, MongoDB Database Tier, using Node.js with MongoDB, using Node.js with MySQL, Server-side rendering.

Module III:

9 lecture hours

REPL environment and commands, Backend frameworks: Django, Spring, Express. Docker Containers, GraphQL integration, API documentation: Swagger API, Postman (API testing), REST Principles for API, Containerize APIs with Docker, Serverless computing, DevOps toolkit.

Studio Work / Laboratory Experiments:

Practical experience of server-side programming languages like PHP and Node.js by connecting with database with the help of server-side scripting-PHP and MySQL, Node.js, and MongoDB.

Text Books :

1. Herron David, *Node.js Web Development: Server-side web development made easy with Node using practical examples* (5th ed.), Packt Publishing Limited, 2020. ISBN 978-1838987572.

Reference Books :

1. Zammetti Frank, *Modern Full-Stack Development: Using TypeScript, React, Node.js, Webpack, and Docker* (1st ed.), Apress, 2020. ISBN 978- 1484257371

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET232	Design of Cloud Architectural Solutions	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	3	0	2	4	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To learn stack of cloud architecture and design guidelines of the final product.

CO2: To build the cloud architectural blueprint for deployment.

CO3: To deploy and implement cloud architectural designed solution.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	2	2	2	2	2	2	2	2	2	2	2	3	2	2
CO2	2	3	3	2	2	2	2	2	2	2	2	2	2	3	3
CO3	2	2	3	3	2	2	2	2	2	2	2	2	2	2	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

11 lecture hours

Cloud Architecture Job roles, Solution Architect Certifications, Cloud Solution Architect Competitions, Cloud Computing with characteristics, Core Cloud Services, Loose Coupling, Loose Coupling Strategies, Communication in coupled components, Decoupling of Infrastructure, Scalable Web Application deployment, Multi-Tenancy, Resource Provisioning, Traffic / Demand Monitoring, Capacity Planning and Prediction, Data centre management, Costing and Pricing of public cloud services, Cloud Governing Rules, Cloud Privacy issues, Design Principles for using cloud services, Design Principles for providing cloud services, Cloud architectural design principles, Principles of the Security, Principles of the Reliability Pillar, Principles of the Performance Efficiency Pillar, Cost Optimization, High-Availability Design Patterns, Criteria for cloud service selection, Multi-criteria Decision Making, Cloud Infrastructure security, IAM user, group, roles, and policy.

Modulo II:

11 lecture hours

Cloud compliances, Authentication protocol implementation, Content Delivery, Domain name System, Content delivery network, CDN Demo, Traditional network architecture, SDN architecture, SDN controller, OpenFlow protocol, Demo of SDN, Discussion of related Start-up, Traditional infrastructure, Virtual infrastructure, Converged infrastructure, IoT-Cloud Infrastructure, Autoscaling, Launch configuration, Autoscaling groups, ASG Balancing, Autoscaling policies, Load Balancer, Fault tolerance, High availability, Architectural need of LB, Application LB, Network LB, Listener, Target groups, Targets, Cross zone load balancing.

Module III:

9 lecture hours

Case Study of a federated cloud, Mobile "backend" as a service (MBaaS), Serverless computing, Business Processing as a service (BPaaS), Virtualization Demo vs Containerization Demo, Virtualization and Containerization, Virtualized Operating Systems, Resource to OS-Hypervisor communication, Design of virtualized Database Clusters, Cloud Regional Backbone Networks, Network Troubleshooting, Transcoding and Serving Video Files Example, Stream Processing Example, Sensor Network Data Ingestion and Processing Example, Application Backend Example.

Module IV:**11 lecture hours**

Cloud Monolith Applications, Cloudna-tive Applications, Cloud Enterprise Applications, Challenges towards Hypergrowth of cloudna-tive applications, Evolution of Deployment, Current Cloud Computing System Designs, Modern Cloud Architecture Integration, Structure of a Cloudna-tive Application, Characteristics of Cloudna-tive Application, Bridging DevOps Culture with cloud architectural solution, Software development on cloud PaaS, Microservices, Microservice Architecture, Microservice Architecture v/s Monolith Architecture, Developing Cloudna-tive Applications using Microservices, Flexibility and Scalability in microservice, Pros and Cons of Cloud native Architecture, Microservice Use Cases, Service Decomposition, Microservice Demo, Strategies for Decomposing a System into Microservice, Migrating Monolithic Solutions to Microservice Ecosystem, Minimizing Dependency to Monolithic Solutions, Orchestration / Choreography of Micro-Services, Shared Data and Communication, Case Study on Cloudna-tive Development, Advanced

Studio Work / Laboratory Experiments:

Studio works include Hands-on experience on public industry cloud platform and will provide the cloud-based solution for the business problems. The activities that are mandatory to be completed includes troubleshooting the errors, accessing various AWS/Azure/GCP Support Options, using compute, storage, database, and networking concepts to improve the architecture's performance, identify design principles that can help cloud user to achieve performance efficiency, Evaluate the most important performance metrics for any applications. Research Topics.

Text Books :

1. Goessling Scott and Kevin Jackson, *Architecting Cloud Computing Solutions: Build cloud strategies that align technology and economics while effectively managing risk* (1st ed.), Packt Publishing, 2018. ISBN 978-1788472425.

Reference Books :

1. Bahga Arshdeep and Vijay Madisetti, *Cloud Computing Solutions Architect: A Hands-On Approach* (1st ed.), Vpt, 2019. ISBN 978194997801X .

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET233	Machine Learning with Quantum Computing	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	4	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explore and understand whether quantum computers can speed up the time it takes to train or evaluate a machine learning model.

CO2: To examine the interplay of ideas from quantum computing and machine learning.

CO3: To develop new quantum algorithms and explore the properties of quantum systems.

CO-PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1		2	3									3			3
CO2		2	3									3			3
CO3		2	3									3			3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I

8 lecture hours

Course structure/handout assessment mechanism, why quantum machine learning? Tensor and Tensor Network, Hadamard Transformation, Quantum lambda calculus, Quantum Encode and Decode, Arbitrary StateGenerator, Quantum Density Operator or Density Matrix, Quantum Probability Theory, Quantum Stochastic Process, Quantum Mean, Variance and Envariance, Quantum Way of Linear Regression, Quadratic Unconstrained Binary Optimization, Quantum Topological Data Analysis, Quantum Bayesian Hypothesis, Haar Transform, Quantum Ridgelet Transform, Variational Quantum Eigen-solver.

Module II

6 lecture hours

Quantum Perceptron's, Quorn's, Quantum Auto Encoder, Quantum kernel estimation, Quantum Feature Map, Comparisons between QKE and classical kernels, Quantum Classifier, Variational Models, Quantum Approximate Optimization Algorithm, Quadratic Unconstrained Binary Optimization (QUBO), MaxCut problem.

Module III

8 lecture hours

Quantum Annealing, Photonic Implementation of Quantum Neural Network, Quantum Feed Forward Neural Network, Quantum Boltzman Neural Network, Quantum Neural Net Weight Storage, Quantum Upside Down Neural Net, Quantum Hamiltonian Neural Net, Quantum artificial neural network, Quantum Perceptron Network,

Quantum approaches to CNNs, Quantum convolutional networks, Dissipative quantum neural networks, Barren plateaus in QNNs, Barren plateaus in deep QNNs, Noise-induced Barren plateaus, Superposition based Architectural Learning Algorithm, Quantum Hamiltonian Learning, Compressed Quantum Hamiltonian Learning, Quantum Classification using Principle Component Analysis, Quantum state classification with Bayesian methods, Quantum K-Nearest Neighbour, Quantum K-Means, Quantum Fuzzy C-Means.

Module IV

6 lecture hours

Quantum Support Vector Machine, Quantum Ant Colony Optimization, Quantum-behaved Particle Swarm Optimization, Quantum Inspired Evolutionary Algorithm, Quantum Genetic Algorithm, Quantum Hidden Markov Models, Quantum Annealing Expectation-Maximization, Quantum Estimation Theory, Quantum Statistical Decision Theory, Quantum Hypothesis Testing, Quantum Chi-squared and Goodness of Fit Testing.

Studio Work / Laboratory Experiments:

In-studio work, students will work on various machine learning models and classical and Quantum algorithms using quantum computers.

Text Books:

1. Siddhartha Bhattacharyya, Sourav De, Indrajit Pan, Ashish Mani, Elizabeth Behrman and Susanta Chakraborti, *Quantum Machine Learning* (1st ed.), De Gruyter, 2020. ISBN 9783110670704.
2. Eric R. Johnston, Nic Harrigan and Mercedes Gimeno-Segovia, *Programming Quantum Computers* (1st ed.), O'Reilly, 2019. ISBN 9781492039659 .

Reference Books:

1. Klaus-Robert Müller, Kristof T. Schütt, Stefan Chmiela, Alexandre Tkatchenko, Koji Tsuda *Learning Meets Quantum Physics* (1st ed.), Springer, 2020. ISBN 9783030402452

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET234	Drone Remote Sensing	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	4	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate the fundamentals of UAV Remote Sensing.

CO2: To learn techniques and deployment scenarios of UAVs, alongside the challenges and potential solutions to data acquisition, mapping, and processing.

CO3: To construct multirotor, agricultural precision remote sensing UAV from scratch using various sensors, cameras, and flight controllers.

CO – PO /PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	2	3										3		
CO2	1	2	3										3		
CO3	1	2	3										3		

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I

8 lecture hours

Sensing and Control; Application Scenarios and Challenges; Radio Controlled Vehicles to UAVs; Sensors: Sensor Types and Applications; Regulations and Guidelines; Attitude Estimation: Sensors, Complimentary Filters, Kalman Filters; UAV Imagery: Accuracy Requirements, Operational Restrictions.

Module II

7 lecture hours

Image Sensors; Image Sensor Selection; Computing Image; Imagery Collection; UAV Remote Sensing; Coverage Control; Georeferencing; Feature-Based Stitching; Position and Attitude Based Stitching; Consensus-Based Formation Control; Consensus Algorithms; Profile Measurement and Estimation; Wind Profile Measurement.

Module III

7 lecture hours

Mission Planning: Surveyed Geography, Alien Terrain; GPS Data Collection; Measurement; Mapping; UAV Path Planning; Path Refinement; Multi-UAV Path; Obstacle Avoidance; Focus of Expansion; Time to Collision; Semi-Autonomous Control; Radio Control; Automatic Control; Control Switching: Radio and Automatic.

Module IV

6 lecture hours

Drone Sensing and Imagery: Riverine, Aquatic, Agriculture, Vegetation and Forest Ecosystem, Oil and Gas, Surveillance, Search and Rescue; UAS Ground Control; Human Machine Interface; Computer Telemetry; UAV Launch System; UAV Recovery System.

Studio Work / Laboratory Experiments:

In this course, students will start with basic remote sensing deployments, develop an understanding of various sensors and cameras, and understand the principles of data acquisition and aggregation. components, and assembly precautions. Then finally design and develop a fully functional agricultural precision remote sensing Multicopter UAV/Drone.

Text Books:

1. Serge A. Wich and Lian Pin Koh, *Conservation Drones: Mapping and Monitoring Biodiversity* (1st ed.), OUP Oxford, 2018. ISBN 0198787618.

Reference Books:

1. Amy Frazier and Kunwar Singh, *Fundamentals of Capturing and Processing Drone Imagery and Data* (1st ed.), CRC.
2. David R. Green, Billy J. Gregory and Alexander Karachok, *Unmanned Aerial Remote Sensing* (1st ed.), CRC Press, 2020. ISBN 1482246074..

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET235	Digital Bots Development	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	4	4
Pre-requisites/Exposure	-				

Course Outcomes (Cos)

On completion of this course, the students will be able to:

CO1: To understand the underlying logic/structure related to robotic process automation.

CO2: To develop digital bots to automate the processes.

CO3: To implement the transactions in business processes using robotic process automation.

CO – PO /PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	POS3
CO1			3									1	2		3
CO2			3									1	2		3
CO3			3									1	2		3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I

06 lecture hours

Course structure/handout Assessment mechanism, Automation 360, Using Discovery Bot for Process Documentation, Creating Discovery Bot Users, Use Case: Discovering the Customer ID Validation, Create Process, Record Process, Review Recordings and Create Opportunities, Convert Opportunity to Bot Prototype, Resilient Bots, Run time challenges, HandlingUnpredictable Exceptions, Handling Predictable Exceptions, Modularity and reusability are the key to scalability, Creating scalable Bots, Universal recorder Excel automation.

Module II

08 lecture hours

Integration of Java Scripts with Automation Anywhere, Invoking of Java Script using automation 360, Integration of VB Script in Automation Anywhere, Code execution of VB Script using Automation Anywhere, Integration of Python script with Automation Anywhere, Running Python Script in Automation Anywhere. Automation anywhere Robotic Interface with back office business process use case, Creating form Creation of the Bots, Creation of Process, Automation anywhere Robotic Interface with front office business process use case, Creating form, Creationof the Bots, Creation of Process, Automation anywhere Robotic Interface process creation, Automation anywhere Robotic Interface task creation.

Module III

08 lecture hours

AAE Client in RPA Development, Smart recorder, Screen recorder, Web recorder, Editing a recorded Bot Workbench components, Leveraging workbench commands, Integrating RPA with Cognitive

Solutions Understanding IQ Bot as a Cognitive solution, Utilizing the IQ Bot Portal, Following the IQ Bot Workflow Creating an Instance and Triggering Document analysis, Reading Instance Details, Editing an Instance.

Module IV

06 lecture hours

Understanding RPA Analytics, Understanding Operational Analytics, Understanding Business Analytics Roles Generating Business Analytics Understanding the CoE Dashboard, Customizing and Comparing CoE Dashboards, Publishing a CoE Dashboard, Using the RPA Mobile App, Recording of task with AISense recorder, Edit a task recorded using AISense, Advanced Topics in AA, Use of variable anchor, Enable debug logs for AISense recorder.

Studio Work / Laboratory Experiments:

In this course, students will learn task organization for automation using RPA Tools.

Text Books:

1. Mullakara, Nandan, and Arun Kumar Asokan, *Robotic process automation projects: Build real-world RPA solutions using UiPath and Automation Anywhere* (1st ed.), Packt Publishing Ltd, 2020. ISBN 9781839217357.
2. Husan Mahey, *Robotic Process Automation with Automation Anywhere: Techniques to fuel business productivity and intelligent automation using RPA* (1st ed.), Packt Publishing Ltd, 2021. ISBN missing.

Reference Books:

1. Husan Mahey, *Robotic Process Automation with Automation Anywhere* (1st ed.), Packt Publishing Ltd, 2021. ISBN 9781839215650.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET236	IoT Networks and Protocols	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	4	4
Pre-requisites/Exposure	-				

Course Outcomes (COs):

On completion of this course, the students will be able to:

CO1: To articulate the protocols and standards designed for IoT.

CO2: To examine the basic protocols IoT and WSN.

CO3: To implement IoT solutions using sensors, actuators, and Devices.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	L	M	H									L			H
CO2	L	M	H									L			H
CO3	L	M	H									L			H

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I

09 lecture hours

Internet of Things, Sensors, Hysteresis Error, Quantization error, aliasing error, Actuator, Actuators, IoT Networking, Functional components of IoT, IoT dependencies, IoT service-oriented architecture, IoT categories, IoT gateways, Associated technologies, technical derivation from the regular web, IoT protocols, MQTT, SMQTT, CoAP, XMPP, IEEE802.15.4, AMQP.

Module II

08 lecture hours

6LoWPAN, LoRAWAN and Reference model, Integration of devices using LoRAWAN, Security in LoRAWAN, Zigbee, 3GPP, NB-IoT, Wireless HART, RFID, ISA100, Z-Wave, WSN, Cluster formation of sensors in WSN, Routing algorithms in WSN, UAV Network, UAV Navigation, 5G based communication among UAVs.

Module III

06 lecture hours

Machine to Machine communication, Architecture, and components for M2M, Standardization Effort for M2M, Interoperability in IoT, IoT Architecture for Interoperability, Industry Standards, SDN Origins and Evolution, Centralized and Distributed Control, Data Planes, Genesis of SDN, API in SDN, Control mechanism, Switch Deployment, Controller configuration software, SDN for WSNs, SDN-WISE Sensor Nodes, SDN-WISE Protocol Structure, Topology in SDN-WISE.

Module IV

05 lecture hours

Software-Defined WSN Prototype, Situation-Aware Protocol Switching in SDN, Performance Analysis of Software Defined Networks, Sensor cloud, Architecture, Service life cycle model, Layered structure, Management issues in sensor cloud, Optimal composition of virtual sensors, Formation of virtual sensor group.

Studio Work / Laboratory Experiments:

Studio works aim to provide hands-on experience of IoT devices to understand the communication part between the devices. Here, we are sending the data using different communication devices such as WiFi, Zigbee and Bluetooth. There are a set of experiments over web layer protocols such as MQTT, HTTP and CoAP. This course is also having projects to solve a real-life problem by using IoT networks and protocols.

Text Books:

1. David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Robert Barton and Jerome Henry, *IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things* (1st ed.), Cisco Press, 2017. ISBN 1587144565.

Reference Books:

1. S. Misra, A. Mukherjee and A. Roy, *Introduction to IoT* (1st ed.), Cambridge University Press, 2021. ISBN 1108959741.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET237	VR and 360 VideoProduction	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	2	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To learn how to create a VR use flow.

CO2: To use Tools and Gaming Platforms to build VR experiences and 360 Video production.

CO3: Make next steps to start building a VR experience using latest tools and technologies.

CO – PO /PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	POS3
CO1		2	3									1			3
CO2		2	3									1			3
CO3		2	3									1			3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I

10 lecture hours

Intro to design: Understand basic VR design principles such as iteration, user testing, and documentation, Design foundations, ergonomics: Create a VR User Persona, design an ergonomicVR experience, VRText, VR Testing Scene.

Module II

12 lecture hours

Introduction to Immersive Media: principles of immersive 360 media such as spherical projection by critically reviewing professional 360 content, workflow of creating 360 Video, Metadata to 360video and publishing, Analyze a variety of existing 360 video content, Analyze footage for the various techniques, tips, and tricks, Scripting and Planning: Create a script for 360 content, Plan a 360 shoot, Edit scripts to match equipment and budget, 360 cameras: Exploration of existing 360 cameras, Differentiate between monoscopic and stereoscopic capture, different equipment and techniques used for audio capture, lighting techniques for 360 video.

Module III

12 lecture hours

Stitching: Stitch 360 video, Blend, weight, synchronize, and stabilize footage, Editing Basics: Edit footage, Create transitions in 360 video, Advanced Editing: Color correct 360 footage, Add pointsof interest, Advanced Stitching: edit and refine stitching using control points and masks, Blending.

Module IV

08 lecture hours

Spatial Audio: Cut spatialized audio, Place sound in 3D space, Interactivity with Gaming platform: Create an interactive 360 experience using Gaming platform's video player, player controls and branching storylines using a custom 360 video player.

Studio Work / Laboratory Experiments:

The experiments to create 360 video with metadata, capture and use 3D audio to 360 videos. Adding interaction control, adding transition in 360 videos, 360 video testing and performance Evaluation.

Text Books:

1. Jesse Glover and Jonathan Linowes, *Complete Virtual Reality and Augmented Reality Development with Unity* (1st ed.), Packt Publishing Limited, 2019. ISBN 9781838644865

Reference Books:

1. Erin Pangilinan, Steve Lukas and Vasanth Mohan, *Creating Augmented on and Virtual Realities* (1st ed.), O'Reilly Publishers, 2019. ISBN 978- 1492044192.
2. Dong Hwa Choi, Judi Estes and Amber Dailey-Hebert, *Current and Prospective Applications of Virtual Reality in Higher Education* (1st ed.), IGI Global, 2020. ISBN 9781799849619

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET239	Product Design Architecture andDelivery	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	3	0	2	4	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate product design architecture and communication in designing.

CO2: To examine all mandatory models by developing the prototype.

CO3: To make use of development methods that can be exploited in the delivery of the product.

CO – PO /PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	3									1		3	
CO2	2	2	3									1		3	
CO3	2	2	3									1		3	

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I

10 lecture hours

What Is Product Architecture? Types of Modularity, Implications of the Architecture, Product Change, Product Variety, Component Standardization, Product Performance, Manufacturability, Product Development Management, Establishing the Architecture, Create a Schematic of the Product, Cluster the Elements of the Schematic, Create a Rough Geometric Layout, Identify the Fundamental and Incidental, Platform Planning, Differentiation Plan, Commonality Plan, Defining Secondary Systems, Creating Detailed Interface Specifications, Industrial Design, Assessing the Need for Industrial Design ,Expenditures for Industrial Design, Industrial Design to a Product, User Experience Needs ,Aesthetic Needs, The Impact of Industrial Design, Is Industrial Design Worth the Investment? How Does Industrial Design Establish a Corporate Identity? The Industrial Design Process, Investigation of Customer Needs, Conceptualization, Preliminary Refinement, Further Refinement and Final Concept Selection, Control Drawings or Models, Coordination with Engineering, Manufacturing and External Vendors, Management of the Industrial Design Process, Timing of Industrial Design Involvement, Assessing the Quality of Industrial Design Usability, Emotional Appeal, Ability to Maintain and Repair the Product, Appropriate Use of Resources, Product Differentiation.

Module II

10 lecture hours

Design for Environment, Environmental Impacts, Herman Miller's Journey toward Design for Environment, The Design for Environment Process, Design for Environment Guidelines, Design for Manufacturing and Supply Chain, Requires a Cross-Functional, Overview of the DFM Method, Strategic Sourcing Decisions, Estimate the Manufacturing Costs, Cost of Goods, Fixed Costs versus Variable Costs, The Bill of Materials, Estimating the Costs of Standard Components, Estimating the

Costs of Custom Components, Estimating the Costs of Assembly, Estimating the Overhead Costs, Reduce the Costs of Components, Understand the Process Constraints and Cost Drivers, Redesign Components to Eliminate Processing Steps, Choose the Appropriate Economic Scale for the Part Process, Standardize Components, Adhere to the “Black Box” Component, Procurement, Reduce the Costs of Assembly and Integrate Parts, Maximize Ease of Assembly, Consider Customer Assembly, Reduce the Costs of Supporting, Production, Minimize Systemic Complexity, Error Proofing, Reduce the Costs of Logistics, guidelines for minimizing the volume, Materials Costs, Component Manufacturing Costs, Assembly Costs, Cost Structures, Impact of DFM Decisions: The Impact of DFM on Development Time, The Impact of DFM on Development Cost, The Impact of DFM on Product Quality, The Impact of DFM on the Larger Enterprise.

Module III

14 lecture hours

Prototyping, Understanding Prototypes, Types of Prototypes, What Are Prototypes Used For? Principles of Prototyping, Analytical Prototypes, Physical Prototypes, Prototyping Technologies, 3D Printing, Planning for Prototypes, Define the Purpose of the Prototype, Establish the Level of, Approximation of the Prototype, Outline an Experimental Plan, Create a Schedule for Procurement, Construction, and Testing, Planning Milestone Prototypes, Robust Design, Design of Experiments, Robust Design Process, Identify Control Factors, Noise Factors, and Performance Metrics, Formulate an Objective Function, Develop the Experimental Plan, Experimental Designs, Testing Noise Factors, Run the Experiment, Conduct the Analysis, Computing the Objective Function, Computing Factor Effects by Analysis of Means, Select and Confirm Factor Setpoints, Reflect and Repeat Caveats, Patents and Intellectual Property, What Is Intellectual Property? Overview of Patents, Utility Patents, Preparing a Disclosure, formulate a Strategy and Plan, Timing of Patent Applications, Type of Application, Scope of Application, Study Prior Inventions, Outline Claims, Write the Description of the Invention Figures, Writing the Detailed Description, Defensive Disclosure, Refine Claims, Writing the Claims, Guidelines for Crafting Claims, Pursue Application, Reflect on the Results and the Process, Advice to Individual Inventors, Service Design Product-Service Systems, In What Ways Are Services and Products Different? Service Design Process, The Service, Development at Zipcar, Service Process Flow Diagram, Subsequent Refinement, Downstream Development Activities in Services, prototyping a Service, Growing Services, Continuous Improvement.

Module IV

08 lecture hours

Product Development Economics, Elements of Economic Analysis, Quantitative Analysis, Qualitative Analysis, Economic Analysis Process, Build a Base-Case Financial Model, Estimate the Timing and Magnitude of Future Cash Inflows and Outflows, Perform Sensitivity Analysis, Development Cost Example, Development Time Example, Understanding Uncertainties, Use Sensitivity Analysis to Understand, Trade-Offs, Potential Interactions, Trade-Off Rules, Limitations of Quantitative Analysis Projects Interact with the Firm, the Market, and the Macro Environment, Carrying Out Qualitative Analysis, Time Value of Money and the Net Present Value Technique, Modeling Uncertain Cash Flows Using Net, Present Value Analysis, Project Management, The Design Structure Matrix, Gantt Charts PERT Charts, Critical Path, Baseline Project Planning, Contract Book, Project Task List, Team Staffing and Organization, Project Schedule, Project Budget, Project Risk Plan, Modifying the Baseline Plan, Accelerating Projects, Project Execution, Coordination Mechanisms, Assessing Project Status Corrective Actions, Postmortem Project Evaluation.

Studio Work / Laboratory Experiments:

In the studio work, students perform the architectural design of the product and deliver the product design prototype.

Text Books:

1. Karl T. Ulrich, Steven D. Eppinger and Maria C. Yang, *Product Design and Development* (7th ed.), McGraw Hill, 2020. ISBN 9390113237.

Reference Books:

1. Desai, Anoop, and Anil Mital, *Sustainable Product Design and Development* (1st ed.), CRCPress, 2020. ISBN 978-0367343217.
2. Ali Jamnia, *Introduction to Product Design and Development for Engineers* (1st ed.), Taylor and Francis, 2018. ISBN 978-1138554214.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET242	Advance Swift Programming	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	4	4
Pre-requisites/Exposure	-				

Course-Outcomes (COs):

On completion of this course, the students will be able to:

CO1: To articulate features of Object-Oriented Programming concepts in Swift.

CO2: To implement object-oriented features of Swift using Playgrounds.

CO3: To solve real-world problems using Swift concepts.

Course Contents:

Module I

6 lecture hours

Closure Expressions, Inferring Type From Context, Implicit Returns from Single- Expression Closures, Shorthand Argument Names, Operator Methods, Trailing Closures, Capturing Values, Escaping Closures. Enumeration: Enumeration, Enumeration with Switch Statement, Iterating Enumeration Cases, Associated Values, Raw Values, Recursive Enumerations.

Module II

6 lecture hours

Definition Syntax, Structure and Class Instances, Accessing Properties, Memberwise Initializers for Structure Types, Value types or Reference Types. Properties, Stored Properties, Lazy Stored Properties, Computed Properties, Property Observers. Global and Local Variables, Type Properties, Type Property Syntax, Querying and Setting Type Properties.

Module III

8 lecture hours

Methods, Instance Methods, self-Property, Mutating Method, Type Methods Inheritance: Base class, types of Inheritance, Subclassing, Overriding: Accessing Superclass Methods, Properties, and Subscripts, Overriding Methods, Overriding Properties, Overriding Property Getters and Setters, Preventing Overrides.

Module IV

8 lecture hours

Initializers, Default Property Values, Customizing Initialization, Initialization Parameters, Parameter Names and Argument Labels,_INITIALIZER Parameters Without Argument Labels, Optional Property Types, Default Initializers,_INITIALIZER Delegation for Value Types, Class Inheritance and Initialization,_INITIALIZER Inheritance and Overriding, Automatic_INITIALIZER Inheritance, Failable Initializers, Failable Initializers for Enumerations, Overriding a Failable_INITIALIZER.

Studio Work/ Laboratory Experiments:

Students will gain practical experience with the implementation of different swift concepts. The lab work consists of swift programming using Playground and Xcode.

Text Books:

1. Matthew Mathias, John Gallagher, Swift Programming: The Big Nerd Ranch Guide edition, 2015.
2. Matt Neuberg, iOS 12 Programming Fundamentals with Swift, O'Reilly; 5th edition. App Development with Swift (as available on iBookStore).

Reference Books:

1. Paris Buttfield-Addison, Jonathon Manning, Tim Nugent Learning Swift: Building App macOS, iOS, and Beyond, O'Reilly Media, Inc., 3rd ed, 2018.
2. Yamacli, Beginner's Guide to iOS 11 App Development Using Swift 4: Xcode, Swift

Name of Program	Bachelor of Technology (Computer Science Engineering)						
CSET246	UI/UX Design for Human Computer Interface	L	T	P	C		
Owning School/Department	School of Computer Science Engineering and Technology	3	0	2	4		
Pre-requisites/Exposure	-						

Course-Outcomes (COs):

On completion of this course, the students will be able to:

CO1: To explain UI design with components and user experience designing.

CO2: To implement UI/UX components and libraries.

CO3: To understand user centered computer interaction

Course Contents:

Module 1

(Contact hours: 11)

Historical evolution GUI, Interactive system design: Concept of usability, HCI and software engineering, GUI design and aesthetics, Prototyping techniques, Heuristic Evaluation, Experimental Design, Importance and benefits of good design, Screen design. Scenarios, Design Process. Characteristics of user interface, Web user –Interface popularity, Psychology and Human factors, Conceptual Models, Mistakes and Error, Cognitive models, Socio-Organizational and stakeholder requirements, Social Computing, Experiments designing concepts and methods, Communication and collaboration models, Mobile Ecosystem: Platforms.

Module 2

(Contact hours: 10)

Application frameworks, Widgets, Applications, Mobile Design: Elements, Tools, Mobile design case studies, Human interaction with computers, design rules, Universal Design Models and Theories, Interface implementation & evaluation, Communication and collaboration models, Power of the Crowd, crowd sourcing, Internet of things in HCI, Experimental Design.

Experiments designing concepts, Exploring design for HCI, GOMS for HCI, Identify Color Guidelines, Stages of action in interaction, Menu types and design, Construction of Prototype, Design Principles for HCI.

Module 3

(Contact hours: 7)

HCI for navigation design, Forms and behavior, Design and data for HCI, One factor test with human subjects, Ubiquitous computing with design analysis, Interface implementation, A/B testing, T-tests, Data assumptions and distributions, Interpretation of non-numeric response, Generalized linear models Case studies.

Module 4

(Contact hours: 14)

Concept to production and publishing of a working game. Preparation for the industry-recognized Unity Certification, Core concepts and techniques of augmented reality (AR) and virtual reality (VR), voice and gesture-based interaction, mobile, animations, haptic, and wearables, as well as voice, and kiosk executions.

STUDIO WORK / LABORATORY EXPERIMENTS:

Students will learn to understand the trouble of interacting with machines and design a system based on user-centered approach. A range of VR/AR/MR experience development tools and skills. Story development, user experience, interface design and implementation are introduced within the context of creating immersive experiences.

TEXTBOOKS/LEARNING RESOURCES:

1. Samit Bhattacharya, Human-Computer Interaction User-Centric Computing for Design (1 ed.), McGraw-Hill, 2019. ISBN 9789353168056.
2. Nirmalya Thakur; Parameshachari B.D, Human-Computer Interaction and Beyond: Advances Towards Smart and Interconnected Environments (Part I), (1 ed.), Bantham Books, 2021. ISBN 9789814998826.
3. "Universal Principles of Design" by William Lidwell, Kritina Holden, and Jill Butler

REFERENCE BOOKS/LEARNING RESOURCES:

1. Helen Sharp, Jennifer Preece, Yvonne Rogers, Interaction Design Beyond Human-Computer Interaction, (1 ed.), Wiley, 2021. ISBN 9781119547358
2. "The Elements of User Experience" by Jesse James Garrett

Elective-I, III,

V

Specialization Electives- Artificial Intelligence

Name of Program	Bachelor of Technology (Computer Science Engineering)						
CSET340	Advanced Computer Vision and Video Analytics	L	T	P	C		
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3		
Pre-requisites/Exposure	-						

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate the key features of Computer Vision to analyze and interpret the visible world around us.

CO2: To build the applications of Deep Learning in Computer vision and video processing.

CO3: Implement the state-of-the-art computer vision and video analytics concepts to different applications.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3	2		2	2	2	2	2	2	2	2	3	3	2
CO2	3	3	2	2	3	2	2		2	2	2	2	3	3	2
CO3	3	2	3	3	3	2	2	2	2	2		2	3	2	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

7 lecture hours

Introduction to Computer Vision, The Four Rs of Computer Vision, Challenges in Computer Vision, Low-level vs High-level processing, Two View Geometry, Binocular Stereopsis: Camera and Epipolar Geometry, Planar Scenes and Homography, Depth estimation and multi-camera views, Robust Correspondence Estimation, 3-D reconstruction, Auto-calibration, DLT and RANSAC, Structure from Motion, Hough Transform, Fourier Transform, Interest Point Detection, Edge Detection, Local Binary Pattern, Convolution and Filtering, Gaussian derivative filters, Gabor Filters, DWT, Pyramids, Visual Matching: Bag-of-words, Pyramid Matching, Part based recognition models, Recognition: Detectors and Descriptors, Optical Flow & Tracking.

Modulo II:

7 lecture hours

Shape from Texture, Color, motion and edges, Face Detection, Feature Tracking, Motion Layers, SIFT & Single Object Recognition, Dense Neural Networks, Backpropagation, Convolutional Neural Networks (CNNs), AlexNet, VGG16, Image Quality Enhancement, Image Restoration, Super resolution, Residual Learning, Visual Saliency detection.

Module III:

7 lecture hours

Evolution of CNN Architectures: AlexNet, MobileNet, InceptionNets, ResNets, DenseNets, 3D CNN for images and videos, Unsupervised image segmentation, Watershed, Level set, Active Contour, GraphCut, Supervised image segmentation, Agglomerative clustering, Segmentation as pixel classification, UNets, FCN, Deep Generative Models, GANs, VAEs, PixelRNNs, naDE, Normalizing Flows, Zero-shot, One-shot, Few-shot Learning, Self-supervised Learning, Reinforcement Learning in Vision, Video Analytics, Spatial Domain Processing, Frequency Domain Processing, Background Modelling, Crowd Analysis, Video Surveillance, Traffic Monitoring, Intelligent Transport System.

Module IV:**7 lecture hours**

Optical Character Recognition, Online Character Recognition, Visual Anomaly Detection, Anomalous action recognition, Post Estimation, Action Recognition, Graph CNN, Shape Recognition, Shape Retrieval, Content based Image retrieval, Visual Instance Recognition, Emotion Recognition from videos, Video Generation.

Studio Work / Laboratory Experiments:

In the lab work, the students will Implement the state-of-the-art computer vision and video analytics concepts to different applications.

Text Books :

1. Rajalingappaa Shanmugamani, *Deep Learning for Computer Vision* (1st ed.), Packt Publishing, 2018. ISBN 9781788295628 .
2. Nedumaan J., Prof Thomas Binford, J. Lepika, J. Tisa, J. Ruby and P. S. Jagadeesh Kumar, *Modern Deep Learning and advanced Computer Vision* (1st ed.), Intel, 2019. ISBN 9781708798641 .

Reference Books :

1. Kar Krishnendu, *Mastering Computer Vision with Tensor Flow* (1st ed.), Packt, 2020. ISBN 9781838826939.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET341	Cognitive Modelling	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate the thought, learning, and mental organization..

CO2: To examine the extraction of brain signals into digital form and develop critical skills to evaluate and assess problems including psychology, philosophy, neuroscience, and computer modelling.

CO3: To implement the models that reflects human cognition and apply to different real-life scenarios.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	3			2	2								3	
CO2	3	3	2	2	3	3			2	2	2	2		3	
CO3	3	2	3		3	3			2	2	2	2		3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

7 lecture hours

Why Cognitive Science, Structure and processes of human cognition, Computational simulation or modelling, Theory and Research in Human Cognition, Object Perception, Object Recognition, Attentional Processes, Concept formation, Visual perception, Acquisition and processing of natural language, Human reasoning and problem-solving.

Modulo II:

7 lecture hours

Memory Introduction, Long Term Memory encoding, Retrieval from Long Term Memory, Memory of general knowledge, Semantic memory basics, Models of semantic memory, Human language skills, Midlevel Vision and Attention, Motion, disparity, depth, and orientation representation, Convolution, Mechanisms underlying attention, Attention at the cognitive/algorithm level.

Module III:

7 lecture hours

Brain Signals and Feature extraction, Types of Brain signals, case study, Feature extraction methods and their analysis, Models of Understanding Cognition or Mind: Neuroscientific Model, Psychological Model, Representational Model, Computational Model, Isomorphic Model, Multiple realizable Model, Multiple Draft Model, Sub personal Model.

Module IV:

7 lecture hours

Thought process and Problem Solving, Applications of cognitive computing in the field of psychology, Applications of cognitive computing in linguistics, Decision Making models, Commercial Applications of Cognitive science/computing, Advanced Topics in Cognitive computing, Applications of cognitive

computing in philosophy, Applications of cognitive computing in computer modelling, Machine Learning models for cognitive computing, Neural Networks for cognitive computing.

Studio Work / Laboratory Experiments:

Implement Long Term Memory encoding approach. Implement Models of semantic memory. Implement Brain Signals and Feature extraction. Implement Neuroscientific Model. Psychological Model. Implement Representational Model. Multiple realizable Model. Implement Isomorphic Model. Multiple Draft Model. Implement Decision Making models. Implement Machine Learning algorithms for cognitive computing. Implement Data Mining algorithms for cognitive computing. Implement Neural Networks algorithms for cognitive computing.

Text Books :

1. Bermúdez José Luis, *Cognitive Science: An Introduction to the Science of the Mind* (3 ed.), Cambridge University Press, 2020. ISBN 978-1108440349.

Reference Books :

1. High Rob and Tanmay Bakshi, *Cognitive Computing with IBM Watson: Build Smart Applications Using Artificial Intelligence as a Service* (1 ed.), Packt Publishing, 2019. ISBN 1788478290.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET342	Probability and Random Processes	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain the probability and random processes.

CO2: To formulate the problems and tackle with probability and random processes.

CO3: To implement the applications of probabilities for a real-world scenario.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	2			2	2									
CO2	2	2	2	2	2	2								2	
CO3	2	2		2	3	2						2		3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

7 lecture hours

Events, sample spaces, and probability, Limitation of classical and relative-frequency-based definitions, Independent events, Conditional probability, Sets and fields, Sample space and events, Axiomatic definition of probability, Joint probability, Conditional probability, Independence probability, Total probability, Stationarity: strict-sense stationary (SSS), wide-sense stationary (WSS) processes, Ergodicity and its importance, Bayes theorem, Combinatorics: Probability on finite sample spaces.

Modulo II:

7 lecture hours

Random variables, Discrete and continuous random variables, Probability mass functions, Moment generating functions – Binomial, Poisson, Geometric, Uniform distribution, Exponential, and Gamma distributions, Normal distributions: definitions, applications, Joint distributions – Marginal and conditional distributions, Covariance: definition, type, applications, Correlation regression, Linear regression, Transformation of random variables, Classification – Stationary process.

Module III:

7 lecture hours

Markov process, Poisson process, Random telegraph process, Bernoulli and Binomial random variables, Geometric random variables, Negative Binomial random variables, Random process realizations, Random process sample paths, Discrete time processes, Continuous time processes, Probabilistic structure of a random process.

Module IV:

7 lecture hours

Mean functions, Autocorrelation functions, Autocovariance functions, Poisson random variables, Hypergeometric random variables, Discrete uniform random variables and counting, Independent continuous random variables, Normal distribution and CLT, Approximate models of continuous uniform distribution, Probability in Spam filtering, Random processes in gambling app design, Probability and random processes in market prediction and risk prediction.

Studio Work / Laboratory Experiments:

Students will learn a practical exposure to implement different probability concepts. The students will design and develop applications of probability for insurance, stock market prediction, gambling etc. During the lab works, students will be able to utilize programming and scientific tools for relevant probabilistic app design.

Text Books :

1. Peebles. P. Z, Probability, *Random Variables and Random Signal Principles* (4 ed.), Tata Mc Graw Hill, 2017. ISBN 978-0070474284.

Reference Books :

1. Peebles. P. Z, Probability, *Random Variables and Random Signal Principles* (4 ed.), Tata Mc Graw Hill, 2017. ISBN 978-0070474284.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET343	AI in Healthcare	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate the factors involved in decision support that can improve business performance across the provider/payer ecosystem.

CO2: To build methods and techniques in order to appropriately apply to pain points using case studies.

CO3: To build methods and techniques in order to appropriately apply to pain points using case studies.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	3			2	2								2	
CO2	3	3	3		3	3			2	2	2	2		3	
CO3	3	3	3		3	3			2	2	2	2		3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

6 lecture hours

History of AI in Medicine, AI for Decision Support, Capabilities and limitations of AI in Healthcare, Automated healthcare system: challenges and opportunities, Biostatistics, Research ethics in AI, Common healthcare data types, Medical data: quality vs quantity, Clinical Data, Clinical decision support systems, Electronic Health Records (EHR).

Module II:

7 lecture hours

Time series and non-time series data, Data Sourcing, Data Enrichment, Handling missing values, Advantages and challenges in observational data, Geographic and demographic variation in medical Data, Classification, regression, clustering for healthcare, Evaluation measures for healthcare applications, Bias and Error in medical data, Analysis of data from IOT body sensors, Automated diagnosis processes, Treatment protocol development.

Module III:

9 lecture hours

Time series and non-time series data, Data Sourcing, Data Enrichment, Handling missing values, Advantages and challenges in observational data, Geographic and demographic variation in medical Data, Classification, regression, clustering for healthcare, Evaluation measures for healthcare applications, Bias and Error in medical data, Analysis of data from IOT body sensors, Automated diagnosis processes, Treatment protocol development.

Module IV:

6 lecture hours

Clinical text, Medical Word Corpus, Text representation, BERT for medical data, PubMed BERT, Question answering systems, Finding similar patients through clustering, Medicine or treatment

recommender systems, Q&A systems for Telemedicine, Personalized medicine recommendation system, Drug development analysis, Drug discovery, Modeling drug-drug interactions, Pandemic spread prediction, Infection pattern identification, Computer Vision systems for physiotherapy, Pose estimation, Gait Analysis.

Studio Work / Laboratory Experiments:

Implementation of program to read tabular, textual, image, signal medical data in different formats. Implement sugar level and blood pressure prediction using regression modelling. Implement an early diabetic prediction system using classification modelling. Implement a insurance risk prediction. Implementation of program to read tabular, textual, image, signal medical data in different formats. Implement sugar level and blood pressure prediction using regression modelling. Implement an early diabetic prediction system using classification modelling. Implement a insurance risk prediction.

Text Books :

1. Bohr A. and Kaveh Memarzadeh, *Artificial Intelligence in Healthcare* (1 ed.), Elsevier Science, 2020. ISBN 978-0128184387.

Reference Books :

1. Bohr A. and Kaveh Memarzadeh, *Artificial Intelligence in Healthcare* (1 ed.), Elsevier Science, 2020. ISBN 978-0128184387.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET344	Image and Video Processing	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain and Analyse images using Histograms and spatial filters. Apply Hough transforms and be familiar with image representation using textures.

CO2: To compute motion using optical flow and understand methods for image description and morphological operations.

CO3: To implement different low-level and high-level image and video processing techniques to wide variety of applications.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3			3	3								3	
CO2	3	3	2		3	2								2	3
CO3	3	3	3		3	3			2	2	2	2		3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

7 lecture hours

Applications of Digital Image Processing, Elements of Digital image processing systems, Sampling and quantization, Neighbours of a pixel, adjacency, connectivity, Regions and Boundaries, Distance measures, Gray scale to Binary image using thresholding, Image Enhancement in the Spatial domain, Gray level transforms, Histogram Processing, Histogram Equalization, Enhancement using Spatial filters, Concept of convolution, Smoothing, Mean, median and Gaussian filters, Edge detection using Prewitt, Sobel, Laplace Filters, Laplace of Gaussian Filter.

Modulo II:

6 lecture hours

Canny Edge Detector, Harris Corner Detector, Color models, RGB, HSV, YCbCr models, Pseudocolor Image Processing, Color Transforms, Color to grayscale conversion, Handling Binary Images, Line detection using Hough transform, Polar form, Circle Detection, Morphological Operations, Dilation, Erosion, Opening, Closing, Boundary detection, Hole filling, connected components, Hit and Miss transform, Shape representation using moments, Texture analysis, Texture from histogram, Texture from GLCM matrices.

Module III:

8 lecture hours

Motion Detection, Concept of Optical Flow, Optical flow equation, Lucas Kanade method, Image Enhancement in the Frequency domain, 1-D and 2-D Fourier Transform and their Inverse, Low pass

and Hi pass Filtering, Ideal and Butterworth and Gaussian Filters, Homomorphic Filtering, Image Compression Fundamentals, Lossless Compression Models, Run length Encoding, Huffman Coding, Lossy Compression, Discrete Cosine Transform, Quantization, Zigzag coding, Color image compression, Text recognition, Feature detection, Integral Image Formation.

Module IV:

7 lecture hours

Face Detection – Viola Jones method, Face Recognition, Principal Component Analysis (PCA), Concept of Eigenface, Feature detection for Machine learning applications, SIFT and HOG parameters, Video Processing, Video formation, Video sampling, Motion estimation, Motion compensated (MC) filtering, Frame-rate conversion, Video Coding, Video Compression, Framebased compression (MPEG), Salient object detection, Human action recognition from videos, Depth cameras– Kinect camera data capture, RGBD data.

Studio Work / Laboratory Experiments:

The lab work will be based on image enhancement, image zooming, image cropping, image restoration, image compression and image segmentation etc. The Lab will use MATLAB and Open CV with python.

Text Books :

1. Szeliski Richard, *Computer Vision: Algorithms and Applications* (2 ed.), Springer, 2022. ISBN 978-3030343712.
2. Jain K., *Fundamentals of Digital Image Processing* (1st ed.), Pearson Education India, 2015. ISBN 978-9332551916.

Reference Books :

1. Bovik, A. C., *Handbook of image and video processing* (1 ed.), Academic press, 2010. ISBN 9780121197902.
2. Kim, B. G., *Digital Signal, Image and Video Processing for Emerging Multimedia Technology. Electronics* (1 ed.), Mdpi AG, 2021. ISBN 978-3039438570.
3. Vyas, A., Yu, S. and Paik, J., *Fundamentals of digital image processing. In Multiscale Transforms with Application to Image Processing* (1 ed.), Springer, 2018. ISBN 978-9811356131.

Name of Program	Bachelor of Technology (Computer Science Engineering)						
CSET345	Information Retrieval and Search Engine			L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology			2	0	2	3
Pre-requisites/Exposure	-						

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain the comprehend types of text analysis, Information retrieval, IR system architecture, query processing models and probabilistic models.

CO2: To build information retrieval systems by performing indexing, compression, information categorization sentiment analysis, network management, search engine optimization, records compliance and risk management.

CO3: Implement different information retrieval approaches for applications in text domain.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3				2	2				2			3		
CO2	3	3	3		3	3		1	2	2	2	2		3	
CO3	2		3		3	3	2		2	1				3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

Text analysis and types, Information retrieval, Text processing, Indexes and query matching, Semistructured text data, Tokenization, Stemming, Lemmatization, Language modelling, Examples of open source IR Systems, Query processing models, Probabilistic models, Binary independence model, Robertson/Spark Jones weighting formula, Two-Poisson model, Relevance feedback, Term selection, Pseudo relevance feedback, Language models, Unigram, Bigram language models, Generating queries from documents, Language models and smoothing, Ranking with language models, Retrieval evaluation measures Normalized Discounted Cumulative Gain (NDCG), Kullback-Leibler divergence, Divergence from randomness, Passage retrieval and ranking, Management of Information Retrieval Systems, Knowledge management, Information management, Digital asset management, Network management.

Modulo II:

8 lecture hours

Search engine optimization, Records compliance and risk management, Version control, Data and data quality, Information system failure, Web retrieval and mining, Semantic web, XML information retrieval, Recommender systems and expert locators, Knowledge management systems, Decision support systems, Geographic information system (GIS), Indexing, Inverted indices, Index components and Index life cycle, Interleaving Dictionary and Postings lists, Index construction, Query processing for ranked retrieval, Compression, General-purpose data compression, Symbolwise data compression, Compressing posting lists, Compressing the dictionary.

Module III:**10 lecture hours**

Information categorization and filtering, Classification, Probabilistic classifiers, Linear classifiers, Similarity-based classifiers, Multi category ranking and classification, Learning to rank, Text Clustering, Partitioning methods, Clustering versus classification, Reduced dimensionality/spectral methods, Lexicons, Corpora, Sentiment Analysis, Document-level, Sentence-level and Aspect-based sentiment analysis, Web crawling, Near duplicate pages, Distributed word representations, Link Analysis, PageRank algorithm, Search engine bias, Personalized searching, Question Answering, Cross-lingual retrieval, Adversarial Information Retrieval.

Studio Work / Laboratory Experiments:

In this lab work, the students will implement problems based on search engines and common opensource software to perform common methods of exploratory and predictive analysis. The students also build application of text analysis techniques for data analysis.

Text Books :

1. Christopher D. Manning, Prabhakar Raghavan and Hinrich Schütze, *Introduction to Information Retrieval* (1 ed.), Cambridge University Press, 2019. ISBN 9781107666392.
2. Mitra Bhaskar and Nick Craswell, *An Introduction to Neural Information Retrieval* (1 ed.), Boston — Delft, 2019. ISBN 9781680835327.

Reference Books :

1. Sarfraz Muhammad, *Critical Approaches to Information Retrieval Research* (1ed.), IGI Global, 2019. ISBN 9781799810232.

Name of Program	Bachelor of Technology (Computer Science Engineering)						
CSET346	Natural Language Processing				L	T	P
Owning School/Department	School of Computer Science Engineering and Technology				2	0	2
Pre-requisites/Exposure	-						

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate natural language processing and importance of word representation.

CO2: To build deep learning model for solving natural language problems such as language modelling, machine translation, POS tagging, Seq2Seq generation.

CO3: To implement state-of-the-art Machine Learning and Deep Learning solutions to NLP problems in Global & Indian context.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2			2	2		2			2		2	3		
CO2	3	3	3	2	2		2							3	
CO3	3	3	3		3	3	3		2	2	2	3		3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

7 lecture hours

natural Language Processing: Need, applications, industry demand, Challenges in NLP: Ambiguity in language, Contextual words and phrases and homonyms, Coreference, Domain-specific language, Low-resource languages, Segmentation, Stemming, Lemmatization, Spelling correction, Synsets, Hypernyms, Tokenization, N-grams, Stops Words, WordNet, WordNet Similarity, Language Corpus, N-gram Language Models, Hidden Markov Models.

Modulo II:

7 lecture hours

NLTK (natural Language Toolkit for Indic Languages), Text normalisation, script normalisation, Parallel Corpus, Handling Code-mix text, Cross Lingual Information Retrieval, Word representation, Sentence representation, Word embedding, Vector space model, Term Frequency, TF-IDF Representation, Distributional representation, Word2vec: CBOW(20), Word embedding for regional language, Word2Vec, GloVe, Document to Vector.

Module III:

7 lecture hours

Neural Networks for text, Recurrent Neural Networks, Vanishing Gradients, Exploding gradient, LSTM (Long sort term memory), GRU (Gated recurrent Unit), Seq2Seq Modelling, Bidirectional Model, Contextual Representations, Transformers, BERT, Multilingual Embedding, Transfer Learning in Word Embeddings, MUSE, POS tagging, named Entity Recognition, Sentiment Analysis, Text Clustering.

Module IV:**7 lecture hours**

Topic Modeling, Latent Semantic Analysis, Statistical Machine Translation, Neural Machine Translation, Self-Attention for Generative Models, natural Language Generation, Attention, Question Answering Bot, 1D-CNN for NLP, Sub-word Models, OpenAI's GPT, Google's ALBERT, ULMFiT, Facebook's RoBERTa, Text Summarization, Extractive, Abstractive Text summarization, Transformer models for Text Summarization.

Studio Work / Laboratory Experiments:

Implementation of text document reading, parsing and applying text pre-processing approaches. Implement wordnet based document representation and word searching. Implement word level and sentence level text representation for searching. Implement one hot vector of the given dataset (Indian language). Design first neural network for text processing.. Implement topic modelling using clustering and LSA. Implement text classification model for sentiment analysis. Implement Part of Speech (POS) tagging and Named Entity Recognition (NER). Implement Machine Translation model. Implement chatbot using sequence to sequence modelling. Implement Text Generation models for social media, news context.

Text Books :

1. Rao Delip and Brian McMahan, *natural Language Processing with PyTorch: Build Intelligent Language Applications Using Deep Learning* (1 ed.), O'Reilly Media, 2019. ISBN 9781491978238.

Reference Books :

1. Eisenstein Jacob, *Introduction to natural Language Processing* (1 ed.), The MIT Press; Illustrated edition, 2019. ISBN 9780262042843.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET347	Social Network Analysis	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To understand human behavior in social networks and related communities.

CO2: To build and apply networking models for understanding social interactions.

CO3: To implement network media graph virtualization and network relationships.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1		2			1	3	1	2	1	2		2		2	1
CO2	3	2	3	1	3	3			2	2	1	2		3	
CO3	3	2	3	2	3	3			2	2	2	2		3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

8 lecture hours

Social network analysis purposes, Phenomenology of social media, Social network monitoring tools , Entity resolution, Types of social networks, Networks Representation, Adjacency Matrix, Weighted and Directed Networks, Hypergraphs, Bipartite Networks, Graph Laplacian, Random Walks, One mode and two mode networks, Heterogeneous Information Networks, Closeness centrality, Betweenness centrality, Eigenvector centrality, Clustering coefficient, Density of graphs, Isomorphic graphs, Reachability, Cliques, Graph colouring problem, Valued graphs, Multi graphs.

Modulo II:

7 lecture hours

Large Scale Structure of Networks, Small World Effect, Degree Distributions, Power Law, Scale Free Networks, Assortative Mixing, Time series analysis in graph, Temporal graph algorithms, Parallel dynamic graph algorithm, Matching theory, Bipartite matching, Konig's theorem, Hall's Matching Theorem, Network flow, Max flow min cut theorem, Menger's theorem, Ford Fulkerson method.

Module III:

7 lecture hours

Ties, Structural holes, Structural balance, Equivalence, Motifs, Random Graphs, Giant Component, Small Components, Configuration Model, Excess Degree Distribution, Vertex Copying network models, Erdős-Rényi model, Barabási-Albert model, Exponential Random Graphs, Percolation

Module IV:

6 lecture hours

Communities in network, Community detection from network, Louvain Method, Overlapping communities, Non-overlapping communities, Information diffusion in social networks, Cascading Behavior in Networks, Link Prediction, Preferential Attachment, Geospatial social networks, Crowdsourcing, Rumours and Deception in social network, Fake News, Spamming, Identify theft.

Studio Work / Laboratory Experiments:

Implementing Random graph Models, finding cliques and Motifs, Exploring and analysing various tools like Tweet Reach, How Sociable, and Addic-to-matic to understand the latest concepts and advancements in social media analysis.

Text Books :

1. Blokdyk Gerardus, *Social Network Analysis A Complete Guide* (3 ed.), 5starcooks, 2020. ISBN 978-1867330097.
2. Beineke L.W., Martin Charles Golumbic and Robin J. Wilson, *Topics in Algorithmic Graph Theory* (1 ed.), Cambridge University Press, 2021. ISBN 9781108492607

Reference Books :

1. Dey Nilanjan, *Social Network Analytics: Computational Research Methods and Techniques* (1 ed.), Academic Press, 2018. ISBN 978-0128154588.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET348	Reinforcement Learning	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Understand key features of Reinforcement Learning (RL).

CO2: Decide, formulate, design, and implement given application as RL problem.

CO3: Implement common RL algorithms and evaluate them using relevant metrics.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3	1	1	2							3	1	3	3
CO2	2	3	3	1	3							3	3	3	3
CO3	2	2	3	3	3							3	3	3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

7 lecture hours

Introduction to RL: Course logistics and overview, Introduction to Reinforcement Learning (RL), Origin and history of RL research, RL and its connections with other ML branches. Linear algebra overview, Probability overview, Sequential Decision Making, Modelling the world, Components of a reinforcement learning agent, Taxonomy of reinforcement learning agents. Introduction to Instance based learning.

Module II:

7 lecture hours

Markov Decision Processes and Bandit Algorithms, Policy Gradient Methods & Introduction to Full RL, Reinforcement Learning Problems, MDP Formulation, Bellman Equations & Optimality Proofs, Markov Processes, Markov Reward Processes, Markov Decision Processes, Bellman Equation, Bandit Algorithms (UCB, PAC, Median Elimination, Policy Gradient), Contextual Bandits.

Module III:

7 lecture hours

Dynamic Programming & Temporal Difference Methods, DQN, Fitted Q & Policy Gradient Approaches, Introduction to Dynamic Programming, Policy Evaluation (Prediction), Policy Improvement, Policy Iteration, Hierarchical Reinforcement Learning, Value Iteration, Generalized Policy Iteration, Hierarchical RL: MAXQ, Asynchronous Dynamic Programming, Efficiency of Dynamic Programming, Temporal Difference Prediction, Why TD Prediction Methods, On-Policy and Off-Policy Learning, Q-learning, Reinforcement Learning in Continuous Spaces, SARSA.

Module IV:

13 lecture hours

Dynamic Programming & Temporal Difference Methods, DQN, Fitted Q & Policy Gradient Approaches, Introduction to Dynamic Programming, Policy Evaluation (Prediction), Policy

Improvement, Policy Iteration, Hierarchical Reinforcement Learning, Value Iteration, Generalized Policy Iteration, Hierarchical RL: MAXQ, Asynchronous Dynamic Programming, Efficiency of Dynamic Programming, Temporal Difference Prediction, Why TD Prediction Methods, On-Policy and Off-Policy Learning, Q-learning, Reinforcement Learning in Continuous Spaces, SARSA Prediction, Monte Carlo Estimation of Action Values, Monte Carlo Control, Monte Carlo Control without Exploring Starts, Incremental Implementation, Policy optimization methods (Trust Region Policy Optimization (TRPO) and Proximal Policy, Optimization (PPO).

Studio Work / Laboratory Experiments:

Implement a RL simulator environment in Python to familiarize with basics. Implement Markov Chain model with random state, action, rewards and perform inference with random input sequence. Implement a simple 2D Atari game playing using CNN based Q learning model. Implement self-driving car system using DQN with input from simulators. Implement a news recommendation system using reinforcement learning techniques. Implement dynamic programming based RL system for board games. Implement a stock prediction and trading system using RL concepts. Implement a multi-document text summarization using Monte Carlo RL. Implement RL based Chatbot for improved customer engagement. Implement a multi-agent RL system for drone navigation using simulator libraries.

Text Books :

1. Sutton R.S. and Andrew G. Barto, *Reinforcement Learning: An Introduction* (2 ed.), MIT Press, 2017. ISBN 978-0262039246..
2. Murphy K.P., *Machine Learning: A Probabilistic Perspective* (1 ed.), MIT Press, 2012. ISBN 978-0262018029.

Reference Books :

1. Bertsekas Dimitri and John G. Tsitsiklis, *Neuro Dynamic Programming*, Athena Scientific (1 ed.), Athena Scientific, 1996. ISBN 978-1886529106. .
2. Sewak Mohit, *Deep Reinforcement learning: Frontiers of Artificial Intelligence* (1 ed.), Springer, 2019. ISBN 978-9811382840..
3. Masashi Sugiyama, *Statistical reinforcement learning: modern machine learning approaches* (1 ed.), Chapman and Hall/CRC, 2015. ISBN 978-1439856895.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET349	Special Topics in Artificial Intelligence	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To familiarize and learn about the latest trends and research in the field.

CO2: To equip themselves with the conceptual and practical experience of few latest methods, tools, technologies or algorithms in Artificial Intelligence.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3			3	2								2	3
CO2	3	3	2	3	3	2								3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

28 lecture hours

This course covers the cutting-edge topics in Artificial Intelligence, and these modules will be chosen by the instructor based on the requirements and relevance at that point of time. These modules need to be relevant to the industry and start-ups will also include related case studies, use cases and implementations scenarios. Students will be working on lab work and projects to get real hands-on experience of these topics and modules.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to Artificial Intelligence.

Specialization Electives: Blockchain

Name of Program	Bachelor of Technology (Computer Science Engineering)						
CSET350	Blockchain Technologies: Platforms & Applications			L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology			2	0	2	3
Pre-requisites/Exposure	-						

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate blockchain platforms that show promise in solving complex business problems.

CO2: To examine the life cycle of a chain code and its components.

CO3: To make use of various blockchain-based enterprise applications.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	2		3	2	2								3	2
CO2	3	3		2	2									3	
CO3	3	3	2	2	2	3			2	2	2	2		3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

8 lecture hours

Blockchain platforms: Ethereum, Hyperledger Project, IBM Blockchain, Multichain, Hydrachain, Ripple, R3 Corda, BigChainDB, IPFS, Building Dapps, Ethereum consensus, Proof of Authority, Ethereum Client, Mist Wallet, Accounts.

Modulo II:

8 lecture hours

Contract creation code, Deploying solidity code, Web3js and RPC protocols, Miners, Transaction and Block in Ethereum, Front End Development, Fabric Model, Identity management in Fabric, MSP in fabric, Policies, Ledgers in fabric, Chaincode, Multiple chaincodes on fabric, Endorsement peers.

Module III:

7 lecture hours

Ordering nodes, committing peers, Anchor peers, creating organizations via fabric, Life cycle of chain code, packing transactions into blocks, Validation and commit, Channels, Solo ordering service, Kafka, Sharing private data, Private data sharing patterns.

Module IV:

5 lecture hours

Key level transaction access control, Key level endorsement, Set up a cluster on fabric model, Set up your CA, Use the CA to create local MSPs, Deploy a production network on HF.

Studio Work / Laboratory Experiments:

This laboratory course enables students to get practical knowledge on Blockchain platforms like Ethereum, Hyperledger, IBM Blockchain, Multichain, Hydrachain, Ripple, R3 Corda, BigChainDB, Open-chain and IOTA. Further, Students will use these Blockchain platforms to solve real-world problems in various sectors like Financial, Digital identity, Education sector, Logistics, Health sector, Insurance, Retail, Agriculture, and Land registrations.

Text Books :

1. Serres Tom, Bill Wagner and Bettina Warburg, *Basics of Blockchain* (1 ed.), Animal Ventures LLC, 2019. ISBN 9781089919441.
2. DiMatteo Larry A., Michel Cannarsa and Cristina Poncib, *The Cambridge Handbook of Smart Contracts, Blockchain Technology and Digital Platforms* (1 ed.), Cambridge University Press, 2019. ISBN 978-1108492560.

Reference Books :

1. Gaur and Nitin, *Hands-On Blockchain with Hyperledger: Building decentralized applications with Hyperledger Fabric an* (1 ed.), Packt Publishing Ltd, 2018. ISBN 978-1788994521.

Name of Program	Bachelor of Technology (Computer Science Engineering)						
CSET351	Smart Contracts and Solidity Programming			L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology			2	0	2	3
Pre-requisites/Exposure	-						

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate smart contracts and construct Decentralized Applications.

CO2: To build solidity programming constructs.

CO3: To implement various smart contracts using solidity.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	2	1	1	3	2								2	
CO2	3	2	3		3	2			2	2	2	2		3	
CO3	3	2	3	2	3	3			2	2	2	2		3	2

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

7 lecture hours

Why Smart Contracts, structure of a smart contract, solidity constructs, global variables in solidity, Life cycle of a solidity contract, Interfaces, External function calls, fall back functions, Payable functions, Revert, assert, Require, Decentralized Autonomous Organization tokens, Maker DAO.

Modulo II:

9 lecture hours

Token based membership, Share based membership, Automated immutable systems, Ethereum with DAOs, Pure functions, View functions, Ethereum Virtual Machine (EVM), Byte Code interpretation, Ethereum mining reward scheme, Gas pricing, Ethereum development.

Module III:

7 lecture hours

Whisper, Swarm, Raiden Network, State Channels, Development with Solidity, Development environments, MIX (The DApp IDE), Ether.camp, Truffle, Sublime, testRPC, Development and reuse of common patterns, modifiers driven development, contract driven development, testing.

Module IV:

5 lecture hours

Smart contract security issues, Common attacks on smart contract, Error Handling in smart contracts, Modifiers, Mutex Pattern, Balance limit pattern, Smart contract security tools, Smart Inspect, GasTap, Smart Check, Solgraph, advance research topic of Smart Contract.

Studio Work / Laboratory Experiments:

This lab enables students to get recognize the architecture and components of Ethereum, Decentralized Applications (DApps). Further, students will learn syntaxes for doing Solidity programming to create and deploy DApps using Ethereum. This laboratory includes tools like Remix, Ganache, MetaMask,

Node JS to build and deploy the smart contracts on Blockchain network. Besides, students will get practical knowledge on Web3.js or Ethereum JS API works, Truffle.JS framework to compile Smart Contracts, automate contract testing and Blockchain based web development.

Text Books :

1. Reed and Jeff, *Smart contracts: The essential guide to using blockchain smart contracts for cryptocurrency exchange* (1 ed.), CreateSpace Independent Publishing Platform, 2016. ISBN 978-1539457442

Reference Books :

1. Antonopoulos Andreas M., Gavin Wood Ph.D, *Mastering Ethereum* (1 ed.), O'Reilly, 2018. ISBN 9781491971916 .

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET352	Digital Currencies and Blockchain	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate blockchain-based digital currencies and data structure of the bitcoin digital currency.

CO2: To examine different approaches of digital currencies.

CO3: To implement the different digital currency wallet types and transactions using different types of wallets.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	2		2	3						3			2	
CO2			2	3	3						3			3	
CO3				2	2				2	2	3	2		2	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

8 lecture hours

Digital currency, cryptocurrency vs fiat currency, web 3.0, transactions in digital currencies, bitcoin model, Bitcoin Structure, Bitcoin implementation, public and private addresses, alternative cryptocurrencies-ETH, Ripple, Litecoin, Matic, Bitcoin network architecture, Data structure of the bitcoin blockchain, Stack for bitcoin blockchain, Linked list representation of bitcoin blockchain, Operations on the bitcoin blockchain, special variables in Bitcoin.

Modulo II:

8 lecture hours

Bitcoin storage, Bitcoin mining, Bitcoin Services and functionality, Bitcoin forks, Users in the Bitcoin system, Applications of Bitcoin, Impact of Bitcoin mining, Bitcoin anonymity, Blockchain verification, Bitcoin consensus, Full nodes vs. SPV (Simplified Payment Verification) nodes, Light nodes.

Module III:

6 lecture hours

Mining economics, Mining infrastructure, the Bitcoin Script language, reverse polish notation, script writing, script execution, bitcoin wallets, types of wallets, security implications of the different type of wallets, Dark web.

Module IV:

6 lecture hours

Socio-political impact, and regulation on crypto currencies, Law enforcement on Bitcoin, sudden growth of Bitcoin, Bitcoin impact on society, issues with bitcoin, Bitcoin impact on environment, Bitcoin application other than cryptocurrency, Bitcoin vs file coin, startups in Digital Currencies.

Studio Work / Laboratory Experiments:

This laboratory course enables students to get practical knowledge on cryptocurrencies and associated wallet creation to transfer of crypto currencies like Bitcoin, Ethers etc. In this course students will learn to use cryptocurrencies to purchase various goods and services using Decentralized Blockchain network. Further, the goal of this laboratory is implementation of “payment,” the initiating process of funds transfer using blockchain technology. The funds transfer will be realized with participation of sending and receiving banks’ in blockchain environment and by their exchange of transfer messages through the environment. Though “clearing” and “settlement” operations follow payment in actual operations.

Text Books :

1. Lewis Antony, *The Basics of Bitcoins and Blockchains: An Introduction to Cryptocurrencies and the Technology that Powers Them (Cryptography, Crypto Trading, Digital Assets, NFT)* (1 ed.), Mango media, 2018. ISBN 978-1633538009.

Reference Books :

1. Tapscott Don and Alex Tapscott, *Blockchain Revolution* (1 ed.), Portfolio, 2018. ISBN 978-1101980149.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET354	Cyber Security with Blockchain	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate the cyberthreat landscape and Security Challenges.

CO2: To build Blockchain-based apps for authentication and for storing DNS entries.

CO3: To implement various decentralized applications using blockchain to provide various security services.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2				2	3								3	
CO2	3	2	3	2	3	2			2	2	2	2		3	3
CO3	3	3	3	2	3	3			3	3	3			3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

8 lecture hours

Cyber Security, Internet Governance – Challenges, Constraints, Threats, Cyber Warfare, Cyber Crime, Terrorism, Espionage, Need for a Cyber Security Policy, Nodal Authority requirement, Requirement of an International Convention on Cyberspace, CIA model, Cyber Security vulnerabilities, Cyber Security attacks.

Modulo II:

8 lecture hours

Security services, Blockchain on the CIA Security Triad, Authentication mechanisms, Two-Factor Authentication with Blockchain, PKI Infrastructure, Deploying PKI-Based Identity with Blockchain, IPNS, Blockchain-Based DNS Security Platform, Deploying Blockchain-Based DDoS Protection, EIP Block for DDoS attacks, Security related issues in smart contracts development, Smart contract testing.

Module III:

6 lecture hours

Exception handling, debugging of applications, Formal verification, smart contracts security Oyente, why3 for smart contracts, Solgraph based formal verification, implications of blockchain technology for digital privacy, implication for Security, Membership and Access control in Fabric, authentication in fabric network.

Module IV:

6 lecture hours

Privacy in Fabric, Channel encryption, Blockchain Security (Fabric SideDB), Security of a ledger, anonymity, pseudonymity, blockchain Implementation Challenges, privacy law applicability, startups in blockchain based cyber security applications.

Studio Work / Laboratory Experiments:

Cyber-Security with Blockchain is to make companies, products, systems, and services as resilient as possible to cyber-attacks, by looking at security from the outset and throughout their entire life cycle. This lab enables students to get practical knowledge on cryptographic primitives, design, and analysis of authentication protocols. Further, this lab mainly focusses on Transaction and communication security, preventing DDOS attacks, preventing data manipulation, and protection from compromised nodes.

Text Books :

1. Gupta R., *Hands-on cybersecurity with blockchain* (1 ed.), Packt Publishing, 2018. ISBN 978-788990189.

Reference Books :

1. Yuan Michael Juntao, *Building Blockchain Apps* (1 ed.), Pearson Education, 2020. ISBN 978-9390168385.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET355	Web Development for Blockchain Applications	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain planning strategy for developing websites to produce functional, flexible, & versatile websites.

CO2: To examine critically assess current & emerging technologies for developing websites.

CO3: To implement the front-end for decentralized applications.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	P3O 34	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3		3									3		
CO2	3	3	3	3									3	3	3
CO3				3									3	3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

8 lecture hours

Why Blockchain with Web Development, Technology stacks, HTML5 & CSS for blockchain-based web development, Chrome dev tools, Functional programming paradigm JavaScript inside a browser, Python data types and basics, communication with peers, building client and server, miner and wallet, building a socket communication utility. Use of Low Code, No Code Tools in the development.

Modulo II:

6 lecture hours

JavaScript enabled blockchain applications, compiling new JavaScript to the old one with webpack, Better CSS with webpack, Code organization in a project, Asynchronous JavaScript code for developing smart contracts, APIs for blockchain solutions.

Module III:

8 lecture hours

Overview of server-side options, Node JS environment for blockchain and its ecosystem, JSON, REST API Using Postman to debug API, Managing server-side application state, web3js for blockchain web applications. Databases and SQL (SQLite, PostgreSQL), data normalization, User authorization and authentication, allowing users to interact.

Module IV:

6 lecture hours

Web security basics, not trusting your clients, why use HTTPS, integrating other software with the server, developing frontend with React, concept of single-page applications, managing client-side application state (Redux), overview of other client JS frameworks, development organization, using git, concept of continuous integration, configuring a production web server with Ubuntu.

Studio Work / Laboratory Experiments:

This studio work enables students to get practical knowledge on core technical components of

blockchains and their overlap with the Web, such as: Blockchain APIs, such as JavaScript, Web3, React or REST APIs, Blockchain primitives such as transaction initiation, key signing, and wallet management. In addition, which focus on creating user-friendly decentralized applications. This can be done with the help of truffle IDE which provides means to compile and unit test smart contracts locally. It also facilitates users to deploy smart contracts over Blockchain network which has several accounts to interact with the smart contract.

Text Books :

1. Eric Traub, *earn Blockchain Programming with JavaScript: Build your very own Blockchain and decentralized netwo* (1 ed.), Packt Publishing Ltd, 2018. ISBN 978-1789618822.

Reference Books :

1. Gabarró Steven A, *Web Application Design and Implementation: Apache 2, PHP5, MySQL, JavaScript, and Linux/UNIX* (1 ed.), Wiley-IEEE Computer Society, 2006. ISBN 978-0471773917.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET356	Security and Privacy for Big Data Analytics	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate security threats in big data analytics.

CO2: To examine the system vulnerabilities, exploitation.

CO3: To implement defence mechanisms on big data analytics.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3	2	2	2							3	2	2	2
CO2	2	3	3	2	2							2	2	2	2
CO3	2	2	3	2	3							2	2	3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

9 lecture hours

Introduction and Basic Concepts, privacy by design, security roadmap, Map Reduce Architecture, Detailed Design of Architecture, Hadoop, Spark, Hive etc. Generic Data Security, Data Theft, Data Manipulations, Privilege Escalations, Big Data Storage based Security Issues, Hardware Failure Impacts, Known Trivial Countermeasures, Failure of Trivial Methods on Big Data.

Modulo II:

9 lecture hours

Case Study of Stolen Data in Big Data Environment, Types of Threats, Distinguished Security Goals for Big Data, Implementing Security on Big Data: Methods and Aim of each Method, Administrative Measures of Big Data, Preventive Measures of Big Data Security using Encryption, Preventive Measures of Big Data Security using Access Control, Detective Measures of Security in Big Data, Auditing, Log Analysis, Data and Network Monitoring Methods and Tools, Best Practice for Securing Big Data, Security Cost of Big Data, Data Anonymization, Data Pseudo Anonymization.

Module III:

10 lecture hours

Differential Privacy in Big Data, Methods for Differential Privacy and Impact of each Method, Homomorphic Encryption, Heteromorphic Encryption, Complexity Analysis of Encryption Techniques, Complexity Analysis of Privacy-based Methods. Malware and their Impact on Big Data, Secure Multiparty Communication in Distributed Environment, Data Protection Laws for Big Data, General Data Protection Regulation, Important changes resulting from the GDPR, Personal Data, Personal data and Big Data applications.

Text Books :

1. Joshi R., *Security, Privacy, and Forensics Issues in Big Data Network Security Essentials* (1st ed.), IGI Global, 2019. ISBN 978-1522597421.

Reference Books :

1. Gupta Brij B. and Ramesh C. Joshi, *Security, Privacy and Forensics Issues in Big Data* (1st ed.), GI Global, 2019. ISBN 9781522597441.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET357	Modern Cryptography	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To comprehend and implement various cryptographic algorithms to protect confidential data.

CO2: To identify network vulnerabilities and apply various security mechanisms to protect networks from security attacks.

CO3: To make use of security tools to locate and fix security leaks in a computer network/software.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3			2								3	2	
CO2	3	3	3			2							3		
CO3	2	3			3								3	2	

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

8 lecture hours

Modular arithmetic, Modular Polynomial Arithmetic, Divisibility and greatest common divisors, Euclidean Theorem, Random Number Generator, Pseudorandom Number Generator, Prime numbers, unique factorization, finite fields, Powers and primitive roots in finite fields, Fermat's theorem, Euler's theorem, Symmetric ciphers (Difference between symmetric and asymmetric), Monoalphabetic ciphers (Caesar cipher, Affine cipher, Additive cipher).

Modulo II:

9 lecture hours

Polyalphabetic cipher (Playfair cipher, Vigenère cipher) hash functions, authentication and key establishment, Message Authentication Codes (MACs), digital signatures, PKI. Block Ciphers (Feistel Ciphers), Numerical of Feistel Ciphers, Data Encryption standards 8-bit, Discrete Logarithms, Logarithms for Modular Arithmetic.

Module III:

11 lecture hours

Data encryption standards 64 bits, Advanced encryption standards, Diffie Hellman Key Exchange Algorithm. Euclid Algorithm, Extended Euclid Algorithm, RSA Algorithm. Message Authentication and Hash Functions. Information Theory, Elliptic curves, Elliptic curves over finite fields, The elliptic curve discrete logarithm problem, Elliptic curve cryptography, Lenstra's elliptic curve factorization algorithm, Hash Algorithm-SHA, MD5. Digital Signature Algorithm and Authentication, Authentication Applications KDC, RSA digital signatures, El Gamal digital signatures, GGH lattice-based digital signatures.

Studio Work / Laboratory Experiments:

Insert malicious shell code into a program file and check its malicious or benign status, create Client Server program to send data across systems as two variants clear text data and encrypted data with different set of encryption algorithms, demonstrate Buffer Overflow and showcase EIP and other register status, perform ARP poisoning, SQL Injection and demonstrate its countermeasure methods, implement stateful firewall using IP Tables, showcase different set of security protocol implementation of Wireless LAN.

Text Books :

1. Stallings William, *Network Security Essentials* (4th ed.), Prentice Hall, 2018. ISBN 978-9352866601

Reference Books :

1. Stallings W., *Cryptography and Network Security* (7th ed.), Prentice Hall, 2017. ISBN 978-9332585225

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET358	Cloud Infrastructure and Services	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain where the wireless technologies can be applicable and evolving wireless technologies and standards.

CO2: To articulate the architecture of various wireless technologies.

CO3: To experiment with various wireless technologies through hands-on projects.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	2	2	1	2	2	1	3		3	1	2	1	1	1
CO2	2	2	2	1	1	2	1	3	1	3	1	2	1	2	1
CO3	2	1	2	1	3			1	1	2	1	1	2	2	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

9 lecture hours

Wireless technologies: Cellular systems, satellite systems, broadcast systems, wireless LANs. Propagation modes, Fading in the Mobile Environment, Characteristics of wireless transmission (signals, antennas, multiplexing, modulation, interference). Signal Encoding Criteria, Digital Data, Analog Signals, Analog Data, Analog Signals, Analog Data, Digital Signals. Frequency Hopping Spread Spectrum, Direct Sequence Spread Spectrum, Code Division Multiple Access, Generation of Spreading Sequences.

Modulo II:

10 lecture hours

Recent trends in Wireless Networking Technology, Layered architecture for Wireless communication, Wireless TCP, Physical and Logical Architecture. Radio and Infrared Communication. Satellite Communications, 802.11 WLAN Standards, 802.11 MAC Layer, 802.11 PHY Layer, 802.11 Enhancements, Other WLAN Standards like 802.11g, 802.11n and 802.11p. Wireless sensor network platforms, Communication architecture and protocols, Energy Management, Area Optimization, Time Synchronization, Duty Cycle Optimization, Sensor Data Acquisition, Processing and Handling.

Module III:

9 lecture hours

Wireless Personal Area Network Standards, Bluetooth (IEEE 802.15.1), Wireless USB, ZigBee (IEEE 802.15.4), Infrared Data Association (IrDA), Gigabit Wireless LANs, 5G Technology: Small Cells, Massive MIMO, mmWave, Intrusion Detection System for WSN, Anomaly and misuse detection, Evolution of cellular systems, 6G Technologies, Starlink wireless, 7G Technology: Intelligent Communication of the Inevitable.

Studio Work / Laboratory Experiments:

Studio work includes best practices of cloud infrastructure principles by integrating cloud services to make it scalable, reliable, and highly available. It also leverages CSP Managed Services to enable greater flexibility and resiliency in an infrastructure. In addition, lab works cover well- architected framework to optimized underlying cloud resources for any workload.

Text Books :

1. C Beard and Stallings W., *Wireless Communication Networks and Systems* (1 ed.), Pearson, 2015. ISBN 978-1292108728.
2. Colbach Gordon, *Wireless Networking: Introduction to Bluetooth and WiFi* (1 ed.), Independently Published, 2017. ISBN 978-1973252115.

Reference Books :

1. Velrajan Saro, *An Introduction to 5G Wireless Networks* (1 ed.), Amazon Digital Services LLC - KDP Print US, 2020. ISBN 979-8643303107.
2. Vannithamby R. and S. Talwar, *Towards 5G: Applications, Requirements and Candidate Technologies* (1 ed.), John Willey & Sons, 2017. ISBN 978-1118979839.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET359	Special Topics in Blockchain	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To familiarize and learn about the latest trends and research in the field.

CO2: To equip themselves with the conceptual and practical experience of few latest methods, tools, technologies or algorithms in Blockchain.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3	2	2	2							3	3	3	3
CO2	2	2	2	2	3							2	2	2	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

28 lecture hours

This course covers the cutting-edge topics in Blockchain, and these modules will be chosen by the instructor based on the requirements and relevance at that point of time. These modules need to be relevant to the industry and start-ups will also include related case studies, use cases and implementations scenarios. Students will be working on lab work and projects to get real hands-on experience of these topics and modules.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to Blockchain.

Specialization Electives: Cyber Security

Name of Program	Bachelor of Technology (Computer Science Engineering)						
CSET360	Malware Analysis for Mobile Devices			L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology			2	0	2	3
Pre-requisites/Exposure	-						

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To possess the skills to carry out independent analysis of modern malware samples.

CO2: To understand and analyse the Mobile application threat landscape.

CO3: To Apply techniques to unpack, extract, decrypt, or bypass in future malware samples.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3	2	2	2							3	3	3	2
CO2	2	3	2	2	2							3	2	2	2
CO3	2	2	3	2	3							3	2	2	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

7 lecture hours

Mobile Operating- System and Threats, Mobile Development Tools, Risky Apps, Looking Closer at Mobile Apps. Malware Threats, Hoaxes, and Taxonomy- FakePlayer, DroidSMS, FakeInst, TapSnake, SMSReplicator, Geinimi, ADRD, Pjapps, AirPush, Boxer, GGSmar, Defender, DriveGenie, Torec.

Modulo II:

6 lecture hours

Open-Source Tools- Locating and Downloading Mobile Packages, Vulnerability Research for Mobile OS, Antivirus Scans, Static Analysis, Linux File Command, APK, Key tool Key and Certificate Management Utility, Sandbox Analysis, Emulation Analysis, native Analysis, Reverse Engineering, Memory Analysis.

Module III:

8 lecture hours

Static Analysis, Collections and Marketplace, Marketplace Mirrors and Cache, Contagio Mobile, File Data, Cryptographic Hash Types and Queries, Metadata, Antivirus Scans and Aliases, Certificate Information, Permissions, Strings, Mobile Malware Evolution, Detecting

malware behaviour, Mobile Malware Trends and Reversing Tactics.

Module IV:

7 lecture hours

Behavioural Analysis, AVD, Component & IPC security, Android app permissions, Network Architecture for Sniffing in a Physical Environment, Traffic analysis and manipulation, Application dynamic runtime analysis, identifying code level vulnerabilities.

Studio Work / Laboratory Experiments:

In-studio works, the student will be able to practically understand how all the security attacks does have happened on mobile devices, as well as learn to recognize and remove common coding errors that lead to vulnerabilities. This lab also gives an outline of the techniques for developing a secure application code by analyzing malware for mobile devices. Setup your kali lab, Install Android Emulator, ADB and Database Isolation, Build your own malware app and steal other app files, Recovering protected secrets and Application patching, Acquire malicious apps, Perform static malware analysis, Perform malware injection, Invoking Internal Activities Using Malicious Intents and – attacking broadcast receivers, Parameter Manipulation Using a Proxy, Bypassing SSL Pinning, Memory dumps and objects analysis, Bypass Application Restrictions without Modifying Any Code, Identifying code level vulnerabilities: security code review.

Text Books :

1. Sikorski M. and A. Honig, *Practical Malware Analysis* (1st ed.), Practical Malware Analysis, 2017. ISBN 978-1593272901 .
2. Mohanta Bhijit an Anoop Saldanha, *Malware Analysis and Detection Engineering a Comprehensive Approach to Detect and Analyze Modern Malware* (1st ed.), Packt Publishing, 2022. ISBN 9781484261927.

Reference Books :

1. Kleymentov Alexey, and Amr Thabet, *Mastering Malware Analysis* (1st ed.), Packt Publishing, 2019. ISBN 9781789614872.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET361	Device Level IoT Security	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate the security issues in IoT.

CO2: To apply the cryptographic techniques in IoT.

CO3: To implement various types of access control mechanism for IoT devices in network environment.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3	2	2	2							3	3	3	2
CO2	2	3	2	2	2							3	2	2	2
CO3	2	2	3	2	3							3	2	3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

6 lecture hours

Brief review of the Internet of Things IoT, IoT in business world, Benefits Applications of IoT, Security Issues with IoT, Basic Architecture of IoT, IoT Attack Surface, OWASP Top 10 for IoT. Concept of Vulnerability management, Quarantine and Prevention.

Modulo II:

8 lecture hours

Security Requirements in IoT Architecture - Security in Enabling Technologies - Security Concerns in IoT Applications. Security Architecture in the Internet of Things- Security Requirements in IoT - Insufficient Authentication/Authorization - Insecure Access Control - Threats to Access Control, Privacy, and Availability - Attacks Specific to IoT. Vulnerabilities - Secrecy and Secret-Key Capacity- Authentication/Authorization for Smart Devices - Transport Encryption - Attack Fault trees.

Module III:

14 lecture hours

Cryptographic primitives and its role in IoT, Encryption and Decryption, Hashes, Digital Signatures, Random number generation, Cipher suites, key management fundamentals cryptographic controls built into IoT messaging and communication protocols, IoT Node Authentication, Identity lifecycle, authentication credentials, IoT IAM infrastructure, Authorization with Publish / Subscribe schemes, access control. Concerns in data dissemination , Lightweight and robust schemes for Privacy protection, Trust and Trust models for IoT, self-organizing Things, Preventing unauthorized access, Cloud security for IoT, Cloud services and IoT, offerings related to IoT from cloud service providers, Cloud IoT security controls.

Studio Work / Laboratory Experiments:

Consists of using network monitoring tools, implementing different types of attacks and some protection schemes of various IoT devices.

Text Books :

1. Saravanan Vijayalakshmi, *Securing IoT and Big Data: Next Generation Intelligence (Internet of Everything (IoE)* (1st ed.), CRC Press, 2020. ISBN 0367432889.

Reference Books :

1. Bhattacharjee S., *Practical Industrial Internet of Things Security* (1st ed.), Packt Publications, 2018. ISBN 978-17888 2687.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET362	Vulnerability Analysis in Network Protocols	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To understand the vulnerabilities of network protocols.

CO2: To examine penetration testing on each network protocol for vulnerability detection.

CO3: To make use of tools for detecting vulnerabilities in OSI layers.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3	2	2	2							2	3	2	2
CO2	2	3	2	2	2							2	2	3	2
CO3	2	2	2	2	3							2	2	3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

6 lecture hours

OSI model and TCP/IP suite, Layer wise security issues, Network Access Layer Vulnerabilities, Wiretaps, Reconnaissance, Hardware Breakage, Voltage Fluctuations, natural Disaster, Misconfigurations and Malfunctioning of Network Interface Cards, Signal Disruption Attacks.

Modulo II:

8 lecture hours

Absence of VLANs, Network Layer Protocol Vulnerabilities, IPv4 Packet Structure and Vulnerabilities, IPv4 Header Based Modification Attack, IPv4 flooding Attack, Man in the Middle Attack by Manipulating IPv4 , IPv6 Security Issues IPsec and its Key Management, ICMP Packet Structure and Working, ICMP Header based Modification Attack, Ping of Death, Countermeasures for Each Attack in Network Layer Protocol, Attacks on Routers and Routing, RIP, OSPF, BGP, Countermeasures of Attacks.

Module III:

7 lecture hours

Transport Layer Protocol Vulnerabilities, TCP Packet Structure and its Working, TCP Header Based Modification Attacks, TCP Incomplete Connection Attack, TCP SYN Flooding Attack, TCP Reset Attack, TCP Session Hijacking Attack, Reverse Shell Attack, Detection and Countermeasures of Each type of TCP Attack, UDP Packet Structure and its Working, UDP Header Based Modification Attack, UDP Generic Flooding Attack, Detection and Countermeasures of Each type of UDP Attack.

Module IV:

7 lecture hours

Application Layer Protocol Attacks, DHCP Packet Structure and Working, Classic DHCP Starvation Attack, Induced DHCP Starvation Attack, DHCP Spoofing Attacks, DHCP Flooding Attacks, Countermeasures for each attack on DHCP, DNS Packet Structure and Working, DNS Cache Poisoning for Phishing, Domain Hijacking DNS Flooding, Reflection and Amplification Attack, Random Subdomain Attack, NXDomain and Phantom Domain Attack, Countermeasures for each attack on DNS, HTTP Packet Structure and its Working, Method based Flooding Attacks, HTTP Half Connection

Attack, HTTP fuzzers and misbehaved fields, Cache bypassing attacks, Countermeasures for each attack on HTTP, Wireless Access Point Firmware Vulnerabilities, SSID Beaconing and Checking for Hidden and Fake Wireless Networks, Brute Force Attack, Evil twin Attack, WiMax vulnerability, Vulnerability of zigbee protocols.

Studio Work / Laboratory Experiments:

In-studio works, the student will be able to practically understand Signal Disruption Attacks , Medium Access Layer Vulnerabilities, Network Layer Protocol Vulnerabilities, Countermeasures for Each Attack in Network Layer Protocol: RIP, OSPF, Network Layer Protocol Vulnerabilities, Countermeasures for Each Attack in Network Layer Protocol: BGP, Transport Layer Protocol Vulnerabilities: UDP, Transport Layer Protocol Vulnerabilities : TCP, Session Layer Protocol Vulnerabilities, Presentation Layer Protocol Vulnerabilities, Application Layer Protocol Vulnerabilities : DHCP, Application Layer Protocol Vulnerabilities :HTTP AND HTTPS DOM- based vulnerabilities, Wireless Access Point Firmware Vulnerabilities.

Text Books :

1. Sanders C. , *Practical Packet Analysis* (3rd ed.), No Starch Press, 2017. ISBN 9781593278020.

Reference Books :

1. Gebali F., *Network Vulnerability Assessment: Identify security loopholes in your network's infrastructure* (1st ed.), Springer, 2018. ISBN 978-1788627252.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET363	Penetration Testing, Auditing and Ethical Hacking	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To identify and analyze the stages an ethical hacker requires to compromise a target system.

CO2: To utilize tools and techniques to carry out a penetration testing.

CO3: To critically evaluate security techniques used to protect system and user data.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3	2	2	2							2	3	2	2
CO2	2	3	3	2	2							2	2	3	2
CO3	2	2	3	2	3							2	2	3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

8 lecture hours

Penetration Testing Concepts, Plan a Pen Test Engagement, Rules of engagement, Communication escalation path, Resources and requirements, Budget, Impact analysis and remediation timelines, Disclaimers, Technical constraints, Support resources, Key legal concepts, Contracts, Environmental differences, Written authorization, Scope and Negotiate a Pen Test Engagement, Prepare for a Pen Test Engagement, Conducting Passive Reconnaissance, Information gathering, Perform Social Engineering Tests, Perform Physical Security Tests on Facilities.

Modulo II:

13 lecture hours

Active Reconnaissance, Scan Networks, Enumerate Targets, Analyzing Vulnerabilities, Exploiting Cross-Site Scripting, Cross-Site Request Forgery, Web Application Exploitation, File upload vulnerability , HTTPS Requests: Get & Post , Using Burp as a Proxy Server, Code Execution Vulnerabilities, Countermeasures, Penetrating Networks, Exploit Network-Based Vulnerabilities, Exploit Wireless and RF-Based Vulnerabilities, Evil twin, Karma attack, Downgrade attack, De-authentication attacks, Fragmentation attacks, Credential harvesting, WPS implementation weakness, Bluejacking, Bluesnarfing, RFID cloning, Jamming, Repeating, Local File Inclusion Vulnerabilities, Getting Shell from LFI Vulnerability, OSINT Tools, Maltego, Shodan, Google Dorks, The Harvester, Metagoofil, TinEye.

Module III:

7 lecture hours

Analyze Pen Test Data, Reporting Pen Test Results, Write and Handle Reports, Normalization of data, Written report of findings and remediation, Risk appetite, Storage time for report, Secure handling and disposition of reports, Conduct Post-Report-Delivery Activity, Post-engagement cleanup, Removing shells, Removing tester-created credentials, Removing tools, Client acceptance, Lessons learned,

Follow-up actions/retest, Attestation of findings, Develop Recommendations for Mitigation Strategies, Solutions, Findings, Remediations.

Studio Work / Laboratory Experiments:

Penetration testing using various tools. Crafting malicious packets for network devices like servers, routers, or switches. Crafting malicious software for system vulnerabilities. Ethical hacking on Wired and Wireless Networks.

Text Books :

1. Harper Allen, *Gray Hat Hacking: The Ethical Hacker's Handbook* (6th ed.), McGraw-Hill Osborne Media, 2022. ISBN 1264268947.

Reference Books :

1. Wallace Connor, *Penetration Testing: Penetration Testing: A Hands-On Guide For Beginners* (1st ed.), Independently Published, 2020. ISBN 979-8614981143.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET364	Forensics and Cyber Law	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To understand the Volatile Data Collection from Windows system.

CO2: To understand cyber activities which are considered as crime in India.

CO3: To apply forensic analysis in Computer Investigations.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	2	2	3	2	2						2	3	2	2
CO2	2	2	1	3	2	2						3	2	2	2
CO3	2	2	1	3	2	2						2	2	2	2

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

8 lecture hours

Cyberspace and Criminal Behaviour, Traditional Problems Associated with Computer Crime, The Emergence of e-Cash: A New Problem for Law Enforcement. Hacking, Intellectual Property, Web-Based Criminal Activity, Theft of Information, Data Manipulation, Web Encroachment, Cyberterrorism, Dissemination of Contraband or Offensive Materials, Threatening and Harassing Communications, Money Laundering, Online Fraud, Identity Theft/Fraud, Victimology, Virtual or Internet-Facilitated Methods, Data Piracy and Counterfeit Goods, Volatile Data Collection from Windows system, Volatile Data Collection from Unix system.

Modulo II:

7 lecture hours

Computer Fraud and Abuse Act of 1986, national Information Infrastructure Protection Act of 1996 (NIIPA), Evolving Child Pornography Statutes, Identity Theft and Financial Privacy Statutes, Law Enforcement Operations and Tools in the United States, Computer-Related Crime of the Council of Europe, Council of Europe's (CoE) Cybercrime Conventions, Law to Child Pornography Statutes, Unlawful Internet Gambling Enforcement Act of 2006, Electronic Communications Privacy Act of 1986, Privacy Protection Act, Electronic Surveillance and Criminal Investigations, Communications Assistance for Law Enforcement Act, Other Questions Regarding Privacy: Peer-to-Peer or File sharing, Internet Service Provider Subscriber Records, Web sites, Cell phones.

Module III:

7 lecture hours

Traditional Problems in Computer Investigations, Forensic Duplication, Forensic Duplicates as Admissible Evidence, Forensic Duplication Tool Requirements, Creating a Forensic Duplicate/Qualified Forensic Duplicate of a Hard Drive, Phase after detection of an incident, Disk Structure and Data Storage, File Systems, Firmware Operating Instructions, Data Integrity, Developing Computer Forensic Science Capabilities, Traditional Problems Associated with Finding Digital

Evidence, Pre-search Activities, On-scene Activities: Knock, Notice, and Document, Securing the Crime Scene, Determining the Need for Additional Assistance, Scene Processing, Locating Evidence, Seizure and Documentation of Evidence, Bagging and Tagging, Interviewing Witnesses, Collecting Network Based Evidence, Scene Departure and Transportation of Evidence to Lab, Forensic Analysis of File Systems, Hard Drives Evidence Handling, Challenges in evidence handling.

Module IV:

6 lecture hours

Email Tracing Internet Fraud, Data Analysis Techniques, Investigating Live Systems (Windows & UNIX), Email Tracing Internet Fraud, Data Analysis Techniques, Investigating Live Systems (Windows & UNIX), Issues related to Data Privacy, Criminal Liability, Electronic Contracts & Digital Signatures, Misappropriation of information, Civil Rights, Evidence. Legal Developments, Security in cyber laws case, Trademarks, Reverse Hijacking, Jurisdiction in Trademark Disputes, Copyright in the Digital Medium, Copyright and WIPO Treaties, Concept of Patent Right (30), Cryptography Laws, Ethical Issues – Cybercrime6.

Studio Work / Laboratory Experiments:

Explore the various tools for cyber forensic investigations. Experiments to show web vulnerabilities using DAMN Vulnerable Web App (DVWA). EnCase Forensics to make and restore image of the hard drive.

Text Books :

1. Britz Marjie T, *Computer Forensics and Cyber Crime: An Introduction* (4th ed.), Pearson Education India, 2022. ISBN 978-0134847528.
2. Johansen Gerard, *Digital Forensics and Incident Response: Incident response techniques and procedures to respond to modern cyber threats* (2nd ed.), Packt Publishing Limited, 2020. ISBN 978-1838649005.

Reference Books :

1. Reddy Niranjana, *Practical Cyber Forensics* (1st ed.), Apress, 2019. ISBN 978-1484244605.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET365	Web Security	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To possess the skills to carry out independent analysis of modern malware samples.

CO2: To understand and analyse the Mobile application threat landscape.

CO3: To apply techniques to unpack, extract, decrypt, or bypass in future malware samples.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	1	1	2	2							2	3	2	2
CO2	2	3	1	2	2							2	3	2	2
CO3	2	1	1	2	2							2	2	2	2

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

5 lecture hours

Web Basics: HTML, CSS, JS, URLs, DOM, Frames, HTTP, navigation, X-Domain communication, web security, Security elements, Implementation of safety assessment, understanding the dangers of an insecure communication channel, Network Attacks & HTTPS, HTTPS deployment, HTTPS impact on your application, Insights into the latest evolutions for HTTPS deployments, Limitations of HTTPS, Cookie Flaws and Server Misconfiguration.

Modulo II:

7 lecture hours

Security of Browser same origin policy, sandbox browser, malicious URL intercept, Rapid development of browser security, cross-site scripting attack, Advanced XSS attack, XSS defence, Cross-Site Request Forgery, Advanced CSRF defence, Clickjacking, HTML5 Securities, other security problems.

Module III:

8 lecture hours

Injection Attacks, SQL injection attacks, Database attacking techniques properly defending against SQL injection and other injection, File Upload Vulnerability, designing secure file upload features, Authentication and session management, Attacks on User Interfaces, Access control, Encryption algorithms and random numbers, Web framework security, Application-layer Denial-of-Service Attacks, PHP security, TCP Reset Attack.

Module IV:

8 lecture hours

Security of Internet Business, Business logic security, How the account is stolen, Internet garbage phishing, User privacy protection ,Security development lifecycle, Security operations, Process of vulnerability patch, security monitoring, Practical ways to secure the authentication process, prevent authorization bypasses and harden session management mechanisms (10) security planning, business

continuity planning, Handling incidents Risk Analysis, Dealing with disaster: privacy on the web, Privacy impacts of emerging technologies, Handling incidents, Risk Analysis, Dealing with disaster: privacy on the web, Privacy impacts of emerging technologies, Browser Design & Flaws.

Studio Work / Laboratory Experiments:

In this Lab, the student will be able to practically understand HTTP Host header attacks, and its countermeasure: Basic password reset poisoning, Password reset poisoning via dangling markup, Constructing a web cache poisoning attack and its countermeasure Cross-site scripting attacks and its prevention, XML external entity (XXE) injection, Cross-site request forgery (CSRF), Server-side template injection, Insecure deserialization, Clickjacking, OS command injection, Business logic vulnerabilities: Excessive trust in client-side controls, High-level logic vulnerability, and Low-level logic flaw, Weak isolation on dual-use endpoint and Authentication bypass via flawed state machine.

Text Books :

1. Hoffman Andrew, *Web Application Security: Exploitation and Countermeasures for Modern Web Applications* (1st ed.), O'Reilly Media, 2020. ISBN 978-1492053118.

Reference Books :

1. McDonald Malcolm, *Web Security for Developers* (1st ed.), No Starch Press, 2020. ISBN 9781593279957.

Name of Program	Bachelor of Technology (Computer Science Engineering)						
CSET366	Special Topics in Information Security			L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology			2	0	2	3
Pre-requisites/Exposure	-						

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To familiarize and learn about the latest trends and research in the field.

CO2: To equip themselves with the conceptual and practical experience of few latest methods, tools, technologies or algorithms in Information Security (IS).

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	2	3	2								3	3		
CO2	2	1	3	1	3							1	3		

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

28 lecture hours

This course covers the cutting-edge topics in Information Security and these modules will be chosen by the instructor based on the requirements and relevance at that point of time. These modules need to be relevant to the Industry and start-ups' will also include related case studies, use cases and implementations scenarios. Students will be working on lab work and projects to get real hands-on experience of these topics and modules.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to IS.

Text Books :

1. Pfleeger Charles P., *Security in Computing*, Fourth Edition, Pearson Education
2. Stallings William, *Network Security Essentials: Applications and Standard*. Prentice Hall.

Reference Books :

1. Stallings William, *Cryptography And Network Security Principles And Practice*, Fourth or Fifth Edition, Pearson
2. Mao Wenbo, *Modern Cryptography: Theory and Practice*, Prentice Hall.

Specialization Electives: Data Science

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET367	Digital Marketing and Trend Analysis	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate online applications based on recent digital marketing trends.

CO2: To examine digital marketing and its applications.

CO3: To make use of state-of-the-art tools for digital marketing and trend analysis.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	1	3	1	3	1							3	1	1	1
CO2	1	2	1	3	1							3	2	1	1
CO3	2	1	3	2	2							2	2	1	2

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

8 lecture hours

Digital Marketing, CRM, Affiliate Marketing, P-O-E-M Framework, Online vs offline retail, Customer profile analysis, Factors influencing consumer behavior, Traffic source analysis, Multi- channel analysis, On-site merchandising analysis, Digital Advertising Market, Growth of e- commerce, Social media monitoring vs listening, Competitor and Website Analysis, Structure of online conversations, Methods of gathering data, Unlocking actionable insights from e-commerce, Adwords, Link Tagging, Exit rate and bound rate, Online advertising optimization, Measuring Performance and Quality Control, Crisis management.

Modulo II:

6 lecture hours

Online content, Content Creation, Types of Content, Content optimization, Content Management & Distribution, Recommendation system for e-commerce, User-User and User Item, Study of popular recommendation platforms, Market basket analysis, Predicting product adoption.

Module III:

7 lecture hours

Conversation modeling, Campaign optimization, Clickstream Analytics, Traffic Analytics, Customer segmentation, Search Engine Optimization techniques, Expertise Authority and Trustworthiness, Featured Snippets, Site architecture optimization, Long-Tail Keyword Phrases, Keyword Planner, Google rankings, Link Building, Steps to optimize the website.

Module IV:

7 lecture hours

Mobile Advertising, Mobile Marketing tool Kit, Mobile Marketing Features, Brand Health Management, Surrogate advertising, Fake news propagation models, Fake news identification, Social Media Monitoring, Digital data treasure, Digital rights.

Studio Work / Laboratory Experiments:

The lab component is meant to analyze e-commerce data and social media data through multiple analytics tools such as Google Analytics, Piwik, Kissmetrics, Crazy Egg, etc.

Text Books :

1. Gary P Schneider, *Electronic commerce* (13th ed.), Cengage Learning, Inc, 2019. ISBN 9781285425436.
2. Kingsnorth Simon , *Digital Marketing Strategy: An Integrated Approach to Online Marketing* (4th ed.), Pearson, 2019. ISBN 0749484225.

Reference Books :

1. Clarke Adam, *SEO 2020* (2nd ed.), Amazon Digital Services LLC - KDP Print, 2019. ISBN 9781712354889.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET368	Structural Equation Modelling	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain the statistical and numerical models used in SEM Framework.

CO2: To build factor analysis to find multiple hidden variables.

CO3: To implement of parameter estimation and regression analysis of practical frameworks.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	1	2	1	2	1							2	3	1	1
CO2	3	1	3	1	1							2	2	3	3
CO3	3	1	3	1	2							1	1	3	2

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

8 lecture hours

Structural Equation Modelling, Representation of Structural Equation Models using Path Diagrams, Multiple Regression, Estimation Parameters, Least Squares estimation, Latent Variable Estimation, Estimation with Non-normal Distributions, Computing Power, Equivalent Models, Causal Foundations of SEM, Representing Interventions, Causal effects, Model Specification, Identification and Estimation, Power Analysis, Categorical Data in SEM Framework.

Modulo II:

6 lecture hours

Model Fitting, Model Selection, Path Modelling, Covariance Structure Analysis, Parameter Estimation, Maximum Likelihood, EM Algorithm, Bootstrapping Standard Errors, Bayesian bootstrap, Smooth bootstrap.

Module III:

8 lecture hours

Factor Analysis, Factor quantification and selection, Exploratory factor analysis, Factor selection and rotation, Velicer's Minimum Average Partial test, Convergence of Tests, Confirmatory Factor Analysis, Investigating Measurement Invariance, Indirect Effects in SEM, Bootstrapping Methods, Monte Carlo Method, Likelihood Ratio Test, Single-Group and Multi-Group Approaches.

Module IV:

6 lecture hours

Latent Interaction Modelling, Quasi-maximum-likelihood (QML), Multilevel Modelling of Networks, Linear and non-linear Multilevel Networks, Growth Mixture Modelling, Multiple-group Growth Curve Modelling, Latent Curve Modelling of Longitudinal Growth Data, Adding Covariates and Multiple Causes Model, Estimating Group Effects, Moderation, Mediation Effect, Indirect effect.

Studio Work / Laboratory Experiments:

Students will be doing Lab experiments using R/Python to implement various Techniques, algorithms and problems related to the respective modules.

Text Books :

1. Thakkar Jitesh J., *Structural Equation Modelling: Application for Research and Practice* (1st ed.), Springer, 2020. ISBN 978-9811537929.

Reference Books :

1. Civelek Mustafa Emre, *Essentials of Structural equation modelling* (1st ed.), Zea Books, 2019. ISBN 978-1-60962-129-2.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET369	Time Series Analysis	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain the working of different components of a time series.

CO2: To build autoregressive models for time series forecasting.

CO3: To implement multiple time series methods on real-world datasets.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	1	2	1	2	1							2	3	1	1
CO2	3	1	3	1	1							2	1	3	3
CO3	3	1	3	2	2							1	1	3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

8 lecture hours

Purpose of Time Series Analysis, Descriptive techniques, Times series plots, Line chart, visualizing multidimensional Time series, Visualizing multiple time series, Histograms, Seasonal effects and trend identification, Transformations, Sample autocorrelation, Correlogram, Time series filtering, Probability models, Stochastic processes, Bernoulli Process, Weiner process, Brownian Motion, Ornstein-Uhlenbeck Process, Stationarity, Second-order stationarity, Autocorrelation.

Modulo II:

7 lecture hours

White noise model, Random walks, moving average, Invertibility, ARIMA Models, Autoregressive processes, Fitting an AR process, Yule–Walker equations, General linear process, Wold decomposition theorem, Time series Forecasting, Exponential smoothing, Holt-Winters, Box- Jenkins forecasting, Optimality models for exponential smoothing, Model selection for time series forecasting.

Module III:

7 lecture hours

Spectral analysis, Sinusoidal model, Wiener-Khintchine theory, Cramer representation, Periodogram analysis, Statistical properties of periodogram, Consistent estimators of spectral density, Bivariate processes, Cross-covariance, Cross-correlation, ARCH, GARCH.

Module IV:

6 lecture hours

Gaussian Process, Gaussian Regression, Vector autoregression models VAR, Structural Form, Reduced Form, Parameter Estimation, Kernel Methods for forecasting, Adaptive filtering mechanism for forecasting, Statistical Testing for stationarity, Augmented Dickey-Fuller, Kwiatkowski–Phillips–Schmidt–Shin Test, Goodness of estimation.

Studio Work / Laboratory Experiments:

The lab component is meant to perform time series analysis, time series forecasting with R, time series prediction with LSTM Recurrent Neural Networks by using Python with Keras etc.

Text Books :

1. Nielsen Aileen, *Practical Time Series Analysis: Prediction with Statistics and Machine Learning* (1st ed.), O'Reilly, 2019. ISBN 1492041653.
2. Mills Terence C., *Applied Time Series Analysis: A Practical Guide to Modelling and Forecasting* (1st ed.), Academic Press, 2019. ISBN 978-0-12-813117-6.

Reference Books :

1. Nielsen Aileen, *Practical Time Series Analysis* (1st ed.), O'Reilly, 2019. ISBN 9781492041629.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET370	Data Visualization and Dashboards	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain best practices in data visualization to develop charts, maps, tables, and other visual representations of data.

CO2: To build an interactive dashboard for a cohesive and seamless visualization.

CO3: To implement different visualization approaches on real-world datasets.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	1	2	1	2	1							2	3	1	1
CO2	3	1	3	1	1							2	1	3	3
CO3	3	1	3	2	2							1	1	3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

8 lecture hours

Gathering data and Business process modelling, Business reference model, Business process integration, Workflow modelling, Business process Management, Visualizing Workflow, Layered graph drawing, Layout Algorithm, Information visualization, Graph Drawing Theory, Layout Methods, Forced Layout, Spectral Layout, Tree Layout, Exploratory Visualization, Visual data exploration, Visual thinking.

Modulo II:

7 lecture hours

Visual Analytics, Visualization from Heterogenous data sources, Reasoning and data representation, Optimal visualization types, Binning, Dashboard development, Design principles, Interactivity, Connected drill-down dashboards, Drill Down and Drill Up Charts, Bidirectional Relationship, Working with stakeholders, Stakeholder categories, Creating analytical products, Receiving feedback, Dashboard retune, Design iterations.

Module III:

7 lecture hours

Performance management, Balanced scorecard, Showing Rank and Magnitude, Bump Chart, Measuring Claims across Multiple Dimensions, KPI wheel, Quantitative KPI, Qualitative KPI, Lagging KPI, Leading KPI, KPI Reporting.

Module IV:

6 lecture hours

Actual versus Potential Utilization, Utilization rollup dashboard, Personalized Dashboards, Demographics Dashboard, Visualizing Trends Across Dimensions, Comparing growth rates, Assessing the quality of dashboards, Measuring success, Dashboard Administration, Designing Colour-Blind-Friendly Visualizations.

Studio Work / Laboratory Experiments:

Students will work on DataSets to use the various features of tools to practice and understand the functionality. Students will be preparing a dashboard for Data Science Case study and will customize. Students will be required to understand the challenges to work on large datasets and will experiment with such challenges.

Text Books :

1. Wexler Steve, Jeffrey Shaffer and Andy Cotgreave, *The Big Book of Dashboards: Visualizing Your Data Using Real-World Business Scenarios* (1st ed.), John Wiley & Sons, 2017. ISBN 978-1119282716.

Reference Books :

1. Kusleika Dick, *Data Visualization with Excel Dashboards and Reports* (1st ed.), ohn Wiley & Sons, 2021. ISBN 1119698723.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET371	Big Data Analytics and Business Intelligence	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate key technologies in data science and business analytics.

CO2: To build Big Data framework: Hadoop (HDFS and MapReduce), Hadoop Ecosystem & spark.

CO3: To make use of cutting-edge tools and technologies to analyze Big Data.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	2	3	2	3	2	3	2		3	2	2	2	3	2
CO2	2	2	3	2	2	2				2	2	1	3	3	2
CO3	3	1	1	1	2								2	2	1

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

8 lecture hours

Big Data Analytics: Data and Relations, Business Intelligence, Business intelligence vs business analytics, Big Data Predictive Analysis, Why what and how BI?, OLTP VS OLAP, Project life cycle, Methodology, Impact Analysis, Testing and release management, Ethics in Business Intelligence, Big Data Technology Component, Real Time Analysis of Big Data, Big Data Architecture, Cost Conscious Design, Storage Conscious Design.

Modulo II:

6 lecture hours

Big Data Warehouse, Functional vs Procedural programming models, Data placement strategies, CAP Theorem, Streaming, Stream Data Types, Stream Computing, Filtering Stream, Estimating Moments, NoSQL, Aggregate Data Models, Document Data Model, Schema Less Databases.

Module III:

6 lecture hours

Hadoop Ecosystem, Hadoop Overview, HDFS, Map-Reduce, Pig Overview, Pig Grunt Shell, Hive, HBase, Architecture of Hive and HBase.

Module IV:

8 lecture hours

HQL, Associations and Joins, Aggregate function, Polymorphic queries, Clauses, Subqueries, Spark, Core, Spark SQL, Spark RDD, Deployment and Integration, Spark GraphX and Graph Analytics, Research Topics in Big Data Analytics.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies like R, HDFS, MapReduce, Hive, pig, NoSQL, tableau.

Text Books :

1. Ghavami Peter , *Big Data Analytics Methods* (2nd ed.), De Gruyter, 2020. ISBN 9781547417951.
2. Acharya and Seema, *Data Analytics using R* (1st ed.), New York: McGraw-Hill Education, 2018. ISBN 9352605241.

Reference Books :

1. Azevedo Ana and Manuel Filipe Santos, *Integration Challenges for Analytics, Business Intelligence, and Data Mining* (1st ed.), Engineering Science Reference, 2020. ISBN 9781799857832.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET372	Advanced Database Management System	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate the concepts of transaction processing, concurrency control techniques, database recovery techniques.

CO2: To build distributed and object-oriented databases and learn to apply concurrency control and recovery algorithms.

CO3: To implement distributed and Object-Oriented DBMS to speed up information processing and storage.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3											3	3		
CO2		3	3	3	3			2	3		2		3	3	3
CO3	3	3	3	3	3			2	3		3		3	3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

9 lecture hours

ER Model, Indexing Dependencies, Transaction processing, Transaction management, Scheduling in Databases, Serial Scheduler, Non-Serial Schedulers, Concurrency Control, Time stamp ordering Protocol, Multi version concurrency control, Validation concurrency control, Recovery Techniques in Databases, Undoing, Deferred Update, Immediate Update, Buffering and Caching in Databases, Design Elements, Cache SQL ResultSet, Cache selected fields and values, Caching serialized applications, Shadow Paging, Data Backup Architecture, Transaction logging, Paging Architecture in Databases, Full Backup, Differential Backup.

Module II:

6 lecture hours

Distributed DBMS, Architecture of a DDBMS, Distributed Design of Relational Database, Variant of the 2 Tier Model, Fragmentation and Replication, Update in DDBMS, Update Strategies, Eager update, Distributed Reliability, Phase commit protocol, Asynchronous Update, Distributed transaction management, Distributed concurrency control, Distributed deadlock management.

Module III:

7 lecture hours

Object-Oriented DBMS Architecture, Capturing Semantics, Message Passing, OODB persistence, Object relational database model, Object relational features, Object tables, Nested tables, Varying arrays, Support for large objects, Database Migration, Migration Architecture, Heterogeneous migration versus homogeneous migration, Migration consistency, Database Transformation,

Differential Querying, Migration Consistency Verification, Order violation, Consistency violation, Local Transactions.

Module IV:

6 lecture hours

Flex Cluster Architecture, Policy-Based Cluster, Grid Infrastructure, Kernel Requirements, Data Redaction, Database Auditing, Database Diagnostic Monitor, Global User Authentication, Label Security, Data Masking, Privilege Analysis, Virtual Private Databases.

Studio Work / Laboratory Experiments:

Setting up of DDBMS environment, Implementation of a case study based on DDBMS, Implementation of a case study on Object Oriented designing approach, Working on On-Line Analytical Processing functions of Oracle, NoSQL tools like Riak, Cassandra, MongoDB and Neo4j.

Text Books :

1. Connolly Thomas and Carolyn Begg, *Database Systems* (6th ed.), Pearson, 2019. ISBN 9789353438913.
2. Sadalage Pramod J and Martin Fowler, *NoSQL Distilled* (1st ed.), Pearson, 2012. ISBN 9780321826626.

Reference Books :

1. Hoffer prescott and Mcfadden, *Modern Database Management* (8th ed.), Prentice Hall, 2008. ISBN 978-8131709481.
2. Bayross Ivan, *SQL and PL/SQL* (4th ed.), BPB Publications, 2010. ISBN 978-8176569644.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET373	Satellite Data Analysis	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Analyse the Fundamental mechanism of GIS and Process spatial and attribute data towards preparation of thematic maps.

CO2: Understand and Interpret the various land features over satellite images.

CO3: Generate and publish the spatial maps derived from satellite data to be used for societal applications.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3	1	3								2		3	2
CO2	3	2	2	1								1		2	3
CO3	3	3	3	2	1							2		2	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

8 lecture hours

GIS Definition, GIS applications, Geospatial data for GIS applications, digital representation of geospatial data. Concepts of mapping, Paper based maps vs digital maps, map scale, map projection, limitations, Cartography – History and Developments, advantages of digital maps. Electromagnetic spectrum, Pre-processing of Satellite Images- Georeferencing and Image enhancement, Ground Control Points collection Electromagnetic spectrum, Pre-processing of Satellite Images-Georeferencing and Image enhancement, Ground Control Points collection.

Modulo II:

8 lecture hours

Key Elements of Satellite Image Interpretation- Colour, Texture, size, shape etc. Raster Based GIS- raster representation of data grid size and resolution, raster data structures, advantages/disadvantages of raster data models, data capture, raster to vector conversion VECTOR BASED GIS, Vector representation of data, advantages/disadvantages of vector data models, data bases Spatial and Non-spatial data, Data Collection, Data Formats, Data Conversion.

Module III:

6 lecture hours

Spatial Database Management – Database Structures, Files; Standard Data Formats, Information Systems, Modelling Real World Features Data Spatial Analysis and Modelling – Proximity Analysis, Overlay Analysis, Buffer Analysis Network Analysis-finding shortest path and Overview of methods of multicriteria analysis for education and health applications.

Module IV:

4 lecture hours

Integration of Elevation data- DTM/DEM with satellite images Generating 3-dimensional view of the land features from satellite images Open source tools for publishing spatial maps-Geoserver.

Studio Work / Laboratory Experiments:

Reading and writing the individual Bands in satellite images, Pre-processing of Satellite Images, Generation of Natural color composite and False color composite images, Measurement from the orthorectified Satellite Images, Digitization, adding attribute information in spatial data, Computation of Various indices for vegetation and water estimation, Publishing spatial layers in Geoserver, Development of Web-GIS applications.

Text Books :

1. Chang, K. T., *Introduction to geographic information systems* (9 ed.), McGraw-Hill Higher Education, 2019. ISBN 978-1260136371.
2. Lillesand, T., Kiefer, R. W., and Chipman, J., *Remote sensing and image interpretation* (7 ed.), John Wiley & Sons, 2014. ISBN 978-1118343289.

Reference Books :

1. Bernhardsen, T., *Geographic information systems: an introduction* (3 ed.), Wiley, 2010. ISBN 978-8126511389.
2. Chris Brunsdon and Lex Comber, *An Introduction to R for Spatial Analysis and Mapping (Spatial Analytics and GIS)* (2 ed.), SAGE Publications Ltd, 2019. ISBN 978-1526428509.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET374	Special Topics in Data Science	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To familiarize and learn about the latest trends and research in the field.

CO2: To equip themselves with the conceptual and practical experience of few latest methods, tools, technologies or algorithms in Data Science.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	1	2	1	2	1							2	2		1
CO2	2	2	2	2	3							3	2		3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

28 lecture hours

This course covers the cutting-edge topics in Data Science, and these modules will be chosen by the instructor based on the requirements and relevance at that point of time. These modules need to be relevant to the industry and start-ups will also include related case studies, use cases and implementations scenarios. Students will be working on lab work and projects to get real hands-on experience of these topics and modules.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to MLDA.

Specialization Electives: Gaming

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET375	VR Gaming	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate the concepts, requirements and processes of VR game development.

CO2: To implement the audio-video development and production process associated with VR games.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3	1	2	1	1						2	3	1	3
CO2	2	1	3	2	2							2	2	3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

Industrial applicability of VR, Five Classic Components of a VR System, navigation and Controllers, Position and Motion Trackers, Inside Out/Outside, Tracker Performance Parameters, Optical - Active and Passive Trackers, Inertial and Hybrid Trackers - HMD Trackers, Magnetic Trackers, Mechanical Trackers, Ultrasonic Trackers, navigation and Manipulation Interfaces, Tracker-Based navigation, Manipulation Interfaces, Three-Dimensional Probes and Controllers, Data Gloves and Gesture Interfaces, Human behind the lenses, Human Perception and Cognition, Human Visual System, Human Auditory System, Human Vestibular System, Camera tracking and 3D Rendering for Immersive Environments, Inside-Out Camera tracking, Depth Sensing, Full-Body tracking, Inverse & Forward Kinematics, Full body inertial tracking , Holographic Video.

Modulo II:

9 lecture hours

Rendering Architecture, Graphics Accelerators, 3D Rendering, Distributed VR Architectures, Multi-pipeline Synchronization, Co-located Rendering Pipelines, Distributed Virtual Environments, Modeling the Physical world, Geometric Modeling, Virtual Architecture, Virtual Object Shape, Virtual Object Appearance, Procedural Textures, Advanced Material Properties, Procedural Objects, Photogrammetry, Kinematics Modeling, Homogeneous Transformation Matrices, Object Position, Transformation Invariants, Object Hierarchies, Scale, Perspective and Perception, Physical Modeling, Collision Detection, Surface Deformation, Force computation, Force Smoothing and Mapping, Haptic Texturing.

Module III:

9 lecture hours

Presence, Agency and Interactivity, Augmenting the sense of Presence, Space and Architecture, Dissolving the Medium, Identity in Immersive Environments, Change of Identity, Transforming the senses, Extending the senses, Agency and Interactivity, Cybernetics, Interactivity within Physical Dimensions, Interactivity beyond Physical restrictions: the Super Hero effect, Sound in Immersive

Environments, Evolution of Sound Systems, From mono to stereo to surround, Object Based Sound, Ambisonics, Sound Design Basics, Sound as Information, Earcons, Impact of Sound in Objects and Actions, natural vs Real Sound, Physical Computing, IoT and sensor networks, Rapid Prototyping.

Studio Work / Laboratory Experiments:

This lab enables students and staff to visualize designs, develop immersive VR environments, and to test new VR technologies. The lab features three-dimensional (3D) projections on the walls and floor, as well as sensory feedback, so participants can manipulate objects like medical tools.

Text Books :

1. Glover Jesse and Jonathan Linowes, *Complete Virtual Reality and Augmented Reality Development with Unity: Leverage the power of Unity and become a pro at creating mixed reality applications* (1st ed.), Packt Publishing Limited, 2019. ISBN 1838648186.

Reference Books :

1. Jerald Jason, *The VR Book: Human-Centered Design for Virtual Reality* Jason Jerald (1st ed.), NextGen Interactions 2016, ACM Books. 2016. ISBN 978-1970001129.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET337	Augmented Reality	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course-Outcomes (COs):

On completion of this course, the students will be able to:

CO1: To articulate alternative 3D compositing techniques using computer vision with applications in interactive interfaces – most notably augmented reality interfaces on mobile devices.

CO2: To implement interactive augmented reality games and understand the practical issues regarding multi-platform reconstruction.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	2	3			1		1		1	1	2	1	1	3
CO2			2		1				2	1			1	1	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I

11 lecture hours

Industrial applicability of AR, AR Systems Overview, Input and Output Devices for AR, Optical vs. Video, See-Through AR, AR system architecture, Senses, Simple sensing/perception model, Human visual system, 3D Vision, Tracking system, Tracking for Augmented Reality, Importance of Accurate Head Tracking, Tracking Problem, Tracking Technologies, Importance of Accurate Head Tracking, Tracking Problem, Tracking Technologies, Head motion prediction, Registration, Calibration – static and dynamic, Real Time Performance, Characteristics – spatial, temporal, system robustness, Scheduling and Fusing Sensor Information, mixed reality UI design, Design of immersive user experience.

Module II

11 lecture hours

Alternative Interface Paradigms, Usability guidelines, immersive environments, Space, Scale, Ergonomics, Physical locomotion techniques, Target based techniques, Steering, Comfort and distress, Gaze direction, comfort range test, Motion Sickness, Simulator Sickness, Cybersickness, AR Interface Design, Properties of AR Environments, Collaborative AR Interfaces, Heterogeneous AR User Interfaces, Tangible and Graspable Interaction, Tracking for Augmented Reality, Augmented Reality Interaction, Augmented Reality Information Browsers, AR Widgets, Graphical Interface Elements, Evaluating AR Interfaces.

Module III

06 lecture hours

Video mixing, optical blending, projection devices, spatially augmented reality, Immersive virtual reality, Desktop metaphor, mobile/wearable computing, Ubiquitous computing, Tangible user interfaces, Distributed graphics.

Studio Work / Laboratory Experiments:

The laboratory of Augmented reality is designed to provide a practical exposure to the students about the concepts and topics taught in the classroom sessions.

Text Books:

1. Alan B Craig, William R Sherman and Jeffrey D Will, Developing Virtual Reality Applications: Foundations of Effective Design (1 ed.), Elsevier Publisher, 2009. ISBN 978-0123749437.
2. Doug A Bowman, Ernest Kuijff, Joseph J LaViola, Jr and Ivan Poupyrev, 3D User Interfaces: Theory and Practice (Usability) (2 ed.), Pearson Education, USA, 2017. ISBN 978-0134034324.

Reference Books:

1. Erin Pangilinan, Steve Lukas, Vasanth Mohan, Creating Augmented and Virtual Realities(1 ed.), Creating Augmented and Virtual Realities (1 ed.), O'Reilly Publishers, 2019. ISBN 978-1492044192.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET377	Game Mechanics and Game	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate different types of bodies and their dynamics, used in-game programming.

CO2: To implement the collision detection and their resolution techniques.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	2	1	3	3	3	3	2	2	1		3	2	2	2
CO2	2	3	1	2	2	1				2		2	1	1	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

11 lecture hours

Game theory and mechanism, Normal form games, Super modular games, Dynamic games, Repeated games, Games with incomplete/imperfect information, Mechanism design, Cooperative game theory, Network games, Linear Algebra, Affine Algebra, Calculus, Physics Concepts, Newton's Law, Forces, Momenta, Energy, Rigid body physics, Rigid body classification, Rigid body kinematics, Newtonian dynamics, Lagrangian dynamics, Equations of motion for a particle, system of particles, continuum of mass, Constraints, Interpretation of equations of motion, Euler's equations of motions, Soft Bodies Physics, Elasticity, Stress and Strain, Mass- Spring Systems, 1D, 2D and 3D array of masses, Arbitrary configurations, Control point deformation, B-Spline curves, B-Spline surfaces, Surfaces built from curves.

Modulo II:

11 lecture hours

Free- form deformation, Implicit surface deformation, Level Set Extraction, Isocurve extraction in 2D images, Isosurface extraction in 3D images, Numerical integration methods, Euler's Method, High order Taylor methods, Runge-Kutta methods, Multistep methods, Predictor-corrector methods, Extrapolation methods, Verlet Integration, Numerical Stability, Stiff Equations, Quaternion: Rotation matrices, Classical and Linear algebraic approach, From rotation matrices to quaternion, Interpolation of quaternion, Collision Detection: pipeline, Broad phase collision detection, Bounding volume hierarchies, spatial partitioning, Generation contacts: Collision geometry, Contact generation Algorithms, Separating Axis tests, Coherence.

Module III:

6 lecture hours

Collision Resolution: Impulse, Torque, Collision Impulses, Resolving interpenetration, The collision resolution process, Resting Contacts and Friction: Resting forces, Micro collisions, Types of friction, Stability, Optimizations, Digital Physics: Physical modelling, Accelerometer, Location based gaming, Pressure sensors, load cells.

Studio Work / Laboratory Experiments:

To implement basic and advanced concepts using a 3D physics engine like bullet or Open Dynamics Engine (ODE). The objective is to design an active physics controller for a virtual creature.

Text Books :

1. Bourg David M and Bryan Bywalec, *Physics for Game Developers* (2nd ed.), O'Reilly Media, 2013. ISBN 978-1449361037.

Reference Books :

1. Schell Jesse, *The Art of Game Design: A Book of Lenses*, (3rd ed.), A K Peters/CRC Press, 2019. ISBN 1138632058.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET378	Game Programming with HTML5	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	3	0	0	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain the fundamentals of Game Programming in HTML5 and understand the elements of programming in the two-dimensional environment for the creation of Games).

CO2: To create the games for the desktop machine and Internet using the different tools available in a major programming language like HTML5.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3			2	2							3		
CO2	3			3	2	2			2	3	3		3		
CO3	3	2			2	3			2	3	1			3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

12 lecture hours

HTML5 Multimedia, Game framework, FrameRate class, Creating custom rendering thread, Creating an active rendered thread, Changing the display mode, Active rendering in full-screen display mode, Handling keyboard input, Keyboard improvements, Handling mouse Input, Relative mouse movement, Points and vectors, Transformations, Transformation using the vector 2f class, Transformation using polar coordinates, Matrix transformations, Row-major vs column-major matrices, Matrix3x3f class, Affine transformation in time and space, Calculating time delta, Screen mapping, Viewport ratio, Cannon physics.

Modulo II:

12 lecture hours

Intersection testing, Point in polygon testing, Testing using AABB, Testing using circles for intersections, Testing using the separating axis method, Testing using the line-line overlap method, Testing using the rectangle-rectangle overlap method, Optimizing the tests, Game development setup, Exploring the Game Library, XML, Deploying an Applet, Executable JAR, Deployment of game on multiple applications, Swing animation, Component animator, RepaintCollector, LoopGovernor, Animated component, Animation library, ComponentPainter, ComponentUpdater, ComponentAnimator. Advanced Graphics, Hardware-Accelerated Images, Multi-Buffering and Multi-Threading, Full-Screen Exclusive Mode, Persistent Data, Persistent data formats, Persistence mechanisms, Data Integrity, Game architecture, Interfaces and inheritance, Model-View-controller, Component View MVC, A* Algorithm, GridCartographer and GradientCartographer.

Module III:

6 lecture hours

HTTP Tunneling, Reusable Client and server-side code, Game specific client and server side code, HTTP Polling, HTTP Pulling, HTML5 file upload and download, File API and Ajax / XHR2, Drag

and drop files, Indexed DB, HTML5 Multimedia, Web Components, Web Workers, The Orientation and Device Motion APIs.

Text Books :

1. Wright Timothy M., *Fundamental Game Programming with Java* (1 ed.), Cengage Learning PTR, 2014. ISBN 978-1305076532.

Reference Books :

1. Millington IAN and John Funge, *Artificial Intelligence for games* (2 ed.), CRC Press, 2009. ISBN 978-0123747310.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET379	AI for Games	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To comprehending an interconnection between the techniques of AI and their usage in Games.

CO2: To Implement the idea of AI in Games for developing games programmatically.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3		3									3		
CO2	3	3	3	3	3								3	3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

11 lecture hours

Artificial Intelligence in Games, Ways of AI in Games, Deterministic Versus Nondeterministic AI, Established Game AI, Chasing and Evading, Agent Movement, Two-Dimensional Movement, Statics, Kinematics, Steering Behaviors, Combining Steering Behaviors, Randomness in game, FSM with probability, Dynamic AI, Coordinated Movement, Motor Control, Movement in the Third Dimension, Pathfinding, Breadcrumb Pathfinding, Path Following, Wall Tracing, Waypoint navigation, A* Pathfinding, World Representations, Defining Search Area, Starting Search, Scoring, Finding Dead End, Hierarchical Pathfinding, Continuous Time Pathfinding.

Modulo II:

9 lecture hours

Decision Making: Finite State Machines, Rule based systems, Decision trees, Fuzzy Logic in gaming, Markov Systems in gaming, Goal-Oriented Behavior, Rule-Based AI Systems, Fighting Game Strike Prediction, Action Execution, Decisions Under Uncertainty-Bayesian Techniques, Bayesian Network, Tactical and Strategic AI, Waypoint Tactics, Tactical Analyses, Terrain Analysis, Learning with Tactical Analyses, Structure for Tactical Analyses, Map Flooding, Tactical Pathfinding, Cost Function, Tactic Weights and Concern Blending, Modifying Pathfinding Heuristic, Tactical Graphs for Pathfinding.

Module III:

8 lecture hours

Coordinated Action, Multi-Tier AI, Emergent Cooperation, Scripting Group Actions, Military Tactics, Learning mechanism, Online or Offline Learning, Intra-Behavior Learning, Inter-Behavior Learning, Over-Learning, Minimax algorithm, Negamax algorithm, Parameter Modification, Parameter Landscape, Hill Climbing in gaming, Extensions to Basic Hill Climbing, Annealing, Action Prediction, Left or Right, Raw Probability, String Matching, N-Grams, Neural Networks in gaming, Chasing and Evading with Brains, Reinforcement learning in gaming.

Studio Work / Laboratory Experiments:

This course lab aims to use the power of AI in Gaming design and development using modern Game and AI resources.

Text Books :

1. Bourg David and Glenn Seemann, *AI for Game Developers: Creating Intelligent Behaviour in Games* (1 ed.), O'Reilly Media, 2020. ISBN 978-0596005559.

Reference Books :

1. Yannakakis Georgios N. and Julian Togelius, *Artificial Intelligence and Games* (1 ed.), Springer, 2018. ISBN 978-3319635194.

Name of Program	Bachelor of Technology (Computer Science Engineering)						
CSET394	Modern and Contemporary Application in Computer Science	L	T	P	C		
Owning School/Department	School of Computer Science Engineering and Technology	3	0	0	3		
Pre-requisites/Exposure	-						

Course-Outcomes (COs):

On completion of this course, the students will be able to:

CO1. Develop the skills to identify and apply modern and contemporary technologies to solve real-world computer science problems.

CO2. Become familiar with programming languages and software tools related to Niche technologies, Analyse the advantages and disadvantages of various technologies and how they impact the world

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	2		2		2							3		2
CO2					3		2	1			1	1		3	1

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I

lecture hours: 42

This course will provide an overview of modern and contemporary applications of computer science. We will cover topics such as artificial intelligence, machine learning, robotics, data science, cloud computing, and more. We will explore the various technologies and techniques used in each area, as well as their real-world applications. We will also look at the ethical implications of certain applications and the importance of responsible use of computer science technology. The course will also feature programming projects and assignments to give students a practical understanding of the concepts discussed. Learning outcomes include an understanding of the fundamentals of computer science, knowledge of the various applications of computer science, and an ability to apply these concepts to real-world problems.

Name of Program	Bachelor of Technology (Computer Science Engineering)						
CSET395	Latest advances in Engineering and Technology	L	T	P	C		
Owning School/Department	School of Computer Science Engineering and Technology	3	0	0	3		
Pre-requisites/Exposure	-						

Course-Outcomes (COs):

On completion of this course, the students will be able to:

CO1. Gain an understanding of the latest engineering and technology advances, their implications, and applications.

CO2. Develop the skills necessary to analyze and apply the latest engineering and technology advances to solve engineering and technology-related problems.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	2	2		2		2	1				2	3	2	2
CO2	2	3	3	2	2	2		1	2	1	1	2	2	3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I

lecture hours: 42

This course will cover the latest advances in engineering and technology, including topics such as artificial intelligence, machine learning, robotics, control systems, Internet of Things, big data, cloud computing, and more. Students will learn about the newest technologies, their applications, and their implications for engineering and technology. They will also explore case studies and gain a better understanding of how current technologies are being used in the industry today. Students will develop their knowledge of the fundamentals of engineering and technology and learn how to apply them to solve real-world problems. Additionally, students will develop their ability to think critically and strategically about the current state of engineering and technology.

Name of Program	Bachelor of Technology (Computer Science Engineering)						
CSET393	Android Application Using Kotlin			L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology			2	0	2	3
Pre-requisites/Exposure	-						

Course-Outcomes (COs):

On completion of this course, the students will be able to:

CO1: To explain best practices in data visualization to develop charts, maps, tables, and other visual representations of data.

CO2: To build an interactive dashboard for a cohesive and seamless visualization.

CO3: To implement different visualization approaches on real-world datasets.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1			3					1					3	1	1
CO2			2				2			2			2	1	1
CO3	2	2		1							2	3	1	1	1

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I

03 lecture hours

Kotlin basics, functions, classes and objects, OOPs, extensions

Module II

03 lecture hours

Build your android application, basic app anatomy, layout: linear layout, add user interactivity, constraint layout, data binding basics, application navigation

Module III

06 lecture hours

Android app architecture, activity and fragment lifecycle, view model, live data, live data transformation, persistence, connect to the internet, repository pattern and work manager

Module IV

02 lecture hours

App designing, styles and themes, material design, dimension and colors

Studio Work / Laboratory Experiments:

Students will develop the android application using latest tools of kotlin.

Text Books:

1. The Android Development with Kotlin course is already published online for self-paced learning.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET380	Animation and Rendering Techniques	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To define viewpoint and common rollouts-editing splines.

CO2: To understand different concepts of light.

CO3: To make use of V-Ray in animation and create animation by using rendering technique.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3		3									3		
CO2	3	3		3									3		
CO3	3	3	3	3	3								3	3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

6 lecture hours

3D View, Viewports, Standard primitives, Transformations, File formats and operations, Selection, Cloning, Group, Ungroup, Alignment, Splines: common, Rollouts-editing splines, Architectural tools, Max scene files, Modifiers, World space modifiers, Object space modifiers, Modifier stack, Instanced modifier.

Module II:

8 lecture hours

Compound objects and types, Modelling, mesh modelling, converting objects to editable mesh, sub objects: vertex, edge, face, polygon, element, Edit mesh modifier, Editable poly modelling, Material Editor, Assigning materials to objects, Material editor options, Material properties, Material types, Concepts of Light, omni lights, Spotlight, target lights, free lights, directional light, area lights, Mental ray, Skylight, Creating max basic lights, lights parameters, Positioning lights, Creating max light in exterior environment, Creating max light in interior environment.

Module III:

8 lecture hours

Introduction to Network Security, Network-based Attacks, Active and Passive Attacks, Phishing and Its Types, Tailgating, Impersonation, Dumpster Diving, Shoulder Surfing, Hoax, Waterhole Attacks, Denial of Service Attacks and its Types, Man in the Middle Attack, Buffer Overflow Attack, Cross-Site Scripting, SQL Injection Attack, Privilege Escalation, Man in the Browser, Zero-Day Attack, SQL Injection Attack, Privilege Escalation, Man in the Browser.

Module IV:

6 lecture hours

Animating modifiers, animating elements, animation helpers, using dummy objects, dynamics and reaction, Dynamic Objects: damper, Spring, simulation mass FX.

Studio Work / Laboratory Experiments:

The experiments to create an image of an object with the different lighting condition. Light the scene with direct and indirect light, Rendering out Animation. Creation of a short-animated movie.

Text Books :

1. Watt Alan H. and Mark Watt, *Advanced Animation and Rendering Techniques Theory and Practice* (1 ed.), ACM Press, 1992. ISBN 9780201544121, 0201544121.

Reference Books :

1. Möller Tomas Akenine, *Eric Haines and naty Hoffman, Real-Time Rendering* (3 ed.), CRC Press, 2019. ISBN 9781315362007, 1315362007.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET440	Pixel and Poly Arts for Games	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain the tools and knowledge required for creating simple art for video games made in a game engine.

CO2: To implement whole pipeline from creation to setting up in a game engine.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	1	2		3	2								2	
CO2	3	2	3	1	3	2		2	2	2	2			3	2

Course Contents:

Module 1

8 lecture hours

Introduction to pixel art, Pixel art environment, Environment Design, Pixel art characters, Pixel art character design, Pixel Art Animation, Animation in Photoshop, Animating an Idle Animation, Animating a Run Cycle, Low Poly Game Art, Objects and Tools, Modeling Tools, Sub-Division Modeling, Rigging and Animation, One point vs twopoint perspective, Modeling, Understanding UVs and Mapping, Texturing.

Module II

10 lecture hours

Low Poly Environment Art, Rock designing, Island, wall, and gate designing. Low Poly Character Art, Body modeling (head, torso, legs, arms, and hands), Model costume, Model cleanup, Unwrapping, UV layout, Animating Low Poly Art, Skeleton, Skinning, Walk Cycle, Idle Animation, 3D Game Prop Production, Essentials Normals and Baking, Essential Modeling Tips and tools, Essentials Substance Painter and ZBrush, Initial Block-In.

Module III

10 lecture hours

Mid-Poly Block-In Model, Mid-Poly Objective Lens, Mid-Poly Eye Piece Swivel, Mid-Poly Eye-Piece Viewer, Mid-Poly Body, Mid-Poly Swivel and Strap Mount, High-Poly Objective Lens, High-Poly Eye Piece Swivel, HighPoly Eye-Piece Viewer, High-Poly Body, High-Poly Floaters, PBR Texturing, Baking Texture Maps, Creating Initial Textures in Substance Painter, Adding Height-Map Details, Creating and using Texture Stencil, Texturing Surface Noise, Dirt, & Wear, Rendering in Marmoset, Compositional Studies for Concept Art, Introduction to 3D for Concept Art (The Drawing, Modifying, and Offset Tools), Setting Up the Layers, Lighting the Foreground, Managing the background and object, Finishing.

Studio Work / Laboratory Experiments:

In this course lab students will create pixel art assets for games, create low-poly 3D assets for games, understand how to study composition, create environment concept painting, create a current gen photo-realistic game prop.

Text Books:

1. Schell, Jesse. *The Art of Game Design: A book of lenses*. CRC press, 2008. ISBN 9781466598645, 1466598646

Reference Books:

1. Dawe, Jennifer, and Matthew Humphries. *Make Your Own Pixel Art: Create Graphics for Games, Animations, and More!*. No Starch Press, 2019. ISBN 9781593278878, 159327887X.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET441	Game Design, Development and Programming	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate the workflow for creating video games and usage of subcomponents of game engines such as graphics, physics and audio engines.

CO2: To develop realistic scenes and environments. Design, write and deploy 2D and 3D games.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	3	3	2			2	2				3	
CO2	3	3	3	2	3	3			3	3	3	2		3	3

Course Content:

Module 1

9 lecture hours

Game design, Aspects of game design, Components in a game, Game engines, Geometric primitives, 2D and 3D linear transformation, Game state, Interactive Storytelling, Game mechanics, Rules and Discovery, Game engine essentials, Interface and Assets, Game engine Interfaces, Prototyping and Scripting Basics.

Module II

9 lecture hours

Terrain, Terrain Assets, Camera, Layer, Game Physics, Collider, Gravity simulation, Rigid body interaction, Collisions, Collision Detection, resolution, Optimization Application and Techniques, Deployment methods, Character scripting, Object modelling, Texture to game objects, Sprites, Animation.

Module III

10 lecture hours

Association of audio, Audio assets, Different audio formats, Audio mixing, visual continuity in tiles, Adding objects to scene, Prefabs, Lighting, RGB space, Transparency, Level design, Scenes, Tiles, Combinatorial games, Twoperson zero-sum games, Zero-sum games on graphs, General-sum games, Adaptive decision making, Artificial Intelligence Agents, Navigation and pathfinding.

Text Books:

1. Schell, Jesse. *The Art of Game Design: A book of lenses*. CRC press, 2008. ISBN 978- 1138632059.

Reference Books:

1. Buttfield-Addison, Paris, Jon Manning, and Tim Nugent. *Unity game development cookbook: essentials for every game*. O'Reilly Media, 2019. ISBN 978-1491999158.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET442	Special Topics in Animation and Gaming	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To familiarize and learn about the latest trends and research in the field.

CO2: To equip themselves with the conceptual and practical experience of few latest methods, tools, technologies or algorithms in animation and gaming.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	POS3
CO1	2	3		3	2									3	3
CO2	3	2	3	2	3	2						2		3	3

Course Content:

Module 1

28 lecture hours

This course covers the cutting-edge topics in animation and gaming, and these modules will be chosen by the instructor based on the requirements and relevance at that point of time. These modules need to be relevant to the industry and start-ups will also include related case studies, use cases and implementations scenarios. Students will be working on lab work and projects to get real hands-on experience of these topics and modules.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to animation and gaming.

Specialization

Electives:

Robotics

Process

Automation

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET443	Digital Process Automation Architecture	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	1	0	4	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To understand the architecture and modelling behind design and development of digital bots.

CO2: To examine the sustainability of digital workforce.

CO3: To enable for understanding the process to automate the extensive business processes.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2				3								3		
CO2	1					1	2							3	
CO3							1			1		2			3

Course Content:

Module1

12 lecture hours

Bot editor for creating bots, Action palette content for bot creation, Build advanced bots and packages, Package software development kit, Build a bot using REST web services and JavaScript actions, Post to Salesforce through custom app with OAuth 2.0, Parsing of JSON response, Use of JavaScript to build a bot to take user input, Bot store submission checklist, Building reusable bots, Troubleshooting and debugging of bot, Bot agent log files, Management of the activities, Workload management, Create workload queues, Bot with queue, Manage work items and development of workload guidelines, Robotic Interface with back-office business process use case, Creating form, Creation of the Bots, Creation of Process, SAML user access to private bot store, Bot lifecycle management, Guidelines for exporting and importing bots, Export bots, import bots, Audit API, Authentication API, Bot scheduler API, Bot insight API, Create credential & locker, Credential vault email notifications, External key vault integration cyberark, Bot store, Configure the control Room, Bot lifecycle management API, Credential vault APIs.

Module II

6 lecture hours

Bot editor for creating bots, Action palette content for bot creation, Build advanced bots and packages, Package software development kit, Build a bot using REST web services and JavaScript actions, Post to Salesforce through custom app with OAuth 2.0, Parsing of JSON response, Use of JavaScript to build a bot to take user input, Bot store submission checklist, Building reusable bots, Troubleshooting and debugging of bot, Bot agent log files, Management of the activities, Workload management, Create workload queues, Bot with queue, Manage work items and development of workload guidelines,

Robotic Interface with back-office business process use case, Creating form, Creation of the Bots, Creation of Proces, SAML user access to private bot store, Bot lifecycle management, Guidelines for exporting and importing bots, Export bots, import bots, Audit API, Authentication API, Bot scheduler API, Bot insight API, Create credential & locker, Credential vault email notifications, External key vault integration cyberark, Bot store, Configure the control Room, Bot lifecycle management API, Credential vault APIs.

Text Books:

1. Mullakara, Nandan, and Arun Kumar Asokan. *Robotic process automation projects: build real-world RPA solutions using UiPath and automation anywhere*. Packt Publishing Ltd, 2020. ISBN 9781839217357.
2. Mahey, Husan. *Robotic Process Automation with Automation Anywhere: Techniques to fuel business productivity and intelligent automation using RPA*. Packt Publishing Ltd, 2020. ISBN 9781839215650.

Reference Books:

1. Husan Mahey, *Robotic Process Automation with Automation Anywhere* (1st ed.), Packt Publishing, 2021, ISBN 9781839215650.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET444	Enterprise ProcessAutomation Frameworks	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To understand the underlying logic/structure related to robotic process automation.

CO2: To deploy the digital bots to automate the processes using RPA Frameworks.

CO3: To implement the transactions in business processes using robotic process automation.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2				3								3		
CO2		2	1							2				3	
CO3			1				2					1			3

Course Content:

Module 1

10 lecture hours

Robotic Process Automation; Transactions and Types of Processes; Dispatcher and Performer (Producer and Consumer) Model; RPA Framework Template; Robot stops on the first unexpected error; Application crash stops the automation; Invalid transaction data stops processing all other transactions; Hardcoded and/or sparse values for File Paths, emails; Difficult to troubleshoot an error in production; Large automation projects are difficult to maintain; Automation Project and the Main Workflow; Initialization State and Configuration File; Get Transaction Data State; Process Transaction and End Process States; Execution and Logging; System and Business Exception Handling.

Module II

11 lecture hours

Concurrent Queue Consumption; RPA Framework Elements; RPA Framework with Queue Items Development Checklist; Whiteboarding: Automation Project Workflows; Project Configuration; Applications Used: Open/Close/Kill; Get Transaction Data and Process; Transaction States; Running the Project; Testing RPA Framework Project with Queues; Changing the TransactionItemData Type; Building An Execution Report; RPA Framework with Different Data; Workflow analyser.

Module III

7 lecture hours

Various state machines; Invoke Method; Invoke Code; Use of Activity Project Settings; Advanced Topics in RPA; RPA Framework; Transaction, Queue, Machine, Variable declaration.

Text Books:

1. Murdoch, Richard. *Robotic Process Automation: Guide to Building Software Robots, Automate Repetitive Tasks & Become an RPA Consultant*. Richard Murdoch & RPA Ultra, 2018. ISBN 978-1983036835.
2. Taulli, Tom. "The robotic process automation handbook." *The Robotic Process Automation Handbook*. <https://doi.org/10.1007/978-1-4842-5729-6> (2020). ISBN 9781484257289.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET446	Server-Side Development Frameworks: Express and Spring	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate the Express which is a minimal and flexible Node.js web application framework that provides a robust set of features for web and mobile applications.

CO2: To examine server-side development of web applications using Spring framework using Java language.

CO3: To make use of the interaction with backend databases such as mongo dB, MySQL, PostgreSQL with Express and Spring applications.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	POS3
CO1	3	3		3									3		2
CO2	3	3	3	3	3									3	3
CO3	3	3	3	3	3									3	3

Course Contents:

Module 1

7 lecture hours

Introduction to Full Stack Development, Server-side Development: Node, Node modules and the Node HTTP server, Third party modules: mongoose, angular, react, express, npx, npm, brew, http server usage and options, Express framework, REST (REpresentation State Transfer (REST)) API server with Express, Get, post, put, delete, Node server with Express, Postman for API Testing Automation, Mongo dB database.

Module II

7 lecture hours

Interacting with MongoDB from a Node application, Install mongo dB using npm, Import mongo client, Connect to client, Query execution from script file (node sample.js), Mongo and Mongoose, authentication and session-based authentication, Npm express-session, Token-based authentication with the support of JSON web tokens and the Passport module, Modules: express, passport, jsonwebtoken, Fundamental concepts of Spring Core: Dependency Injection (DI) and Inversion of Control (IoC), Automated Java-Based Configuration: @Component, @Autowired, @ComponentScan, Constructor Injection, Field Injection, Setter Injection.

Module III

6 lecture hours

Spring Core application with Maven, Using Eclipse, Contents of pom.xml, Spring Framework support for Model, Controller and View pattern Directory Structure of Spring MVC using Maven, Required Jar files or Maven Dependency, Entry of controller in the web.xml file, Define the bean in the xml file, Displaying the message in the JSP page, Java, Spring MVC, Spring Security, Algorithm, Hibernate, Maven, Mysql Integration Application, Java Persistence API (management of relational data in the Java applications) and Hibernate (Object-Relational Mapping (ORM)), JPA (javax.persistence), Hibernate (org.hibernate), JPA (EntityManagerFactory), Hibernate (SessionFactory), JPA (Java Persistence Query Language (JPQL)), Hibernate (Hibernate Query Language (HQL)).

Module IV

8 lecture hours

CRUD paradigm in java application, RESTful web service: Create (POST), Read (GET), Update (Put), Delete (DELETE), Basic PostgreSQL data storage and management capabilities [create a database, drop a database, select database, select table, update a record, create a table, delete record, drop table, triggers, functions, insert the record, procedures, cursors], Spring SecurityBasics, Spring Security Features, Spring Security Authentication, Spring JDBCTemplate Introduction, RowMapper, Builder, Spring Boot, JDBCTemplate using Postman API Hands on Session.

Text Books:

1. Brown, Ethan. *Web development with node and express: leveraging the JavaScript stack*. O'Reilly Media, 2019. ISBN 978-1492053507.

Reference Books:

1. Kumar, Akshi. *Web technology: theory and practice*. CRC Press, 2018. ISBN 9781138550438.
2. Moseley, Ralph. *Developing web applications*. John Wiley & Sons, 2007. ISBN 9788126538676.
3. Lowe, Doug, and Joel Murach. *Murach's Beginning Java 2 JDK 5*. Murach, 2005. ISBN 9781890774295.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET447	Special Topics in RPA	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To familiarize and learn about the latest trends and research in the field.

CO2: To equip themselves with the conceptual and practical experience of few latest methods, tools, technologies or algorithms in RPA.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1		2				3							1		
CO2	1	2	1							2		3		3	2

Module 1

28 lecture hours

This course covers the cutting-edge topics in RPA, and these modules will be chosen by the instructor based on the requirements and relevance at that point of time. These modules need to be relevant to the industry and start-ups will also include related case studies, use cases and implementations scenarios. Students will be working on lab work and projects to get real hands-on experience of these topics and modules.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to RPA.

Specialization Electives: DevOps

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET448	Build and Release Management in DevOps	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain the build and release management cycle in DevOps.

CO2: To examine the Assess packages, repositories, needs, and usage.

CO3: To experiment with building a management tool for dependency management.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3		3									3		
CO2	3	3		3	2								3	1	3
CO3	3	3	3	3	3								3	3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

6 lecture hours

Build and Release Management Overview: Overview of Build Management, Build Abstraction, Overview of Packaging Management, Packaging repositories, Package Registries using the Package, Package Manager Vs Package Management, Package Management Tools.

Module II:

7 lecture hours

Build and Release Concept and Process, release management, Overview of Release and Deployment Management, Objective and Benefits, Key terminologies used in Release Management, working of Release Management, Release Management Process Components, Release Lifecycle Management, Change management, Need of change management, Change Management Vs Release Management.

Module III:

10 lectures hours

Declarative Dependency Management, Dependency Management and Build Tools, Repositories, Dependency Identification, Transitive Dependencies, Dependency Scope, Tools (Maven, Ant and Gradle) introduction, Apache Ant Fundamental, Apache Maven Fundamental, Apache Maven Advance, MsBuild Fundamental, Gradle Fundamental.

Module IV:

5 lectures hours

Documentation and Reporting: Documentation Vs Technical Documentation, Different Types of Documentation, General Practice in Documentation, Agile and Waterfall Approaches, Tools used for Documentation, System Vs User Documentation, Discuss Some samples/Templates used in General for Software Documentation.

Studio Work/Laboratory Experiments:

Organizations that have adopted agile software development are seeing much higher quantities of releases. With the increasing popularity of agile development, a new approach to software releases known as Continuous delivery is starting to influence how software transitions from development to a release. One goal of Continuous Delivery and DevOps is to release more reliable applications faster and more frequently. The movement of the application from a build through different environments to production as a release is part of the Continuous Delivery pipeline. The lab work for this subject deal with the concept that how the students can build and release their software.

Text Books :

1. Blokdyk Gerardus, *Change and Release, Management A Complete Guide* (1 ed.), 5STARCook, 2021. ISBN 978-1867310242.

Reference Books :

1. Arora Tarun, and Utkarsh Shigihalli, *Azure DevOps Server* (1 ed.), Packt Publishing, 2019. ISBN 9781788830148.
2. Zaal Sjoukje, Stefano Demiliani and Amit Malik, *Azure DevOps Explained* (1 ed.), Packt Publishing, 2020. ISBN 9781800568657.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET449	Cloud Services Development & Operations	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate service API Gateway and using of serverless computing.

CO2: To build, test and integrate cloud services.

CO3: To deploy, operate and maintain the secure cloud service solution.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1		3	2		3								3	3	2
CO2	2	3	3	2	3								3	3	
CO3	2	2	3		2								2	3	

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

Systems Development Lifecycle, Workflow for developing cloud application on public clouds, Cloud Software Development Kit (SDKs), SDK Terminology, Cloud Testing Service, Errors and Exceptions, Cloud Service developing tools, Storage Solutions with public cloud, Developing NoSQL Solutions with public cloud platform, Caching Information for Scalability.

Module II:

9 lecture hours

Cloud notification services, Message Queueing Service, Events Processing with Serverless computing service, Implementation and Authentication using Serverless computing, Serverless best Practices, Serverless Step Functions, Secure network Connections, manage applications secrets, Authenticate with security token, Cognito for user sign-up and sign-in to the web and mobile app, Container, Container Management.

Module III:

9 lectures hours

Configure auto-scaling, launch auto-scaling, Self-healing Kubernetes cluster, Deploy the microservices using Kubernetes cluster, Cloud service registration, Cloud service discovery, Configure scaling and self-healing for each service, best practices for debugging, Microservices, Monitoring and Logging for microservices, Case Study of resilience cloud service.

Studio Work/Laboratory Experiments:

The lab objectives are to test serverless applications, minimize risk of vendor lock-in, create multiple stages of our API and improve security and observability of our application on cloud platform.

Text Books :

1. Stephen Orban, *Ahead in the Cloud: Best Practices for navigating the Future of Enterprise IT (1st ed.)*, CreateSpace Independent Publishing Platform, 2018. ISBN 9781981924318. .
2. Wadia and Yohan, *AWS Administration–The Definitive Guide (1st ed.)*, Packt Publishing Ltd, 2016. ISBN 9781782173755..

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET450	Source and Version Control in DevOps	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate the technical aspects of Source and Version Control systems in DevOps

CO2: To examine the version control systems for tracking, branching, merging, and managing code revisions.

CO3: To make use of the workflows of various version control systems like Git, Sun and Mercurial.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	3			3								3	3	
CO2	3	3	2	2	3								3	3	
CO3	2	2			3									3	

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

8 lecture hours

Source Version Control (SVC), SVN (Sub Version Systems), Mercury and Git, Single Source repository, Creation, Commit, Builds, Cloud Computing, Version Control Over Cloud, Implementation of Version control over cloud, copy-merge and merge -update algorithms, Practices and rules of version control systems, Industries using version control system (Case study).

Module II:

9 lecture hours

Distributed Version Control System (DVCS), DVCS Vs CVS, Local Repository, Advantages of Distributed version Control System, Local repository operations (Resetting Local Environment, Revert - Cancelling Out Changes, Private Workspace), Multiple Repositories Models, Repositories for Different Services, Merging Repositories, Branching Repositories, Automation of SVC, BitBucket SVC on cloud, BitBucket SVC on local system.

Module III:

7 lectures hours

Continuous Integration and Continuous Deployment (CICD), CI Models, CI Practices over SVC, Operations of CI over SVC, Automated CI over SVC, CD over SVC, CD Models, CD Practices over SVC, Operations of CD over SVC, Automated CD over SVC, CICD Tool.

Module IV:

4 lectures hours

Git Basics and Remote Repositories, BitBucket Server and its working, Security over BitBucket, Creation of Projects over BitBucket.

Studio Work/Laboratory Experiments:

The lab components of this course demonstrate how to implement GitHub on desktop to manage local and remote repository and explores the functionality of Software Version Control Systems with hands-on experience of concepts taught in the lecture.

Text Books :

1. Gerardus Blokdyk, *Distributed Version Control System A Complete Guide (1st ed.)*, 5STARCook, 2021. ISBN 978-1867331193.
2. Leornardo Christian, *Git: A fast and easy guide to version control (1st ed.)*, Independently published, 2020. ISBN 979-8642660034.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET451	Software Craftmanship in DevOps	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1 : To articulate the concept of Software Craftmanship

CO2 : To examine the concept of software documentation, structure, testing and validation.

CO3 : To implement the rules and principles of software craftmanship.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	2											2		
CO2			3	2	3								3	3	2
CO3		3	3										3	3	

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

8 lecture hours

Software Craftmanship, Four Dimensions of Quality, Software Ethics, Clean code, Craftmanship Vs Engineering, Frameworks and Tools, Design, Structure, Formatting, and Documentation of Code, Types of Designs/Models, Design Structure Matrix (DSM), Product-Service System using DSM.

Module II:

8 lecture hours

Process Documentation, Product Documentation, Different Phases of Design, Requirement of Documentation, Advantages of Documentation, Technical and Non-Technical Challenges in Code Documentation, Content Authoring, Formatting, Styling, Issue Tracking, Testing, Publishing, Documentation: Testing, Debugging, Refactoring Improving Structure.

Module III:

7 lectures hours

Testing and Validation, Different Types of Testing, Properties of Testing, Customizable, Extendable, Link Validity, Component Checking, Semantic, and Syntax Parsing.

Module IV:

5 lectures hours

Frameworks, Tools, and the Programming Process, DevOps Frame Definition, Agile Framework, Scaled Agile Framework, Adoption Framework, Industry Practices.

Studio Work/Laboratory Experiments:

The lab work of this subject deals with documentation as well as designing projects using different models. The projects should also go through different testing and validation. Also, define appropriate boundaries and layers, and organize components and services.

Text Books :

1. *Fowler Martin, Refactoring: Improving the Design of Existing Code (2nd ed.), Addison- Wesley, 2019. ISBN 978-0134757599.*
2. *Martin Series Robert C., Software Craftsman, The: Professionalism, Pragmatism, Pride 1st Edition (1st ed.), Addison Wesley, 2020. ISBN 978-0134052502.*

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET452	Software Containerization in DevOps	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1 : To articulate the software containerization with virtual machines.

CO2 : To examine the need for Manipulation, Coordination and Orchestration Tools for a large production environment.

CO3 : To implement Containers and Image Operations along with Container Networking.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	1		2		1		1					1		2	
CO2	2				3	1	3				2	2		3	2
CO3	3	2	2	3	3		1					3			1

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

5 lecture hours

Containerization and Virtualization, Virtual Machine (VM), Hypervisor and its types, Architecture of VM, Architecture of Container, Docker Engine, Difference between Containers and Virtual Machines.

Module II:

7 lecture hours

Container Manipulation, Container's Providers Examples, Cloud Platforms using Containers, Multiple, Scalability, Portability of Containers, Orchestration, Architecture of Orchestration, Container Orchestration, Cloud Platforms using container orchestration, Container Operations, Image Operations Running Multiple Containers, Custom Image in Containers.

Module III:

9 lectures hours

Creation of Custom Image, Need and Benefits of Custom Image, Running Container using Custom Image, Publishing of Custom Image, The Chroot System, FreeBSD Jails, Linux Containers, Docker architecture, Docker Daemon (Container Platform), Docker Rest API, Industries using Dockers, Networking, Container Networking, Architecture of Container Network, Types of Container Networking, Use of Container networking in Web-Scale Applications.

Module IV:

7 lectures hours

OpenShift Enterprise, OpenShift Features, OpenShift in DevOps, Providers of OpenShift, OpenShift in DevOps, Providers of OpenShift, Container and Microservices, Kubernetes, Kubernetes on cloud, Docker Swarm and Kubernetes, Deployment of Containers, Monitoring of containers, Elastic Container Service providers (AWS, Azure).

Studio Work/Laboratory Experiments:

The lab work demonstrates orchestration tools like Docker, Swarm, and Kubernetes with hands-on experience of concepts taught in the lecture. Students will gain hands-on experience using real accounts on AWS, Azure, Google Cloud Platform.

Text Books :

1. Panthofer Mark, *Mastering Docker Enterprise: A companion guide for agile container adoption* (1 ed.), Packt Publishing,, 2019. ISBN 978-1789612073.

Reference Books :

1. Rice Liz, *Container Security: Fundamental Technology Concepts that Protect Containerized Applications* (1 ed.), O'Reilly Media, 2020. ISBN 978-1492056706.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET453	System Provisioning and Configuration Management in DevOps			L	T P C
Owning School/Department	School of Computer Science Engineering and Technology			2	0 2 3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain the system provisioning and configuration management.

CO2: To examine the various configuration tools and their suitability.

CO3: To implement and use of configuration management tools.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	2	1										3		
CO2	2	3			3								3	2	
CO3	2	2			3								2	3	

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

5 lecture hours

Overview of Provisioning: Introduction and Significance of Provisioning. Need and Role of Provisioning under DevOps Model, Provisioning Vs Configuration, Infrastructure Provisioning and its Tools, Environment Provisioning., Automated Provisioning.

Module II:

7 lecture hours

Overview of Configuration Management: Introduction of Configuration Management, Basic Terminologies of Configuration Management, Server, Nodes, Workstation, Workstation Setup, Organization Setup, Test Node Setup, Benefits of Configuration Management Servers, Configuration Management Tools, Features and Comparison of Configuration Management Tools, Selection Criteria of Tools.

Module III:

9 lectures hours

Configuration Management & Tools (Puppet, Chef, Ansible, and Saltstack): Puppet Definition, Working, Architecture, Master and Agents, Installation and Configuration of Puppet, Ansible Definition, Ansible Requirement Specification, Ansible Components, Ansible Strategy, Saltstack Introduction, Important Features of Saltstack, SaltStack Enterprise and Salt Open-Source Software Versions, Salt vs. Ansible, Puppet.

Module IV:

7 lectures hours

Application Configuration: Introduction to ConfigMaps and Secrets, Application Configuration with ConfigMaps and Secrets, Creating Config Map, Environment Specific Config, Creating Secrets, Setting Environment for Secrets. Pods Overview: Introduction to Pod, Pod Specification, Pod Lifecycle, Launching Pods, Pods Operation, Pods and Containers, Pods and Controllers, Pods Template, Resource Sharing and Communication: Storage and Networking in Pods, Static Pod.

Studio Work/Laboratory Experiments:

The lab work of this subject deals with Design, development, and solving real-world automation and orchestration problems by unlocking the automation capabilities of Configuration tools (Puppet, Chef, Ansible, and Saltstack) used for DevOps. A small web application-based software is to be developed using any of the tools of DevOps.

Text Books :

1. *Freeman James and Jesse Keating, Mastering Ansible: Effectively automate configuration management and deployment challenges with Ansible 2.7 (1st ed.)*, Packt Publishing, 2019. ISBN 978-1789951547.

Reference Books :

1. *Verona Joakim, Practical DevOps (2nd ed.)*, Packt Publishing, 2018. ISBN 978-1788392570

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET454	Test Automation in DevOps	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate the various testing approaches.

CO2: To examine the automation testing technologies through tools.

CO3: To make use of various automation test cases under DevOps environments.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3		3									3	3	1
CO2		3	3	3	3							3	3		2
CO3	1	3	3	3	3							3	3		1

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

8 lecture hours

Test Automation Manifesto and Testing: Test Automation Use cases, Principles, SDLC vs STLC, Testing Life Cycle, Usability Testing, Functional Testing, non-functional testing, End to End Testing, Compatibility Testing, GUI Testing, API testing, Usability Testing.

Module II:

6 lecture hours

Testing Approaches and Test Cases Design: Manual Testing, Automation Testing, Unit Testing, Integration Testing, System Testing, Acceptance Testing, Smoke-Sanity Testing, Regression Testing, Test Scenario, Test Case Design, Test Basis, Traceability Matrix.

Module III:

7 lectures hours

DevOps Testing Best Practices, Business Test Automation, Technical Test Automation, Non-Functional Test Automation, Functional Test Automation, Different Test Automation Tools(Kobiton, Ranorex, Zeuz, etc.).

Module IV:

7 lectures hours

Mobile Automation and Cloud Testing: Mobile Automation and its Requirement, Application and Role of DevOps in Mobile Automation, Cloud Testing, Cloud Testing Setup, Intelligent Test Agents. Continuous Testing, Continuous Testing Requirement and its Advantages, Continuous Testing Strategy, Continuous Delivery, Continuous Testing Vs Continuous Delivery.

Studio Work/Laboratory Experiments:

The lab component of this course demonstrates test cases design and illustrates automation testing and continuous integration with tools like Agile and Jenkins. The practical work will help to identify all the stages of your release along with the gates and requirements necessary in a

build's journey to production. Also, all the operational procedures, services, and actions involved with a release cycle will be elaborated.

Text Books :

1. Wolfgang Platz and Cynthia Dunlop, *Enterprise Continuous Testing: Transforming Testing for Agile and DevOps (1st ed.) (1 ed.)*, CreateSpace Independent Publishing Platform, 2019. ISBN 978- 1699022948.

Reference Books :

1. Arnon Axelrod, *Complete Guide to Test Automation (1 ed.)*, Apress, 2019. ISBN 978-1484238318.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET456	Special Topics in DevOps			L	T
Owning School/Department	School of Computer Science Engineering and Technology			P	C
	2	0	2	3	
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To familiarize and learn about the latest trends and research in the field.

CO2: To equip themselves with the conceptual and practical experience of few latest methods, tools, technologies or algorithms in DevOps.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	<u>3</u>	<u>3</u>		<u>3</u>									<u>3</u>		<u>1</u>
CO2	<u>3</u>	<u>3</u>	<u>3</u>	<u>3</u>	<u>3</u>	<u>3</u>						<u>3</u>	<u>3</u>	<u>3</u>	<u>3</u>

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Unit/Module I:

28 lecture hours

This course covers the cutting-edge topics in DevOps, and these modules will be chosen by the instructor based on the requirements and relevance at that point of time. These modules need to be relevant to the industry and start-ups will also include related case studies, use cases and implementations scenarios. Students will be working on lab work and projects to get real hands-on experience of these topics and modules.

Studio Work/Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to DevOps.

Specialization Electives: Full Stack

Name of Program	Bachelor of Technology (Computer Science Engineering)						
CSET457	Front-End Web UI Frameworks and Tools:Bootstrap	L	T	P	C		
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3		
Pre-requisites/Exposure	-						

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate grids and responsive design, Bootstrap CSS, and JavaScript components.

CO2: To examine Bootstrap Grid and display of responsive web page design.

CO3: To implement Bootstrap components and Plug-Ins.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3		3									3		
CO2	3	3	3	3									3		
CO3	3	3	3	3	3								3	3	3

Course Content:

Module 1

16 lecture hours

Responsiveness, Features of Bootstrap, Mobile-First Strategy, Setting up Environment, Bootstrap to Applications, Bootstrap Grid, Setting up Git, Apply Bootstrap Grid, Display responsive Images, Use readymade themes, Customize Bootstrap's components, Less, Variables, jQuery plug-in, Bootstrap Typography, Use Typography, Bootstrap Tables, Bootstrap Form Layout, Bootstrap Button, Images in different styles: Circle shape, Display text like muted and warning, Carets Classes, Hide or show the text in Bootstrap. Bootstrap Components: Advantages, Types, Glyphicons Component: Use, Navigation Pills & Tabs Components, Navbar Component, build a Responsive Navbar, Forms and Navbar controls, Navbar Position, Breadcrumb, Component, Pagination Component, Labels/Badge Components, Jumbotron / Page Header Components, Thumbnail Component, Alerts & Dismissible Alerts, Progress Bar, Media Objects Component, Use Media Objects Component, Bootstrap List Group Component, Bootstrap Panel Component.

Module II

12 lecture hours

Bootstrap Plug-Ins, Use Bootstrap Plug-Ins, Transition Plug-in, Modal Dialog Box, Properties, Methods and Events of Model Dialog Box, Scrollspy Plug-In, Tab Plug-in, use Tab Plug-in, Drop Down Plug-in, Tooltip Plug-in, Use Button Plug-in, Methods and events of Tooltip Plug-in, Popover Plug-in, alert and Button Plug-ins, Collapse Plug-in: Properties, Collapse Plug-in: Methods, Events of Collapse Plug-in, Carousel Plug-in, Affix Plug-in.

Studio Work / Laboratory Experiments:

Studio work consists of UI Design and Prototyping using Front-End Web UI Frameworks and Tools of Bootstrap 4 and advanced CSS and JavaScript. Here, building responsive design and Bootstrap Grid System, Learning Git, and utilizing Bootstrap CSS Components for displaying content is also targeted.

Text Books:

1. Chiaretta, Simone. *Front-end Development with ASP. NET Core, Angular, and Bootstrap*. John Wiley & Sons, 2018. ISBN 9781119243963.
2. Moreto, Silvio, Matt Lambert, Benjamin Jakobus, and Jason Marah. *Bootstrap 4–Responsive Web Design*. Packt Publishing Ltd, 2017. ISBN 9781785288876.

Reference Books:

1. Duckett, Jon. *Javascript and jquery: Interactive front-end web development*. Wiley Publishing, 2014.. ISBN 9781118531648.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET458	Front-End Web UI Frameworks and Tools: React Native	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To understand of JavaScript, react and native components.

CO2: To design and display of web app and mobile app.

CO3: To implement native components and Plug-Ins.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3		3									3		3
CO2	3	3	2	2	3								3	3	3
CO3	3	3	2	2	3								3	3	3

Course Content:

Module 1

7 lecture hours

Course handout assessment mechanism, Front end web development, JavaScript: Arrow Functions, React, spread operator, Default params, Let and const variables, React Components: State and Props, composing components, adding states, React Components: Lifecycle, Event handling, Intermediate React: Rendering lists, Form elements, Forms: Controlled, Uncontrolled, Functional Components. React Virtual DOM, The Flux Architecture, Redux, Redux form validation, Brief Representational State Transfer (REST).

Module II

7 lecture hours

React native, Animation API, Components: Basic, Stateful, Stateless, Custom fonts, Component API, Higher-Order Components: Keys, Fragments, Constructor, Animations, HTTP Requests: Syntax, XMLHttpRequest, Web Socket with Socket.io, Images module: Image source, image path, Network image, Firebase integration, ReactNative ListView, Authentication in ReactNative Using Firebase, Layouts, ListView Navigators, Bottom Navigation, Push Notifications: Simple and Navigation based.

Studio Work / Laboratory Experiments:

Studio work consists of React library for JavaScript based front-end application development. It uses of JavaScript ES6, JSX and React framework for building user interfaces in front-end application development. Developing web Applications using React is to be focused.

Text Books:

1. Banks, Alex, and Eve Porcello. *Learning React: functional web development with React and Redux*. " O'Reilly Media, Inc.", 2017. ISBN 9352135636.
2. Stefanov, Stoyan. *React: Up & Running*. " O'Reilly Media, Inc.", 2021. ISBN 1491931825.

Reference Books:

1. Vipul, A. M., and Prathamesh Sonpatki. *ReactJS by Example-Building Modern Web Applications with React*. Packt Publishing Ltd, 2016. ISBN 1785289640.
2. Gackenhimer, Cory. *Introduction to React*. Apress, 2015. ISBN 1484212460.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET459	Front-End UI Frameworks and Tools: Flutter and Django	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (Cos)

On completion of this course, the students will be able to:

CO1: To understand of UI design with flutter components.

CO2: To design the flutter app and understand Django.

CO3: To implement Flutter and Django components and libraries.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3		3									3		
CO2	3	3	3	3	3								3	3	3
CO3	3	3	3	3	3								3	3	3

Course Content:

Module 1

6 lecture hours

Basics of Flutter and Dart Programming, Dart Programming – Syntax: Variables, Data type, Dart conditional operators: If-else, For, while do-while Dart Functions: structure, return type, expression, OOP: classes, object, Methods, constructor, Inheritance, Abstract class, Flutter widgets: Scaffold, image, container, Icon, Flutter Layouts: card. Stateful and stateless widget, Navigation: button widget-floating raised, flat, dropdown, outline, Flutter Webview widget, Visual, Behavioral, and Motion-Rich Widgets, List View, Data View, Date Picker, Time Picker, Dialogs, alerts.

Module II

8 lecture hours

Django structure, URL dispatcher Requests and Responses, GET and POST methods, MVC model and Generic Views, Images, Forms Alert Handling, Django REST, Generic Views in Django Rest, Building RESTful APIs, Database- Add tables, Dynamic progress bar Deploying Web APIs, Advanced Django for Web and Automation.

Text Books:

1. Zaccagnino, Carmine. "Programming Flutter: Native, Cross-Platform Apps the Easy Way." *Programming Flutter* (2020): 1-275.. ISBN 1680506951.

Reference Books:

1. Payne, Rap. *Beginning app development with flutter: Create cross-platform mobile apps*. Apress, 2019.. ISBN 1484251806.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET460	Special Topics in Full Stack	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To familiarize and learn about the latest trends and research in the field.

CO2: To equip themselves with the conceptual and practical experience of few latest methods, tools, technologies, or algorithms in Full Stack.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3		3									3		3
CO2	3	3	3	3	3								3	3	3

Module 1

28 lecture hours

This course covers the cutting-edge topics in Full Stack, and these modules will be chosen by the instructor based on the requirements and relevance at that point of time. These modules need to be relevant to the industry and start-ups will also include related case studies, use cases and implementations scenarios. Students will be working on lab work and projects to get real hands-on experience of these topics and modules.

Specialization Electives: Cloud Computing

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET461	Cloud System Administration & Operations	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To get fundamentals of support and maintain cloud workloads as per the cloud architecture.

CO2: To perform operations by using the cloud management console, CLI and SDK.

CO3: To implement architectural requirements with monitoring, logging, and troubleshooting.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3		3	2									3		
CO2	3	3	2	3	3								3	2	3
CO3	3	3	3	3	3								3	3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

9 lecture hours

Need of cloud administration and operations, Roles and responsibilities of cloud Admin, Inter-cloud Resource Management, Resource Deployment and Provisioning, Identifying the steps to provision cloud resources, Identify, and remediate deployment issues, Storage and Data Management Connectivity services of public/private cloud Automation and Optimization, Manage and assess resource utilization, Employ cost optimization strategies, Automate manual or repeatable process, Minimize management overhead, Monitoring and Reporting, Maintain metrics and alarms utilizing, Recognize and differentiate performance metrics, Availability metrics, Manage security policies on cloud, Access controls when using cloud, Shared responsibility model.

Module II:

12 lecture hours

Data Centre Management Tools Integration, Service and Resource Management, implement scalability and elasticity, highly available and resilient environments on cloud, automate snapshots, Data Lifecycle Manager, Data Retention policy, Restore databases versioning, Lifecycle rules, Disaster recovery procedures, Cost optimization strategies, Networking and Content Delivery, Performance optimization strategies.

Module III:

7 lectures hours

OS boot process (Win/Linux) and Troubleshooting: Memory management, Memory pages, Buffer and Caches, System date/time management, Network time protocol, Network Logs Auditing, Managing Users and groups, File permissions, Assessment/Buffer Lecture, managing software, Managing system services, Background processes, Windows start-up tasks Integrated Internet-aware network

troubleshooting, Network tools to report the problem, Network address translation, troubleshooting in common VPC, Troubleshooting in On-premises to VPC, System performance – DISK, System performance – Memory, System performance – Network System Login issues, System booting issues System logs, Network connectivity issues

Studio Work/Laboratory Experiments:

Studio work focuses on seven cloud sub domains that are as; 1. Monitoring and Reporting, 2. High 3. Availability, 4. Deployment and Provisioning, 5. Storage and Data Management 6. Security and Compliance, 7. Networking and Automation and Optimization.

Text Books :

1. S., Digby, G., Fitch, C., Friedberg, S., Qualheim, S., Rhoads, J and Sundrud, B, *AWS Certified SysOps Administrator Official Study Guide: Associate Exam (1 ed.)*, John Wiley & Sons, 2017. ISBN 9781119377429.
2. Chandra Rajasekharaiah, *Cloud-Based Microservices: Techniques, Challenges, and Solutions (1 ed.)*, Apress, 2020. ISBN 9781484265642.

Reference Books :

1. Jackson, K. L., & Goessling, *Architecting Cloud Computing Solutions: Build cloud strategies that align technology and economics while effectively managing risk (1 ed.)*, Packt Publishing Ltd, 2017. ISBN 9781788472425.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET462	Cloud Security & Compliances	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To learn and examine the security breaches of IaaS, PaaS, and SaaS.

CO2: To apply various data encryption methods and security mechanisms to get the administrative control using IAM service.

CO3: To create a secure production environment using cloud security features and services.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3		3									3		
CO2	3	3	2	3	3								3	2	3
CO3	3	3	3	3	3								3	3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

Security Patterns for Cloud Computing – Network Security, Identity & Access Management & Trust, Secure On-Premises Internet Access, Secure External Cloud Connection, Cloud Denial-of- Service Protection, Cloud Traffic Hijacking Protection, Automatically Defined Perimeter, Cloud Authentication Gateway, Federated Cloud Authentication, Cloud Key Management, Trust Attestation Service, Collaborative Monitoring and Logging, Independent Cloud Auditing

Module II:

11 lecture hours

End-to-end security, Shared responsibility in the cloud, Backup and restore, Detect and mitigate threats, Plan for ransomware, recovering from systemic identity compromise, Threat protection, Securing Workloads, Security technical capabilities, Physical security, Components and boundaries, Operations and Monitoring, Platform integrity and security, Secure Data Protection Law: national and International, Firmware security, Code integrity, Secure Boot, Secure Isolation of Physical & Logical Infrastructure, Compute, Network, Storage, Common attack vectors and threats, Secure Isolation Strategies, Multitenancy, Virtualization strategies.

Module III:

7 lectures hours

Data Protection for Cloud Infrastructure and Services. Understand the Cloud based Information Life, Cycle, Data protection for Confidentiality and Integrity, Encryption, Data Redaction, Tokenization, Obfuscation, PKI and Key, Management, Assuring data deletion, Data retention, deletion and archiving procedures for tenant data, Data Protection Strategies.

Studio Work/Laboratory Experiments:

Learner will be able to examine Public Clouds Shared Responsibility Model, Identify the security and compliance benefits of using the AWS/Azure/GCP cloud, Describe the basic features for access control and management, Describe AWS/Azure/GCP data encryption methods, Describe AWS/Azure/GCP services used to protect network security, Describe the basic steps to ensure strong governance of your AWS resources, examine legal and regulatory issues related to cybersecurity.

Text Books :

1. *Chen, Lei, Hassan Takabi, and Nhien-An Le-Khac, Security, privacy, and digital forensics in the cloud (1 ed.), John Wiley & Sons, 2019. ISBN 13: 9781119053286.*
2. *Russell, Brian, and Drew Van Duren, Practical Internet of Things Security: Design a security framework for an Internet connected ecosystem (2 ed.), Packt Publishing Ltd, 2018. ISBN 13: 9781788625821.*

Reference Books :

1. *Anthony, Mastering AWS Security: Create and maintain a secure cloud ecosystem (1 ed.), Packt Publishing Ltd, 2017. ISBN 13: 9781788293723.*
2. *Tevault, Donald A, Mastering Linux Security and Hardening: Secure your Linux server and protect it from intruders, malware attacks, and other external threats (1 ed.), Packt Publishing Ltd, 2018. ISBN 1788620305.*

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET463	AWS Cloud Support Associate	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To identify the possible scope of bug, error, and glitch in system.

CO2: To analyse and solve problems for AWS client/user.

CO3: To get expertise in AWS cloud solutions to solve technology challenges.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3		3									3		2
CO2	3	3		3									3	2	3
CO3	3	3	3	3	3								3	3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

Setting up AWS accounts, Accessing Methods of AWS Services, AWS Organizations, AWS Cost Explorer, AWS Technical Support, Cloud Billing and Support, AWS Well-Architected Framework Well-Architected Design Principles, Choosing a Region, Selecting Availability Zones, Virtual Private Cloud, Dividing VPCs and Subnets, Default VPCs and Default Subnets, Controlling VPC Traffic, Connecting Multiple VPCs, Integrating On-premises Components, Load Balancing and Fault Tolerance, High Availability Across Regions, Connections Outside of Amazon VPC, Designing for High Availability CLOUD Solution, Scaling Data Stores, AWS Lambda and Event-Driven Scaling, Manual AWS Environment Configuration, Infrastructure as code on AWS, Grouping resources in a template, Automating AWS Infrastructure.

Module II:

11 lecture hours

Loose Coupling, Loose Coupling Strategies, Communicating Easily and Reliably Among Components, Communicating with Loose Coupling and Amazon DynamoDB, Amazon API Gateway, Serverless Architectures, Decoupling Infrastructure, Storing Web-Accessible Content with Amazon, Caching with Amazon CloudFront, Managing NoSQL Databases, Storing Relational Data in Amazon RDS, Designing Web-Scale Media, Scalable Web Application, Operational Excellence Pillar, Security Pillar,

Reliability Pillar, Performance Efficiency Pillar, Cost Optimization, High- Availability Design Patterns, Stream Processing Example, Sensor Network Data Ingestion and Processing, Application Backend, Transcoding and Serving Video Files Example, Dedicated Instances and Dedicated Hosts, Trusted Advisor, Optimizing Costs with Caching, AWS Cost Calculation Tools, Amazon CloudWatch, Amazon CloudWatch Monitoring, Amazon CloudWatch Events, Amazon CloudWatch Logs, AWS CloudTrail, AWS Service Integration with Amazon Athena, AWS Config, Elastic Load Balancing, Amazon EC2 Auto Scaling.

Module III:

7 lectures hours

Amazon Route 53, Amazon Route 53 Routing, Configure EC2 Auto Scaling, Configure Failover Routing with Amazon Route 53, Continuous Integration and Continuous Deployment on AWS, AWS Developer Tools and Services, Database Cluster on Amazon Aurora, Amazon DynamoDB, AWS Database Migration Service (AWS DMS), Migrate to Amazon RDS, Tagging, AWS Trusted Advisor, Managing Resources, Optimize AWS Resource Utilization, Features of AWS Support plans, creating support cases and case management, Monitoring, resolving, and reopening the ticket/case, Access permissions for AWS Support, changing your AWS Support plan, Using AWS Support with an AWS SDK.

Studio Work/Laboratory Experiments:

By performing lab work on AWS cloud platform, students will have the knowledge of the Design principles and best practices of the Reliability pillar, select compute, storage, database, and networking resources to improve performance, evaluate the most important performance metrics for your applications, follow best practices to eliminate unneeded costs or suboptimal resources, troubleshoot common errors.

Text Books :

1. Piper, Ben, and David Clinton, *AWS Certified Solutions Architect Study Guide: Associate SAA-C02 Exam (1 ed.)*, John Wiley & Sons, 2020. ISBN 978-1119138556.

Reference Books :

1. S Wadia and Yohan, *AWS Administration–The Definitive Guide* (2 ed.), Packt Publishing Ltd, 2016. ISBN 9781782173755

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET464	Developing Solutions for Microsoft Azure	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate Azure IaaS, PaaS, and SaaS solutions.

CO2: To develop the applications using Azure services integration.

CO3: To deploy, operate, and monitor Azure solutions.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3		3									3		
CO2	3	3	3	3	3								3	2	3
CO3	3	3	3	3	3								3	3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

Setting up Azure accounts, Azure Console Walkthrough, Microsoft Azure Services, Azure CLI, Azure PowerShell, Configuring Azure CLI, Azure Resources & Subscriptions, Azure Resource Manager, Azure Tags, Azure Storage Account & its types, Azure Blob Storage Azure Content Delivery Network (CDN), Azure Files Storage, Azure File Sync, Azure Table Storage, Azure Queue Storage, Azure Storage Explorer, Azure Shared Access Signature (SAS), Azure Databox, Azure Storage Replication Data replication options, Azure Import/Export Service, Azure virtual machines, Data disks in Azure Azure VMs and interfaces, ARM templates, VHD templates, Custom images of Azure VM, Virtual machine scale sets, Virtual machine availability sets, App Service Web App for containers, App Service plan, Networking for an App Service, Deployment slots, Container image.

Module II:

10 lecture hours

Azure Kubernetes Service, Azure Container Registry, Azure virtual networks, Azure VNetComponents, Azure VNet subnets, Azure Network Interface Cards (NIC), Network Security Groups (NSGs), Route tables, Service tags, Azure DNS, Private DNS, Application Gateway, Azure Front door Service, Azure Traffic Manager, Application Security Groups, Azure Load Balancer, Azure,

ExpressRoute, ExpressRoute Circuits, ExpressRoute Peering, Azure Firewall, Azure Bastion, Network Watcher, Identity and Access Management in Azure, Role-based Access Control (RBAC), Role definitions, Role assignment in Azure resources, Azure users and groups, RBAC policies.

Module III:

8 lectures hours

Microsoft azure active directory, Azure Monitoring and Autoscaling, Azure Data Platform, Azure Migration, Server-side programming model, Triggers, User-defined functions (UDFs), Table API, Cosmos DB graph database, populating a graph, Writing Gremlin queries, Azure Table Storage, integrate caching and content delivery, Configure cache and expiration policies for Azure Redis Cache Connections, App Configuration Azure Key Vault, Key Encryption, expiration.

Studio Work/Laboratory Experiments:

Studio work covers accessing Azure cloud services with handling data migration based on provided architectural requirements using Azure migrate service. Learners would be able to create and migrate Azure virtual machines and create storage accounts and application gateways for aiding Azure Migrate. They will also develop the applications using Azure services by using cloud architecture.

Text Books :

1. *Orban and Stephen, Ahead in the Cloud: Best Practices for navigating the Future of Enterprise IT (1 ed.)*, CreateSpace Independent Publishing Platform, 2018. ISBN 9781981924318.
2. *Modi, Ritesh, Azure for Architects: Implementing cloud design, DevOps, containers, IoT, and serverless solutions on your public cloud (2 ed.)*, Packt Publishing Ltd., 2019. ISBN 9781788398732.

Reference Books :

- 1 *Benjamin Perkins and William Panek, Microsoft Azure Architect Technologies, and Design Complete Study Guide: Exams AZ-303 and AZ-304 (1 ed.)*, Wiley Publication, 2020. ISBN 9781119559573.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET465	Google Associate Cloud Engineer	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1 : To articulate the Google cloud services and platform

CO2 : To integrate cloud resources and services using Google cloud.

CO3 : To develop, deploy and monitor secured cloud-based solution on the GCP Platform

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	<u>1</u>	<u>2</u>		<u>2</u>	<u>2</u>							<u>1</u>	<u>3</u>		
CO2		<u>3</u>		<u>3</u>	<u>1</u>							<u>2</u>	<u>1</u>	<u>3</u>	
CO3	<u>2</u>	<u>2</u>	<u>3</u>	<u>2</u>	<u>3</u>							<u>3</u>	<u>1</u>	<u>1</u>	

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

Setting up cloud projects and accounts, creating projects in GCP, Viewing Google Cloud Platform Status Cloud Console, Cloud Shell, Cloud SDK, Configuring access and security, Assigning users to predefined IAM roles within a project, Managing users, groups, and roles in Cloud Identity, Enabling APIs within projects, Provisioning one or more Stackdriver workspaces, Managing billing configuration, Creating one or more billing accounts, Linking projects to a billing account, Establishing billing budgets and alerts, Setting up billing exports to estimate daily/monthly charges, Billing Services, Configuring the command line interface (CLI), Accessing services using CLI, Working with GCP SDK, Integration of GCP Platform with other Cloud Services.

Module II:

10 lecture hours

Planning and estimating GCP product use, Pricing Calculator, Planning and configuring compute resources, Selecting appropriate compute choices for workload, Compute Engine (20) Google Kubernetes Engine, App Engine, Power Shell, Cloud Functions, Deploying and implementing Compute Engine resources, deploying a Google Kubernetes Engine cluster, Planning and configuring network resource, Load balancing options, Identifying resource locations in a network for availability, Configuring Cloud DNS, Deploying a

container application to Google Kubernetes Engine, Auto Scaling in Google Kubernetes, Handling Fault Tolerance in Google Kubernetes.

Module III:

8 lectures hours

Deploying and implementing data solutions, Cloud SQL, Cloud Datastore, BigQuery, Cloud Spanner, Cloud Pub/Sub, Cloud Bigtable, Cloud Dataproc, Cloud Dataflow, Cloud Storage, Loading data, Deploying and implementing networking resource, Monitoring and logging, Creating Stackdriver alerts based on resource metrics, Creating Stackdriver custom metrics, configuring log sinks to export logs to external systems, Viewing and filtering logs in Stackdriver, Viewing specific log message details in Stackdriver, Cloud diagnostics, Cloud Trace data, Cloud Debug to view an application point, Case Study.

Studio Work/Laboratory Experiments:

Studio work focuses on the Google Cloud platform of deploying, monitoring, and maintaining projects on Google Cloud.

Text Books :

1. Legorie Rajan, Steven Porter, and Ted Hunter, *Building Google Cloud Platform Solutions (1 ed.)*, Packt Publishing, 2019. ISBN 97818386474382.

Reference Books :

1. Legorie Rajan, Steven Porter, and Ted Hunter, *Building Google Cloud Platform Solutions (1 ed.)*, Packt Publishing, 2019. ISBN 97818386474382.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET466	Special Topics in Cloud Computing	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To familiarize and learn about the latest trends and research in the field.

CO2: To equip themselves with the conceptual and practical experience of few latest methods, tools, technologies or algorithms in Cloud Computing.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	<u>2</u>	<u>3</u>	<u>2</u>	<u>3</u>	<u>1</u>							<u>2</u>		<u>2</u>	
CO2	<u>3</u>		<u>3</u>	<u>3</u>	<u>3</u>							<u>3</u>		<u>3</u>	

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

28 lecture hours

This course covers the cutting-edge topics in Cloud Computing, and these modules will be chosen by the instructor based on the requirements and relevance at that point of time. These modules need to be relevant to the industry and start-ups will also include related case studies, use cases and implementations scenarios. Students will be working on lab work and projects to get real hands-on experience of these topics and modules.

Studio Work/Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to Cloud Computing.

Text Books :

1. Judith S. Hurwitz and Daniel Kirsch., *Cloud Computing for Dummies*. Hoboken (1 ed.), John Wiley & Son, 2020. ISBN 978-1119546658.
2. Chandra Rajasekharaiah, *Cloud-Based Microservices: Techniques, Challenges, and Solutions* (1 ed.), Apress, 2020. ISBN 978-1484265642

Specialization

Electives:

Drones

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET467	Drone Communication	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To understand components, attributes, and challenges associated with the developing field of aerial communications.

CO2: To configure aerial vehicles using MAVLink and UAVCAN.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	1	1	1								2	1		
CO2	2	2	3	2	1							2	2	3	3

Module 1

7 lecture hours

UAV Miniaturization: Challenges and Opportunities; Unique Network Characteristics; Aerial Mobility Models; Aerial Communication: Radar Identification, Beyond Radar Measurements, Aerial Voice Communication; Single Antenna Systems; Multi-Antenna Systems; Communication Design Requirements: Vehicular Environment, 3D Tracking, Spontaneous Mobility, Timing Constraints, Device Autonomy, Mission Autonomy.

Module II

9 lecture hours

Data Link Functions; Data Link Attributes; Data Link Margin; AJ Margin; Propagation; Data Rate; Data Link Trade Off; Aerial Channel Modelling: Free Space Path Loss, Ray Tracing, Long Distance Path Loss, Shadowing, Line of Sight Probability, Atmospheric Interventions; MAVLink Protocol; MAVLink Message Types; MAVLink; Message Format; MAVLink Heartbeat; MAVLink Commands; MAVLink Security Requirements; MAVLink Security Threats.

Module III

8 lecture hours

UAV for Coverage; UAV for Data Collection; UAV Assisted Wireless/Cellular Networks; Cooperative Communication; UAV Assisted Ground Networks; Disruption-Tolerant Airborne Networks; UAV Security; UAV Safety; Drone Communication: Smart Cities; Under Water; UAS Safety Deployments; Security in Smart Cities; Internet of Drones.

Module IV**4 lecture hours**

Mobile Technology Enabled Drones; LTE; Integrating UAS into the NAS: Civil Aviation, Legislation; 3GPP Standardization on UAV.

Studio Work / Laboratory Experiments:

In lab works, students will configure and deploy UAVs using MAVLink and UAVCAN protocols.

Text Books:

1. Saad, Walid, Mehdi Bennis, Mohammad Mozaffari, and Xingqin Lin. *Wireless communications and networking for unmanned aerial vehicles*. Cambridge University Press, 2020.
2. Fahlstrom, Paul G., Thomas J. Gleason, and Mohammad H. Sadraey. *Introduction to UAV systems*. John Wiley & Sons, 2022.

Reference Books:

1. Namuduri, Kamesh, Serge Chaumette, Jae H. Kim, and James PG Sterbenz, eds. *UAV networks and communications*. Cambridge University Press, 2017.

Name of Program	Bachelor of Technology				
CSET468	Drone Application, Components and Assembly	L	T	P	C
Owning School/Department	Computer Science Engineering	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To understand UAVs, UAV application, needs, deployments, and advancements.

CO2: To construct multirotor UAVs from scratch using various flight controllers.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	3	1	2	3	2	2	2	2	3	2	2	2	2	2
CO2	3	3	2	1			2	3		3		3	2	2	2

1=weakly related 2= moderately related 3=strongly related

Course Contents:

Module I:

06 lecture hours

Technological Advancements; Unmanned Aerial Vehicles; Unmanned Aerial Systems; UAV Concept of Operations (CONOPs); UAV Chemical, Biological, Radiological and Nuclear (CBRN); UAV Classification: Wing Geometry Based, Payload Based, Size Based, Mission-Based, UAV Anatomy, Mission Planning.

Module II:

08 lecture hours

Wing Configuration and Thrust; Airframe Configuration; Aerodynamic Configuration; Vibration and Noise; UAV Propulsion; Propeller Modelling; Motor Modelling; UAV Communication; Radio Wave Communication; Microwave Communication; Line of Sight Communication; Beyond Line-of-Sight Communication; Waypoint Tracking Navigation.

Module III:

07 lecture hours

Sensor Calibration; Tracking using GPS; UAV Stability; UAV Controllability; Flight Control; Autonomous Control; Semi-Autonomous Control.

Module IV:

07 lecture hours

Drone Sensing and Imagery: Riverine, Aquatic, Agriculture, Vegetation and Forest Ecosystem, Oil and Gas, Surveillance, Search and Rescue; UAS Ground Control; Human Machine Interface; Computer Telemetry; UAV Launch System; UAV Recovery System.

Studio Work / Laboratory Experiments:

In Lab work, students will start with basic Multicopter design principles, components, and assembly precautions. Then finally design and develop a fully functional Multicopter UAV/Drone using off- the-shelf Flight Controllers.

Text Books :

1. Quan, Quan. *Introduction to multicopter design and control*. 1st ed. Singapore: Springer, 2017.

Reference Books :

1. Barnhart, R. Kurt, Douglas M. Marshall, and Eric Shappee, eds. *Introduction to unmanned aircraft systems*. 2nd ed. Crc Press, 2021.
2. Quan, Quan, Xunhua Dai, and Shuai Wang. *Multicopter design and control practice: a series experiments based on MATLAB and Pixhawk*. 1st ed. Springer Nature, 2020.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET469	UAV Simulation	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To understand modeling platforms and 3D simulations of real-time UAS mission objectives.

CO2: To model UAV missions using emulators and gaming engines

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	1	1	1								2	1		
CO2	2	2	3	2	1							2	2	3	3

Course Contents:

Module 1

9 lecture hours

Experimental Process; Software Platform Configuration; Hardware Configuration; Simulink Based Control Design; UAV Scenarios; UAV Bindings; UAV Platform; Fixed Wing; Quadcopter; Attitude; Representation; Coordinate Frame; UAV Waypoints; External Attitude Control in Copter; External Attitude Control in Fixed Wing UAVs.

Module II

6 lecture hours

Sensor Calibration; Sensor Data Aggregation; State Estimation and Filter Design; Complementary Filter; Kalman Filter; Extended Kalman Filter; Trajectory Generation; Axis Angels; Euler Angels; Quaternions; Rotation Matrices; Point Clouds.

Module III

6 lecture hours

Waypoint Following; Waypoint Trajectories; Tuning Waypoints; UAV Path Management; UAV Obstacle Avoidance; Moving Obstacle Avoidance; High Fidelity UAV Guidance Model; Transition from Low to High Fidelity Models.

Module IV

7 lecture hours

Motion Planning using RTT; Motion Planning using Light Detection and Ranging; MAVLink; MAVLink Parameter Protocol; MAVLink Data Exchange; Flight Log Analysis.

Studio Work / Laboratory Experiments:

In the studio work, students will learn to model Pixhawk and ArduPilot systems using MATLAB and Unreal engines

Text Books:

1. Quan, Quan, Xunhua Dai, and Shuai Wang. *Multirotor design and control practice: a series of experiments based on MATLAB and Pixhawk*. Springer Nature, 2020.

Reference Books:

1. Yasmina Bestaoui Sebbane, Multi-UAV Planning and Task Allocation (1 ed.), CRC Press, 2020. ISBN 9781000049981

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET470	Drone Swarming	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To understand, model, implement and configure naturally occurring swarming behaviors to the hardwired aerial robots.

CO2: To Construct a fleet of autonomous aerial vehicles that work together to achieve a common goal.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	1	1	1								2	1		
CO2	2	2	3	2	1							2	2	3	3

Course Contents:

Module 1

8 lecture hours

Introduction to UAV Swarms; Swarm Performance; Self-Organization and Feedback; Self-Organizing Systems; Emergence; Homogeneous Swarms; Heterogeneous Swarms; Cognitive Mapping; Aggregation; Clustering; Dispersion; Self-Assembly; Collective Construction; Collective Transport; Collective Manipulation; Flocking; Collective Motion; Foraging.

Module II

9 lecture hours

Modelling Swarm Systems; Decision Making; Collective Decision Making; Collective Motion as Decision Making; Collective Decision-Making Models; UAV Networking Architecture; UAV Communication Topology; UAV Networking Scenarios; Object Detection and Tracking using Deep Learning; Obstacle Avoidance using Deep Learning; Autonomous Landing using Deep Reinforcement Learning; Autonomous Landing using Fast Reinforcement Learning; UAV Formation Architecture; UAV Formation Control Models; Consensus-Based Formation Control Models.

Module III

5 lecture hours

UAV Mobility: Spatiotemporal Predictions; Multivariate Spatiotemporal Predictions; Deep Spatiotemporal Residual Networks; UAV Mobility Models for Reconnaissance Swarms; Search and Rescue; Independent Mobility Models.

Module IV

6 lecture hours

UAV Routing Topologies; Computation Applications; Directional Airborne Networks; Low Probability Detection; Security Threats; Security Requirements; Jamming Characteristics; Security Protocols; Resilient UAV Networks.

Studio Work / Laboratory Experiments:

In project work, students are required to develop a UAV mission that replicates biologically occurring swarms to achieve predefined objectives using custom programmed flight controllers.

Text Books:

1. Bouffanais, Roland. *Design and control of swarm dynamics*. Vol. 1. Singapore: Springer, 2016.

Reference Books:

1. Hu, Fei, DongXiu Ou, and Xin-lin Huang, eds. *UAV swarm networks: models, protocols, and systems*. CRC Press, 2020.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET471	UAV Mission Planning and Deployments	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To understand various parameters and environmental considerations for efficient deployment of Unmanned Aerial Systems.

CO2: To perform site survey and operational characteristic analysis and develop mission deployment accordingly.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3		3	3								3		2
CO2	3	3		3	3							2	3		3

Course Contents:

Module 1

9 lecture hours

Mission Planning and Control Station; Physical Configuration; Planning; Navigation; Target Location; Air Vehicle; Payload Control; Remote Pilot; Assisted Auto Pilot; Complete Autopilot; Payload; Reconnaissance/Surveillance Payloads; Target Detection; Recognition; Identification; Searching Stable Line of Sight.

Module II

7 lecture hours

Weapon Payloads; Capacity; Structure; Interference; Radar; Chemical, Nuclear and Meteorological Sensors; Fixed Wing UAV Launch Systems; Vertical Take-off and Landing; Conventional Landing; Vertical Net Systems; Parachute Recovery; VTOL UAVs; Mid Air Retrieval; UAV Launch Method Trade-offs; UAV Recovery Trade-offs.

Module III

7 lecture hours

Aerial Photography; Photogrammetry; Radiometric Errors and Effects; Radiometric Correction; Geometric Errors and Effects; Georeferencing and Geometric Correction; Georeferenced Point Clouds; Digital Elevation Models; Image Enhancement; Image Classification; UAS Imagery Mission Planning; Product Specification; Accuracy Requirements; Operational Site Restrictions.'

Module IV

5 lecture hours

Digital Sensors; Geometry of Vertical Imagery; Aerial Imagery Collection; Flight Plan and Layout Design; Aerial Cinematography; Camera Angle; Navigation; Narration; Platform Accessories; Cinematography Technique.

Studio Work / Laboratory Experiments:

In project work, students will be developing independent surveillance missions based on underlying geography and geometry.

Text Books:

1. Barnhart, R. Kurt, Douglas M. Marshall, and Eric Shappee, eds. *Introduction to unmanned aircraft systems*. Crc Press, 2021.

Reference Books:

1. Fahlstrom, Paul G., and Thomas J. Gleason. *Introduction to UAV systems*. UAV Systems, Incorporated, 1993.
2. Sebbane, Yasmina Bestaoui. *Multi-UAV Planning and Task Allocation*. CRC Press, 2020.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET472	Auto Pilot and Flight Control	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Prerequisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To understand Autopilot and autonomous mission design.

CO2: To construct autonomous UAV missions using programming Autopilot hardware.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3		3	3								3		2
CO2	3	3	3	3	3							2	3	3	3

Course Contents:

Module 1

6 lecture hours

Auto Pilot; Flight Control: Linear Flight Control, Non-Linear Flight Control, Adaptive Flight Control; Open-Pilot; Pixhawk; Paparazzi; ArduPilot; DroneKit-Python.

Module II

7 lecture hours

MAVLink: Message Format, High-Level Message Flow; Auto-Pilot Data Channelling; MAVLink Routing; CAN Bus; UAVCAN Protocol; ROS: Robot Operating System; MAVROS; MAVProxy.

Module III

8 lecture hours

Mission Planner Firmware; Waypoints and Events; Camera Control; Mission Planning Setup; Mission Planning Calibration; Flight Setup; Autonomous Missions; Vehicle Calibration; Vehicle Attitude; Vehicle Positioning.

Module IV

7 lecture hours

Terrain Recognition; Mapping; Surveillance; Video and Image Tracking; Failsafe; Vehicle Health.

Studio Work / Laboratory Experiments:

In-studio work, students will start with ArduPilot and Pixhawk flight controllers, components, and learn to create autonomous UAV missions using programming MAVLink and UAVCAN protocols.

Text Books:

1. Mendoza-Mendoza, Julio Alberto, Victor Javier Gonzalez-Villela, Gabriel Sepulveda-Cervantes, Mauricio Mendez-Martinez, and Humberto Sossa-Azuela. *Advanced Robotic Vehicles Programming: An Ardupilot and Pixhawk Approach*. Apress, 2020.

Reference Books:

1. Ty Audronis, , *Designing Purpose-Built Drones for Ardupilot Pixhawk 2.1: Build drones with Ardupilot* (1 ed.), Packt, 2017. ISBN 1786469162.
2. Audronis, Ty. *Designing Purpose-built Drones for Ardupilot Pixhawk 2.1: Build Drones with Ardupilot*. Packt Publishing Ltd, 2017.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET473	Special Topics in Drone Technology	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To familiarize and learn about the latest trends and research in the field.

CO2: To equip themselves with the conceptual and practical experience of few latest methods, tools, technologies or algorithms in Drone Technology.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3		3	2							2	3		
CO2	3	3	2	3	3							2	3	3	3

Course Contents:

Module 1

28 lecture hours

This course covers the cutting-edge topics in Drone Technology, and these modules will be chosen by the instructor based on the requirements and relevance at that point of time. These modules need to be relevant to the industry and start-ups will also include related case studies, use cases and implementations scenarios. Students will be working on lab work and projects to get real hands-on experience of these topics and modules.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to Drone Technology.

Specialization

Electives:

Quantum Computing

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET474	Quantum Computing for Data Analysis	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate Quantum Statistical knowledge helps an individual use the proper methods to collect the data and analyze it.

CO2: To employ the correct analyses, and effectively present the results.

CO3: To develop analytical ability and critical thinking in the selection of statistics and their use in making interpretation meaningful and productive.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3		3	3							2	3		
CO2	3	3	2	3	3								3	3	3
CO3	3	3	3	3	3			2		2		3	3	3	3

Course Content:

Module 1

12 lecture hours

Statistics, Density Operators, Pure and Mixed States, Quantum Probability Theory, Kolmogorovian Theory, Quantum Measurement Problem, Intuitionistic Logic, Heyting Algebra, Quantum Filtering, Paradoxes Quantum Stochastic Process, Double Negation, Quantum Stochastic Calculus, Hamiltonian Calculus, Quantum Ito's Formula, Quantum Stochastic Differential Equations (QSDE), Quantum Stochastic Integration, Itô Integral Quasiprobability Distributions, Quantum Wiener Processes, Quantum Statistical Ensemble, Quantum Density Operator or Density Matrix, Gibbs Canonical Ensemble, Quantum Mean, Quantum Variance, Envariance Polynomial Optimization, Quadratic Unconstrained Binary Optimization.

Module II

9 lecture hours

Quantum Gradient Descent, Quantum Based Newton's Method for Constrained Optimization, Quantum Based Newton's Method for Unconstrained Optimization, Quantum Ensemble, Quantum Topology, Quantum Topological Data Analysis, Quantum Bayesian Hypothesis, Quantum Statistical Decision Theory, Quantum Minimax Theorem, Quantum Hunt-Stein Theorem, Quantum Locally Asymptotic

Normality, Quantum Ising Model, Quantum Metropolis Sampling, Quantum Monte Carlo Approximation, Quantum Bootstrapping, Quantum Bootstrap Aggregation.

Module III

7 lecture hours

Quantum Decision Tree Classifier, Quantum Outlier Detection, Cholesky-Decomposition for Quantum Chemistry Quantum Statistical Inference, Asymptotic Quantum Statistical Inference, Quantum Gaussian Mixture Model Quantum t-design, Quantum Central Limit Theorem, Quantum Hypothesis Testing, Quantum Chi-squared and Goodness of Fit Testing, Quantum Estimation Theory, Quantum Way of Linear Regression, Asymptotic Properties of Quantum, Outlier Detection in Quantum Concepts.

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Reference Books:

1. Luger, G. F., and W. A. Stubblefield. "Introduction to AI and Expert systems." (1989). ISBN 978-9332551947.
2. Bishop, Christopher. "Pattern recognition and machine learning." *Springer google schola* 2 (2006): 5-43. ISBN 978-0387310732.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET475	Quantum Information Theory	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate the concepts of bit and entropy and Shannon's theory to the quantum.

CO2: To articulate the concepts of bit and entropy and Shannon's theory to the quantum.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3		3	2	2						2	3		2
CO2	3	3		3	2	2						2	3		2

Course Content:

Module 1

10 lecture hours

Classical information theory. Classical Shannon Theory: Data Compression, Channel Capacity. Classical Information and Entropy: Entropy of a Random Variable, Conditional Entropy, Joint Entropy, Mutual Information, Relative Entropy, Conditional Mutual Information, Entropy Inequalities, Near Saturation of Entropy Inequalities, Classical Information from Quantum Systems. Concepts in Quantum Shannon Theory: Overview of the Quantum Theory, The Emergence of Quantum Shannon Theory.

Module II

11 lecture hours

Quantum Information and Entropy: Quantum Entropy, Joint Quantum Entropy, Potential yet Unsatisfactory Definitions of Conditional Quantum Entropy, Conditional Quantum Entropy, Coherent Information, Quantum Mutual Information, Conditional Quantum Mutual Information, Quantum Relative Entropy, Quantum Entropy Inequalities, Continuity of Quantum Entropy.

Module III

7 lecture hours

The Noiseless Quantum Theory: Reversible Evolution, Schmidt Decomposition, Schumacher Compression, Entanglement Manipulation. The Noisy Quantum Theory: Noisy Quantum States, Classical Communication, Entanglement-Assisted Classical Communication, Coherent Communication with Noisy Resources, Private Classical Communication, Quantum Communication. The Purified Quantum Theory.

Studio Work / Laboratory Experiments:

Implementation and calculation of entropy, conditional entropy and mutual information, Implementation and calculation of channel capacity, Quantum entropy and mutual information implementation, Conditional Quantum Mutual Information and Quantum Relative Entropy, Rényi Information Measure and Proof of the Recoverability Theorem, Refinements of Quantum Entropy Inequalities, Mutual Information of a Classical Channel and Private Information of a Wiretap Channel, Schumacher Compression, Entanglement Manipulation, Entanglement-Assisted Classical Communication.

Reference Books:

1. Vedral, Vlatko. *Introduction to quantum information science*. Oxford University Press, USA, 2006. ISBN 9780191526985, 0191526983.
2. Brass, Dagmar, and Gerd Leuchs, eds. *Quantum Information, 2 Volume Set: From Foundations to Quantum Technology Applications*. John Wiley & Sons, 2019. ISBN 9783527805792, 3527805796.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET476	Quantum Computer Architecture	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To get how to construct and operate the physical systems behind quantum computers.

CO2: To examine problems of different computational complexity and explain why certain problems are rendered tractable by quantum computation concerning the relevant concepts in quantum theory.

CO3: To develop new applications of quantum technologies to potential real-world problems.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3		3	3								3		
CO2	3	3	2	3	3	2				2		2	3	2	3
CO3	3	3	3	3	3	2				2		2	3	3	3

Course Content:

Module 1

14 lecture hours

Basic quantum mechanics, Qubits: superconducting Transmon qubits, Silicon spin qubits, diamond NV center qubits, and topological qubits. The Quantum Circuit Model, Quantum Gates, Qubit Gates, Controlled-U Gates, Universal Sets of Quantum Gates

Module II

8 lecture hours

Quantum Factoring and Finding Hidden Structures, Hamiltonian Simulation Algorithms, Quantum Algorithms for Linear Algebra, Quantum Error Mitigation Strategies, Variational Quantum Algorithms, Analog Quantum Algorithms.

Module III

6 lecture hours

Quantum Computing as a Service (QCaaS), QCaaS, speed, and trust, Quantum Error Correction: Shor code, Theory of Quantum Error –Correction, Constructing Quantum Codes, Stabilizer codes, Basic properties of Entropy, Von Neumann

Studio Work / Laboratory Experiments:

Lab work consists of designing the quantum circuit, Quantum Circuit Model, Qubit Gates, Quantum Error Mitigation, QCaaS, speed, and trust, Quantum Error Correction, Constructing Quantum Codes and Stabilizer codes.

Text Books:

1. Metodi, Tzvetan, and Arvin I. Faruque. *Quantum computing for computer architects*. Springer Nature, 2022. ISBN 9781608456215, 1608456218.

Reference Books:

1. National Academies of Sciences, Engineering, and Medicine. "Quantum computing: progress and prospects." (2019). ISBN 9780309479721, 030947972X.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET477	Quantum Cryptography	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To design and analyse the security of quantum cryptographic protocols.

CO2: To implement tasks with strong security of quantum cryptography.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3		3	3	2						2	3	3	3
CO2	3	3	2	3	3	2		3				2	3	3	3

Course Contents:

Module 1

11 lecture hours

About Quantum Computing, Quantum money and attacks on it, Density matrices and CPTP maps, Quantum encryption, Notions of security, No-Cloning Theorem, Wiesner's scheme for quantum money, SWAP Operator, Quantum Cryptography Protocols, The quantum one-time pad, Measuring randomness, Nonlocal games and Guessing games, Distributing keys, Distribution over a noisy channel, Quantum key distribution protocols, BB84 protocol: Assumptions, the protocol, BB84: relaxing assumptions, BB84: summary and length of the final key, Warmup: Security against a classical eavesdropper, Security of BB'84 using guessing games, The issue with the distance: error correction and privacy amplification.

Module II

9 lecture hours

Eavesdropping strategy, E91 Protocol: purifying protocols using entanglement, Quantum cryptography beyond keydistribution, Quantum cryptography using the untrusted device, Security of device-independent quantum key distribution against collective attacks, Two-party cryptography: bit commitment and oblivious transfer, Impossibility of bit commitment, The BBM92 protocol, The Ekert91 protocol, Quantum relays and repeaters, Device-Independent Quantum Key Distribution, Classical fingerprinting, Application of such fingerprints, Cryptographic hash function, Quantum fingerprinting, Error correction codes, Fingerprinting protocols.

Module III

8 lecture hours

Goal of public-key encryption, Quantum public-key and their usefulness, Conventional digital signatures, Quantum digital signatures, Quantum randomness, Quantum Coin Flipping, Quantum money: WIESNER's scheme, BOZZIO et al.'s scheme, Semi-quantum money, Electronic voting, Quantum voting.

Studio Work / Laboratory Experiments:

Studio work includes Wiesner's quantum money implementation of attacks on it, Implementation of the quantum one-time pad and its use in cryptography, Implementation of BB84 Quantum Key Distribution, Implementation and analyzing the security of BB'84 using guessing games, Quantum Error Correction: Phase flip code, Quantum Error Correction: Bit flip code, BB92 protocol implementation, Ekert91 protocol implementation, Implementation of Quantum-Safe Signatures, Quantum fingerprinting implementation.

Text Books:

1. Grasselli, Federico, and Federico Grasselli. "Quantum Conference Key Agreement." In *Quantum Cryptography: From Key Distribution to Conference Key Agreement*, pp. 55-70. Springer International Publishing, 2021. ISBN 978-3-030-64360-7.
2. Van Assche, Gilles. *Quantum cryptography and secret-key distillation*. Cambridge University Press, 2006. ISBN 9780511617744.
3. Chaubey, Nirbhay Kumar, and Bhavesh B. Prajapati, eds. *Quantum Cryptography and the Future of Cyber Security*. IGI Global, 2020. ISBN 9781799822559.

Reference Books:

1. Kollmitzer, Christian, and Mario Pivk, eds. *Applied quantum cryptography*. Vol. 797. Springer, 2010. ISBN 978-3-642-04831-9.
2. Kumar, Neeraj, Alka Agrawal, Brijesh K. Chaurasia, and Raees Ahmad Khan, eds. *Limitations and future applications of quantum cryptography*. IGI Global, 2020. ISBN9781799866793.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET478	Special Topics in Quantum Computing	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs):

On completion of this course, the students will be able to:

CO1: To familiarize and learn about the latest trends and research in the field.

CO2: To equip themselves with the conceptual and practical experience of few latest methods, tools, technologies or algorithms in Quantum Computing.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3		3	3								3		1
CO2	3	3		3	3	2						2	3	2	3

Course Content:

Module 1

28 lecture hours

This course covers the cutting-edge topics in Quantum Computing, and these modules will be chosen by the instructor based on the requirements and relevance at that point of time. These modules need to be relevant to the industry and start-ups will also include related case studies, use cases and implementations scenarios. Students will be working on lab work and projects to get real hands-on experience of these topics and modules.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to Quantum Computing.

Specialization Electives: IoT and Robotics

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET479	Robotics: Dynamics and Controls	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain algorithmic approaches, mathematical models, and motion control methods applicable to robotic manipulators.

CO2: To examine computational systems concerning manipulators.

CO3: To implement motion control techniques to the robotic manipulators.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3		3	3	2							3		
CO2	3	3		3	3							2	3	3	3
CO3	3	3	3	3	3	2						2	3	3	3

Module 1

8 lecture hours

Robot mechanics, Description of position and orientation of Manipulator, Transformation matrix, Compound Rotations, DH parameters, Frame arrangement and DH Representation, Forward and Inverse kinematics, Differential kinematics, Velocity propagation model for serial manipulators.

Module II

8 lecture hours

Jacobian matrix for manipulator, Euler's angles, Manipulator Statics, Workspace singularities, Manipulator dynamics, Lagrange-Euler method for manipulator Dynamics, Newton-Euler method for Manipulator Dynamics, Manipulator motion in state-space.

Module III

6 lecture hours

Trajectory generation using smooth functions, Trajectory generation schemes for serial manipulators, Robot motion control, Kinematic control.

Module IV

6 lecture hours

Dynamic control, Cascade control, DH description of Mobile robot, Kinematic and dynamic models of a mobile robot.

Studio Work / Laboratory Experiments:

This course will cover robot dynamics, trajectory generation, motion planning, and nonlinear control, and develop real-time planning and control software modules for robotic systems. This course will give you the basic theoretical tools and enable you to design control algorithms. Using MATLAB, you will apply what you have learned through a series of projects involving real-world robotic systems.

Text Books:

1. Craig, John J. *Introduction to robotics*. Pearson Education, 2006. ISBN 0133489795.
2. Corke, Peter. *Robotics and control: fundamental algorithms in MATLAB®*. Vol. 141. Springer Nature, 2021. ISBN 3030791785.

Reference Books:

1. Mittal, R. K., and I. J. Nagrath. *Robotics and control*. Tata McGraw-Hill, 2003. ISBN 9780070482937.
2. Kurdila, Andrew J., and Pinhas Ben-Tzvi. "Dynamics and control of robotic systems." (2019). ISBN 9781119524830.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET480	IOT Analytics	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate visualization, processing, and storage of device data in a time-series data store.

CO2: To examine collect, process, and analyze data from IoT Cloud.

CO3: To make use of IoT devices that are integrated with IoT Analytics services.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3		3	3								3		
CO2	3	3		3	3	2				2		2	3	3	3
CO3	3	3	3	3	3	2				2		2	3	3	3

Course Content:

Module 1

9 lecture hours

Need of IoT analytics, IoT analytics challenges, Business value concerns, IoT Analytics for the Cloud, Building elastic analytics, PUBLIC CLOUD overview, PUBLIC CLOUD key concepts, PUBLIC CLOUD key core services, PUBLIC CLOUD key services for IoT analytics, Creating an PUBLIC CLOUD Cloud, Analytics Environment, Collecting Data from PUBLIC CLOUD Cloud, Data processing for analytics, Applying big data technology to storage: Hadoop, HBase, Amazon DynamoDB, Amazon S3, Apache Spark for data processing.

Module II

7 lecture hours

Adding External and Internal datasets, IoT Dashboards, Sensor data visualizing, Feature engineering with IoT data, Dealing with missing values, centering, and scaling, Machine learning (ML) Model Representation, Evaluation, Optimization, Time series data handling, Validation methods, Cross-validation, Test set, Precision, Recall, Specificity.

Module III

12 lecture hours

Bias–Variance trade off, Random Forest models, Gradient Boosting Machines (GBM), Anomaly detection, Forecasting using ARIMA, Deep Learning Analysis with PUBLIC CLOUD, Strategies to Organize Data

for Analytics, Linked Analytical Datasets, linking together datasets, Managing data lakes, Data retention strategy, The Economics of IoT Analytics, Cost considerations for IoT analytics, Geospatial Analytics to IoT data.

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Text Books:

1. Minteer, Andrew. *Analytics for the internet of things (iot)*. Packt Publishing Ltd, 2017. ISBN 978-1787120730.

Reference Books:

1. Geng, Hwaiyu, ed. *Internet of things and data analytics handbook*. John Wiley & Sons, 2017. ISBN 978-1119173649.

Studio Work / Laboratory Experiments:

This course will cover robot dynamics, trajectory generation, motion planning, and nonlinear control, and develop real-time planning and control software modules for robotic systems. This course will give you the basic theoretical tools and enable you to design control algorithms. Using MATLAB, you will apply what you have learned through a series of projects involving real-world robotic systems.

Name of Program	Bachelor of Technology				
CSET420	IoT: Security and Attacks	L	T	P	C
Owning School/Department	Computer Science Engineering	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Explain Security in IoT, IoT Vulnerabilities, Authentication, Authorization.

CO2: Examine attack Models of IoT and Secure Communication Links in IoTs.

CO3: Implement solutions of public cloud IoT security.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	3	1	2	3	2	2	2	2	3	2	2	2	2	2
CO2	3		2	1	2	3	3	1		1		3	3	3	3
CO3	3		1	2		1	2	3	2	2	3	2	2	2	2

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

08 lecture hours

Architecture of IoTs, IoT Security Requirements Privacy preservation, Device security, Authentication, Confidentiality, Integrity of IoT devices, Benefits of IoT, IoT Attack Surface, Industrial Standards of IoT device security, Gateway security, IoT Privacy Concerns, Privacy by Design, Conducting a Privacy Impact Assessment, Case Study: The Connected Barbie.

Module II:

10 lecture hours

IoT Vulnerabilities, Secret-Key, Authentication/Authorization for Smart Devices, Constrained, System Resources, Device Heterogeneity, Fixed Firmware, AttackModels, Layer-wise Attack model, Attacks to Sensors in IoTs, Attacks to RFIDs in IoTs, Attacks to Network Functions in IoT attacks to Back-end Systems, Security in Front-end Sensors and Equipment. IoT Attacks - Side-channel Attacks, Spoofing Attack, Sniffing Attack, Rogue Attack, Devices Attack, Man-in- Middle Attack, DDoS Attack.

Module III:

05 lecture hours

Securing internet of things environments, Networking Function Security, IoT Networking Protocols, Layering Architecture, Secure IoT Lower Layers, SecureIoT Higher Layers, Secure Communication Links in IoTs, Back-end Security, Secure Resource Management, Secure IoT Databases, IoT Hardware Test, Test Device Range, Latency and Capacity.

Module IV:

05 lecture hours

Manufacturability Test, Secure from Physical Attacks, IoT Software -Trusted IoT Application Platforms, Secure Firmware Updating, Network Enforced Policy, Secure Analytics Visibility and Control.

Studio Work / Laboratory Experiments:

This course lab will articulate possible attacks on IoT devices and networks. In this course lab students will experiment to prevent attacks on IoT devices. Multiple tools such as Contiki, Wireshark will be used for doing all the experiments.

Text Books :

1. Liyanage, Madhusanka, An Braeken, Pardeep Kumar, and Mika Ylianttila, eds. *IoT security: Advances in authentication*. 1st ed. Wiley Telecom, 2020.

Reference Books :

1. Chantzis, Fotios, Ioannis Stais, Paulino Calderon, Evangelos Deirmentzoglou, and Beau Woods. *Practical IoT hacking: the definitive guide to attacking the internet of things*. 1st ed. No Starch Press, 2021.

Name of Program	Bachelor of Technology				
CSET361	Device Level IoT Security	L	T	P	C
Owning School/Department	Computer Science Engineering	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate the security issues in IoT.

CO2: To apply the cryptographic techniques in IoT.

CO3: To implement various types of access control mechanism for IoT devices in network environment.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3	2	2	2							3	3	3	2
CO2	2	3	2	2	2							3	2	2	2
CO3	2	2	3	2	3							3	2	3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

06 lecture hours

Brief review of the Internet of Things IoT, IoT in business world, Benefits Applications of IoT, Security Issues with IoT, Basic Architecture of IoT, IoT Attack Surface, OWASP Top 10 for IoT. Concept of Vulnerability management, Quarantine and Prevention.

Module II:

08 lecture hours

Security Requirements in IoT Architecture - Security in Enabling Technologies - Security Concerns in IoT Applications. Security Architecture in the Internet of Things- Security Requirements in IoT - Insufficient Authentication/Authorization - Insecure Access Control - Threats to Access Control, Privacy, and Availability - Attacks Specific to IoT. Vulnerabilities - Secrecy and Secret-Key Capacity- Authentication/Authorization for Smart Devices - Transport Encryption - Attack Fault trees.

Module III:

14 lecture hours

Cryptographic primitives and its role in IoT, Encryption and Decryption, Hashes, Digital Signatures, Random number generation, Cipher suites, key management fundamentals cryptographic controls built into IoT messaging and communication protocols, IoT Node Authentication, Identity lifecycle, authentication credentials, IoT IAM infrastructure, Authorization with Publish / Subscribe schemes, access control. Concerns in data dissemination , Lightweight and robust schemes for Privacy protection, Trust and Trust models for IoT, self-organizing Things, Preventing unauthorized access, Cloud security for IoT, Cloud services and IoT, offerings related to IoT from cloud service providers, Cloud IoT security controls.

Studio Work / Laboratory Experiments:

Consists of using network monitoring tools, implementing different types of attacks and some protection schemes of various IoT devices.

Text Books :

1. Saravanan, Vijayalakshmi, Alagan Anpalagan, T. Poongodi, and Firoz Khan, eds. *Securing IoT and big data: next generation intelligence*. 1st ed. CRC Press, 2020. ISBN 0367432889.

Reference Books :

1. Bhattacharjee, Sravani. *Practical Industrial Internet of Things security: A practitioner's guide to securing connected industries*. 1st ed. Packt Publications, 2018. ISBN 978-17888 2687.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET421	Humanoids	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate the bipedal mechanism of waking and navigation of humanoids.

CO2: To examine visual, gesture, cognitive-based controlling action of humanoid.

CO3: To implement stable gait generation of walking and localization and navigation of humanoid.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3		3									3		
CO2	3	3		3	3					2		2	3		3
CO3	3	3	3	3	3			2				2	3	3	3

Course Content:

Module 1

9 lecture hours

Humanoid, Bipedal Humanoid, Kinematics, Dynamics, Stability criteria of Bipedal robot, Zero moment point, Limit Cycle, Poincare Map, 3D Gait generation, Inverted pendulum model of Humanoid, Arm Movement with stability, Dynamics of Arm movement, Walking robot pattern generation, Central pattern generation, Torque control mechanism, Impact recovery, Push Recovery.

Module II

8 lecture hours

Touch sensor, sound sensor, Vision sensor, Tactile Sensing, Robot Localization, Monte Carlo localization, Landmark based localization, Detect Landmark, Update landmark, Kalman filter Localization, Tracking: Known initial position, Global Localization: Unknown initial position, Re-Localization: Incorrect known position, Obstacle avoidance, Potential Field, Simultaneous Localization and Mapping (SLAM), Path Planning, Graph Construction.

Module III

5 lecture hours

Vision based Edge detection for navigation, Vision based object recognition for Grasping, Shift Algorithm, Human Robot Interactions using Gestures, Hidden Markov model, Gesture creation (using If THEN Rules), Gesture executions.

Module IV

6 lecture hours

Cognitive robotics, Subsumption Architecture, Potential field-based architecture, Human activity recognition using vision, Motion capture/Learning, mimicking behavior from demonstration.

Studio Work / Laboratory Experiments:

Studio works is to make use of Choreograph Simulator for Nao Robot. In this course students will implement Stability Criteria during the walking using choreograph data of Nao walking, Joint Angle (hip, Knee, Ankle) generation for walking pattern. It will help students to experiment for Navigation of Humanoid robot using vision, Object recognition and grasping, Gesture based interaction with Nao.

Text Books:

1. Goswami, Ambarish, and Prahlad Vadakkepat. *Humanoid robotics: a reference*. Springer Publishing Company, Incorporated, 2018. ISBN 9400760450.
2. Nenchev, Dragomir N., Atsushi Konno, and Teppei Tsujita. *Humanoid robots: Modeling and control*. Butterworth-Heinemann, 2018. ISBN 978-0128045602.

Reference Books:

1. González Aguirre, David Israel. "Visual perception for humanoid robots: environmental recognition and localization, from sensor signals to reliable 6D poses." (*No Title*). ISBN 331997839.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET422	Mobile IoT Networks	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate the concepts of Mobile IoT Network systems.

CO2: To examine the routings of mobile IoT Network.

CO3: To implement the applications for the mobile IoT networked embedded systems.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3		3									3		
CO2	3	3		3	3							2	3		3
CO3	3	3	3	3	3			2		2		2	3	3	3

Course Content:

Module 1

9 lecture hours

Internet Working, Hosts Communication, Protocol Stacks, IoT Protocols, Intra-domain vs. Inter-domain, X0 Communications Backbone, Network Virtualization, Delivery Methods, Wired Networking, Reference Architectures for IoT, Layer Networking for IoT, Networked Embedded Systems, Components of networked embedded systems, Real-time embedded systems, Centralized and distributed embedded systems, Physical sensor, Passive sensor, Semi-passive, Active sensors, Soft sensors, Sensor nodes.

Module II

8 lecture hours

Mobile sensor network, Sensor networks with mobile nodes, Power management, Mobile node discovery, Data transfer to mobile nodes, Mobile nodes routing, Distributed Algorithms for IoT, Hierarchical Addressing and Routing, Geographic Addressing and Routing, Delay-Tolerant Networking, Optimized Link State Routing (OLSR), Dynamic Source Routing (DSR).

Module III

5 lecture hours

Sockets API Programming, Packet Forwarding, Network Design, IoT forwarding, IoT processing, IoT Isolation for devices, Participatory sensing of Networked Embedded Systems.

Module IV

6 lecture hours

Vehicular Networked Embedded Systems, Intra -vehicular Network Embedded Systems, Event Triggered Networked Embedded Systems System, Time Triggered Networked Embedded Systems System, Intelligent transportation systems.

Studio Work / Laboratory Experiments:

In this course lab students will Configure Networked Embedded Systems and Sensor Networks. Implement the network to acquire sensor data from the wireless sensor board and from external sensors. Students will be building various applications for smart cities using networked embedded sensor and actuators.

Text Books:

1. Hernández-Callejo, Luis, and Sergio Nesmachnow, eds. *Mobility and IoT for the Smart Cities*. MDPI-Multidisciplinary Digital Publishing Institute, 2020. ISBN 3039430505.
2. Mahmood, Zaigham. "Connected vehicles in the internet of things." *Springer Nature SwitzerlandAG* (2020). ISBN 3319888870.

Reference Books:

1. Lea, Perry. *IoT and Edge Computing for Architects: Implementing edge and IoT systems from sensors to clouds with communication systems, analytics, and security*. Packt Publishing Ltd, 2020. ISBN 1839214805.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET423	Special Topics in IoT	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To familiarize and learn about the latest trends and research in the field.

CO2: To equip themselves with the conceptual and practical experience of few latest methods, tools, technologies or algorithms in IoT.

CO – PO /PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3		3	2								3		
CO2	3	3	3	3	3					2		2	3	3	3

Module 1

28 lecture hours

This course covers the cutting-edge topics in IoT, and these modules will be chosen by the instructor based on the requirements and relevance at that point of time. These modules need to be relevant to the industry and start-ups will also include related case studies, use cases and implementations scenarios. Students will be working on lab work and projects to get real hands-on experience of these topics and modules.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to IoT.

**Specialization
Electives:
Augmented &
Virtual Reality**

Name of Program	Bachelor of Technology				
CSET425	Augmented Reality and ARCore	L	T	P	C
Owning School/Department	Computer Science Engineering	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To identify different types of AR experiences, tools and platforms used in the AR landscape.

CO2: To break down complex AR concepts to make them easy to understand.

CO3: To start building an AR experience using ARCore and other tools.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3		3	3									3	
CO2	3	3		3	3									3	3
CO3	2	2	3	2	3						2		3	3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

06 lecture hours

AR today: smartphone vs. standalone, Google AR, AR for shopping and retail, AR for business, AR for social media, AR for gaming, AR for education, AR for healthcare, AR for non-profits, AR reality design guidelines, Physical environment, virtual environment, environmental limitations, movements, types of movements, accessibility, Tracking in AR, outside-in tracking, inside-out tracking, motion tracking.

Module II:

08 lecture hours

Light estimation, Anchors, Interface issues and lack of UI metaphors, Constraints of occlusion and shading, AR required vs AR optional, Runtime considerations, Performance considerations, Run AR Apps in Android Emulator, Camera configuration, Buffering Camera frames, Sharing camera access to ARCore, Working with Anchors, Creating play instant app, ARCore as input for Machine Learning models.

Module III:

08 lecture hours

Enable call logging for Android, Disable call logging for Android, Display call logs, Recording and playback in AR, Add custom data while recording, Depth in ARCore, Distance and depth of field, Depth images, Use of Raw depth, Lightening estimation, Environmental HDR mode, Ambient intensity mode.

Module IV:

06 lecture hours

Augmented Images APIs, Augmented Images APIs' Capabilities, Requirements of Augmented Images APIs, Creation of image database, Image tracking, Cloud Anchors on Android, Augmented Faces on Android, custom textures and 3D models for Augmented Faces, ARCore SDK for Unity, ARCore SDK for Unreal.

Studio Work / Laboratory Experiments:

Experiments using Scene Viewer to display interactive 3D models in AR from an Android app or browser, AR Dragon using ARCore, AR Drone using ARCore, AR Core drawing, Creation of augmented faces, Drawing a virtual dancing character with space sound.

Text Books :

1. Lanham, Micheal. *Learn ARCore-Fundamentals of Google ARCore: Learn to build augmented reality apps for Android, Unity, and the web with Google ARCore 1.0*. 1st ed. Packt Publishing, 2018. ISBN 9781788833639.

Reference Books :

1. Linowes, Jonathan, and Krystian Babilinski. *Augmented reality for developers: Build practical augmented reality applications with unity, ARCore, ARKit, and Vuforia*. 1st ed. Packt Publishing, 2017. ISBN 9781787288041.

Name of Program	Bachelor of Technology				
CSET380	Animation and Rendering Techniques	L	T	P	C
Owning School/Department	Computer Science Engineering	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To define viewpoint and common rollouts-editing splines.

CO2: To understand different concepts of light.

CO3: To make use of V-Ray in animation and create animation by using rendering technique.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3		3									3		
CO2	3	3		3									3		
CO3	3	3	3	3	3								3	3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

06 lecture hours

3D View, Viewports, Standard primitives, Transformations, File formats and operations, Selection, Cloning, Group, Ungroup, Alignment, Splines: common, Rollouts-editing splines, Architectural tools, Max scene files, Modifiers, World space modifiers, Object space modifiers, Modifier stack, Instanced modifier.

Module II:

08 lecture hours

Compound objects and types, Modelling, mesh modelling, converting objects to editable mesh, sub objects: vertex, edge, face, polygon, element, Edit mesh modifier, Editable poly modelling, Material Editor, Assigning materials to objects, Material editor options, Material properties, Material types, Concepts of Light, omni lights, Spotlight, target lights, free lights, directional light, area lights, Mental ray, Skylight, Creating max basic lights, lights parameters, Positioning lights, Creating max light in exterior environment, Creating max light in interior environment.

Module III:

08 lecture hours

Rendering, rendering techniques, Photorealistic rendering, Shading, V-RaySun, V-Ray Light, Ray Tracing, Creating Animations: particles and dynamics, rendering the animation, automating the output of multiple still images, rendering a shadow study, creating a walkthrough, output options, particle systems: creating hierarchies.

Module IV:

06 lecture hours

Animating modifiers, animating elements, animation helpers, using dummy objects, dynamics and reaction, Dynamic Objects: damper, Spring, simulation mass FX.

Studio Work / Laboratory Experiments:

The experiments to create an image of an object with the different lighting condition. Light the scene with direct and indirect light, Rendering out Animation. Creation of a short-animated movie.

Text Books :

1. Watt, Alan H. and Mark Watt. *Advanced Animation and Rendering Techniques Theory and Practice*. 1st ed. ACM Press, 1992. ISBN 9780201544121.

Reference Books :

1. Akenine-Moller, Tomas, Eric Haines, and Naty Hoffman. *Real-time rendering*. 3rd ed. CRC Press, 2019. ISBN 9781315362007.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET376	3D Models for Augmented and Virtual Reality	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To establish and cultivate a broad and comprehensive understanding of this rapidly evolving and commercially viable field of Computer Science.

CO2: To prepare for participating in the production of highly integrative immersive applications, immersive social platforms, cross-disciplinary academic research projects, and leading developments in 3D user interfaces.

CO3: To create avatars: the virtual representation of other players, and agents: computer-controlled NPC characters.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3		3									3		
CO2	3	3		3	2								3	2	3
CO3	3	3	1	3	2								3	2	3

1=weakly related 2= moderately related 3=strongly related

Course Contents:

Module I:

7 lecture hours

Virtual reality and environment, The historical development of VR and AR, Scientific landmarks in Computer Graphics, Real-time computer graphics, Flight simulation, Virtual environments, Requirements for VR and AR, benefits of Virtual reality.

Modulo II:

7 lecture hours

SOFTWARE TECHNOLOGIES: Database - World Space, World Coordinate, World Environment, Objects - Geometry, Position / Orientation, Hierarchy, Bounding Volume, Scripts and other attributes, VR and AR Environment – VR and AR Database, Tessellated Data, LODs, Cullers and Occluders, Lights and Cameras, Scripts, Interaction - Simple, Feedback, Graphical User Interface, Control Panel, 2D Controls, Hardware Controls, Room / Stage / Area Descriptions, World Authoring and Playback, VR toolkits, Available software in the market.

Module III:

7 lecture hours

3D INTERACTION TECHNIQUES: 3D Manipulation tasks, Manipulation Techniques and Input Devices, Interaction Techniques for 3D Manipulation, Design Guidelines - 3D Travel Tasks, Travel Techniques, Design Guidelines - Theoretical Foundations of Wayfinding, User Centered Wayfinding Support, Environment Centered Wayfinding Support, Evaluating Wayfinding Aids, Design Guidelines - System Control, Classification, Graphical Menus, Voice Commands, Gestural Commands, Tools, Multimodal System Control Techniques, Design Guidelines, Case Study: Mixing System Control Methods, Symbolic Input Tasks, symbolic Input Techniques, Design Guidelines, Beyond Text and Number entry.

Module IV:

7 lecture hours

ADVANCES IN 3D USER INTERFACES: 3D User Interfaces for the Real World, AR Interfaces as 3D Data Browsers, 3D Augmented Reality Interfaces, Augmented Surfaces and Tangible Interfaces, Agents in AR, Transitional AR-VR Interfaces - The future of 3D User Interfaces, Questions of 3D UI Technology, 3D Interaction Techniques, 3D UI Design and Development, 3D UI Evaluation and Other Issues. VR and AR APPLICATIONS: Engineering, Architecture, Education, Medicine, Entertainment, Science, Training.

Studio Work / Laboratory Experiments:

Explore and setup the various AR/VR devices. Creating Google cardboard and designing VR database and objects. Learn how to create characters for VR.

Text Books :

1. Bowman Doug A, Ernest Kuijff, Joseph J LaViola, Jr and Ivan Poupyrev, *3D User Interfaces, Theory and Practice* (2 ed.), Pearson Education, 2019. ISBN 978-0134034324.
2. Glover Jesse and Jonathan Linowes, *Complete Virtual Reality and Augmented Reality Development with Unity: Leverage the power of Unity and become a pro at creating mixed reality applications* (1 ed.), Packt Publishing, 2019. ISBN 9781838644865.

Reference Books :

1. Pangilinan Erin, Steve Lukas and Vasanth Mohan, *Creating Augmented and Virtual Realities* (1 ed.), O'Reilly Publishers, 2019. ISBN 978-1492044192.

Name of Program	Bachelor of Technology				
CSET426	Building Social VR with 3D Characters	L	T	P	C
Owning School/Department	Computer Science Engineering	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To understand the psychology of social interaction and the practical skills to implement it.

CO2: To make characters that can respond to players' speech and body language.

CO3: To create avatars: the virtual representation of other players, and agents: computer-controlled NPC character.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3	3	3										3	
CO2	3	3		3	2									3	2
CO3	3	3	3	3	3									3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

06 lecture hours

Social VR and Virtual Characters, Virtual Characters Realism, Agents and Avatars, Applications of Virtual Characters, Character Animation, Cristina, Designing Virtual Characters, Importing a Virtual Character to Unity, Embodiment in Unity using Mobile VR, Skeletal Animation, Motion Capture, Capturing posture, gesture, and body language, Applying Motion Capture Data onto a Virtual Character, Applying Speech interaction in character.

Module II:

08 lecture hours

Markerless motion capture, Bodily Communication, Psychology for facial expression, Facial Expression in Unity, Blinking and Lip Sync Animation, Gaze in Social Interaction, Gaze Interaction, Other Facial Features and Their Psychological Effect, Psychology Behind Social Interaction, Challenges in Human-Avatar Interaction, Challenges in Human-Agent Interaction, Evaluation of Social Interaction in VR, Human-avatar: Social VR apps, Embodiment - implementation and psychological effect, Natural Interaction, Magic Interaction, Active and Passive Interaction, Affordances.

Module III:

07 lecture hours

VR Interaction Theory, Standard HMDs, Navigation in VR, Real Walking, Redirected Walking, Walk-in-Place, Virtual Navigation, Objects Interaction, Object Interaction Within Reach, Hyper-Natural Interaction, Magic Interaction, Evaluation Metrics, Physics Interaction, Physics Objects, Physics Objects in Unity,

Physics in VR, Physics Interaction in Unity with NewtonVR, State Machines, State Machine Interactions in Unity.

Module IV:

07 lecture hours

Case Study: Popular interaction-based products, Graphical User Interfaces in VR, Abstract Interfaces, Diegetic and Non-Diegetic UI, Gestural Interaction, Issues in VR Interaction, UX of VR, ImmersedVR, Spatial.io, EngageVR, Mozilla Hubs, Unity User Interfaces, Unity gesture plugin, Bespoke VR, VFairs, INXPO, AltspaceVR, High Fidelity, Sansar, Rec room, and Webaverse.

Studio Work / Laboratory Experiments:

The experiment in Unity: loading designed character, Skelton animation, movements, and interaction.

Text Books :

1. Pangilinan, Erin, Steve Lukas, and Vasanth Mohan. *Creating augmented and virtual realities: theory and practice for next-generation spatial computing*. 1st ed. O'Reilly Publishers, 2019. ISBN 978-1492044192.

Reference Books :

1. LaViola Jr, Joseph J., Ernst Kruijff, Ryan P. McMahan, Doug Bowman, and Ivan P. Poupyrev. *3D user interfaces: theory and practice*. 2nd ed. Pearson Education, USA, 2017. ISBN 978-0134034324.

Name of Program	Bachelor of Technology				
CSET427	Virtual Reality: Interface, Application and Design	L	T	P	C
Owning School/Department	Computer Science Engineering	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To demonstrate an understanding of fundamental techniques, processes, technologies, and equipment used in immersive virtual reality.

CO2: To explore the materials and processes used in immersive virtual reality.

CO3: To show a basic awareness and understanding of historical and theoretical contexts relevant to immersive virtual reality and demonstrate an understanding.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	3	1	2	3	2	2	2	2	3	2	2	2	2	2
CO2	2		2	1			2	3		2		3	3	2	2
CO3	1		2	2		3	2	3	2	2	3	2	2	2	2

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

08 lecture hours

VR Goals and definitions, Historical perspective, Birds-eye view, Geometry of Virtual Worlds, Geometric modelling, transforming models, Matrix algebra and 2D rotations, 3D rotations and yaw, pitch, and roll, Axis-angle representations, Quaternions, Converting and multiplying rotations, Homogeneous transforms, the chain of viewing transforms, Eye transforms, Canonical view transform, Viewport transform. Hardware Technologies for 3D user Interfaces: Visual Displays Auditory Displays, Haptic Displays, Choosing Output Devices for 3D User Interfaces. 3D user Interface Input hardware: Input device characteristics, Desktop input devices, Tracking Devices, 3D Mice, Special Purpose Input Devices, Direct Human Input, Home - Brewed Input Devices, Choosing Input Devices for 3D Interfaces.

Module II:

08 lecture hours

Light and Optics: Three interpretations of light, Refraction, Simple lenses, Dioptres, Imaging properties of lenses, Imaging properties of lenses, Lens aberrations, Optical system of eyes. Visual Physiology: Photoreceptors, Enough resolution for VR, Light intensity, Eye movements, Eye movement issues for VR, Neuroscience of vision. Visual Perception: Depth perception, Motion perception, Frame rates and displays.

Module III:

08 lecture hours

Tracking Systems: Orientation tracking, Tilt drift correction, Yaw drift correction, Tracking with a camera, Perspective n-point problem, Filtering, Lighthouse approach. Visual Rendering: Visual Rendering-overview, Shading models, Rasterization, Pixel shading, VR-specific problems, Distortion shading, Post-rendering image warp.

Module IV:

04 lecture hours

Physics and physiology, Auditory perception, Auditory localization, Rendering, Spatialization and display, Combining other senses. Interfaces: Interfaces - overview, Locomotion, Manipulation, System control, Social interaction, Evaluation of VR Systems.

Studio Work / Laboratory Experiments:

The laboratory of virtual reality is designed to provide a practical exposure to the students about the concepts and topics taught in the classroom sessions. Implementing the learnt concepts using Python will help the students to have a better understanding of the subject. 3D Modelling and surfacing, Bill Fleming, Elsevier (Morgan Ka VR Programming-I: Introducing Java 3D, loading and manipulating external models, using a lathe to make shapes, Programming-II: 3D Sprites, animated 3D sprites, particle systems.

Text Books :

1. Sherman, William R., and Alan B. Craig. *Understanding virtual reality: Interface, application, and design*. 2nd ed. Morgan Kaufmann, 2018.
2. Gregory, Josh. *Virtual reality*. 1st ed. Cherry Lake Publishing, 2018.

Reference Books :

1. Tacgin, Zeynep. *Virtual and Augmented Reality*. 1st ed. Routledge, 2022.
2. Greengard, Samuel. *Virtual reality*. 1st ed. Mit Press, 2019.

Name of Program	Bachelor of Technology					
CSET428	Special Topics in AR and VR	L	T	P	C	
Owning School/Department	Computer Science Engineering	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To familiarize and learn about the latest trends and research in the field.

CO2: To equip themselves with the conceptual and practical experience of few latest methods, tools, technologies or algorithms in AR and VR.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3		3									3		
CO2	3	3	3	3	3								3	3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

28 lecture hours

This course covers the cutting-edge topics in AR and VR, and these modules will be chosen by the instructor based on the requirements and relevance at that point of time. These modules need to be relevant to the industry and start-ups will also include related case studies, use cases and implementations scenarios. Students will be working on lab work and projects to get real hands-on experience of these topics and modules.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to AR and VR.

Specialization

Electives:

Product

Design

Technology

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET434	Art, Craft and Design	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To Art emphasizes ideas, feelings, and visual qualities.

CO2: To Craft emphasizes the right use of tools and materials.

CO3: To Design emphasizes planning, problem-solving, and completion, using drawing as a means of thinking.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1		2	2	1	2							2	3		
CO2		1	1		3							2		3	
CO3	2	2	3	2	3							3		1	

Module 1

13 lecture hours

Express oneself orally in Art, craft, and design, Professional terminology of the subject, Subject conversation, Drawing, understanding of pictorial Space, Form & Planes and Drawing based worksTwo-Dimensional Art Design, Develop Sense of Structure, Subject presentation and discussions , Writing Art, craft, and design, Visual competence of facts, ideas and attitudes, Investigating, interpreting and using information from texts and aesthetic expressions from images, Architecture, design, and the use of formatting, Latest mediums, archives, collections, libraries, and experience from work, Monochrome, Charcoal, Black, White distribution, Pen & Ink (Chinese Ink), Dry Brush, Pastel, Water Colour, Poster Colour.

Module II

15 lecture hours

Drawing, images, and symbols in visual expression, Image-making, Painting, using software ‘Word Art’, Corel Draw, Photoshop, (Design part), Catalogue / Card – design, Printmaking Elements and development, Photography, Three-Dimensional Design, Lettering Graphic Design, Image Making and Photography, Photo-Image transferring simple, Working with Digital Art, Skills in numeracy in Art, Calculating standard measurements, Product Forms, Proportions Rules and Guidelines, Volumes and dimensions for product development, Calculation in connection with geometry, Drawing and axonometry, Digital tools in Art, craft, and design, Visual communication, Layouts, Presentations , Documentation.

Studio Work / Laboratory Experiments:

Studio Work aims to design principles and practice the creative, technically proficient, and financially savvy also Designing Profits focuses on the product running is the business aspects.

Text Books:

1. Fiell, Charlotte, and Peter Fiell. *Robert Welch: Design: Craft and Industry*. Laurence King Publishing, 2015. ISBN 978-1780676050.

Reference Books:

1. Klingelhofer, Robert. *The craft and art of scenic design: Strategies, concepts, and resources*. Taylor & Francis, 2016. ISBN 978-1138937642.
2. Nunes, Morris, and Andrew Pressman. *Designing Profits: Creative Business Strategies for Design Practices*. Routledge, 2015. ISBN 9781317560562, 1317560566.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET437	Product Design Technologies and Tools	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	0	0	6	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate modern tools and technologies in the design process that are recognized globally.

CO2: To get the design sense of the object pattern during product design.

CO3: To implement the various methods and design patterns in Product design using technology.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	POS3
CO1	1		3		3							2		3	
CO2		1	3	1								2			
CO3	2	2	3	1	3							3	2	3	3

Course Content:

Module 1

18 lecture hours

Course Introduction and Assessment Plan, Modern Product Design Tools and Techniques, User tracking and analysis tools, Roadmapping software (such as ProductPlan), Customer survey tools (such as SurveyMonkey or Typeform), Presentation software, Project management tools, Flowcharting tools, Epilog Laser Cutting, 3D Printer, Color LaserJet Printer, Vertical Plastic Injection Molding, Universal Wood Working.

Module II

9 lecture hours

Vacuum Forming, Chroma Studio, VR Kit, Eye Tracking System, Heart Rate Monitor.

Module III

8 lecture hours

Bicycle ergometer, Knowledge of wood Material, Metal, Polymer, Fabrics and composite fabrication, Advance Mac and Windows workstations equipped with various professional 3D and 2D software etc., Set up an industry-level environment for design process execution, User tracking and analysis tools, Road mapping software (such as Product Plan).

Studio Work / Laboratory Experiments:

In-studio work, students will investigate, explore, and experiment with an extensive bucket of new technologies and tools for designing a Project/Product.

Text Books:

1. Knapp, Jake, John Zeratsky, and Braden Kowitz. *Sprint: How to solve big problems and test new ideas in just five days*. Simon and Schuster, 2016. ISBN 978150114080.

Reference Books:

1. Penty, Jane. *Product Design and Sustainability: Strategies, Tools and Practice*. Routledge, 2019. ISBN 9780203732076.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET438	Special Topics in Product Design Technology	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To familiarize and learn about the latest trends and research in the field.

CO2: To equip themselves with the conceptual and practical experience of a few latest methods, tools, technologies, or algorithms in Product Design Technology.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1		3		3								2			2
CO2	1	2	2	1	3							3			3

Module 1

28 lecture hours

This course covers the cutting-edge topics in Product Design Technology, and these modules will be chosen by the instructor based on the requirements and relevance at that point of time. These modules need to be relevant to the industry and start-ups will also include related case studies, use cases and implementations scenarios. Students will be working on lab work and projects to get real hands-on experience of these topics and modules.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to Product Design Technology.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET433	Design and Manufacturing for Digital Products	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To shift from paper-based processes to digital processes in the manufacturing industry.

CO2: To identify opportunities for increased quality and efficiency of the product.

CO3: To make ready the digital product after the design.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1			2		3		2					1		1	
CO2		2		2			3					2		3	
CO3			3	1								2		3	

Course Content:

Module 1

15 lecture hours

Designing Products, Not Pixels, The UX design community, Design the information architecture, Difference between design and craft—and that process, Designing suits for individual organizations, Product Design Team and its Skills, Design and Craft, Design decisions during execution, Fine arts, applied arts, and design, Design and Production, Manufacturing Digital Products, Digital product development uses, Modern industrial setup for Product Design, Hardware, Firmware, and some embedded software, Product Manufacturability, Maintainability, and sustainability, Product sustainable methods, Digital product-design process, Ecosystem Design, Outputs of Ecosystem Development, Product Audience, Existing solutions or competing products, Objectives of the organization, Users’ needs, Success measures, Product platforms, Product data needs, Product data architectures, Product good ideas and bad ideas, Concept Design, The design looks, Prototype, The output of Concept Design, High-level task flow, High-level data architecture for service design, Conceptual UI and interaction design, Operational Planning, Target Scope of Work, Data model, Proposed software design, Plan or design for APIs, services, and data storage, Identity management, access control, and security.

Module II

13 lecture hours

Domains, Platforms, Frameworks, Technical methods, Plan and operations Technically and Operationally, output of Operational Planning, Sustainable Design, Standards and Reusable Components, Success Measures, Sustainability Planning, Planning for efficient design, Design standards, Style guide,

Communication Standards, Brand Standards, The output of Planning for component development, Final goals, Objectives, Success measures, Test planning, Scheduling and logistics, Choosing and installing analytics tools, Scheduling customer input and communications, Planning for the product's end of life, Production Design, Ongoing Concept Design, Analyze and Specify the Designing screen, The output of Production Design, Technical specifications, UX specifications, Test plans and test criteria for each feature, Production Engineering Understanding, Proof of Concept Production Code, Iteration's scale, The output of Production Engineering, Test, Validate, and Fix, Check the Work, Fix the Problems, The output of Testing and Iterating, Recurring Features, Designing a Process for Organization.

Studio Work / Laboratory Experiments:

In the studio work, students work on innovative, mobile, and professional digital manufacturing for education or prototyping in engineering, manufacturing, and other fields. It also focuses on a digital-manufacturing-lab-for education Professional 3D Scanner for capturing complex objects.

Text Books:

1. Cross, Nigel. *Engineering design methods: strategies for product design*. John Wiley & Sons, 2021. ISBN 9780470519264.

Reference Books:

1. Tao, Fei, Meng Zhang, and Andrew Yeh Chris Nee. *Digital twin driven smart manufacturing*. Academic press, 2019. ISBN 9780128176313.

Name of Program	Bachelor of Technology (Computer Science Engineering)						
CSET435	Fashion Design Technology	L	T	P	C		
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3		
Pre-requisites/Exposure	-						

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate and demonstrate the significance of technology in fashion design.

CO2: To explain the concepts of productivity, quality tools, and how to use them to promote productivity and Quality in the workplace.

CO3: To create and design Garments and Accessories using computer-based tools and commands.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2		2								2	2			
CO2	2	1	2								2		3		
CO3		2	3	2	3						3	3		3	

Course Content:

Module 1

11 lecture hours

Course Handouts, Fashion product ethics, Fashion and Technology ERGONOMICS, Designing Tools & equipment, Dress & Culture, Dress Codes, Body Rituals, Cultural Identity, Dress as a Non-verbal Communication, Fashion as a Dynamic Process, Fashion System and its impact, Change in fashion scenario, Ethnicity & its influence, Body Image, Appearance of Gender, Dress in Workplace, Study of Global Fashion Markets, Global Sourcing & Merchandising, Global Retailing, Policies & Strategies, Fashion Auxiliary Services, IPR & Fashion Definition, Concept, IPR & law, Trademark & Copyright, Process of filing for Copyright, Drawing tools and Techniques, Design Materials, Elements of design, Principles and concept of design, Understanding of colors, Color Theories and color scheme, Textures shapes and forms, Selection of Dresses according to (age, occasion, climate, personality), Drawing Texture Fabric rendering, Motifs (enlargement and reduction), Sources of design inspiration& conceptualization, Designs Using Elements and principles, Design through sketching, Design through Editor and Computer Tools.

Module II

9 lecture hours

Design 3D models, Design through VR tools, Working with special effects, Creating Fabric Designs Creating , Rendering & Draping Accessories Designing, Edit Patterns, Sew Garments, Sew using the Free Hand Tool Add Textures, Use Cut and Sew to add details, Tools for Editing.

Module III

8 lecture hours

Garment Patterns, Add Quality Details using Particle Distance, Export your Garment PNG images, Animate your Clothing, CAD systems, Determining main functions of CAD, Creating an avatar, Animate the Clothing, Create a Dress Models, Create Design of Garment (image, animation, 3D, video), Virtual Fashion Show.

Studio Work / Laboratory Experiments:

The lab part focuses on the techniques, skills, resources, and modern engineering and IT tools that are required in practical application.

Text Books:

1. Faerm, Steven. "Fashion design course: principles, practice and techniques: the ultimate guide for aspiring fashion designers." (*No Title*) (2010). ISBN 978-1438011073.
2. Hopkins, John. *Fashion Design: The Complete Guide*. Bloomsbury Publishing, 2021. ISBN 978-2940411528.

Reference Books:

1. Alvarado, Jane. *Computer aided fashion design using gerber technology*. Fairchild Books, 2007. ISBN 1563674327.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET436	Furniture Design	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To understand how the choice of process and the qualities of materials affect design.

CO2: To get the designing sense the design pattern of modern furniture and draw sketches as well as working drawings.

CO3: To build the various sustainable design using craftsmanship in different materials and communication through exhibition and presentation.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	POS3
CO1						1	2		1			1	2		
CO2	2	1	3	2	1							2		3	
CO3			3		3				2	3		3		3	

Course Content:

Module 1

11 lecture hours

Wood Joinery & Furniture in Interior Design, Wood and metals used in Furniture, Types of wood based products, Methods of care and maintenance, Economics of furniture, durability and usability, Revival of roman style, Types of furniture product, Gothic furniture and décor, Timber Expression, Study of shapes, Environmental conditions influencing furniture designs: Climatic, Social, Economic, Availability of materials Construction techniques, Forms finishes on furniture, Cabinets used in furniture making, Techniques of finishing the Surfaces, Wood and Metal Paints, Wood Cutting Tools, Saws used for woodworking, Filing, Planing, Sanding Supplies Used In Woodworking, Assembly Tools For Woodworking, Woodworking Tools For Measurement and Angles, Polishes and varnishes, Hand painting, Brush painting, Roller, Spray, Study of Anthropometric and ergonomic data in relation to various furniture, Diagrammatic representation through charts, Furniture presentations, Analysis of existing furniture: Drawings, elevation, sections and details, Joinery as applied to furniture, Redesigning of existing furniture.

Module II

9 lecture hours

Upholstery Materials: Types of foams, springs and other materials, Techniques of upholstering and various hardware used for the same, Use of Interior and Exterior Spaces awareness of Different types of plants –

Indoor & Outdoor Plants, Designing of different Interior spaces, Furniture Layout, Drawing – Plan, Elevation 3D view/Perspective View, Furniture detailing using, Colour schemes, Accessories, Electronics Equipment based layout, Electrical layout, Furniture Design Catalogue 2D, Furniture Design Catalogue 3D and VR.

Studio Work / Laboratory Experiments:

In the studio, students will learn the design journey, concept development of wooden and other different products choice for the multipurpose modern interior of the workspace.

Text Books:

1. Lawson, Stuart. *Furniture Design: An Introduction to Development, Materials, Manufacturing*. Laurence King Publishing, 2013. ISBN 1780671202.
2. Rohrbacher, Gary, Anne Filson, Anna Kaziunas France, and Bill Young. *Design for CNC: Furniture Projects and Fabrication Technique*. Maker Media, Inc., 2017. ISBN 978-1457187421.

Reference Books:

1. Kries, Mateo, and Jochen Eisenbrand, eds. *The Atlas of Furniture Design*. Weil am Rhein:: Vitra Design Museum, 2019. ISBN 3931936996.

Specialization Electives: Mobile Technologies

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET396	iOS Application Development	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course-Outcomes (Cos)

On completion of this course, the students will be able to:

CO1: To Define key programming terms relevant to Swift and iOS programming.

CO2: To gain an understanding of iOS application development.

CO3: To gain knowledge for best programming practices in Swift for designing an iOS app.

Course Contents:

Module I

6 lecture hours

Introduction to iOS, Mobile application development, Overview of iOS platform, setting up Xcode & tools, MVC design pattern. Interface Builder Basics: Common system views, Interface Builder Storyboards, project options, default project, create a new project with a label and a greet function.

Module II

6 lecture hours

Common system views configuration, Label(UILabel), Image view, Textview, Scroll view, Table view, Toolbars(UIToolbar), Navigation bars, tab bars, Controls, Button, Segmented controls, Sliders, Switches, Date pickers, UIKit User Interface Catalog, Displaying data: Content mode, Unexpected Clipping.

Module III

8 lecture hours

Layout for multiple sizes, Why Auto Layout?, Create alignment constraints, create size constraints, Resolve constraint issues, Safe area layout guide, resolve constraint warnings, Constraints between siblings, Stack views, stack view attributes, Size classes.

Module IV

8 lecture hours

App life Cycle, break down the delegate, Protocols methods: Did Finish Launching, Will Resign Active, Did Enter Background, Will Enter Foreground, Did Become Active, Will Terminate. View Controller life Cycle: viewDidLoad, viewWillAppear, viewDidAppear, viewWillDisappear.

Studio Work/ Laboratory Experiments:

Students will gain practical experience with the implementation of different swift concepts. The lab work consists of swift programming using Playground and Xcode .

Text Books:

1. Develop in Swift Data Collections. (as available on Apple Book Store) Develop in Swift Fundamentals.
2. Jon Hoffman, Mastering Swift 5: Deep dive into the latest edition of the Swift

Reference Books:

1. Paris Buttfield-Addison, Jonathon Manning, Tim Nugent Learning Swift: Building App macOS, iOS, and Beyond, O'Reilly Media, Inc., 3rd ed, 2018.
2. Yamacli, Beginner's Guide to iOS 11 App Development Using Swift 4: Xcode, Swift

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET397	Mobile Application Development	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course-Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To describe the advanced tools and techniques to develop an iOS application.

CO2: To illustrate the advanced concepts of application development for iOS with programming language.

CO3: To design complex applications and user's interaction for iOS applications.

Course Contents:

Module I

6 lecture hours

Protocols: Custom String Convertible, Equatable, Comparable, Codable, Creating a protocol, Delegation
Model-View-Controller: Pattern, Model objects, Model object Communication, Views, Views
Communication, Controllers: View controllers, Model controllers, Helper controllers; Controller
Communication , Project organization.

Module II

6 lecture hours

Scroll Views: UIScrollView, Scroll views Scroll views in Interface Builder : Define scroll view frame, Add
constraints, Define content view using stack view, Programmatic constraints, Keyboard issues, Content
insets, Scroll indicator ,Scroll View family. Table Views: UITableView, Types, Table view controllers,
Table view styles, Table view cells, UITableViewCell properties, Index paths, Arrays and table views, Cell
dequeuing, Table view protocols, Table view data source, Table view delegate, Reload data, Intermediate
Table Views.

Module III

8 lecture hours

Apple system view controllers, Activity view controller, SFSafariViewController, Alert controllers. Access
the camera: UIImagePickerController, Permission, Send email: Verify whether mail can be sent,
MFMailComposeViewController, mailComposeController.

Module IV

8 lecture hours

Closures: Passing closures as arguments, Syntactic sugar, Collection functions using closures: Map, Filter,
Reduce, Adding computed properties, Adding instance or type methods, Organizing code. Generic Data:

Array, Dictionary, Type constraints, Functions, and methods, Associated types. Dynamic Data: Search Controllers, Handling Data Changes, Diffable Data Sources.

Studio Work/ Laboratory Experiments:

Students will gain practical experience with the implementation of different swift concepts. The lab work consists of swift programming using Playground and Xcode.

Text Books:

1. “Pro Design pattern in Swift” Adam Freeman. 2019
2. Design Patterns in Swift 5: Learn How to Implement the Gang of Four Design Pattern Using Swifts. Karoly Nyisztor. 2018

Reference Books:

1. Design Patterns in Swift: A Different Approach to Coding with Swift - iOS. Vamshi Kris
2. Test iOS Apps with UI Automation, O'Reilly 2019

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET439	Design Patterns and UI Testing	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course-Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To impart knowledge of Swift and Apple iOS application design, development and testing.

CO2: To provide students with the skills necessary to develop an iOS App from scratch to develop on the Apple Store.

CO3: To provide UI test tests a specific user interaction with the application's UI.

Course Contents:

Module I

6 lecture hours

Understanding design patterns, getting used to Xcode, creating the sports Store app, creating design pattern: object template pattern, prototype pattern, singleton pattern, object pool pattern, builder pattern

Module II

6 lecture hours

Structural Patterns: Adapter pattern, Bridge pattern, Decorator pattern, Composition pattern, Facade pattern, Flyweight pattern, Proxy pattern.

Module III

7 lecture hours

MVC Pattern: the Model/View/Controller Pattern.

Module IV

9 lecture hours

Testing basics, Performance testing, UI testing, app and Library tests, XCTest—the Xcode Testing Framework. Writing Test Classes and Methods, Running Tests and Viewing Results. Test Debugging Workflow, Test Specific Debugging Tools, Code Coverage, Enable Code Coverage, Automating the Test Process.

Studio Work/ Laboratory Experiments:

Students will gain practical experience with the implementation of different swift concepts. The lab work consists of swift programming using Xcode Testing Framework .

Text Books:

1. “Pro Design pattern in Swift” Adam Freeman. 2019
2. Design Patterns in Swift 5: Learn How to Implement the Gang of Four Design Pattern Using Swifts. Karoly Nyisztor. 2018
3. iOS Unit Testing by Example Book by Jon Reid. 2018
4. Test-Driven iOS Development with Swift Book by Dominik Hauser.

Reference Books:

1. Design Patterns in Swift: A Different Approach to Coding with Swift - iOS. Vamshi Kris
2. Test iOS Apps with UI Automation, O'Reilly 2019

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET494	iOS Machine Learning	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course-Outcomes (COs):

On completion of this course, the students will be able to:

CO1: Become familiar with the basic concepts of machine learning and various ML models.

CO2: Implement ML models using iOS apps.

CO3: Learn latest iOS frameworks for ML.

Course Contents:

Module I

6 lecture hours

Introduction to Machine Learning, Types of learning, Regression versus Classification. Different types of classification algorithms.

Module II

6 lecture hours

Introduction to Neural Networks, perceptron model, activation functions, multi-layer perceptron model, backpropagation, deep learning, deep learning models: CNN, RNN.

Module III

8 lecture hours

How to add ML model in Xcode project, Properties and picker delegate methods, Pickerview data source methods, iOS ready machine learning models, ML models with images, Vision framework: Programming the VN request callback method, Testing different ML models.

Module IV

8 lecture hours

Finding custom ML models, Converting ML models get Anaconda IDE, Python libraries for core ML, Caffe tools for core ML conversion.

Studio Work/ Laboratory Experiments:

Students will gain practical experience with the implementation of ML models with iOS ML frameworks. The lab work consists of CoreML and Apple NLP Framework.

Text Books:

1. Machine Learning with Swift, Alexander Sosnovshchenko, February 2018, Publisher(s): Packt Publishing ISBN: 9781787121515.
2. Machine Learning with Core ML 2 and Swift, Karoly Nyisztor, 2019.

Reference Books:

1. Practical Artificial Intelligence with Swift, From Fundamental Theory to Development of AI-Driven Apps, Mars Geldard, O'reilly, 2019.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET495	Backend in iOS	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course-Outcomes (COs):

On completion of this course, the students will be able to:

CO1: Understand concept of backend programming using cocoapods for iOS.

CO2: Understand the accessibility and authorization using firebase for iOS.

CO3: Apply the Firebase project to multimedia storage and download.

Course Contents:

Module I

6 lecture hours

Introduction to Backend Programming, Introduction to Cocoapods. Installation of Cocoapods, creating project as a pod file.

Module II

6 lecture hours

Introduction to Firebase, Creating a Firebase account, Integrating the account with the project, setting up auth of Firebase, Polishing the Auth UI, Connecting Auth UI with Firebase

Module III

8 lecture hours

Creating a snapchat clone app-snapfun, Allow the user to add images, Handling the images on Firebase, Use real time database, creating a snap, setting up the view of snap, deletion of snap.

Module IV

8 lecture hours

Live project development.

Studio Work/ Laboratory Experiments:

Students will gain practical experience with the implementation of the Firebase project with Cocoapods. The lab work consists of authorization, accessibility, and functionality of UI with backend.

Text Books:

1. Learn Swift by Building Applications: Explore Swift programming through iOS app development, Emil Atanasov, May 2018, Publisher(s): Packt Publishing ISBN: 9781787121515.
2. iOS 15 Programming Fundamentals with Swift: Swift, Xcode, and Cocoa Basics, Matt Neuburg, 2021.

Reference Books:

1. iOS and macOS Performance Tuning: Cocoa, Cocoa Touch, Objective-C, and Swift (Developer's Library) s, Marcel Weiher, O'reilly, 2016.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET496	Special Topics in Mobile Technologies	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course-Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To familiarize and learn about the latest trends and research in the field.

CO2: To equip themselves with the conceptual and practical experience of few latest methods, tools, technologies or algorithms in Mobile Technologies.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	3				1				2		1			3
CO2		2	2	2	3									2	

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I

28 lecture hours

This course covers the cutting-edge topics in Mobile Technologies, and these modules will be chosen by the instructor based on the requirements and relevance at that point of time. These modules need to be relevant to the industry and start-ups will also include related case studies, use cases and implementations scenarios. Students will be working on lab work and projects to get real hands-on experience of these topics and modules.

Studio Work/ Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to Artificial Intelligence

Specialization

Electives:

UI/UX

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET313	User Centered Design	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course-Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Articulate appropriateness of potential design methods such as contextual design, prototyping, ideation given a problem setting.

CO2: Examine the issues and challenges to achieving a user-centered design process.

CO3: Implement design methods at a basic level of competence: interviews, personas, storyboarding, sketching, and evaluation.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1															
CO2															

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module 1

(8 hours)

User Centred Analysis (UCA), Mental models, knowing how the user works, Learnability, User model vs System Model, Which UCA steps to perform, Obstacles of user-centred analysis, Return on Investment (ROI), Justifying the right process, Creating a Design Strategy: Components of a design strategy, Site strategy drives design strategy, Where to get strategy information, Mining existing documentation, Working with brand objectives, Design techniques: Sketching, Scenarios, Storyboards, Design patterns.

Module 2

(9 hours)

Profiles and Personas, Value of profiles and Personas, User profiles, Task profiles, Environment profiles, Data gathering methods, Choosing the right method, Elements of field interview, Developing interview questions, Good and bad interview technique, Conducting a user observation. Complementary Data Gathering Methods: Value of complementary methods, Focus groups, User group meetings, Usability roundtables, Complementary Data Gathering Methods, Facilitated workshops, JAD sessions, Using surveys and other indirect methods, Online surveys, Using multiple methods.

Module 3

(11 hours)

Primary Noun Architecture, Describing primary nouns, Primary noun views, Primary noun details, Primary nouns to navigation, Information Architecture, Costs of poor organization, Basic organization schemes, Hybrid schemes, Shallow vs. deep structures, Labelling systems, Information Architecture, Affinity diagrams, Card sorting techniques, Card sorting tools, Getting sign-off on the contract for design, Using concept sketches to drive out requirements, Setting usability criteria, Scenario and Task Analysis: The power of a scenario, Scenarios

vs. use cases, Determining the level of detail, Scenarios driven priorities. Scenario and Task Analysis: Identifying functions and tasks, Common errors and challenges in task analysis, Characterizing the new task design, Primary Noun Architecture: Value of primary nouns, Identifying primary nouns, Domain Analysis, Requirement analysis.

STUDIO WORK / LABORATORY EXPERIMENTS:

Students will gain practical experience with the implementation and use of various user interface design methods and functionalities User interface prototyping and Agile Model Driven Development (AMDD). Moreover, students will have exposure to the tools for designing the User Interface to improve the User experience. The labwork will give and exposure to Android mobile development platform and other open-source software tools.

TEXTBOOKS/LEARNING RESOURCES:

1. Travis Lowdermilk, *User centred design: A Developers Guide* (1st ed.), O'Reilly Media, 2013. ISBN 978-1449359805.
2. "The User Experience Team of One" by Leah Buley
3. "Interviewing Users: How to Uncover Compelling Insights" by Steve Portigal

REFERENCE BOOKS/LEARNING RESOURCES:

1. Krug Steve, *don't make me think Revisited: A common sense approach to web usability* (3rd ed.), Berkeley CA: New Riders Publishing, 2020. ISBN 978- 0321965516.
2. Travis, D., and Hodgson, P., *Think Like a UX Researcher: How to Observe Users, Influence Design, and Shape Business Strategy* (1st ed.), CRC Press, 2019. ISBN 978-1138365292.

Name of Program	Bachelor of Technology (Computer Science Engineering)						
CSET314	Strategic UI/UX Design: From Concept to Market Success	L	T	P	C		
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3		
Pre-requisites/Exposure	-						

Course-Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Strategic Design Thinking

CO2: This course introduces Adaptive Design for Emerging Technologies.

CO3: Mastering Design Collaboration

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1															
CO2															

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module 1

(9 hours)

Understanding business goals, Aligning design with organizational strategy, Design sprints and strategic design frameworks, Techniques for increasing user conversions, A/B testing and data-driven design decisions, Optimizing user flows for conversions, Advanced prototyping tools and techniques, Incorporating interactivity for engaging experiences, Prototyping for user testing and feedback.

Module 2

(9 hours)

Designing for brand consistency, Building emotional connections through design, Adaptive Design for Emerging Technologies, Creating a cohesive brand experience, Designing for brand consistency, Building emotional connections through design, Designing for AR, VR, and mixed reality, UI/UX considerations for AI-driven interfaces, Navigating challenges in adaptive design, Advanced prototyping tools and techniques, Incorporating interactivity for engaging experiences, Prototyping for user testing and feedback

Module 3

(10 hours)

Utilizing analytics for informed design decisions, User behavior analysis and heatmaps, Iterative design based on user data, Mobile UX best practices, Designing for various mobile platforms, Mobile app usability testing, Applying game design principles to user interfaces

Gamification for user engagement, Balancing challenge and usability, Ethical considerations in UI/UX design, Responsible use of data and technology, Designing for inclusivity and diversity.

STUDIO WORK / LABORATORY EXPERIMENTS:

Students will gain hands-on experience on Unity tool. A range of VR/AR/MR experience development tools and skills. Story development, user experience, interface design and implementation are introduced within the context of creating immersive experiences.

TEXTBOOKS/LEARNING RESOURCES:

1. "Lean UX: Designing Great Products with Agile Teams" by Jeff Gothelf and Josh Seiden
2. "Strategic Writing for UX: Drive Engagement, Conversion, and Retention with Every Word" by Torrey Podmajersky
3. "UX Strategy: How to Devise Innovative Digital Products that People Want" by Jaime Levy

REFERENCE BOOKS/LEARNING RESOURCES:

1. "Inspired: How To Create Products Customers Love" by Marty Cagan
2. "Value Proposition Design: How to Create Products and Services Customers Want" by Alexander Osterwalder, Yves Pigneur, Gregory Bernarda, and Alan Smith

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET315	UX Research and Data Analytics	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course-Outcomes (COs)

On completion of this course, the students will be able to:

CO1: This course introduces Creating research plans

CO2: Qualitative and quantitative research analysis

CO3: Find user's needs and create surveys.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1															
CO2															

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module 1

(6 hours)

Defining analysis objectives, Organizing data, Investigation, Clusterization, Results and insights identification. Check the methodology: Respondents, Tasks, Logistics, Data analysis.

Module 2

(6 hours)

Creating research plan: Define the problem statement, Get stakeholders' buy-in, Identify your objectives, Choose the right research method, Recruit participants, Prepare the brief Establish the timeline, Decide how you'll present your findings.

Module 3

(8 hours)

Quantitative research data analysis, the success rate of a specific task, how long a user takes to complete a task, the bounce rate of a webpage, user's demographic profile. Thematic Analysis, gather all the data, Read all the data, Code the text, Create new codes that encapsulate potential themes, Take a break, Evaluate the themes.

Module 4

(8 hours)

Qualitative research data analysis, what do the users like the most about the product, What do they like the least, Which features are more valuable, Why do some users react differently from others, Did they have an emotional response? When, Are they satisfied with the product? Why.

STUDIO WORK / LABORATORY EXPERIMENTS:

Analytics tools provide product managers and UX/UI designers with robust capabilities to measure and evaluate product performance. Tools like Google Analytics, Mixpanel, and Hotjar offer a wealth of data on user engagement, conversion rates, and user journeys.

TEXTBOOKS/LEARNING RESOURCES:

1. "Measuring the User Experience: Collecting, Analyzing, and Presenting Usability Metrics" by Thomas Tullis and William Albert
2. "Quantifying the User Experience: Practical Statistics for User Research" by Jeff Sauro and James R. Lewis
3. "Storytelling with Data: A Data Visualization Guide for Business Professionals" by Cole Nussbaumer Knaflic

REFERENCE BOOKS/LEARNING RESOURCES:

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1. "Quantifying User Experience: Practical Statistics for User Research" by Jeff Sauro
2. "UX Research: Practical Techniques for Designing Better Products" by Brad Nunnally and David Farkas

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET316	UI/UX Design Testing and Evaluation	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	0	0	6	3
Pre-requisites/Exposure	-				

Course-Outcomes (COs)

On completion of this course, the students will be able to:

CO1: This course introduces software designing process and modeling (Software Development Life Cycle, Design Handoff to Software Teams).

CO2: Design Analysis & Outcomes

CO3: Designing for Mobile Applications (ios/Android) and Progressive Web Applications

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1															
CO2															

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module 1

(8 hours)

User Experience Metrics, Conversion Rate Optimization and A/B testing. Methods of usability testing, conducting UX research, analyzing user feedback. Performing usability tests; using tools for analytics and feedback collection.

Module 2

(10 hours)

Type and font size, colors, Icon styles, Visual inconsistencies, Required fields, Data type errors, The width of the fields, On-screen instructions, Progress indicators, Confirmation of action (save, change, delete), Drop-down lists, Error messages, Shortcuts.

Module 3

(10 hours)

Navigation through the website (or application), navigation through the website (or application), order placement process must be effortless, performance of the site (or app), Automatic record of screen activities of the user sessions, Heat maps, User questionnaires to evaluate the product, Analysis of the competitors' products.

STUDIO WORK / LABORATORY EXPERIMENTS:

Students will gain hands-on experience on tools such as:

1. Maze
2. Lookback
3. UserTesting
4. Optimal Workshop

5. UsabilityHub
6. Loop11
7. Userfeel
8. Trymata
9. Hotjar
10. UserZoom
11. Qualaroo
12. GetFeedback
13. CrazyEgg
14. Userlytics
15. Userbrain

TEXTBOOKS/LEARNING RESOURCES:

1. "The Handbook of Usability Testing: How to Plan, Design, and Conduct Effective Tests" by Jeffrey Rubin and Dana Chisnell
2. "Usability Testing Essentials: Ready, Set...Test!" by Carol M. Barnum
3. "Usability Testing for Survey Research" by Emily Geisen and Jennifer Romano Bergstrom

REFERENCE BOOKS/LEARNING RESOURCES:

1. "Effective UI: The Art of Building Great User Experience in Software" by Jonathan Anderson and John McRee
2. "Validating Product Ideas: Through Lean User Research" by Tomer Sharon

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET317	Product Management for UX Designers	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course-Outcomes (COs)

On completion of this course, the students will be able to:

CO1: This course introduces Product Strategy and UX Design Strategy

CO2: Agile Product Development

CO3: Tracking Performance & Product Roadmap

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1															
CO2															

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module 1

(6 hours)

Creation and management of a digital product. Working on real projects operational skills in various domains such as marketing, tech/IT, design, business. Successful product managers possess a diverse skill set that combines business acumen, technical understanding, and design sensibility. Excel in strategic thinking, market analysis, and project management. Deep understanding of UX and UI design principles, enabling effective collaboration with designers and engineers to translate user requirements into compelling product experiences.

Module 2

(6 hours)

Product managers and UX/UI designers work hand in hand to shape the digital product's UX/UI design. Collaboration involves frequent communication, knowledge sharing, and joint decision-making. Product managers bring their market and business insights, while UX/UI designers contribute their expertise in user research, information architecture, interaction design, and visual aesthetics. This collaborative approach ensures a harmonious blend of user needs, business goals, and design excellence.

Module 3

(8 hours)

A well-defined product development process is vital for successful digital product UX/UI design. This module provides an overview of the product development lifecycle, including the integration of UX and UI design activities. Agile and Waterfall Methodologies

Two commonly used methodologies in product development are Agile and Waterfall. Agile emphasises iterative and flexible development, allowing for continuous feedback and adjustments throughout the process. Waterfall, on the other hand, follows a linear approach with distinct phases, where each phase is completed

before moving to the next. Both methodologies have their merits, and product managers need to choose the most suitable approach based on project requirements.

Module 4

(8 hours)

Designing for Next-gen technologies

- Immersive Experience Design: AR/VR Design
- Intelligent Product Design: Design with AI, ML, and IoT technologies
- Integration of Emerging Technologies into Legacy/Existing Systems

STUDIO WORK / LABORATORY EXPERIMENTS:

Analytics tools provide product managers and UX/UI designers with robust capabilities to measure and evaluate product performance. Tools like Google Analytics, Mixpanel, and Hotjar offer a wealth of data on user engagement, conversion rates, and user journeys.

TEXTBOOKS/LEARNING RESOURCES:

1. "Product Leadership: How Top Product Managers Launch Awesome Products and Build Successful Teams" by Richard Banfield, Martin Eriksson, and Nate Walkingshaw
2. "Lean UX: Designing Great Products with Agile Teams" by Jeff Gothelf and Josh Seiden
3. "Lean Product and Lean Analytics" by Ben Yoskovitz and Alistair Croll

REFERENCE BOOKS/LEARNING RESOURCES:

1. "Inspired: How To Create Products Customers Love" by Marty Cagan
2. "User Story Mapping: Discover the Whole Story, Build the Right Product" by Jeff Patton

General Electives (Elective II, IV, VI)

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET344	Image and Video Processing	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain and Analyse images using Histograms and spatial filters. Apply Hough transforms and be familiar with image representation using textures.

CO2: To compute motion using optical flow and understand methods for image description and morphological operations.

CO3: To implement different low-level and high-level image and video processing techniques to wide variety of applications.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3			3	3								3	
CO2	3	3	2		3	2								2	3
CO3	3	3	3		3	3			2	2	2	2		3	3

1=weakly related 2= moderately related 3=strongly related

Course Contents:

Module I:

7 lecture hours

Applications of Digital Image Processing, Elements of Digital image processing systems, Sampling and quantization, Neighbours of a pixel, adjacency, connectivity, Regions and Boundaries, Distance measures, Gray scale to Binary image using thresholding, Image Enhancement in the Spatial domain, Gray level transforms, Histogram Processing, Histogram Equalization, Enhancement using Spatial filters, Concept of convolution, Smoothing, Mean, median and Gaussian filters, Edge detection using Prewitt, Sobel, Laplace Filters, Laplace of Gaussian Filter.

Modulo II:

6 lecture hours

Canny Edge Detector, Harris Corner Detector, Color models, RGB, HSV, YCbCr models, Pseudocolor Image Processing, Color Transforms, Color to grayscale conversion, Handling Binary Images, Line detection using Hough transform, Polar form, Circle Detection, Morphological Operations, Dilation, Erosion, Opening, Closing, Boundary detection, Hole filling, connected components, Hit and Miss

transform, Shape representation using moments, Texture analysis, Texture from histogram, Texture from GLCM matrices.

Module III:

8 lecture hours

Motion Detection, Concept of Optical Flow, Optical flow equation, Lucas Kanade method, Image Enhancement in the Frequency domain, 1-D and 2-D Fourier Transform and their Inverse, Low pass and Hi pass Filtering, Ideal and Butterworth and Gaussian Filters, Homomorphic Filtering, Image Compression Fundamentals, Lossless Compression Models, Run length Encoding, Huffman Coding, Lossy Compression, Discrete Cosine Transform, Quantization, Zigzag coding, Color image compression, Text recognition, Feature detection, Integral Image Formation.

Module IV:

7 lecture hours

Face Detection – Viola Jones method, Face Recognition, Principal Component Analysis (PCA), Concept of Eigenface, Feature detection for Machine learning applications, SIFT and HOG parameters, Video Processing, Video formation, Video sampling, Motion estimation, Motion compensated (MC) filtering, Frame-rate conversion, Video Coding, Video Compression, Framebased compression (MPEG), Salient object detection, Human action recognition from videos, Depth cameras– Kinect camera data capture, RGBD data.

Studio Work / Laboratory Experiments:

The lab work will be based on image enhancement, image zooming, image cropping, image restoration, image compression and image segmentation etc. The Lab will use MATLAB and Open CV with python.

Text Books :

1. Szeliski Richard, *Computer Vision: Algorithms and Applications* (2 ed.), Springer, 2022. ISBN 978-3030343712.
2. Jain K., *Fundamentals of Digital Image Processing* (1st ed.), Pearson Education India, 2015. ISBN 978-9332551916.

Reference Books :

1. Bovik, A. C., *Handbook of image and video processing* (1 ed.), Academic press, 2010. ISBN 9780121197902.
2. Kim, B. G., *Digital Signal, Image and Video Processing for Emerging Multimedia Technology. Electronics* (1 ed.), Mdpi AG, 2021. ISBN 978-3039438570.
3. Vyas, A., Yu, S. and Paik, J., *Fundamentals of digital image processing. In Multiscale Transforms with Application to Image Processing* (1 ed.), Springer, 2018. ISBN 978-9811356131.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET320	Information Retrieval	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Comprehend types of text analysis, Information retrieval, IR system architecture, query processing models and probabilistic models.

CO2: Manage information retrieval systems by performing indexing, compression, information categorization sentiment analysis, network management, search engine optimization, records compliance and risk management.

CO3: Implement different information retrieval approaches for applications in text domain.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3				2	2				2			3		
CO2	3	2			2	3		1	2	2	2				
CO3	2	1		2	2	2			2	1					2

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

7 lecture hours

Text analysis and types, Information retrieval, Text processing, Indexes and query matching, Semistructured text data, Tokenization, Stemming, Lemmatization, Language modelling, Examples of open source IR Systems, Query processing models, Probabilistic models, Binary independence model, Robertson/Spark Jones weighting formula, Two-Poisson model, Relevance feedback, Term selection, Pseudo relevance feedback, Language models, Unigram, Bigram language models, Generating queries from documents, Language models and smoothing, Ranking with language models, Retrieval evaluation measures.

Modulo II:

7 lecture hours

Normalized Discounted Cumulative Gain (NDCG), Kullback-Leibler divergence, Divergence from randomness, Passage retrieval and ranking, Management of Information Retrieval Systems, Knowledge management, Information management, Digital asset management, Network management, Search engine optimization, Records compliance and risk management, Version control, Data and data quality, Information system failure, Web retrieval and mining, Semantic web, XML information retrieval, Recommender systems and expert locators, Knowledge management systems, Decision support systems, Geographic information system (GIS).

Module III:**9 lecture hours**

Indexing, Inverted indices, Index components and Index life cycle, Interleaving Dictionary and Postings lists, Index construction, Query processing for ranked retrieval, Compression, General-purpose data compression, Symbol-wise data compression, Compressing posting lists, Compressing the dictionary, Information categorization and filtering, Classification, Probabilistic classifiers, Linear classifiers, Similarity-based classifiers, Multi category ranking and classification, Learning to rank.

Module IV:**7 lecture hours**

Text Clustering, Partitioning methods, Clustering versus classification, Reduced dimensionality/spectral methods, Lexicons, Corpora, Sentiment Analysis, Document-level, Sentence-level and Aspect-based sentiment analysis, Web crawling, Near duplicate pages, Distributed word representations, Link Analysis, PageRank algorithm, Search engine bias, Personalized searching, Question Answering, Cross lingual retrieval, Adversarial Information Retrieval.

Studio Work / Laboratory Experiments:

Problems based on search engines and common open-source software to perform common methods of exploratory and predictive analysis. Application of text analysis techniques for data analysis. Performing sentiment analysis.

Text Books:

1. Buttcher Stefan, Charles L. A. Clarke and Gordon V. Cormack, *Information Retrieval: Implementing and Evaluating Search Engines* (1 ed.), The MIT Press, 2016. ISBN 978-0262528870.
2. Croft W. Bruce, Donald Metzler and Trevor Strohman, *Search Engines - Information Retrieval in Practice* (1 ed.), Pearson, 2009. ISBN 978-0136072249.

Reference Books :

1. Bates, M. J., *Understanding information retrieval systems: management, types, and standards* (1 ed.), Auerbach Publications, 2011. ISBN 978-1439891964.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET321	User Centered Design	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Articulate appropriateness of potential design methods such as contextual design, prototyping, ideation given a problem setting.

CO2: Examine the issues and challenges to achieving a user-centered design process.

CO3: Implement design methods at a basic level of competence: interviews, personas, storyboarding, sketching, and evaluation.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	1	2	3	3								1	1		
CO2	2	2	2									2	2	3	
CO3	2	3	2	3	3				3	3		2	1	2	

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

8 lecture hours

User Centred Analysis (UCA), Mental models, knowing how the user works, Learnability, User model vs System Model, Which UCA steps to perform, Obstacles of user-centred analysis, Return on Investment (ROI), Justifying the right process, Creating a Design Strategy: Components of a design strategy, Site strategy drives design strategy, Where to get strategy information, Mining existing documentation, Working with brand objectives, Design techniques: Sketching, Scenarios, Storyboards, Design patterns.

Modulo II:

9 lecture hours

Profiles and Personas, Value of profiles and Personas, User profiles, Task profiles, Environment profiles, Data gathering methods, Choosing the right method, Elements of field interview, Developing interview questions, Good and bad interview technique, Conducting a user observation. Complementary Data Gathering Methods: Value of complementary methods, Focus groups, User group meetings, Usability roundtables, Complementary Data Gathering Methods, Facilitated workshops, JAD sessions, Using surveys and other indirect methods, Online surveys, Using multiple methods.

Module III:

11 lecture hours

Primary Noun Architecture, Describing primary nouns, Primary noun views, Primary noun details, Primary nouns to navigation, Information Architecture, Costs of poor organization, Basic organization schemes, Hybrid schemes, Shallow vs. deep structures, Labelling systems, Information Architecture, Affinity diagrams, Card sorting techniques, Card sorting tools, Getting sign-off on the contract for design, Using concept sketches to drive out requirements, Setting usability criteria, Scenario and Task Analysis: The power of a scenario, Scenarios vs. use cases, Determining the level of detail, Scenarios driven priorities. Scenario and Task Analysis: Identifying functions and tasks, Common errors and challenges in task analysis, Characterizing the new task design, Primary Noun Architecture: Value of primary nouns, Identifying primary nouns, Domain Analysis, Requirement analysis.

Studio Work / Laboratory Experiments:

Students will gain practical experience with the implementation and use of various user interface design methods and functionalities User interface prototyping and Agile Model Driven Development (AMDD). Moreover, students will have exposure to the tools for designing the User Interface to improve the User experience. The lab work will give and exposure to Android mobile development platform and other open-source software tools.

Text Books :

1. Lowdermilk Travis, User centred design: A Developers Guide (1 ed.), O'Reilly Media, 2013. ISBN 978-1449359805.

Reference Books :

1. Steve Krug, *Don't make me think Revisited: A common sense approach to web usability* (3 ed.), Berkeley CA: New Riders Publishing, 2020. ISBN 978-0321965516.
2. Travis, D., and Hodgson, P., *Think Like a UX Researcher: How to Observe Users, Influence Design, and Shape Business Strategy* (1 ed.), CRC Press, 2019. ISBN 978-1138365292.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET322	Secure Coding	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain secure programming philosophy, design principles, and its methods.

CO2: To examine typical mistakes done during programming and the methods of their handling.

CO3: To implement the typical threats in programming and understand concepts of implementing the secure codes.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	2	2	2							2	2	2		
CO2	3	3	2	3								2	2		
CO3	1		3	3	3							3	3		

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

9 lecture hours

Secure Programming, Robust vs. Secure Programming, Security Policies and Procedures, Checking Design and Implementation, Where to Look for Vulnerabilities, Classification of Security Flaws, Landwehr's Taxonomy, Fortify Taxonomy, Protection methods at different layers, PreDeCo matrix of software security. Input Validation in Programming, Improper Error and Exception Handling, Code Injection and Mitigation, Broken Authentication..

Modulo II:

6 lecture hours

Secure Programming Design Principles, Principle of Least Privilege, Fail-Safe Defaults, Principle of Economy of Mechanism, Principle of Complete Mediation, Separation of Privilege Principle, Principle of Open Design, Principle of Least Common Mechanism, Principle of Least Astonishment, Control Hijacking Attacks and Defences, Attacks Using Virtual Machines, Static and Dynamic Analysis, Language-based Security Models, Isolation Techniques.

Module III:

13 lecture hours

XML External Entity (XXE), Cross-Site Scripting (XSS), Insecure Deserialization, LFI (Local File Inclusion) and RFI (Remote File Inclusion) vulnerabilities, Unvalidated File Upload vulnerability, Buffer

Overflow vulnerabilities, Client Side Security, JavaScript Security, Click Jacking, Ajax Security, HTML5 Security, Java Secure Socket Extension (JSSE), Common Coding Errors and Vulnerability, Automation and Testing for secure coding, Research Issues in Secure Coding.

Studio Work / Laboratory Experiments:

Lab experiments on JAVA and C/C++ to understand the threats and implement secure coding. Implementation of input validation, improper error and exception handling, code injection and the techniques of its mitigation, demonstration of Broken authentication, and different types of threats, e.g., XXE Attack, XSS software attack, insecure deserialization, and user-interface redress attack.

Text Books :

1. Richardson T. and Thies C. N., *Secure Software Design*, Jones & Bartlett Learning (1 ed.), Jones & Bartlett, 2012. ISBN 978-1449626327.

Reference Books :

1. Codings Zach, *Computer Programming and Cyber Security for Beginners* (1 ed.), Independently published, 2020. ISBN 978-1801091442.
2. Seacord Robert C., *Secure Coding in C and C++* (2 ed.), Pearson Education, 2013. ISBN 978-0321822130.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET323	Compiler Construction	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate major phases of compilation.

CO2: To examine various parameters passing scheme, explain memory management of a programming languages and perform code optimization.

CO3: To Implement lexical analyzers, parsers, and small compilers by using general purpose programming languages.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1		2	2									1	3		
CO2	2	1	1	3	2							2	3		
CO3		3	3	2	2							3		2	

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

Compiler, Code Analysis, Interpreter, Single Pass, Two pass, Multi Pass compiler, Preprocessor, Macros, Phases of compiler, Symbol table manager, Operations on symbol table, Error handling, Bootstrapping and cross compiler, Lexical Analysis, Tokens, Regular expression, Generation of lexical analysis from DFA, Syntax Analysis, Parser, Context Free Grammar, Conversion Rule for Ambiguous To Unambiguous Grammar, Non-Deterministic & Deterministic Grammar, Left Recursive And Right Recursive Grammar, Parsing: Top down and Bottom up, Backtracking and their automatic generation, LL (1) Parser, LR Parser, LR (0) items, SLR (1), LALR (1), Canonical Parsing..

Modulo II:

9 lecture hours

Error Analysis, Error Classification, Error detection, Error Detection in LL and LR parsers, Error recovery, Panic mode error recovery, Static semantic, Intermediate code generation, static semantic analyses in declaration processing, name, and scope analysis, S-attribute, Semantic analysis through S-attribute grammar, L-attribute, Type checking, Language features influencing run time memory management. Parameter passing mechanism, Division of memory into code, stack, heap and static, Activation record, Garbage collection, Code generation for expressions, issues in efficient code generation.

Module III:**9 lecture hours**

Sethi Ullman algorithm, optimal code generation, Retargetable code generation, Code generation for control structures, Code Optimization, Local and global optimization, Control flow analysis, Data flow analysis, Global optimizations, Graph colouring in optimization, Live ranges of run time values.

Studio Work / Laboratory Experiments:

Construct a lexical analyzer using Flex. Construct a parser using Prison Bison. Build simple compilers from parsing to intermediate representation to code generation and simple optimization.

Text Books :

1. Sunitha, K. V. N., *Compiler construction* (1 ed.), Pearson Education India, 2013. ISBN 978-9332500297.
2. Thain Douglas, *Introduction to Compilers and Language Design* (2 ed.), Lulu.com, 2019. ISBN 978-0359142835.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET324	Software Project Management	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate and determine the purpose and importance of basic processes involved in project management from the perspectives of planning, tracking and completion of project.

CO2: To examine different organization structures and project structures.

CO3: To implement programs to manage project management, project schedule, expenses and resources with the applications of project management tools.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1		2		1		3					3	1		2	
CO2		1	1	2	1	3			3	3	3	2		3	
CO3	1	3	3	3	2	3			2	2	2	3		3	

1=weakly related 2= moderately related 3=strongly related

Course Contents:

Module I:

10 lecture hours

Project Management: Characteristics of software projects, Objectives, Stakeholders, Feasibility Study, Cost-benefit Analysis, Planning, Project Execution, Project and Product Life Cycles, Role of project manager, Knowledge areas, Tools & Techniques, System view of project management, Agile software, Iterative steps for planning, Project Plan documentation methods, Software Requirement Specification, Measurement and Control, Reviews, feedback and reporting mechanisms, revisiting the plan, Scope Planning & Scope management plans, Function point calculation, Scope definitions & project scope statement, Project time management, Activities sequencing, Network diagrams, Activity resource estimation, Activity duration estimation, Schedule development, Gantt Charts, Critical path method, Program evaluation & review technique (PERT) and CPM, Principles of cost management, Cost estimating, Type of cost estimate, Cost estimate tools & techniques, COCOMO.

Modulo II:

7 lecture hours

Putnam/SLIM model Estimating by Analogy, Cost budgeting, Cost control, Earned value management, Project portfolio management, Project Quality Planning, Quality Assurance, Quality control, Tool & techniques for quality control, Pareto Analysis, Six Sigma, CMM, ISO Standards, Juran Methodology, Project Human resource planning, Project organisational charts, Responsibility assignment metrics, Acquiring project team, Resource assignment, resource loading, Resource levelling, Team structures, Project Communication Planning, Performance reporting, Managing stakeholders.

Module III:**11 lecture hours**

Project Risk Management planning, Common sources of risk, Risk identification techniques, Qualitative risk analysis, Expert judgement, Decision trees, Expected monetary value, Simulation, sensitivity analysis, Risk response planning, Risk monitoring & control, Project Procurement management plans, Contract statement of work. Planning contracts, Requesting seller responses, Selecting sellers, Administrating the contract, Closing the contract, Software Configuration Management, Retaining versions, Software Configuration elements (SCI's), Change Control and Management.

Studio Work / Laboratory Experiments:

Using Function Point calculation tools for estimation, comparing with COCOMO estimates, Implementation of various exercises using PERT, CPM methods, preparing schedule, resource allocation etc. using MS Project or Fissure. sim or VENSIM or Visual Paradigm can also be used, Preparing an RMMM Plan for a case study, Preparing Project Plan for a Software Project for Lab Project or case study. Exploring about PMBOK (Project Management Body of Knowledge) and SWEBOK (Software Engineering Body of Knowledge) from related website, Implementation of software project management concepts using related tools and technologies.

Text Books :

1. Pressman Roger and Bruce Maxim, *Software Engineering: A practitioner's approach* (9 ed.), Tata McGraw Hill, 2020. ISBN 978-1259872976.

Reference Books :

1. Aggarwal Manuj and TetraTutorials Team, *The Art of Project Management for Software and IT Engineers* (1 ed.), Packt Publishing, 2018. ISBN 978-1789804768.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET325	Digital Communication	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain digital communication systems with emphasis on digital modulation techniques.

CO2: To implement information theory and channel coding/decoding.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	1	1	1								1	2		
CO2	1	3	1	1	1							2	3		
CO3	3	1		2								3	3		

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

9 lecture hours

Digital Communication, Elements of Digital Communication, Classification of Signals, Deterministic and Random Signals, Periodic and Non-Periodic Signals, Analog and Discrete Signals, Sampling Theorem, Methods of Sampling, Random Signal Analysis, Random Variables, Limit theorems for sum of random variables, Random variable functions, Information Theory, Mathematical Models for Information Theory, Logarithmic Measure of Information, Coding of Information Sources, Coding Algorithms, Lossy Data Compression, Statistical Signal processing, Linear models of random signals, Estimation Theory, Quantization, Uniform Quantization, Nonuniform Quantization, Statistics of Speech Amplitudes.

Modulo II:

8 lecture hours

Pulse Code Modulation (PCM), Sampling in PCM, Quantization in PCM, Encoding, Logarithmic PCM, Differential PCM, Delta Modulation, Delta Sampling, Delta Quantization, Baseband Processing, Orthogonality in digital signal, Signal Noise, Signal-Space Analysis, Likelihood Functions, Correlation Receiver Structure, Bandpass Modulation, Amplitude Shift keying (ASK), Frequency Shift Keying (FSK), Binary Phase Shift Keying (BPSK), Quaternary Phase Shift Keying (QPSK).

Module III:

11 lecture hours

Differential Encoding, Performance of BPSK and QPSK, Multiplexing, Frequency division multiplexing, Time division multiplexing, Channel Coding, Types of Error Control, Structured Sequences, Block Codes, Linear Block Codes, Error Correction, Convolutional Encoding, Coding Modulation Trade-off, Spread

Spectrum, Direct Sequence Spread Spectrum Systems, Frequency Hopping Spread-Spectrum, Spread-Spectrum as Multiple Access, Code-Division Multiple Access, Time-Division Multiple Access.

Studio Work / Laboratory Experiments:

Hands-on performed using kit to understand the concepts of modulation, synchronization, etc.

Text Books :

1. Proakis J.G. and Masoud Salehi, *Digital Communication* (1 ed.), McGraw Hill, 2018. ISBN 978-9353163020.

Reference Books :

1. Lathi B. P., Zhi Ding, *Modern Digital and Analog Communication* (5 ed.), Oxford University Press, 2018. ISBN 978-0190686840.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET326	Soft Computing	L	T	P	C
Owning School/Department	Computer Science Engineering	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To comprehend the fuzzy logic and the concept of fuzzy set theory in soft computing.

CO2: To examine Neuro-Fuzzy and Genetic Algorithm expert system.

CO3: To implement applications on different soft computing techniques like Fuzzy, Multi Objective optimization and Genetic Algorithm (GA).

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	2											2		
CO2	3		2	1								2		3	2
CO3	3	1	3	2	3							3		2	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

7 lecture hours

Soft Computing, Key characteristics and applications of soft computing, Soft vs. Hard computing, Fuzzy logic, Fuzzy set, Crisp vs. Fuzzy Set, Fuzzy Set Properties, Operations on Fuzzy set: Union, Intersection, Complement, Sum and Difference, Equality and Power, Cartesian Product, Fuzzy IfThen Rules – Fuzzy Reasoning, Fuzzy membership functions, Gaussian membership function, Sigmoid membership function, Triangular membership function, Trapezoidal membership function, Fuzzy proposition, Fuzzy interferences, Fuzzy relations- Max-Min Approach, Fuzzification – Defuzzification, Fuzzy Logic Controller, Neuro-Fuzzy modelling.

Modulo II:

7 lecture hours

Multi Objective Optimization, Multi-Objective Evolutionary Algorithm, Pareto based Approach, Non-Pareto based approaches, Genetic Algorithm (GA), GA working architecture, Genetic representations, GA Encoding and Selection Techniques, Survival of the Fittest, Fitness Computations, GA Crossover Techniques, GA Mutation, Reproduction, Rank method, Rank space method, GA Case Studies: Optimisation of traveling salesman problem using Genetic Algorithm, Genetic algorithm-based Internet Search Techniques.

Module III:

11 lecture hours

Evolutionary Algorithm, Ant system, Ant Colony Optimization, Max-Min Ant System, Ant Miner, Snake-Ant Algorithm, Particle Swarm Optimization, Artificial Bee Colony, Cuckoo Search Algorithm, Working

architecture, Co-evolution, Plasticity and life-time learning, Lamarckian learning, “No free lunch” theorem, Hybrid fuzzy controller, Fuzzy Logic Controlled Genetic Algorithms.

Module IV:

6 lecture hours

Genetic Algorithms–Neural Networks, Neural Networks Fuzzy Logic, Extreme Learning Machine, Training SLFN using ELM, Extreme Learning Machine, Variants of ELM, Applications of ELM, Extended ELM.

Studio Work / Laboratory Experiments:

1. Fuzzy Logic, Multi Objective Optimization, GA, and hybrid fuzzy controller, implementation on MATLAB/PYTHON.

Text Books :

1. Kaushik Saroj and Sunita Tiwari, *Soft Computing, Fundamentals, Techniques and Applications* (1 ed.), McGraw-Hill Education, 2018. ISBN 978-9353160678.
2. Sivanandam, S. N., and Deepa, S. N, *Principles of soft computing* (1 ed.), John Wiley & Sons, 2011. ISBN 978-8126527410.

Reference Books :

1. Buontemp F., *Genetic Algorithms and Machine Learning for Programmers* (1 ed.), Pragmatic Bookshelf, 2019. ISBN 978-1680506204.
2. Gridin I., *Learning Genetic Algorithms with Python* (1 ed.), BPB Publications, 2021. ISBN 978-8194837756.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET327	Embedded Systems	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain the functional difference between general computing systems and Embedded systems.

CO2: To articulate the architecture of the ATOM and ARM processor and its programming aspects (assembly level).

CO3: To examine partition among HW and SW a high-level functionality with emphasis on the interaction with the network and operating system. Model, design and verify a complex digital device.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3												2		
CO2	3				2							2		3	
CO3	2	1		2	3										1

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

Components of Embedded systems, Application areas of Embedded System, Characteristics of an Embedded System, Hardware and Software building blocks of Embedded Systems, Core of Embedded system, Embedded system vs general Computing system, Architecture of an Embedded System, Embedded Product Development Life Cycle (EDLC), Analysis, Design , and Implementation of embedded system, Pros and cons of an Embedded System, Classification of Embedded systems Based on Performance and Functional Requirements, IA Architecture and features, ATOM processor and its addressing modes, Advanced RISC Machines (ARM) processor, Registers and Memory accesses, Memory map and Instruction set, Segmentation and Task switching, Paging, Hyper-threading, Caches, and TLB, Execution pipeline, Optimization in Embedded Systems.

Modulo II:

9 lecture hours

Case Study: Multimedia handsets and DVB-MHP Set top Boxes, Middleware for Chip Integration, System Middleware for Runtime Environment Abstraction, Firmware for Embedded Systems, Embedded Computing Software Environment, Interrupts, VT. FSB Architecture, Chipset overview and function, Steps before Boot sequence, BOOT up sequence, Embedded Computing Hardware Architecture, Platform-based Design Concept, FPGA-based System Design.

Module III:**11 lecture hours**

Embedded Processor Cores, Hardware and Software Interfaces, HW-SW Interface Design, Bus Protocols (I2C, AMBA, Core Connect, and Wishbone), Hardware-Software Co-Design, System Models and Formalisms, System Co-design Flow, Verification and Debugging of a Co-Designed System, Embedded systems using Atom processors, Complex Embedded Systems and Embedded Multimedia System Co-Design.

Studio Work / Laboratory Experiments:

In lab students will implement program in various embedded system then compare and Identify the best architecture of an embedded system. Moreover, student will perform partitioning among S/W and H/W a high-level functionality

Text Books :

1. Shibu K. V., *Introduction to Embedded Systems* (2 ed.), Tata McGraw, 2009. ISBN 978-9339219680.
2. Kamal Raj, *Embedded Systems Architecture, Programming, and Design* (2 ed.), Tata McGraw Hill, 2008. ISBN 978-9332901490.

Reference Books :

1. Barry Peter and Patric Crowley, *Intel architecture for Embedded system* (1 ed.), Morgan Kaufmann, 2012. ISBN 978-0123914903.
2. Wolf Wayne, *Computers as Components: Principles of Embedded Computer System Design* (4 ed.), Elsevier, 2020. ISBN 978-0128053874.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET328	Distributed Computing	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate the models, architectures, and the concept of virtual clock and clock synchronization.

CO2: To examine the algorithms for mutual exclusion, deadlock detections, and termination detection.

CO3: To implementing the concepts of distributed computing on Google File System , Hadoop Distributed File System (HDFS), and sensor networks.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3												1	1	1
CO2		3										2	1	2	3
CO3		2	3	3	2	1							3	2	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

8 lecture hours

Introduction, process communication, Message Passing, Leader Election, Leader election algorithm, Distributed Models, Causality and Logical Time, Size of Vector Clock, Matrix Clocks, Virtual Time and Physical Clock Synchronization, Global State and Snapshot Recording Algorithms, Distributed Mutual Exclusion and Non-Token based Approaches, Quorum Based Distributed Mutual Exclusion Approaches: Maekawa's Algorithm.

Modulo II:

7 lecture hours

Distributed Mutual Exclusion-Token based approaches, Consensus & Agreement, Checkpointing & Rollback Recovery, Deadlock Detection in Distributed Systems, Approaches, Algorithms for deadlock detection: Path-Pushing, Edge Chasing, Diffusion Computation, and Global state detection Distributed Shared Memory, Features and advantage, Distributed Minimum Spanning Tree.

Module III:

7 lecture hours

Termination Detection, Huang's algorithm, Message Ordering & Group Communication, Fault Tolerance and Self-Stabilization, Distributed Randomized Algorithms, Distributed Hash Tables and Peer to Peer Computing.

Module IV:**13 lecture hours**

Case Studies: Google File System and HDFS, Distributed Execution using Map Reduce, Introduction to Spark, Introduction to Sensor Networks, Distributed Algorithms for Sensor Networks: Coverage and Connectivity, Topology Discovery, LEACH – Cluster based Low Power Algorithm, Authentication in Distributed Systems, Security in Distributed Systems and Block Chain.

Text Books :

1. Maarten Van Steen and Tanenbaum Andrew S., *Distributed Systems* (3 ed.), Amazon Digital Services, 2017. ISBN 978-1543057386.

Reference Books :

1. Aspnes James, *Notes on Theory of Distributed Systems*, Yale University, 2021.<http://www.cs.yale.edu/homes/aspnes/classes/465/notes.pdf>.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET329	Engineering Optimization	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate, Identify and develop operational research models from the verbal description of the real system.

CO2: To examine the mathematical tools & software that are needed to solve optimization problems.

CO3: To analyze the results to resolve resource optimization.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	1	1	1	1	2	1						3	2	3
CO2	3	2	3	2	2		3						1	1	2
CO3	1	1	1	3	2									2	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

7 lecture hours

Need of Optimization, Development of Optimization, Engineering Applications of Optimization, Design Vector, Design Constraints and Constraint Surface, Objective Function, Objective Function Surfaces, Stationary Points: Functions of Single and Two variables, Convexity and Concavity of Functions of One and Two Variables, Optimization of Functions of Single and Two Variables, Multiple Variables Subject to Equality Constraints, Lagrangian Function, Hessian Matrix Formulation, Eigen Values Kuhn-Tucker Conditions.

Modulo II:

11 lecture hours

Linear Programming (LP) Problem, Canonical Form of LP Problem, Assumptions in LP Models, Elementary Operations, Graphical Method for Two Variable Optimization Problem, Branch-and-Bound Method, Sequential Linear Discrete Programming, Generalized Penalty Function Method, Geometry and Formulation of LPP, Graphical Solution, Duality Theory, Duality in LP, Primal- dual Relations, Dual Simplex Method, Post Optimality Analysis, Karmarkar's Projective Scaling Method, Network Models, Shortest-Route Problem, Maximal Flow Model, CPM and PERT, Transportation Model, Nontraditional Transportation Models, Transportation Algorithm, Hungarian Method, Probability Theory, Random Variables and Probability Density Functions, Stochastic Linear Programming, Stochastic Nonlinear

Programming, Objective Function and Constraints, Stochastic Geometric Programming, Heuristic Programming, Local Search Heuristics, Tabu Search Algorithm, Simulated Annealing Algorithm.

Module III:

10 lecture hours

Genetic Algorithms, Differential Evolution, Particle Swarm Optimization, Ant Colony Optimization, Crow Search Algorithm, Firefly Optimization Algorithm, Harmony Search Algorithm, Teaching-Learning-Based Optimization, Honey Bee Swarm Optimization Algorithm, Reduced Basis Technique, Design Variable Linking Technique, Incremental Response Approach, Basis Vector, Derivatives of Static Displacements and Stresses, Multi-objective Optimization, Parallel Processing, Utility Function Method, Inverted Utility Function Method, Global Criterion Method, Bounded Objective Function Method, Lexicographic Method, Goal Programming Method, Goal Attainment Method, Game Theory Approach, Inventory Problem: A Supply Chain Perspective, Static Economic-Order-Quantity Models, Dynamic EOQ Models, Sticky Issues in Inventory Modeling.

Studio Work / Laboratory Experiments:

Lab will be conducted on the MATLAB, TORA and LINGO.

Text Books :

1. Singiresu S. Rao, *Engineering Optimization Theory and Practice* (5 ed.), John Wiley & Sons, 2019. ISBN 978-1119454717.1

Reference Books :

1. Taha H.A., *Operations Research: An Introduction* (10 ed.), Pearson Global Edition, 2017. ISBN 978-0134444017.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET330	Wireless Networks	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain where the wireless technologies can be applicable and evolving wireless technologies and standards.

CO2: To articulate the architecture of various wireless technologies.

CO3: To experiment with various wireless technologies through hands-on projects.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	2	1	1	2			3	1	2		2	1	2	2
CO2	1	2	2	1	3	1	1	2		2	1	2	1	1	3
CO3	2	1	2	2	1			2		1		1	1	2	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

9 lecture hours

Wireless technologies: Cellular systems, satellite systems, broadcast systems, wireless LANs. Propagation modes, Fading in the Mobile Environment, Characteristics of wireless transmission (signals, antennas, multiplexing, modulation, interference). Signal Encoding Criteria, Digital Data, Analog Signals, Analog Data, Analog Signals, Analog Data, Digital Signals. Frequency Hopping Spread Spectrum, Direct Sequence Spread Spectrum, Code Division Multiple Access, Generation of Spreading Sequences.

Modulo II:

10 lecture hours

Recent trends in Wireless Networking Technology, Layered architecture for Wireless communication, Wireless TCP, Physical and Logical Architecture. Radio and Infrared Communication. Satellite Communications, 802.11 WLAN Standards, 802.11 MAC Layer, 802.11 PHY Layer, 802.11 Enhancements, Other WLAN Standards like 802.11g, 802.11n and 802.11p. Wireless sensor network platforms, Communication architecture and protocols, Energy Management, Area Optimization, Time Synchronization, Duty Cycle Optimization, Sensor Data Acquisition, Processing and Handling.

Module III:

9 lecture hours

Wireless Personal Area Network Standards, Bluetooth (IEEE 802.15.1), Wireless USB, ZigBee (IEEE 802.15.4), Infrared Data Association (IrDA), Gigabit Wireless LANs, 5G Technology: Small Cells, Massive MIMO, mmWave, Intrusion Detection System for WSN, Anomaly and misuse detection, Evolution of cellular systems, 6G Technologies, Starlink wireless, 7G Technology: Intelligent Communication of the Inevitable.

Studio Work / Laboratory Experiments:

Simulation a wireless sensor network, creating an Ad-hoc Network, illustration of security threats in wireless networks and handling them. Cisco Aironet AP antenna configuration, Configuration of a site-to-site wireless link. Configuring LEAP/EAP using Local RADIUS Authentication for secure client authentication.

Text Books :

1. Beard C and Stallings W., *Wireless Communication Networks and Systems* (1 ed.), Pearson, 2015. ISBN 978-1292108728.
2. Gordon Colbach, *Wireless Networking: Introduction to Bluetooth and WiFi* (1 ed.), Independently Published, 2017. ISBN 978-1973252115.

Reference Books :

1. Velrajan Saro, *An Introduction to 5G Wireless Networks* (1 ed.), Amazon Digital Services LLC - KDP Print US, 2020. ISBN 979-8643303107.
2. Vannithamby R. and S. Talwar, *Towards 5G: Applications, Requirements and Candidate Technologies* (1 ed.), John Willey & Sons, 2017. ISBN 978-1118979839.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET331	Combinatorics	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain knowledge of combinatorics, numbers, and probability theory.

CO2: To articulate combinatorial problems, extract and interpret descriptive statistics from social networks, search engine and hypergraphs.

CO3: To implement and design various quantitative properties of large combinatorial structure.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	2	2	2	1	1		1		1		1	1	1	2
CO2	3	2	2	1	3	2			2	2	2	2	2	3	3
CO3	2	1	2	1	1	2						3	2	2	2

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

8 lecture hours

Combinatorics, Counting and product principle, Counting over counting and sum principle, Function and bijection principle, Relations and equivalence principle, Pigeonhole principle, Subsets, Partitions, Subsets of fixed size, Subset of multiset, Subset of combination, Binomial theorem and Pascal's triangle, Binomial Coefficients, Congruences, Congruence of binomial coefficients, Permutations, One-line notation, Two-line notation, Canonical cycle notation, Estimates for factorials, Stirling's Approximation, Ramanujan's factorial approximation, Selections, Equivalence and order, Finite topologies, 0 or 1 points, 2 points.

Modulo II:

7 lecture hours

Compactness and countability, Connectivity, Separation axioms, Cayley's Theorem on trees, Algebraic topology, Generating Combinatorial Objects, Generating Subsets, Variable Size Decrease Algorithms, PageRank (PgRk) algorithm for searching engine, Kernel Networks for pattern analysis, Partially ordered sets or Posets, Graded poset, Lattices, Metroid, Linear extensions of posets, Distributive lattices, Propositional logic, Chains and antichains, Products and dimensions.

Module III:

7 lecture hours

Mobius function of poset, Famous Number Families, Multinomial Coefficients, Fibonacci Numbers, Lucas Numbers, Stirling Numbers, Integer Partition Numbers, Bell numbers, Recurrence Relation and Generating Functions, First order recurrence relation, Second order recurrence relation, Non-homogeneous recurrence relation.

Module IV:

6 lecture hours

Combinatorics on graphs, Infinite Combinatorics and Graphs, Counting trees, Minimal spanning Trees, Chromatic polynomial, Manifold method for non-linear dimensionality reduction, Spectral methods to solve differential equations, Turan Problem, Littlewood-Offord Problem, Catalans Numbers, Isoperimetric Problems.

Studio Work / Laboratory Experiments:

To implement discrete mathematics models- tree, graph, optimization problems, with several Combinatorics models such as The Analytic Hierarchy Process (AHP), PageRank (PgRk), Kernel Networks.

Text Books :

1. Bona M Miklos, *A Walk-Through Combinatorics: An Introduction to Enumeration and Graph Theory*(4th ed.), WSPC, 2016. ISBN 978-9813148840.

Reference Books :

1. Mladenović Pavle, *Combinatorics: A Problem-Based Approach* (1st ed.), Springer, 2019. ISBN 978-3030008307

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET332	Mobile and Networked Embedded Systems	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate the concepts of mobile and networked embedded systems.

CO2: To explain the architecture of networked embedded systems.

CO3: To design and develop networking systems for automated applications for smart cities, building, parking.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	2	3	1					2		2	1	1	2	3
CO2	2	1	2	1	2	2	1	3	2		2	1	3	2	3
CO3	2	2	2	2		1	1		2	1		2	2	1	2

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

12 lecture hours

Smart Environments, Paradigms for pervasive networking, Networked Embedded Systems, Wireless Embedded Networking, Applications, Network Topology, Real-time embedded systems, Components of networked embedded systems, Centralized and distributed embedded systems, Physical sensor, Passive sensor, Semi-passive, Active sensors, Soft sensors, Sensor nodes, Hardware architecture. Operating systems for sensor nodes, Mobile sensor network, Sensor networks with mobile nodes.

Modulo II:

8 lecture hours

Power management and mobile node discovery, Data transfer to mobile nodes, Routing to mobile nodes, Sensor networks with all mobile nodes, Participatory sensing, Vehicular Networked Embedded Systems, Embedded Networks for Car Domains, Intra -vehicular Network Embedded Systems, Event Triggered Systems, Time Triggered Systems, Inter-Vehicular Network Embedded Systems.

Module III:

8 lecture hours

Applications for smart cities (pollution monitoring), Applications for smart cities (smart lighting, context-aware applications), Smart mobility (parking area management), Smart mobility (intelligent transportation systems), Smart buildings (home/building automation, energy efficiency), Social sensing applications.

Studio Work / Laboratory Experiments:

In lab students will develop applications - based on smart networked embedded system - in several application domains, including smart cities, smart mobility, smart buildings, smart grid, etc.

Text Books :

1. Chattopadhyay S., *Embedded System and Design* (2nd ed.), Prentice Hall India Learning Private Limited, 2013. ISBN 978-8120347304.

Reference Books :

1. Lyla B. Das, *Embedded Systems: An Integrated Approach* (1st ed.), Pearson Education India, 2012. ISBN 978-9332511675.
2. Peckol, *Embedded Systems* (2nd ed.), Wiley, 2019. ISBN 978-1119457497.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET333	Problem Solving using C	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain various concepts in C programming language and understand the problem-solving aspect.

CO2: To Implement concept of pointer and perform I/O operations in files and make use of concepts in finding solutions to real-life problems.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3	3	2	2	1	2		2	1	1	2	1	2	1
CO2	2	2	2	1	2	1	1					2	2	3	1

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

8 lecture hours

Program structure, Communications with the Operating System, Library Functions, Pre-processor Directives, Debugging and Efficiency, Memory Models, Data Types-Simple C data types, Integer Data types, Floating Point types, Derived Data types, The ASCII Character Set, Compilation and Linking, Types of compilers, Decision making, Loops, Switch statement, Goto statement, Null statement, Comma operator, setjmp functions, longjmp functions, Storage classes: automatic variable, external, static, register, Arrays, Array Indexing, Using Array names as Pointers, Character Arrays.

Modulo II:

10 lecture hours

Advanced data types: #define statement, Variable length array, Flexible array members, Complex number type, Type Qualifiers: Const, Volatile, Restrict, Functions, Pass by value, Pass by reference, Command Line Arguments, Structures, Arrays of Structures, Structures of Arrays, Structures of Structures, Bit Fields in Structures, offsetof(), Macro, unions, Typedef, Typecasting, Implicit type casting, Explicit typecasting, Pointers- Declaration, NULL pointers, Indirection, Pointer to array, Pointer to function, Pointer to structure, Arrays of Pointers, Passing pointers to functions.

Module III:

10 lecture hours

File handling, Recursive functions, Memory allocation in a recursive method, Dynamic memory allocation, Global Memory versus Local Memory, Error handling in C, Global Variable Errno, perror() and strerror(),

Exit status, Divide by zero errors, Interfacing C with Python code: Calling C functions from Python, Calling python functions from C, Threads: Creating a thread, Passing arguments and returning values, Common thread functions, Thread synchronization concepts, Mutexes, condition variables.

Module IV:

13 lecture hours

Generic Security Software's, Windows Firewalls, Linux Firewalls, Access Control Lists of Firewalls, Types and Examples of Firewalls, Network-based Honeypots and Trapdoors, Virtual Private Network and its Implementation, Network Intrusion Detection System, Network Intrusion Prevention System, Router Security, Switch Security, Proxy Server and its Configurations, Load Balancers, IPv6 and IPv6 Security, Secure Forwarding in Overlay Networks.

Studio Work / Laboratory Experiments:

Students will implement the following programs in C language on a Linux based platform using GCC compiler.

Text Books :

1. Gottfried Byron, *Outline of Programming with C* (4th ed.), McGraw Hill Education, 2018. ISBN 978-0070145900.

Reference Books :

1. Balaguruswamy E., *Programming in ANSI* (8th ed.), McGraw Hill Education, 2018. ISBN 978-9351343202.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET334	Programming using C++	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain the fundamental programming concepts and methodologies to building C++ programs.

CO2: To implement various OOPs concepts including memory allocation/deallocation procedures and Member functions.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3	3	2	2	1	1	3	2	2	2	1	2	3	3
CO2	2	3	2	2	3	2		3					2	2	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

9 lecture hours

Principles of Object-Oriented Programming, data types, Symbolic constants, Reference by variables, Operators, Operator precedence, Control structures, If-else, Nested If, Switch, break, continue, Functions, main function, Function prototyping, Call by reference, Return by reference, Inline function, Default arguments, Function overloading, Defining a class and member functions, Private member functions, Nesting of member functions.

Module II:

8 lecture hours

Virtual base classes, Abstract classes, File Handling, Throwing Mechanism, Caching Mechanism, Rethrowing an Exception, Constructors in derived classes, Nesting of classes, Exception Handling, Opening and closing a file, Detecting End-of-file, Sequential input and output operations, Generic Programming using template, Class template, Function template, Class member function template, Function overloading, Standard Template Library: Containers, Stack, List, Queue, Algorithms, Iterators.

Module III:

11 lecture hours

Arrays within a class, Arrays of objects, Memory allocation, Static data members, Static member functions, Friendly functions, Objects as function arguments, Returning Objects, Constructors, Default constructor, Parameterized constructor, Copy constructor, Multiple constructors, Constructors with default arguments, Dynamic constructor, Destructors, Rules for overloading, Operator overloading, Unary and binary operator

overloading, Overloading using friends, Type conversion, Inheritance, Defining derived classes, Visibility modes, Single inheritance, Multilevel inheritance, Multiple inheritance, Hierarchical inheritance, Hybrid inheritance, Runtime Polymorphism

Studio Work / Laboratory Experiments:

This lab to implement various OOPs concepts including memory allocation/deallocation procedures and Member functions. This course lab is to strengthen problem solving ability by using object-oriented approach and to design applications using object-oriented concepts

Text Books :

1. Stroustrup Bjarne, *The C++ Programming Language*. (4th ed.), Addison-Wesley Professional, 2013. ISBN 978-0321563842.
2. Balagurusamy E., *Object Oriented Programming with C++* (1st ed.), Tata McGraw Education Hill, 2013. ISBN 978-1259029936.

Reference Books :

1. Ravichandran D, *Programming with C++* (3rd ed.), McGraw Hill Education, 2017. ISBN 978-0070681897.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET335	Deep Learning	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain the fundamentals of deep learning, Convolution neural network.

CO2: To articulate different problem of classification, detection, segmentation, generation and understand existing solutions/ deep learning architectures.

CO3: To implement a solution for the given problem and improve it using various methods transfer learning, hyperparameter optimization.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	2	1	2	1					2	1	2	1	1	2
CO2	2	3	2	1	2				1	2	1	1	2	3	2
CO3	2	2	2	2	2	1	2	1	2	1	1		1	2	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

7 lecture hours

Why Deep Learning?, Machine Learning: features, weights, Artificial Neural Network, loss function, cost function, ANN: forward propagation; Backpropagation, Stochastic Gradient Descent, Batch gradient descent, mini batch gradient descent, Optimizers: Momentum, RMSProp, Adam, Deep Learning Experiments: Datasets, training-validation testing set, evaluation measures: accuracy, precision, recall, f-measure, Model Improvement: Overfitting vs underfitting, Bias vs Variance, Regularization: L1, L2 regularization, Dropout, Early stopping, Data normalization, Batch normalization, Hyper parameter Tuning: random, coarse to fine, Network architecture search.

Modulo II:

11 lecture hours

Imbalance data problem (25) Data Augmentation in image: Cropping, Flipping, Rotation, Brightness, Contrast, Color Augmentation, Saturation, Convolutional Neural Networks: convolution, striding, padding, pooling, Alexnet Architecture, Image classification (ImageNet Challenge), Well known CNN architectures VGG16&19, Residual Block, Resnet50, 1x1 convolution, XceptionNet, EfficientNet, Transfer learning, Object Detection: setup problem and cost function, well known datasets, Evaluation measure: Average precision, Mean average precession, Two stage detector, single stage detector, RCNN, Fast RCNN, Faster RCNN, SSD, YOLO1-4, RetinaNet, EfficientDet, Image Segmentation: setup problem

and cost function, various dataset, Semantic segmentation, Instance segmentation, Evaluation measure: IoU/Jacard Index, Dice score, Mean pixel accuracy, Segnet, Unet, Mask R-CNN.

Module III:

10 lecture hours

Generative Learning, Variational Auto-encoders, Generative Adversarial Neural Networks, GL Applications: Image generation, font generation, video generation, anime face/celebrity face generation, Deep Reinforcement Learning, Markov decision Processing, Deep Q Learning, Exploration vs Exploitation, Value Iteration vs Policy Iteration, RL Applications: Robotics, gaming, Ad Targeting, recommendation system, decision making, Model optimization for Deployment, Pruning, Quantization and binarization, Transferred or Compact Convolutional Filters, Knowledge distillation.

Studio Work / Laboratory Experiments:

Aim of this lab is to focus on gathering, pre-processing tabular, visual, textual and audio data for building deep learning models using standard Python libraries. To train, improve, and deploy deep learning models in different devices. To analyse performance of different deep learning models using speed, accuracy, size trade-offs.

Text Books :

1. Goodfellow Ian, Yoshua Bengio, Aaron Courville and Yoshua Bengio, *Deep learning. Vol. 1 C* (1st ed.), ambridge: MIT press, 2016. ISBN 978-0262035613

Reference Books :

1. Zhang Aston, Zack C. Lipton, Mu Li and Alex J. Smola, *Dive into Deep Learning* (1st ed.), Corwin, 2019. ISBN 978-1544361376.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET336	Bioinformatics and Computational Genomics	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate knowledge of Bioinformatics, Computational Biology, and Genomics.

CO2: Implement algorithms and programs related to sequence database, search and alignments, gene prediction and homology.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3	1	1	2		2			3	1	1	2	2	1
CO2	2	2	3	3	3					3	2	2	2	1	1

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

Dna, Rna, Proteins, Splicing, Gene structure, Medicine as a Data-Driven Science, Human Genome Project, Biomedical Data, Exact Sequence Searches: Z-algorithms, Knuth-Morris, Boyer- Moore, Rabin-Karp, Sequence Analysis: Pairwise Sequence Alignment, Homology, Sequence alignment with Dynamic Programming, Extensions of Pairwise Sequence Alignment, Phylogenetic Tree Reconstruction, Biological and Molecular Databases, BLAST Search Engine, Human Variation Databases and Genome Viewers.

Modulo II:

8 lecture hours

Markov chain, Hidden Markov Model, Viterbi Algorithm, Forward Backward algorithm, HMM for motif finding, Advanced alignment techniques: Linear space, Affine gaps, Banded linear time alignments, Time warping, Burrow Wheeler Index, Next Generation Sequencing, Comparative genomics, Micro arrays, Shotgun sequencing, BAC to BAC sequencing, Phylogeny, Fitch algorithm.

Module III:

10 lecture hours

Multiple sequence alignment, MSA algorithms, Progressive alignment, CLUSTALW, Expectation Maximization, Gibbs Sampling, Genetic Algorithm, Clustering with a Genetic Algorithm, System Biology: Network Analysis, Rational Drug Design, Biomarkers, Human Genomic Variations, Monogenic Diseases, Complex Diseases, Genetic Predisposition to Cancer, Cancer Immunotherapy.

Studio Work / Laboratory Experiments:

Implementing few important algorithms in the field of Genomics: Viterbi Algorithm, BLAST Algorithm. Pattern Matching Algorithms: Rabin Karp and Knuth Morris Algorithm.

Text Books :

1. Akalin Altuna, Vedran Franke, Bora Uyar and Jonathan Ronen, *Computational Genomics with R* (1st ed.), CRC Press, 2020. ISBN 978-1498781862.
2. Baxevanis A. D., Wishart D. S., Bader G. D., *Bioinformatics* (4th ed.), Wiley, 2020. ISBN 978-1119335580.

Reference Books :

1. Singh Dev Bukhsh and Rajesh Kumar Pathak, *Bioinformatics, Methods and Applications* (1st ed.), eBook, 2021. ISBN 978-0323900058.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET338	Computational Geometry	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate the representation of geometric primitives and evaluation of geometric predicates.

CO2: To implement Motion Planning and approximation methods, related problems, and algorithms.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	3	2	1	2		2	1	1	2	2	1	2	2	3
CO2	3	3	1	1	3					2	2	1	1	3	3

1=weakly related 2= moderately related 3=strongly related

Course Contents:

Module I:

11 lecture hours

Convex Hulls: naïve Algorithm, Graham's scan, Quick Hull, Chan's algorithm, Polygon Triangulation and partitioning, Triangulations of point sets, Triangulating a monotone polygon, Ear- cutting algorithm, Art gallery problems, Gift Wrapping algorithm, Polyhedra, Fractional cascading, Priority search trees, Plane-sweep intersection, Line segment intersection, Representation and intersection of planar subdivisions, Trapezoidalization, Voronoi Diagrams, Post Office problem, Divide and conquer algorithm, History graphs, Fortune's algorithm, Connection to Convex Hulls.

Modulo II:

9 lecture hours

Delaunay refinement, Mesh refinement algorithm, Roadmaps, Linear Programming, Half-plane intersection, Arrangements, Duality, Point/line duality, Kirkpatrick's method, Point Location, Convex polygons, Intersection of convex polygons, Extreme point of convex polygon, Search and Intersection, One-dimensional range searching, KD-trees, Range trees, Higher dimensional range trees, Convex Partitioning, Zone theory.

Module III:

8 lecture hours

Motion planning: Shortest paths, Visibility graphs, Approximation Method: Dudley's theorem, Well separated partitioning, Epsilon-Nets, VC Dimension, Geometric set cover, Clustering point sets using Quadtree, Quadtrees epsilon, Construction of epsilon.

Studio Work / Laboratory Experiments:

To introduce geometric algorithms and to give exposure to algorithms and data structures for geometric problems.

Text Books :

1. Berg M. de, M. van Kreveld, M. Overmars and O. Schwarzkopf, *Computational Geometry: Algorithms and Applications (CGAA)* (3rd ed.), Springer, 2008. ISBN 978-3642096815.

Reference Books :

1. Boissonnat, Jean-Daniel, Chazal, Frédéric, Yvinec and Mariette, *Geometric and Topological Inference* (1st ed.), Cambridge University Press, 2018. ISBN 978-1108410892.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET339	Advances in AI	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain new research trends of Artificial Intelligence and interpret advanced topics in natural Language Processing and chat bots.

CO2: To implement the model and use the current trends in Artificial Intelligence.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	2	2	2	2	1	2	2	1	3	1	3	1	1	1
CO2	2	1	1	1		1			2	3	2	2	2	2	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

9 lecture hours

Bootstrapping, semantic draft, zipf distribution, Never Ending Learning (NELL), co-training, multi- task learning, coupled semi-supervised learning, macro reading vs micro reading, Open IE, lexicalized vs unlexicalized, types of semantics, distributional semantics, frame semantics, full semantic parsing, Davidsonian semantics, relation kernels, dependency parsing, combinatory categorial grammar, supervised vs. semi-supervised vs. unsupervised IE, tree kernels, distant supervision, macro vs micro reading, voted perceptron, stochastic gradient descent, multi-instance learning, self-learning, Deep learning for IE, CNNs: subgradient, CNNs: non-linearity (sigmoid, tanh, relu, leaky relu, maxout), CNNs: initialization, CNNs: residual networks, CNNs: regularization, Deep learning for IE: LDA, Bayesian non-parametrics (Chinese restaurant process), Dirichlet distribution, conjugate distribution, Reinforcement Learning: Q-learning.

Modulo II:

9 lecture hours

Knowledge Base Inference, Challenges to Inference in KBs, Horn Clause Inference: Assumptions, Horn Clause Inference: Issues, Horn Clause Inference: Examples, Random Walks Inference, Random Walks Inference: PRA, Path Ranking Algorithm, Link Prediction Task, Data-driven Path finding, Neural models for KB Inference, Matrix factorization (MF) for KB, Tensor factorization (TF) for KB, Evaluation of KB Inference, Joint MF-TF models, Graph-based semi-supervised learning, Text as interlingua, Text for inference, Temporal scoping, Open Question-Answering, Paraphrasing, Parsing, Rewriting, Executing, Latent Variable Perceptron, Semantic Parsing for Question Answering, Semantic Representation of Questions, Answer Generation.

Module III:**10 lecture hours**

Recurrent Neural Network (RNN), Bidirectional RNN, RNN problems, Vanishing/Exploding Gradient Problem, Long Short-Term Memory (LSTM), Attention – Motivation, Highway networks, Maxout networks, additive vs. multiplicative models, attention over attention, Dependency tree RNNs, weighted rank loss, deep feature fusion networks, Squaring probabilities, Synthetic oversampling, Bi-LSTMs, Siamese networks vs. LSTM., Conversation Agents, Open Domain vs. Closed Domain, Challenges in building Chatbots, Taxonomy of Models, Retrieval Based Models, Generative Models, Dialogue Systems, Applications of Dialogue Systems, Traditional Pipeline models, End-To-End models with DL, Generative Dialogue Modelling, Evaluation Metric, Memory Networks, End-to-End Memory Network with Single Hop, End-to-End Memory Network with Stacked Hops, Gated End-to-End Memory Network, End-to-End Memory Networks with Unified Weight Tying, Generative Adversarial Networks, Adversarial Nets Framework, GANs - The Good and the Bad, GANs for NLP: Dialogue systems, GANs for NLP: Dialogue systems Architecture, Teacher Forcing learning, Adversarial Evaluation, Automatic Evaluation vs Human Evaluation.

Studio Work / Laboratory Experiments:

Python experiments to understand new trends in AI.

Text Books :

1. Russel, *Artificial Intelligence: A Modern Approach* (1sted.), Pearson Education India, 2016. ISBN 9781292153964.
2. Rothman Denis, *Artificial Intelligence by Example: Develop machine intelligence from scratch using real artificial* (1st ed.), Packt Publishing, 2018. ISBN 978-1788990547.

Reference Books :

1. Elaine Rich, *Artificial Intelligence* (3rd ed.), McGraw Hill Education, 2017. ISBN 978-0070087705.

Name of Program	Bachelor of Technology				
CSET429	Network Drivers and Protocols	L	T	P	C
Owning School/Department	Computer Science Engineering	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain purpose, types and configuration of various network drivers and deliver in-depth knowledge of the network protocols.

CO2: To develop a network device driver from scratch.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	3	1	2	3	2	2	2	2	3	2	2	2	2	2
CO2		2	2	1			3	3		3		3	2	2	2

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

OSI model, Application Layer Protocols: BOOTP: Bootstrap Protocol, DCAP: Data Link Switching Client Access Protocol, DHCP: Dynamic Host Configuration Protocol, DNS, FTP, HTTP, S-HTTP: Secure Hypertext Transfer Protocol, IMAP & IMAP4, IRCP, LDAP, MIME, Secure MIME, NAT: Network Address Translation, NNTP: Network News Transfer Protocol, NTP: Network Time Protocol, POP and POP3, Rlogin: Remote Login in UNIX Systems, RMON: Remote Monitoring MIBs (RMON1 and RMON2), SLP, SMTP, SNMP (SNMPv1, SNMPv2, SNMPv3, SNTp, TELNET, TFTP, Who is (and RWhois). Presentation Layer Protocols: LPP: Lightweight Presentation Protocol. Session Layer Protocols: RPC: Remote Procedure Call Protocol.

Module II:

09 lecture hours

Transport Layer Protocols, ITOT:ISO Transport Service on top of TCP, RDP: Reliable Data Protocol, RUDP: Reliable User Datagram Protocol, Introduction to transport layer, TCP, UDP, Network Layer Routing Protocols, BGP (BGP-4): Border Gateway Protocol, EGP: Exterior Gateway Protocol, OSPF and RIP, IPv4, IPv6, Mobile IP, NARP and NHRP, RIPng: Routing Information Protocol next generation for IPv6, RSVP, VRRP, Multicasting protocols: MARS, MBGP, MOSPF, MSDP, MZAP, Multicast-Scope Zone Announcement Protocol.

Module III:

09 lecture hours

Network Security Technologies and Protocols: Kerberos, Network Drivers: Bus Infrastructure, Network hardware driver, block drivers and network drivers, Designing the kernel interface for network drivers. Network Simulator NS3, introduction and programming.

Studio Work / Laboratory Experiments:

Explore the various network protocols (HTTP, FTP, TCP, etc.) using Wireshark. Tracing using Wireshark. In-depth 'Socket programming' in Java for communication in a network. UDP socket Programming, TCP socket Programming. Development of a network device driver from scratch. Network device Driver examples contd. Mapping the I/O registers, Hardware reset and initialization. Use of NS-3 for simulating network protocols. NS-3: simulating a various wireless and wired LAN topology.

Text Books :

1. Marks, Kristin B. *Handbook of Server Management and Administration*. 1st ed. CRC Press, 2019.

Reference Books :

1. Raj Rajgopal, *Handbook of Heterogeneous Networking*. 1st ed. CRCPress, 2018

Name of Program	Bachelor of Technology				
CSET430	Digital Signal Processing	L	T	P	C
Owning School/Department	Computer Science Engineering	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To design FIR and IIR filters by hand to meet specific magnitude and phase requirements.

CO2: To perform Z and inverse Z transforms using the definitions, Tables of Standard Transforms and Properties, and Partial Fraction Expansion.

CO3: To design and implement digital filters by hand and by using MATLAB.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3	1	2	3	2	2	2	2	3	2	2	2	2	2
CO2	3	2	2	1				3		3		3	2	3	2
CO3		3	2	2	2	3	2	3		2	3		2	2	2

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

09 lecture hours

Signals and Systems, Discrete time complex exponentials signals, Elements of digital signal processing, Technology for DSP, Building blocks to represent a digital system, Types of DSP processors, Recursive and non-recursive systems, Canonic and Non-Canonic Structures, IIR Filter Realization, Direct form, Cascade realization, Parallel form realization, Ladder structures, Continued fraction expansion, Realization of a ladder structure, FIR Filter Realization, Direct, Cascade, FIR Linear Phase Realization, Infinite Impulse Response Digital (IIR) Filter Design, Low pass Filters, High pass Filters, Band pass Filters.

Module II:

11 lecture hours

Impulse Invariant Transformation, Bi-Linear Transformation, Analog Filters, Butterworth, Chebyshev, Finite Impulse Response Filter (FIR) Design, Windowing and the Rectangular Window, Gibb's phenomenon, Hamming, Hanning, Blackmann, Bartlet, Kaiser, Filter Designs Using Windows, Finite Word length effects in digital filters, Coefficient quantization error, Quantization noise – truncation and rounding, Limit cycle oscillations-dead band effects, Discrete Fourier Transform, Fast Fourier Transform, Circular Convolution, Linear Convolution using Circular Convolution, Decimation in Time (DIT) Algorithm: DFT, FFT, Decimation in Frequency (DIF) Algorithm: DFT, FFT.

Module III:

08 lecture hours

Z-Transform: Definition, Region of convergence (RoC) , Z-transform properties, Inversion methods: partial fraction expansion, Power series, Contour integral method-causality, Contour integral method: stability, Implications on pole locations.

Studio Work / Laboratory Experiments:

Laboratory work involves developing signal processing systems on a personal computer and using them with both real and simulated data. Generally, MATLAB and DSP processor are the working platform.

Text Books :

1. Lathi, Bhagwandas Pannalal, and Roger A. Green. *Essentials of digital signal processing*. 1st ed. Cambridge University Press, 2014. ISBN 978-1107444454.
2. Salivahanan, S. *Digital signal processing*. 4th ed. McGraw-Hill Education, 2020. ISBN 978-9353167431.

Reference Books :

1. Holton, Thomas. *Digital signal processing: Principles and applications*. 1st ed, Cambridge University Press, 2021. ISBN 978-1108290050.

Name of Program	Bachelor of Technology				
CSET431	Special Topics in Computer Science	L	T	P	C
Owning School/Department	Computer Science Engineering	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1 : To familiarize and learn about the latest trends and research in the field.

CO2 : To equip themselves with the conceptual and practical experience of few latest methods, tools, technologies or algorithms in Computer Science.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3	1	2	3	2			2	3	2	1	2	2	2
CO2		3	2	1			2	3		3		3	2	2	2

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

28 lecture hours

This course covers the cutting-edge topics in Computer Science, and these modules will be chosen by the instructor based on the requirements and relevance at that point of time. These modules need to be relevant to the industry and start-ups will also include related case studies, use cases and implementations scenarios. Students will be working on lab work and projects to get real hands-on experience of these topics and modules.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to Computer Science.

Name of Program	Bachelor of Technology				
CSET432	Advanced Microprocessor	L	T	P	C
Owning School/Department	Computer Science Engineering	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1 : To articulate the functionality of RISC architecture to design advanced microprocessors systems.

CO2 : To make of use of new interface techniques principle to access the peripherals.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3	1	2	3	2	2	2	2	3	2	2	2	2	2
CO2	3		2	1			2	3		3		3	2	2	2
CO3		1		2	2	3	2			2	1	2	2	2	2

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

Von Neuman versus Harvard architecture, RISC and CISC concepts, Advanced Microprocessors, Advanced Microprocessors core Architecture, ARM bus technology, Advanced Microcontroller Bus Architecture, Data flow model, Programmer's model, 64-bit general purpose register set and input/output, Current program and save program status register, 64 bit Addressing modes, Exceptions, Interrupts, Vector Table, Thumb Instruction set.

Module II:

11 lecture hours

Parallel processors, Multi-core processors, Supercomputers, Pipelining, Parallel processing concepts, Parallelism algorithm for multiprocessor systems, Super Scalar, Super-pipeline, Multiprocessing, Very Long Instruction Word Architectures, Low-power Architecture, Co-processor 15, Flynn's Taxonomy, Single Instruction Multiple Data (SIMD), Multiple Instruction Multiple Data (MIMD), Vector Processors, Vector Floating Point Processor, Example of vector operation, Advanced Instruction flow techniques.

Module III:

07 lecture hours

Cache memory and Virtual Memory, Coprocessor 15 and caches, Flushing, Cleaning, Lockdown, Memory mapping and management support, MemoryProtection Unit, Tightly Coupled Memory (TCM), Multi-User concept in 64-bit microprocessors, Multitasking, Multi-threading in 64-bit microprocessors.

Studio Work / Laboratory Experiments:

In lab students will perform experiments in assembly language. Students will implement code for RISC and CISC architecture, they also write code to perform parallelism in processor architecture pipelining, super-scalar, super-pipeline. They can create application specific processor from scratch as project.

Text Books :

1. Tahir, Muhammad, and Kashif Javed. *ARM microprocessor systems: cortex-M architecture, programming, and interfacing*. 1st ed. CRC Press/Taylor and Francis, 2017.
2. Kusswurm, Daniel. *Modern Arm Assembly Language Programming: Covers Armv8-A 32-bit, 64-bit, and SIMD*. 1st ed. Apress, 2020.

Reference Books :

1. Hall, Douglas V. *MICROPROCESSORS AND INTERFACING*. 3rd ed. McGraw Hill Education, 2017.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET403	Emerging Topics in Mobile Technology	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1. At the end of the course, students will be able to identify, understand and apply emerging concepts and technologies related to mobile technology and app development using Swift.

CO2. Students will be able to develop and prototype mobile apps using Swift, evaluate the potential of new mobile technologies, and synthesize new solutions for mobile applications.

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PSO 1	PSO2	PSO3
CO1	3	2			3				2		3		
CO2	2	3		3						3	2		
CO3	3	3	3	2	3				2				2

1=weakly related

2= moderately related

3=strongly related

Module 1

lecture hours: 28

The Emerging Topics in Mobile Technology using Swift module will introduce learners to the ever-evolving world of mobile technology. The module will cover the basics of developing applications for mobile devices using the Swift programming language. Learners will be exposed to key concepts such as creating user interfaces, dealing with the device hardware, connecting to the cloud, understanding mobile security, and creating mobile apps for different platforms. The module will include both lectures and hands-on labs to give learners the opportunity to practice their skills. By the end of this module, learners will have a solid understanding of the basics of mobile development and be able to apply their knowledge to their own projects

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to Mobile Technology.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET404	Emerging Topics in Product Design Technology	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1. Develop an understanding of the current trends and issues in product design technology, including digital fabrication, rapid prototyping, design for sustainability, and additive manufacturing.

CO2. Analyze how product design technology can be used to create innovative products that meet consumer needs and create a design that incorporates these concepts.

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PSO 1	PSO2	PSO3
CO1	3	2			3				2		3		
CO2	2	3		3						3	2		
CO3	3	3	3	2	3				2				2

1=weakly related

2= moderately related

3=strongly related

Module 1

lecture hours: 28

This course will explore emerging topics in product design technology. Topics covered include the use of digital prototyping, 3D printing, computer-aided design (CAD), rapid prototyping, and other advanced technologies used in product design. The class will also examine the application of these technologies to various industries, such as consumer goods, medical devices, automotive, and aerospace. In addition, students will be introduced to topics such as sustainability, product lifecycle management, and market analysis. Finally, the course will cover the tools and techniques used to create innovative products, as well as the process for commercializing them.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to Product Design Technology.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET405	Emerging Topics in Augmented & Virtual Reality	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Understand the fundamentals of augmented and virtual reality and its applications in various industries.

CO2: Explore the current trends and developments in augmented and virtual reality technology and its implications for businesses.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PSO 1	PSO2	PSO3
CO1	3	2			3				2		3		
CO2	2	3		3						3	2		
CO3	3	3	3	2	3				2				2

1=weakly related

2= moderately related

3=strongly related

Module 1

lecture hours: 28

Emerging Topics in Augmented and Virtual Reality provides a comprehensive overview of the cutting-edge technologies, innovations, and applications of augmented and virtual reality. The course will cover the technical aspects of virtual reality, its history and development, and the contemporary applications and challenges associated with the technology. The course will also explore emerging trends and topics such as haptics, volumetric capture, gesture interfaces, and spatial computing. This course will equip students with an understanding of the potential of augmented and virtual reality, and the challenges associated with the technology.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to Augmented Reality and Virtual Reality.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET406	Emerging Topics in IoT and Robotics	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1. Gain a comprehensive understanding of the fundamentals of IoT and Robotics, including the potential applications and implications of these technologies in the world today.

CO2. Develop the ability to analyze and evaluate current and emerging trends in IoT and Robotics, and to develop strategies to deploy these technologies in a business context.

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PSO 1	PSO2	PSO3
CO1	3	2			3				2		3		
CO2	2	3		3						3	2		
CO3	3	3	3	2	3				2				2

1=weakly related

2= moderately related

3=strongly related

Module 1

lecture hours: 28

This course will provide an in-depth look at current and emerging topics in IoT and Robotics. We will explore the technological advancements in this field, such as artificial intelligence, machine learning, and deep learning. We will also discuss the ethical implications of these technologies, as well as the challenges and opportunities they present. Additionally, students will gain practical skills in programming, interfacing, and integrating IoT/Robotics components and systems. By the end of the course, students will be well-versed in the fundamentals of IoT and Robotics and have the skills needed to create and implement projects and applications in this field.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to IoT and Robotics.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET407	Emerging Topics in Quantum Computing	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1. Develop an understanding of emerging topics in quantum computing, including entanglement, superposition, and quantum teleportation.

CO2. Gain knowledge of the various applications of quantum computing, such as cryptography, artificial intelligence, and quantum simulation.

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PSO 1	PSO2	PSO3
CO1	3	2			3				2		3		
CO2	2	3		3						3	2		
CO3	3	3	3	2	3				2				2

1=weakly related

2= moderately related

3=strongly related

Module 1

lecture hours: 28

This course will provide an overview of emerging topics in quantum computing, such as quantum algorithms, quantum error correction, quantum simulation, and quantum networks. We will discuss the potential applications of quantum computing and explore the latest research findings in this field. The course will also provide an overview of the theoretical framework underpinning quantum computing and discuss the practical considerations associated with quantum computing hardware and software.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to Quantum Computing.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET408	Emerging Topics in Drone Technology	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1. Develop an understanding of the various emerging applications of drone technology in different sectors, such as military, energy, surveillance, delivery, and more.

CO2. Analyses the advantages and disadvantages of using drones in different contexts and evaluate the legal, ethical, and safety implications of using drone technology.

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PSO 1	PSO2	PSO3
CO1	3	2			3				2		3		
CO2	2	3		3						3	2		
CO3	3	3	3	2	3				2				2

1=weakly related

2= moderately related

3=strongly related

Module 1

lecture hours: 28

Emerging topics in drone technology will include the use of drones in agricultural production and the development of autonomous drones. We will explore topics such as the use of drones in precision agriculture and the development of new drone technologies, such as autonomous drones. Additionally, we will look at the impact of drone technology on economic and social development, as well as the regulatory framework in which drones operate. We will also discuss how drones can be used in various other industries, such as delivery services and media production. Finally, we will explore the ethical implications of drone technology and the implications of its use on global security.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to Drone Technology.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET409	Emerging Topics in Cloud Computing	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1. Students will be able to analyses emerging trends in cloud computing, such as serverless computing, containers, and edge computing.

CO2. Students will be able to evaluate the benefits and challenges of the various cloud computing models and be able to recommend the most appropriate cloud model for a specific application.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PSO 1	PSO2	PSO3
CO1	3	2			3				2		3		
CO2	2	3		3						3	2		
CO3	3	3	3	2	3				2				2

1=weakly related 2= moderately related 3=strongly related

Module 1

lecture hours: 28

Cloud computing is a rapidly evolving field that is constantly introducing new technologies, services, and applications. Emerging topics in cloud computing include edge computing, serverless computing, blockchain, containers, and artificial intelligence. Edge computing is a distributed computing model that brings compute, storage, and networking closer to the data source. Serverless computing is a model of computing that allows developers to deploy code without worrying about provisioning and managing servers. Blockchain is a distributed ledger technology that is being used to create decentralized applications and services. Containers are an isolated, lightweight, and portable environment for running applications in the cloud. Artificial intelligence and machine learning are increasingly being used to automate tasks and provide insights from data. These emerging topics in cloud computing are transforming the way businesses operate and provide new opportunities for developers to create innovative applications and services.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to Cloud Computing

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET410	Emerging Topics in Full Stack	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1. Understand the fundamentals of emerging technologies in Full Stack development, such as serverless computing, microservices, and containerization.

CO2. Develop the skills to deploy applications and services using the most up-to-date tools and techniques for developing robust Full Stack solutions.

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PSO 1	PSO2	PSO3
CO1	3	2			3				2		3		
CO2	2	3		3						3	2		
CO3	3	3	3	2	3				2				2

1=weakly related

2= moderately related

3=strongly related

Module 1

lecture hours: 28

Full stack web development is a broad and rapidly evolving field. As technology advances, new topics are emerging that are becoming increasingly important to full stack developers. These topics include machine learning, artificial intelligence, blockchain, cloud computing, data science, and mobile app development. Machine learning and artificial intelligence are becoming increasingly important for creating applications that can respond to user input and learn from user interactions. Blockchain is a secure and distributed ledger technology that has the potential to revolutionize the way data is stored and used. Cloud computing offers developers a way to store and access data in the cloud, as well as create applications that can be used across multiple devices and platforms. Data science is a method for extracting insights from large datasets and is becoming increasingly important for creating predictive applications. Finally, mobile app development is a key aspect of full stack development, and developers need to be able to create applications that are optimized for different devices.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to Full Stack

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET411	Emerging Topics in DevOps	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1. Develop an understanding of the core principles, practices, and technologies of modern DevOps, including containerization, automation, and cloud-native solutions.

CO2. Identify and implement best practices for aligning DevOps processes and tools with organizational objectives and desired outcomes.

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PSO 1	PSO2	PSO3
CO1	3	2			3				2		3		
CO2	2	3		3						3	2		
CO3	3	3	3	2	3				2				2

1=weakly related

2= moderately related

3=strongly related

Module 1

lecture hours: 28

DevOps is a rapidly growing practice that is changing the way organizations develop, deploy, and monitor software applications. It is a combination of tools, processes, and people that enables organizations to rapidly develop, test, deploy, and monitor applications in a more efficient and cost-effective manner. Emerging topics in DevOps include containerization and microservices, infrastructure automation, distributed computing, cloud computing, and DevOps Security. Containerization and microservices allow for increased scalability, flexibility, and portability, while infrastructure automation reduces the need for manual configuration and accelerates the deployment of applications. Distributed computing enables organizations to make use of multiple nodes for increased scalability and performance, and cloud computing enables organizations to take advantage of the scalability and cost benefits of cloud computing. DevOps Security is a critical element of DevOps that ensures applications are secure and compliant with regulatory requirements. With the increasing adoption of DevOps, organizations are quickly realizing the benefits of improved agility, scalability, cost savings, and security.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to DevOps

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET412	Emerging Topics in RPA	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1. Demonstrate a comprehensive understanding of the principles and potential applications of Robotic Process Automation (RPA).

CO2. Evaluate the most cutting-edge developments and trends in RPA and identify opportunities to apply this technology to improve business processes.

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PSO 1	PSO2	PSO3
CO1	3	2			3				2		3		
CO2	2	3		3						3	2		
CO3	3	3	3	2	3				2				2

1=weakly related

2= moderately related

3=strongly related

Module 1

lecture hours: 28

Robotic Process Automation (RPA) is an emerging field of technology that uses software robots to automate business processes. RPA can be used to automate mundane tasks and complex processes, allowing businesses to reduce costs, increase efficiency, and improve customer service. RPA is being used in a variety of industries, such as finance, healthcare, retail, and manufacturing, to streamline operations and reduce manual labor. Emerging topics in RPA include natural language processing, cognitive computing, and machine learning. Natural language processing allows robots to understand and respond to commands given in natural language, while cognitive computing enables robots to think like humans and make decisions based on data. Machine learning allows robots to learn from their experiences and apply that knowledge to new situations. With these advances, RPA is quickly becoming an essential tool for businesses to gain a competitive edge.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to Robotic Process Automation

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET414	Emerging Topics in Data Science	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1. Develop an understanding of the current trends, tools, and techniques applicable to data science in order to apply them to solve real-world business problems.

CO2. Explore and analyze large datasets using advanced analytics and machine learning techniques to gain insights and develop effective data-driven solutions.

CO-PO/PSO Mapping:

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PSO 1	PSO2	PSO3
CO1	3	2			3				2		3		
CO2	2	3		3						3	2		
CO3	3	3	3	2	3				2				2

1=weakly related 2= moderately related 3=strongly related

Module 1

lecture hours: 28

Data Science is a rapidly growing field with new and exciting topics emerging all the time. Some of the most interesting and emerging topics in Data Science include data visualization, machine learning, natural language processing, data mining, deep learning, big data analytics, and predictive analytics. Data visualization is the process of creating visual representations of data, such as graphs, charts, and maps, to make it easier to understand and interpret. Machine learning is the process of using algorithms to identify patterns and trends in data. Natural language processing makes it possible for machines to understand and interpret human language. Data mining is the process of extracting information from large datasets. Deep learning is a subset of machine learning that is used to create models for more accurate predictions and analysis. Big data analytics is the process of extracting, transforming, and loading large datasets for analysis and decision making. Predictive analytics is the process of using data to make predictions about future events. These topics and more are all part of the ever-growing field of Data Science.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to Data Science

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET415	Emerging Topics in Cyber Security	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1. Develop an understanding of the emerging threats, technologies, and strategies for defending and protecting an organization's cyber security.

CO2. Analyse and evaluate the effectiveness of current cyber security protocols and systems and develop best practices for mitigating and preventing new threats.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PSO 1	PSO2	PSO3
CO1	3	2			3				2		3		
CO2	2	3		3						3	2		
CO3	3	3	3	2	3				2				2

1=weakly related 2= moderately related 3=strongly related

Module I

28 lecture hours

Cyber security is an ever-evolving field as new technologies and threats continue to emerge. In the past few years, there have been several emerging topics in cyber security that have become increasingly important, including cloud security, artificial intelligence, the Internet of Things (IoT), mobile security, blockchain technology, and quantum computing. Cloud security is the practice of protecting cloud-based systems, services, and data from malicious actors. Artificial intelligence is being used to detect and prevent cyber-attacks in real-time, as well as to automate manual security tasks. The Internet of Things (IoT) is a rapidly growing network of connected devices, all of which need to be secured. Mobile security is the practice of protecting mobile devices, including phones, tablets, and laptops, from threats. Blockchain technology is a distributed, secure ledger that is being used to store and share data and transactions, increasing security. Lastly, quantum computing is a new technology that has the potential to revolutionize computing power, with both positive and negative implications for cyber security. All of these technologies and topics must be addressed in order to ensure the security of systems, data, and users.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to Cyber Security

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET416	Emerging Topics in Blockchain	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1. Develop an understanding of emerging trends and technologies in Blockchain and how they are being used to solve business problems.

CO2. Identify potential use cases and applications of Blockchain-based solutions and evaluate the associated risks and benefits of implementation.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PSO 1	PSO2	PSO3
CO1	3	2			3				2		3		
CO2	2	3		3						3	2		
CO3	3	3	3	2	3				2				2

1=weakly related 2= moderately related 3=strongly related

Module I

28 lecture hours

The syllabus on emerging topics in Blockchain will cover the most recent developments in the technology, including topics like distributed ledger technology, decentralized applications, cryptocurrency and smart contracts. We will discuss the different applications of blockchain technology, the benefits and challenges of using this technology, and the various ways it can be implemented. We will also look at various case studies of companies that have successfully implemented blockchain technology, as well as the potential implications for businesses and other organizations. Finally, we will also explore the potential for blockchain to be used to create new forms of digital money, and the impact this could have on the global economy.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to Blockchain.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET417	Emerging Topics in Artificial Intelligence	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Develop an understanding of the current trends and challenges of emerging Artificial Intelligence (AI) technologies.

CO2: Identify and analyse potential applications of AI in the context of real-world problems.

CO-PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO1	3	2			3				2		3		
CO2	2	3		3						3	2		
CO3	3	3	3	2	3				2				2

1=weakly related

2= moderately related

3=strongly related

Module I

42 lecture hours: 42

The Emerging Topics in Artificial Intelligence course will provide an overview of the current state of Artificial Intelligence (AI) research. Students will learn about the major topics in AI such as machine learning, natural language processing, computer vision, robotics, and reinforcement learning. Additionally, students will be exposed to the latest applications of AI in various domains such as healthcare, finance, and autonomous vehicles. They will learn how to apply AI algorithms and explore the ethical implications of AI. Finally, students will get an introduction to the potential future of AI and the implications of these technologies on society.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies related to Artificial Intelligence.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET419	Introduction to Generative Artificial Intelligence	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Understand the foundational principles and concepts of generative artificial intelligence, including the underlying algorithms and techniques used in creative AI systems.

CO2: Apply the basic generative algorithms for creative tasks, such as image generation, text generation, and artistic expression.

CO3: Implement generative AI techniques in real-world scenarios through practical implementation to produce effective solutions.

CO-PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO1													
CO2													
CO3													

1=weakly related 2= moderately related 3=strongly related

Course Contents:

Module I

8 lecture hours

Overview of Artificial Intelligence and its subfields, Introduction to Generative AI and its applications, Understanding creative algorithms in the context of generative systems, Introduction to machine learning and its role in generative models, Types of generative models: GANs, VAEs, and more, Exploring the principles of creative algorithms, Introduction to Large Language Models, Case studies of successful applications in various domains, designing simple creative algorithms

Module II

8 lecture hours

In-depth study of GAN architecture and components, Training and fine-tuning GANs for specific tasks, Implementing a GAN for image generation, Understanding VAE architecture and working principles, Exploring advanced generative models, Developing a VAE for data generation, Overview of generative art and design, Case studies of successful projects

Module III

6 lecture hours

Introduction to NLP and its role in generative models, prompt engineering for generative AI, Text generation using GPT and other language models, Building a text generation model, Industry applications of generative AI in fields like gaming, fashion, and more, Emerging trends and innovations, Exploring ethical challenges and biases in generative models, Responsible AI practices

Module IV

6 lecture hours

Introduction to reinforcement learning, Combining reinforcement learning with generative models, reinforcement learning-based generative model, Applications of generative AI in business strategies, Innovation and entrepreneurship with generative technologies, Developing a business proposal integrating generative AI, real-world generative AI problem, Troubleshooting and refining generative models

List of Experiments:

- 1) Implement a simple generative algorithm for data generation.
- 2) Train a basic GAN model for image generation.
- 3) Implement a VAE for generating diverse samples.
- 4) Develop a simple text generation model.
- 5) Explore image-to-image translation tasks.
- 6) Implement a Pix2Pix model for converting images from one domain to another.
- 7) Investigate style transfer techniques in generative AI.
- 8) Create artistic outputs using neural art concepts.
- 9) Understand the application of generative models in sequences.
- 10) Implement a generative model for sequential data generation.
- 11) Fine-tune a pre-trained generative model for a specific task.
- 12) Implement a generative model with attention mechanisms.
- 13) Implement algorithms that promote diversity in generated outputs.
- 14) Implement a generative model that incorporates reinforcement learning.

Text Books :

1. Smith, J. K. Introduction to Generative Artificial Intelligence: Fundamentals and Applications. Academic Press, 2021.
2. Wang, H., & Chen, Q. Generative AI Explained: A Beginner's Guide to Creative Algorithms. Springer, 2020.
3. Martin Musiol. Generative AI Navigating the Course to the Artificial General Intelligence Future. Wiley, 2023.

Reference Books :

1. A Johnson, M. L. Generative AI Unveiled: A Primer on Creative Algorithms. Academic Press, 2022.
2. Rodriguez, A. R. Exploring Generative Artificial Intelligence: Foundations and Applications. Springer, 2021.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET424	Advanced Topics in Generative Artificial Intelligence	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Understand the intricate workings and structures of advanced generative models.

CO2: Evaluate and analyze the effectiveness of fine-tuning techniques for various generative models.

CO3: Implement advanced generative models in solving complex problems across various domains

CO-PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO1													
CO2													
CO3													

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I

8 lecture hours

Introduction to advanced generative models, Large language models, Exploration of current challenges and opportunities in the field, Survey of state-of-the-art generative models, Understanding Flow-based models, Autoencoders, and Transformer-based architectures, Case studies showcasing the strengths and limitations of different models, Transfer learning principles in generative models, Fine-tuning and adapting pre-trained models for specific tasks

Module II

8 lecture hours

Exploration of attention mechanisms, Application of attention mechanisms in generative models, generative model with attention mechanisms, Prompt Engineering, Understanding image-to-image translation tasks, CycleGAN and Pix2Pix models, image-to-image translation model, Principles of style transfer in generative models, Creating neural art using style transfer techniques, neural art project

Module III

6 lectures

hours

Introduction to video synthesis with generative models, Applications in video generation and manipulation, Implementing a generative model for video synthesis, reinforcement learning in the context of generative models, Reinforcement learning-based generative architectures, Integrating reinforcement learning into a generative model, State-of-the-art language models, Advanced techniques for text generation and manipulation, sophisticated NLP-based generative model

Module IV

6 lectures hours

Applications of generative AI in scientific research, Applying generative models to scientific problems, Ethical considerations in scientific applications of AI, generative AI success stories in various industries, Discussion and analysis of case studies, Challenges of robustness and interpretability in generative models, Techniques for improving model interpretability and robustness, Exploration of the latest research trends in generative AI

List of Experiments:

- 1) Implement transfer learning techniques on pre-trained generative models for specific tasks.
- 2) Fine-tune a pre-trained generative model and evaluate its performance on a novel dataset.
- 3) Design and train a generative model with attention mechanisms for a given application.
- 4) Explore the principles of image-to-image translation using generative models.
- 5) Develop and test a generative model for converting images from one domain to another.
- 6) implement style transfer techniques.
- 7) Create unique neural art using generative models and style transfer algorithms.
- 8) applications of generative AI in video synthesis.
- 9) Develop a generative model capable of generating or manipulating video content.
- 10) implement integration of reinforcement learning with generative models.
- 11) Implement a generative model that incorporates reinforcement learning principles.
- 12) Develop a sophisticated NLP-based generative model for creative text synthesis.
- 13) Collaborate on a project applying generative models to solve scientific problems.
- 14) Develop strategies to enhance the robustness and interpretability of a generative model.

Text Books :

1. Husain, Amir. Generative AI for Leaders: Atlantic Press, 2023.
2. Smith, J. K., & Johnson, L. M. Advances in Generative Artificial Intelligence: Cutting-Edge Techniques and Applications. New York, NY: Springer, 2023.
3. Anand Nayyar, Arun Solanki, Mohd Naved. Generative Adversarial Networks for Image-to-Image Translation. Elsevier Science, 2021.

Reference Books :

1. Foster David, Friston Karl. Generative Deep Learning: Teaching Machines To Paint, Write, Compose, and Play. Second Edition. O'Reilly, 2023.
2. Chen, Q., & Wang, H. Deep Generative Models: Theory, Algorithms, and Applications. Cambridge, MA: MIT Press, 2022.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
SCSE3042	Generative AI and Large Language Models on AWS	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Understand the principles of Deploy and Manage Generative AI Models on AWS.

CO2: Apply skills to Mastery of AWS Tools for Model Fine-Tuning and Optimization.

CO3: Implement GenAI and LLM skills to address real-world challenges, with expertise in securing and automating LLM workflows

CO-PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO1													
CO2													
CO3													

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I

8 lecture hours

Overview of LLMOps: Concepts, significance, and challenges of deploying and maintaining large language models (LLMs). Key differences between MLOps and LLMOps. Computational challenges: Cost, scaling, and memory issues. Introduction to AWS ecosystem: Amazon SageMaker, EC2, Lambda, and Elastic Kubernetes Service (EKS). AWS managed services for LLM deployment and scaling. Introduction to Cloud Service Model for AI, Cloud Deployment Model for AI, Benefits of Cloud Computing,

Module II

8 lecture hours

AWS Cloud Adoption Framework for AI, Development Environments for AI, MLOps Challenges and Opportunities in Rust and Python, Generative AI Workflow for Rust, Python for Data Science in the Era of Rust and GenAI, Emerging Rust LLMOps Workflows, Getting Started with Code Catalyst for Rust, Getting Started with Sagemaker Editor, Getting Started Lightsail for Research, Diagram of Serverless with AWS Bedrock Service,

Module III

6 lectures hours

AWS Bedrock Knowledge Agent, AWS Bedrock CLI, Diagram Serverless Options for Rust, Diagram Rust Axum Greedy Coin Microservice Components, Demo Rust Axum Greedy Coin. Exploration Prompt Engineering Workflows, Summarizing Text with Claude, Code Whisperer for Rust in Cloud, Install and Configure Code Whisperer CLI, Using Code Whisperer CLI, Building Bash CLI, Bash Functions,

Module IV

6 lectures hours

Key Components of Amazon Bedrock, Exploring the Boto Bedrock Client Python SDK, Exploring the Cargo Rust SDK, Invoking Python List Models, Invoking Rust List Models, Invoking Claude via Bedrock

Runtime API, What is Amazon Bedrock, Agents for Amazon Bedrock. Introduction to Cargo Lambda, Building Rust Add Function for AWS Lambda, Cargo Lambda, Challenge Lab-Bedrock Agents. Strategies for model optimization and cost management on AWS.

List of Experiments:

- 1) Setting Up an AWS Environment for LLM Development
- 2) Fine-Tuning a Pre-Trained Generative Model (BERT/GPT)
- 3) Deploying an LLM for Real-Time Inference
- 4) Optimizing LLM Inference with AWS Elastic Inference
- 5) Building a CI/CD Pipeline for LLM Deployment
- 6) Monitoring and Logging Model Performance
- 7) Implementing Data Security for LLMs on AWS
- 8) Optimizing Model Size with Model Compression Techniques
- 9) Handling Large-Scale Data with Amazon S3 and Athena
- 10) Building a Serverless Application for LLM Inference Using AWS Lambda
- 11) Leveraging Transfer Learning for Domain Adaptation in LLMs
- 12) Implementing A/B Testing for Model Performance Evaluation
- 13) Creating Custom LLMs with AWS SageMaker JumpStart
- 14) Analyzing Model Interpretability and Bias in Generative AI

Text Books :

- 1) Webber, Emily. Pretrain Vision and Large Language Models in Python: End-to-end Techniques for Building and Deploying Foundation Models on AWS. Apress, 2024.
- 2) Auffarth, Ben. Generative AI with LangChain: Build Large Language Model (LLM) Apps with Python, ChatGPT, and Other LLMs. O'Reilly Media, 2023.

Reference Books :

- 1) Fregly, Chris, Antje Barth, and Shelbee Eigenbrode. Generative AI on AWS: Building Context-Aware Multimodal Reasoning Applications. O'Reilly Media, 2024.
- 2) Marr, Bernard. Generative AI in Practice: How to Use AI Tools to Improve Your Business. Wiley, 2023.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
SCSE3043	Prompt Engineering for Generative AI	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Understand the principles of prompt engineering and how it applies to various AI models.

CO2: Apply skills to optimize prompts for text and image generation.

CO3: Implement prompt engineering skills to solve real-world problems in multiple domains.

CO-PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO1													
CO2													
CO3													

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Unit/Module I: Foundations of Prompt Engineering

8 lecture hours

What is Generative AI?, Overview of models: GPT, DALL·E, and other generative models, The role of prompt engineering in AI interactions, Applications of prompt engineering in various domains; Understanding the structure of prompts: clarity, specificity, context, How models interpret and generate responses based on prompts, Exploring common mistakes and pitfalls in prompt creation; The architecture of language models (e.g., GPT-3, GPT-4), How models generate text based on input, Experimenting with different styles of prompts for various outputs; Overview of image-generative models (DALL·E, Stable Diffusion), Designing prompts for images, Exploring multimodal models (text + image).

Unit II: Designing and Refining Effective Prompts

8 lecture hours

Exploring elements of a good text prompt, Techniques to optimize prompts for various tasks (summarization, storytelling, Q&A), Refining prompts for creative and technical text generation; Crafting Prompts for Image Generation and Multimodal Systems, Detailed design of prompts for image generation (e.g., landscapes, objects, abstract art), Using multimodal models to create integrated outputs (text + image), Case studies of multimodal applications; Advanced Prompt Refinement Techniques, Iterative refinement: improving model responses, Conditional prompting: guiding AI behavior with specific conditions, Multi-step prompts: designing chains of prompts for complex tasks.

Unit III: Advanced Prompt Engineering Techniques and Applications

6 lectures hours

How to create prompts for specific domains (legal, healthcare, finance), Real-world applications in these domains (e.g., AI chatbots, content generation); Using External Data and APIs in Prompts, Integrating external data into prompts for retrieval-augmented generation, How to use APIs and datasets to enhance AI outputs; Advanced Techniques: Controlling AI Behavior, Prompt templates for complex AI behaviors, Multi-layered prompts for handling ambiguity and variability in outputs.

Unit IV: Ethics, Limitations, and Future of Prompt Engineering

6 lectures hours

Discussing AI-generated content: bias, misinformation, and hallucinations, Ensuring responsible use of generative AI through prompt design, Case studies: Unintended consequences of poorly designed prompts; Limitations of AI models in understanding complex prompts, Strategies for overcoming challenges in specific use cases; Future Trends in Prompt Engineering, The evolving role of prompt engineering in AI development, Emerging techniques and tools for refining and optimizing prompts.

List of Experiments:

- 1) Exploring and Modifying Prompts with GPT Models to Change Text Generation
- 2) Generating Images with DALL·E Using Descriptive Prompts.
- 3) Create prompts that combine text and image generation to explore multimodal models and their capabilities.
- 4) Generate text in multiple languages by crafting multilingual prompts, analyzing the model's language understanding and output.
- 5) Create effective prompts for text summarization of articles, books, or reports.
- 6) Create prompts for generating creative stories or narratives, experimenting with tone, style, and genre.
- 7) Refine image generation prompts to see how slight changes can produce different image variations.
- 8) Create domain-specific prompts (e.g., legal, medical) and evaluate how well the model generates accurate and relevant content.
- 9) Design a chain of prompts to generate a sequence of outputs for complex tasks (e.g., first generating an outline, then expanding each point).
- 10) Design prompts that allow the creation of a simple chatbot capable of answering specific questions with conditions. - chatbot platforms (e.g., Rasa, Dialogflow).
- 11) Integrate external data into prompts to generate context-specific responses (e.g., using APIs to fetch live data). - APIs (e.g., OpenWeather, news APIs).
- 12) Develop prompt templates to automate the creation of specific types of content, such as social media posts, product descriptions, or blog articles.
- 13) Create prompts that explore potential biases in the model's responses, analyzing outputs for racial, gender, or cultural bias.
- 14) Create prompts that intentionally produce harmful or unethical content, then refine them to prevent such outputs and ensure responsible AI use.

Text Books :

1. David Scott Bernstein. Prompt Engineering for Everyone: A Comprehensive Guide to Unlock the Potential of ChatGPT and AI-Language Models. The Passionate Programmer, 2023.
2. James Phoenix, Mike Taylor. Prompt Engineering for Generative AI. O'Reilly Media, Inc., 2024.

Reference Books :

1. **James Phoenix, Mike Taylor.** Prompt Engineering for Generative AI: Future-Proof Inputs for Reliable AI Outputs. O'Reilly Media, 2023.
2. **Ian Khan.** The Quick Guide to Prompt Engineering: Generative AI Tips and Tricks for ChatGPT, Bard, Dall-E, and Midjourney. Packt Publishing, 2023.

Key Tools and Platforms:

- OpenAI GPT models: (ChatGPT, GPT-3, GPT-4)
- DALL·E or Stable Diffusion: For image generation tasks.
- Hugging Face: For various models and multimodal tasks.
- Google Colab/Jupyter Notebooks: For API-based integrations and running experiments.
- APIs: (OpenWeather, news APIs) for real-time data integration in prompts.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET498	Computational topo-geometrical data analysis	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1 : Understand the basics of computational geometry and topology

CO2 : Understand the basics of algebraic topology and differential topology and their applications

CO3 : Apply computational topo-geometrical methods to real world data analysis

CO4 : Gain knowledge of algorithm and libraries for topological data analysis and scientific visualization

CO-PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO1													
CO2													
CO3													

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I: Algebraic topology and integral geometry: theory .

11 lecture hours

Basics of set theory, group theory. Abelian groups, algebraic topology, homology, chains, cycles boundaries, boundary operator, fundamental lemma of homology, Betti numbers, short and long exact sequences, Betti numbers, hierarchical topology, persistent homology, differential topology, Morse theory, Morse-Smale complex segmentation, Integral geometry, Weyl's Tube formula, Minkowski functionals, Lifshitz-Killing curvature

Unit II: Computational framework and algorithms.

11 lecture hours

Simplices, simplicial complex, cech complex, alpha complex, simplicial topology, triangulation data structure, Delaunay triangulation, Voronoi diagrams, incremental algorithm of Betti numbers, boundary matrix, boundary matrix reduction, filtration, excursion sets, level sets, superlevel sets, persistent homology algorithm, MS complex algorithm, Voronoi diagram data structure, alphashapes

Unit III: Applications

06 lectures hours

Cosmology, cosmological principle, standard cosmological model, cosmic microwave background (CMB) CMB data analysis, Healpix library, Computational geometric and algorithmic library (CGAL), Topology toolkit library, visualization toolkit (VTK), paraview,

List of Experiments (Laboratory/projects)

Coding with CGAL, HEALPix Libraries, Writing own codes to manipulate triangulation datasets, create filtrations, boundary matrix, homology and persistent homology computation. Implement and visualize MS complex of datasets, coding with TTK and VTK. All of the lectures and practical will be geared towards the analysis of real datasets from various scientific and commercial disciplines.

Text Books :

1. Edelsbrunner, Herbert and Harer, John, Computational topology: An introduction, American Mathematical Society, reprint, 2022
2. Munkres James, Topology, Pearson Education Limited, 2014

Reference Books :

1. Dey, Tamal K. and Wang, Hsu, Computational topology for Data Analysis, Cambridge university press, 2016

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET498	Computational topo-geometrical data analysis	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: *Relate* and explain the ethical considerations in the development and deployment of AI.

CO2: *Analyze* and evaluate the social and ethical impacts of AI on various stakeholders and society as a whole.

CO3: *Extend* propose ethical decision-making models relevant to AI applications.

CO4: *Make use of* the implications of AI on privacy, data protection, bias, fairness, transparency, and accountability.

CO5: *Explain* and address ethical challenges in AI research, development, and governance.

CO6: *Develop* and discuss the ethical responsibilities of AI practitioners, policymakers, and organizations.

CO-PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO1													
CO2													
CO3													

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

6 lecture hours

Introduction to Ethics and Social Implications of AI

Introduction to Ethics and AI, Definition of Morality and ethics in AI, Historical and philosophical foundations of ethics, Ethical theories and frameworks ,Ethical decision-making models Impact of AI on society Ethical considerations in AI development and deployment Privacy and data protection in AI, Impact on Psychology. Bias ,fairness, and accountability in AI Transparency and explainability in AI systems Ethical challenges in AI research.

Module II:

6 lecture hours

Ethical Issues in AI Governance and Policy

AI governance frameworks and initiatives, Ethical considerations in AI regulation and policy-making, Intellectual property and AI, Ethical implications of AI patents. Ethical issues in AI transparency and auditability, Algorithmic accountability and responsibility, Ethical considerations in AI procurement and use by governments. AI ethics committees and their role, Ethical challenges in AI governance and policy, International perspectives on AI ethics and regulation.

Module III:**6 lectures hours****AI and Human Rights**

AI and privacy rights, Ethical considerations in AI surveillance technologies, AI and freedom of expression. Ethical implications of AI in law enforcement and criminal justice, AI and discrimination in employment and hiring, AI and social inequality. Ethical issues in AI-powered decision-making systems, AI and the right to access information, Ethical considerations in AI-mediated communication, AI and the right to a fair trial

Module IV:**4 lectures hours****AI and Workforce Ethics**

AI and the future of work, Ethical implications of AI in job displacement and automation, AI and job creation. Ethical considerations in AI-based hiring and recruitment, AI and workplace surveillance, Bias and discrimination in AI-based employment systems. Ethical challenges in AI-driven skill assessment and training, AI and worker well-being, AI and ethical implications for professional responsibilities, AI and labor rights,

Module V:**6 lectures hours****Ethical AI Development and Deployment**

Ethical considerations in AI system design and development, Ethical use of data in AI, responsible AI research and innovation. Ethical implications of AI in healthcare, AI and autonomous systems ethics, AI and environmental sustainability, Ethical considerations in AI for social good. AI and the ethical challenges in autonomous vehicles, AI and ethical implications in education and decision making role in Industries, National and International Strategies of AI and the future of humanity.

Recent Case Study of Ethical Initiatives in Healthcare, autonomous vehicles and defense.

Text Books :

1. Paula Boddington, —Towards a Code of Ethics for Artificial Intelligencel, Springer, 2017
2. Markus D. Dubber, Frank Pasquale, Sunit Das, —The Oxford Handbook of Ethics of AII, Oxford University Press Edited book, 2020.

Reference Books :

1. Wallach, W., & Allen, C, —Moral machines: caching robots right from wrong, Oxford University Press, 2008.
2. Bostrom and E. Yudkowsky. —The ethics of artificial intelligencel. In W. M. Ramsey and K. Frankish, editors, The Cambridge Handbook of Artificial Intelligence, Cambridge University Press, Cambridge, 2014.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
SCSE4041	Prompt Engineering	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	0	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Understand foundational and advanced concepts in prompt engineering.

CO2: Apply prompt engineering techniques to a variety of AI-driven tasks, including automation, summarization, and content generation.

CO3: Implement ethical AI principles, addressing issues such as bias and data privacy.

CO4: Utilize industry-standard tools and platforms in prompt engineering applications.

CO-PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO1													
CO2													
CO3													

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

8 lecture hours

Introduction to Prompt Engineering

Overview of prompt engineering: significance, evolution, and industry applications. Basics of prompt design: structure, components, types, and tools for basic prompt generation (e.g., OpenAI Playground), Crafting effective prompts: techniques for clarity, specificity, and contextual appropriateness. Key challenges in prompt engineering: handling ambiguity, ensuring consistency, and managing prompt length. Latest advancements: interactive AI, multimodal inputs, and their implications for prompt engineering.

Module II:

8 lecture hours

Advanced Prompting Techniques

Contextual prompts: incorporating conversation history, topic adaptation, and refining user intent. Dynamic, adaptive, and conditional prompts: using prompts that adapt based on responses, conditional branching, and multi-turn prompts. Multimodal prompts: integrating text with images, audio, or video to enhance user interaction. Prompt chaining and task decomposition: breaking down complex tasks into a sequence of prompts to improve model responses. Evaluation and iteration: methodologies for systematic prompt testing, evaluation metrics, and automated prompt refinement.

Module III:**6 lectures hours****Practical Applications and Ethical Considerations**

Real-world applications: prompt engineering for content generation, automated customer support, data extraction, and educational applications. Ethical and legal considerations: bias mitigation, ensuring user privacy, and compliance with industry standards and regulations. Case studies on ethical dilemmas in prompt engineering, such as biases in training data and limitations of model interpretability. Advanced ethical techniques: fairness evaluation, transparency, and accountability in AI-driven interactions. Cross-cultural and accessibility considerations in prompt design for diverse user demographics.

Module IV:**6 lectures hours****Prompt Engineering Tools and Platforms**

Introduction to AI platforms: an overview of OpenAI, Hugging Face, Cohere, and Anthropic tools. Using OpenAI's API for prompt experimentation: practical exercises on prompt design and optimization. Fine-tuning with Hugging Face: customizing models for specific applications with transformers and datasets. Prompt templates and libraries: exploration of prompt engineering resources, reusable templates, and case studies. Debugging prompts and using evaluation tools: practical guidance on resolving prompt-related issues.

Text Books :

- *"AI and Machine Learning for Coders"* by Laurence Moroney - O'Reilly Media, 2020.
- *"Prompt Engineering for Large Language Models: A Practical Guide"* by Lewis Tunstall, Leandro von Werra - Packt Publishing, 2024 (anticipated).

Reference Books :

- *"Generative Deep Learning: Teaching Machines to Paint, Write, Compose, and Play"* by David Foster - O'Reilly Media, 2019.
- *"Artificial Intelligence: A Guide for Thinking Humans"* by Melanie Mitchell - Farrar, Straus and Giroux, 2019.
- *Prompt Engineering Guide* by Prompt Engineering Hub (online resource).
- OpenAI API Documentation, Anthropic AI Documentation, and Hugging Face Platform Tutorials.

**Minor in AI (I,
II, III, IV)**

Name of Program	Bachelor of Technology				
CSET291	Data Structure Fundamentals	L	T	P	C
Owning School/Department	Computer Science Engineering	4	0	0	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1 : To explain the operations of arrays, linked list, stacks and queues.

CO2 : To examine the binary trees, binary search trees, heaps and AVL trees.

CO3 : To implement preliminary concepts array, linked list, stacks and queues, tree, graph.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	1	3				1				2		1	2		
CO2	3	3	2	2	2	1				2		1	3		
CO3	3	3	2	2	2	1	2			2	2	1	3		

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

14 lecture hours

Data Structures (What, Why, Where), Arrays, (Search, insert, delete, merge), Linked list, (Search, insert, delete, merge), bubble sort, selection sort, counting sort, radix sort.

Module II:

10 lecture hours

Circularly and Doubly linked list (Traversal, Insertion, Deletion), Stacks (Traversal, Insertion, Deletion), Simple Queue (Insertion, Deletion), Circular Queue (Insertion, Deletion, Traversal), Tree Data Structures, Height, Complete, Full, Perfect Trees.

Module III:

16 lecture hours

Tree Preorder traversal, Postorder traversal, Inorder traversal, Binary search trees, BST searching, BST insertion, BST deletion, Heap, Max heap, min heap, Heapify, Heap sort, Priority queues, Priority queues using max heap and min heap.

Module IV:

07 lecture hours

AVL trees, AVL trees traversal, AVL trees insertion, AVL trees deletion, Hashing, Hash Functions, Hash Tables, Hashing Collision Resolution Strategies, Separate Chaining, Open Addressing, Double Hashing, Graphs, Adjacency matrix, Adjacency list, Incidence matrix, Graph traversal, BFS, DFS.

Text Books :

1. Balagurusamy, E. *Data Structures Using C*. 1 ed. McGraw Hill Education, 2019. ISBN 978-9353161835.

Reference Books :

1. Jagtap, Amol M., and Ajit S. Mali. *Data Structures Using C: A Practical Approach for Beginners*. 1 ed. CRC Press, 2021. ISBN ISBN 978-1000470741.

Name of Program	Bachelor of Technology				
CSET292	Introduction to Algorithm Design and Development	L	T	P	C
Owning School/Department	Computer Science Engineering	4	0	0	0
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain time and space complexity of algorithms.

CO2: To examine the divide and conquer approach for making algorithms

CO3: To Implement greedy, dynamic, and backtracking algorithm approaches

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	1	2	3									1	3		1
CO2	1	2	3									1	3		1
CO3	1	2	3									1	3		1

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

12 lecture hours

Algorithms need (Why, what where), Analysing control statement, Loop invariant and the correctness of the algorithm, Analysis of Algorithm: Amortized analysis, Asymptotic Notations: Big Oh, Theta, Omega, Average, Best and worst-case analysis of Time Complexity and Space Complexity, Iterative method complexity, Recurrence relations, Master's Theorem.

Module II:

12 lecture hours

Divide and Conquer Algorithm, Multiplying large Integers Problem, Binary Search, Merge Sort, Quick Sort, Complexity analysis of quick sort, Matrix Multiplication (Strassen's matrix multiplication), Complexity analysis of Matrix Multiplication.

Module III:

12 lecture hours

Greedy Algorithms, Disjoint sets union, find, Spanning Tree, Minimum Spanning Tree, Kruskal's algorithm, Prim's algorithm, Analysis of Minimum spanning trees algorithms, Single source shortest path in graphs, Dijkstra's algorithm, The Knapsack Problem, Analysis of Knapsack problem.

Module IV:

20 lecture hours

Dynamic Programming, The Principle of Optimality, 0-1 knapsack problem, Matrix chain multiplication, All Points Shortest path (Floyd Warshall algorithm), Exploring Graphs: BFS,

Applications of DFS- bi-connectivity, Backtracking and Branch and Bound, The Eight queen's problem, Graph Coloring.

Text Books :

1. Cormen, Thomas H., Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein. *Introduction to algorithms*. 4th ed. The MIT Press, 2022. ISBN 978-0262367505.
2. La Rocca, Marcello. *Advanced Algorithms and Data Structures*. 1st ed.), Manning, 2021. ISBN 978-1617295485..

Reference Books :

1. Karumanchi, Narasimha. *Algorithm Design Techniques: Recursion, Backtracking, Greedy, Divide and Conquer, and Dynamic Programming*. 1st ed. Career Monk Publications, 2018. ISBN 978-8193245255.

Name of Program	Bachelor of Technology				
CSET391	Artificial Intelligence	L	T	P	C
Owning School/Department	Computer Science Engineering	4	0	0	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain the applications of artificial intelligence, categorize various problem domains and basic knowledge representation.

CO2: To examine Machine Learning, Deep Learning, and Neural Networks algorithms.

CO3: To Implement Supervised and Unsupervised Learning approaches for real-life problems.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3			3	2			2					3	3
CO2	3	3	3	2	3				2					3	
CO3	3	3	3	2	2									3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

18 lecture hours

Artificial Intelligence, Intelligent Agents, Agent structure, Agent Types, Environments and its properties, Environment programs, Types of Environments (25), Search Strategies for problem solving, Search Strategies Evaluation Criteria, Uninformed Search, Uniform cost search, Depth-First Search (DFS), Breadth First Search (BFS), Depth Limited search, Iterative deepening depth first search, Bidirectional search, Informed Search, Heuristic Functions, Best-FirstSearch method, A* search.

Module II:

18 lecture hours

Machine learning, Regression- Linear, Linear regression with gradient descent, Batch gradient descent, Polynomial regression, Decision tree regression, Formalizing the Learning Problem, Classification, Types of Classification techniques and its applications, Logistic regression hypothesis representation, Decision boundary, Regularization, Overfitting and underfitting, Gradient descent with regularization, Support Vector Machine, Random Forest, Unsupervised Learning, K-means, C-means.

Module III:

10 lecture hours

Artificial Neural Networks (ANN), Network structures, Optimal network structure, Activation Functions and types, Perceptron, Multi-layer Perceptron, Irrelevant and Redundant Features, linearly separable functions, Multilayer Feed-Forward Networks, Back propagation concept, Back-propagation as gradient descent search, Bayesian Methods for Learning.

Module IV:**10 lecture hours**

Convolution Neural Network (CNN), Layers in a CNN, Recurrent Neural network (RNN), Working of RNN, FeedForward NNs vs RNNs, Measures of Classification Performance, Confusion Matrices and its applicability, ROC Curves, Area Under ROC Curves.

Text Books :

1. Rich, Elaine, Kevin Knight, and S. B. Nair. *Artificial Intelligence*. 3 ed. McGraw Hill Education, 2017. ISBN 978- 0070087705.
2. Alpaydin, Ethem. *Introduction to machine learning*. 4 ed. MIT Press, 2020. ISBN 978-0262043793.

Reference Books :

1. Sabouret, Nicolas. *Understanding artificial intelligence*. 1 ed.), Chapman and Hall/CRC, 2020. ISBN 978-1000284158.

Name of Program	Bachelor of Technology				
CSET392	Applied AI	L	T	P	C
Owning School/Department	Computer Science Engineering	4	0	0	4
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain Supervised and Unsupervised AI models for solving real-life problems.

CO2: To examine the need for Artificial Intelligence and the limitations of conventional methods.

CO3: To implement the ideas behind selection of various AI methods.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	3	3			2							3		2
CO2	2	3	3	2	3								3	3	
CO3	2	2	3	2	3									2	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

16 lecture hours

The AI basics, Problems, The core assumptions of AI, Need of AI over the traditional methods, AI Techniques, The level of the Model, Criteria for success and failure of the models, AI in Natural language processing, Syntactic Processing, Semantic analysis, Discourse and pragmatic processing, Spell checking, Measures of WordNet similarity, Sentiment analysis and opinions on the web, Sentiment analysis Use case, Exploring sub-discipline of AI: Machine Learning, Supervised learning, Unsupervised learning, Reinforcement learning, Statistical Reasoning: Probability and Bayes' Theorem, Bayesian Network.

Module II:

14 lecture hours

AI for knowledge representation, Rule-based knowledge representation, Procedural and declarative knowledge, Application of AI in Healthcare, Data preparation for healthcare modelling: image data, tabular data, text data, Case study: Early detection of Type 2 diabetes, AI in Digital Marketing, Various digital marketing platforms, Impact of AI on those platforms, Artificial Intelligence impact on Web Design/UI, CHATBOTS, Benefits & its impact on Digital Marketing, Churn prediction using AI, Use case for Digital Marketing.

Module III:

10 lecture hours

Detecting anomalies in online social networks use case, Recommender Systems, Content Filtering, Collaborative Filtering, Matrix Factorization method, Recommender systems Case study, Overview of ML and DL Applications in Agriculture, Principal Component Analysis and Regression Applications in

Agriculture, Predicting crop yield use case, Classification and Clustering Methods in Agriculture, ML And DL for Soil and Crop Image Processing.

Module IV:

12 lecture hours

Time series analysis, Random Processes; Stationarity & Ergodicity, Auto- and cross-correlation functions; Partial correlation functions, Linear random processes; Auto-regressive, moving average and ARMA models, Models for non-stationary processes; Trends, heteroskedasticity and ARIMA models, Machine learning for Time series forecasting, Use case for time series forecasting.

Text Books :

1. Yao, Mariya, Adelyn Zhou, and Marlene Jia. *Applied artificial intelligence: A handbook for business leaders*. 1 ed. TOPBOTS, 2018. ISBN 978- 0998289027.
2. Géron, Aurélien. *Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow*. " 2nd ed. O'Reilly Media, 2019. ISBN 978-1491962299.

Reference Books :

1. Goodfellow, Ian, Yoshua Bengio, and Aaron Courville. *Deep learning. Adaptive computation and machine learning*. 1 ed. MIT Press, 2017. ISBN 978-0262035613.
2. Liu, Yuxi Hayden. *Python machine learning by example: implement machine learning algorithms and techniques to build intelligent systems*. 1 ed. Ingram short title, 2017. ISBN 978-1783553112.

**CS Open
Electives-I, II,
III, IV**

Name of Program	Bachelor of Technology				
CSET381	Applications of AI	L	T	P	C
Owning School/Department	Computer Science Engineering	3	0	0	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Know various AI Agents and AI search algorithms (uninformed, informed, heuristic, constraint satisfaction, genetic algorithms).

CO2: Demonstrate working knowledge of reasoning in the presence of incomplete and/or uncertain information.

CO3: Apply knowledge representation, reasoning, and machine learning techniques to real-world problems.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3			2	2							3		
CO2	3			3	2	2			2	2	2		3		
CO3	3	2			2	3			2	2	1			3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

12 lecture hours

Introduction: Introduction to Artificial Intelligence, Foundations and History of Artificial Intelligence, Introduction to Search: Searching for solutions, Uniformed search strategies, Informed search strategies, Local search algorithms and optimistic problems, Adversarial Search, Search for games, Alpha - Beta pruning, Applications of Artificial Intelligence. Markov decision processes, Intro to neural nets.

Module II:

12 lecture hours

Reasoning with uncertainty, Probabilistic reasoning over time Learning Gaming: Movement, Decision Making, Strategy, Infrastructure, Agent-Based AI.

Module III:

06 lecture hours

Hacks and Heuristics. Vision Systems: fundamentals of image formation, camera imaging geometry, feature detection and matching, Multiview geometry.

Module IV:

09 lecture hours

Motion estimation and tracking, and classification, Action recognition Color spaces and Segmentation.

Text Books :

1. Russell, Stuart J., and Peter Norvig. *Artificial intelligence a modern approach*. 3 ed. Prentice Hall, 2010.

Reference Books :

1. Forsyth, David A., and Jean Ponce. *Computer vision: a modern approach*. 2 ed. Prentice Hall, 2011.
2. Millington, Ian, and John Funge. *Artificial intelligence for games*. 2 ed. CRC Press, 2009.

Name of Program	Bachelor of Technology				
CSET382	Web Technologies	L	T	P	C
Owning School/Department	Computer Science Engineering	3	0	0	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain the development and transition of the web.

CO2: To learn creating the web pages and apply the styles.

CO3: To implement web-based application and deploy it.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3			2	2						2	3		
CO2	3		3		2							2	3		
CO3			2			3						2		3	

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

08 lecture hours

Introduction to Web Technologies, History of the Web, OSI Reference Model, Understanding Web System Architecture, understanding 3-Tier Web Architecture, Layers in the TCP/IP Model, Web Browsers, Overview of HTTP, Exploring Web Technologies, Introduction to Web Services, About IIS, Services Supported by IIS7, Installation of IIS7, Administer Web Server Remotely, Creating Web Sites.

Module II:

09 lecture hours

Introducing HTML Document Structure, Creating Headings on a Web Page, Working with Links, Creating a Paragraph, Working with Images, Working with Tables, Working with Frames, Introduction to Forms and HTML Controls, Introducing Cascading Style Sheets, Introducing DHTML, Introducing JavaScript, Client-Side Benefits of using JavaScript over VB Script, Embedding JavaScript in an HTML Page, Handling Events, Using Variables in JavaScript, Using Array in JavaScript, Creating Objects in JavaScript, Using Operators, Working with Control Flow Statements, Working with Functions.

Module III:

09 lecture hours

Introducing PHP , Working with Variables and Constants, Controlling Program flow, Working with Functions, Arrays, Files, and Directories, Working with Forms and Database, Databases : Basic command with PHP examples, Connection to server, creating database, selecting a database, listing database, listing table names creating a table, inserting data, altering tables, queries, deleting database, deleting data and tables, PHP myadmin and database bugs, Exploring Cookies, Sessions, and PHP Security.

Module IV:**08 lecture hours**

Getting Started with Web Applications in Java, Working with JavaBeans, Working with Servlet Programming, Working with JSP, Java Database Programming, ASP.NET 3.5 Essentials, Developing a Web Application , Application Structure and State, Web Forms: Standard Controls, Navigation Controls: Tree View, Menu, and Site Map Path, Validation Controls, Introducing Web Parts Controls, Working with Database Controls, Introducing Login Controls, Inside Master Pages and Themes, Managing Web Applications, Application Globalization.

Module V:**08 lecture hours**

Introduction to XML, Basic and advance XML, Java API for XML Processing, Document Object Model (DOM), XML Linking Mechanism, Evolution of Web Application, Understanding JavaScript for AJAX, Asynchronous data transfer with XML Http Request, Implementing AJAX Frameworks, Integrating PHP and AJAX, Consuming Web Services in AJAX .application in IT sector

Text Books :

1. Kumar, Akshi. *Web technology: theory and practice*. 1st ed. CRC Press, 2018.
2. Meloni, Julie C. *Sams teach yourself HTML, CSS, and JavaScript all in one*. 3rd ed. Sams Publishing, 2019.

Reference Books :

1. Robbins, Jennifer Niederst. *Learning web design: A beginner's guide to HTML, CSS, JavaScript, and web graphics*. 1st ed. Springer InternationalPublishing, 2018.

Name of Program	Bachelor of Technology				
CSET383	Semantic Technology	L	T	P	C
Owning School/Department	Computer Science Engineering	3	0	0	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain the modelling of information and knowledge.

CO2: To examine RDF and OWL Technologies.

CO3: To Implement sophisticated querying approaches for facilitate distributed information retrieval and aggregation.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	1	2	1								1	2	2	
CO2	2	1	2	1						1		2	2	3	
CO3	2	2	3	2	3						2			3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

09 lecture hours

Evolution of web technologies; Shortcomings of the traditional web; General scheme of knowledge representation models; Brief historical information on the development of models; Knowledge-based systems and examples; Types of logic models; General terms and definitions; Formal (Aristotle's) logic: Names, Statements, Proof, and rebuttal procedures; Mathematical implementation of formal logic; Methods of automatic theorem proof (calculation of predicates). Fuzzy sets; Definition of concepts: ontology, concept, attribute, axioms. Examples of ontologies.

Module II:

08 lecture hours

The concept of Semantic Web; Multilevel representation; Semantic Web applications; Main tendencies of development of Internet technologies; Electronic commerce; Auctions; Information collection and management; Personal assistants; Scientific and educational information environments; Electronic tourism; E-government; Bioinformatics; Semantic Grid; Business process management; Semantic Web SPARQL query language; Simple queries; Themas; Literals; Variables; List of predicates-objects.

Module III:

12 lecture hours

Anonymous nodes; RDF collection; Samples of triplets; Sample solutions; Multiple comparisons; Working with RDF literals; Comparison of RDF literals; Limitations of values; Samples of graphs; Combination of samples; RDF data sets; RDF data set queries; Description of RDF datasets; Solutions and result forms; Selection of variables; Building the resulting graph; Resource descriptions; Explicit IRI. Resource

identification; Functions and operators of SPARQL; Description of resources in RDF language; OWL ontology description language; Standard metadata views; FOAF technology; Intelligent agents; Multi-agent technologies.

Module IV:

13 lecture hours

Data processing algorithms in Semantic Web; Semantic Web services; Ontologies of web services modelling; Service description: profile, process model, interaction (grounding); Stages of work with web services: annotation, detection, handling, composition; Monitoring of service performance; Specifications for semantic web services: WSMO, WSML; WSMX, OWL-S, SWSF; IRS-III, WSDL-S; Methods, Algorithms, and tools to detect and compose web services; Examples of service descriptions; Options for using discovery and service compositions in an enterprise B2B system.

Text Books :

1. Yu, Liyang. *Introduction to the semantic web and semantic web services*. 1st ed. Chapman and Hall/CRC, 2019.

Reference Books :

1. Alor-Hernández, Giner, José Luis Sánchez-Cervantes, Alejandro Rodríguez-González, and Rafael Valencia-García, eds. *Current trends in Semantic Web technologies: Theory and practice*. 1st ed. Springer, 2019.

Name of Program	Bachelor of Technology				
CSET384	IT Support Technologies	L	T	P	C
Owning School/Department	Computer Science Engineering	3	0	0	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Understand the need for IT Support Technologies.

CO2: To articulate network security and firewall concepts.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	1	2			3	1			1	2	2	1	3		
CO2	3	3	3	2	3	2		2	2	2	1	3	3		

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

13 lecture hours

Introduction to IT Support; Need for IT Support; Software Support; Hardware Support; Network Support; The Human Factor; Assumptions; Language Barrier; Understanding End User; Staff Training; Structuring Training and Education; Support Methodology; Flow Logic and Troubleshooting; Querying Users; Understanding IT System Issues; IT System Structure; Peripheral Issues; Understanding Environmental Factors; Documentation and Reporting; Creating Troubleshooting Guides; First-Line Support; Paperwork; Second- and Third-Line Support Paperwork; Engineer Paperwork; Administrative Tools; Performance Monitoring Tools; Custom Views; Task Manager; Error and Status Information; Event Details; Event Logs; Error Logs; Remote Support Tools; Remote Desktop; Remote Assistant.

Module II:

15 lecture hours

Evolution of Computers; Aging Technology; Interface Standards; USB; Firewire; Serial; Parallel; Unix Event Catalogue; Linux Systems; Windows NT; Windows Next; LANs; WAN; SD-WAN; Types of WAN; Packet Switching Networks; Circuit Switching Networks; Overlay Networks; Packet over SONET/SDH; Asynchronous Transfer Mode; Frame Relay; Metropolitan Area Network; TCP/IP Protocol Stack; Cabling; Hubs; Switches; Routers; Repeaters; Topologies; Cloud Services; Cloud Software; Cloud Platforms; Software as a Service (SaaS); Platform as a Service (PaaS); Infrastructure as a Service (IaaS); Server Virtualization; Types of Server virtualization; Advantage of Server Virtualization.

Module III:

14 lecture hours

Security Concerns in IT; Network Security; Security Threats; Network Attack; Types of Network Attack; Cryptography; Confidentiality; Data Integrity; Authentication; Non-Repudiation; Symmetric Key; Symmetric Key Algorithms; Applications and Drawbacks; Asymmetric Key; Asymmetric Key

Algorithms; Applications and Drawbacks of Asymmetric; Key Algorithms; Digital Signatures; Digital Signature Algorithms; Notions of Security; Firewalls; Developing Security Policy; Firewall Configuration Strategies; E-mail; E-mail Protocols; Secure E-mail; Importance of Email Security; Best Practices for Email Security; IPsec; Uses of IPsec; Component of IPsec; SSL; TLS; SSL certificate; Trends and Challenges in IT Support.

Text Books :

1. Richard, Deal. *Cisco ASA configuration*. 1st ed. Tata McGraw-Hill Education, 2009.
2. Stallings, William. *Data and computer communications*. 9th ed. Pearson Education India, 2010.

Reference Books :

1. Buyya, Rajkumar, James Broberg, and Andrzej M. Goscinski, eds. *Cloud computing: Principles and paradigms*. 1st ed. MIT Press, 2011.
2. Forouzan, Behrouz A. *Data communications and networking*. 5th ed. McGraw Hill, 2007.
3. Tanenbaum, Andrew S. *Computer networks*. 5th ed. Prentice Hall, 2010.

Name of Program	Bachelor of Technology				
CSET385	Software Engineering	L	T	P	C
Owning School/Department	Computer Science Engineering	3	0	0	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate Software Engineering as an iterative and systematic process.

CO2: Make use of development life cycle through the IDE, UML, and Git.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	1	3	1	1		1			1	2	3	1		3	
CO2	3	3	3	2	3	2		1	2	2	2	2		3	

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

12 lecture hours

Importance of Software Engineering, Phases of software development lifecycle, SDLC case study, Software Process Model, Waterfall model, Prototyping model, Incremental model, RAD model, Spiral model, Version Control System.

Module II:

10 lecture hours

Check-in and check-out code in repository, Create branch and merging branch, Need of agile, Agile manifesto, Agile principles, Agile development methods, Extreme programming (XP), XP principles, Test first development, Refactoring.

Module III:

10 lecture hours

Requirement engineering, User stories, Acceptance criteria, Requirement validation and verification, UML, Behavioral UML diagrams, Structural UML diagrams.

Module IV:

10 lecture hours

Software quality assurance and testing, Designing test cases, Black Box testing, White box testing, Black box vs. white box testing, Control flow testing technique.

Text Books :

1. Pressman, Roger S. *Software engineering: a practitioner's approach*. 7th ed. McGraw Hill International, 2014.
- 2.

Reference Books :

1. Sommerville, Ian. *Software Engineering*. 10th ed. Person Publications Publishing Company, 2015.

Name of Program	Bachelor of Technology				
CSET386	Programming Languages	L	T	P	C
Owning School/Department	Computer Science Engineering	3	0	0	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain the requirement for the interdisciplinary application of programming languages.

CO2: To build the logic for the given problem.

CO3: To develop programs and debug.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	1	2		2		2						2	3	1	1
CO2		2	3	1			2					1		3	3
CO3			3	1								1		3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

Programming, programming language. Types of programming languages, high level vs low level, compilers, interpreters, assemblers, binary code. Datatypes, variables, keywords, identifiers, Importance of comments in programming. Implementation of scope rules. Static scoping and dynamic scoping.

Module II:

11 lecture hours

Programming constructs: conditional statements if statements, else statement, if-else statement. For loop, while loop, nesting loop. Do-while loop, infinite loops, break, continue statements. Arrays, one dimensional array, multidimensional array.

Module III:

10 lecture hours

Functions, in-built functions vs user-defined functions, importing libraries for using in-built functions. Pointers and its types, arrays of pointers, pointers, and functions. Passing parameters to functions, returning values from functions, recursion. Dynamic arrays, string, string variables, string handling functions.

Module IV:

11 lecture hours

Object-oriented paradigm. Fundamental concepts, objects, classes, encapsulation, and inheritance. Errors and warnings, rectifying errors and debugging. File handling, opening and closing file, input / output operations on file.

Text Books :

1. Bansal, Arvind Kumar. *Introduction to programming languages*. 1st ed.), Chapman and Hall/CRC, 2017.
2. Kanetkar, Yashavant, and Aditya Kanetkar. *Let Us Python Solutions: Learn by Doing-the Python Learning Mantra*. 1st ed. BPB Publications, 2020.

Reference Books :

1. Klabnik, Steve, and Carol Nichols. *The Rust programming language*. 1st ed. No Starch Press, 2023.

Name of Program	Bachelor of Technology				
CSET387	Artificial Intelligence for Creative Expression	L	T	P	C
Owning School/Department	Computer Science Engineering	3	0	0	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain the creative expression possibilities of artificial intelligence.

CO2: To create artistic experiments and prototypes, in a variety of output forms like visual, textual, musical.

CO3: To examine the implication of AI in the field of creative expressions, which was human territory till now.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	1	1	1			1	2					3	1	1	1
CO2	1	1	3	2	1		2					2	3	3	2
CO3		3	1	3	1		1					2	3	2	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

14 lecture hours

What is machine learning? Playing around with Wekinator and Tensorflow.js to start making some interactive work and get familiar with concepts of training the machine with some data, running a trained model and seeing interactive output. What are neural networks? Looking inside neural networks and how neural networks are trained, understanding what terms like weights of a network, activation function etc. mean. Convolutional neural networks and how that leads to Deep dream. Experiments with deep dream to visualize layers.

Module II:

14 lecture hours

Discussions around how AI is like a photography moment for art to give some art historical context to AI. Style transfer- real-time style transfer from pre-trained models. Run style transfer experiments through webcam input. Discussions around how artists are using style transfer.

Module III:

07 lecture hours

Recurrent neural networks- Text Modeling, Sketch (vector drawing) Modeling, Music Modeling. Introduction to sequential datasets in the context of art, like quickdraw, text corpuses and musical datasets and what can be done with such sequential data. Ethical considerations around widespread use of AI. Conceptual art pieces that comment on this issue.

Module IV:

07 lecture hours

Dimensionality reduction algorithms like TSNE, and how they can be used to form interactive ways of exploring data. Example art pieces using this. Generative models- Generative Adversarial Networks (GANs). Understanding the basic formulation of a GAN. Pipeline of how to train a GAN for art. GAN latent space arithmetic.

Text Books :

1. Marcus Du Sautoy, *The Creativity Code, Art and Innovation in the Age of AI*. 1st ed. Harvard University Press, 2020

Reference Books :

1. Kevin Ashley, *Make Art with Artificial Intelligence, Make and Sell Your Art with AI, Blockchain and NFT*. 1st ed. Independently published, 2021

Name of Program	Bachelor of Technology				
CSET388	AI and Public Policy	L	T	P	C
Owning School/Department	Computer Science Engineering	3	0	0	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain the policy frameworks and strategy reports by different countries and organizations.

CO2: To articulate the arguments for AI laws related to Autonomous and Robotics systems.

CO3: To make use of legal framework and challenges concerning AI and Data science.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	1	2	2	3				1	3		2		2	2
CO2	2	3	2	2	2					3		2			2
CO3	2	2	2	2	2					2		2		2	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

11 lecture hours

AI: An opportunity and a risk, Comparing and interpreting the strategy and reports of different countries on Artificial Intelligence: Singapore, USA, UK, Germany, India, China, Blockchain: Japan, China, USA, Switzerland, Singapore, India, Robotics: Singapore, Japan, USA, India, International AI Strategies: European Union, United Nations, AI Agreement between UAE and India, International Study Group of AI.

Module II:

13 lecture hours

AI Policies in India: NITI Aayog, AI initiatives by Ministry of Electronics and IT, AI Initiative by Ministry of Commerce and Industry, National AI based portal, AI Academia/ Institutes and Centers in India, AI standardization in India: Bureau of Indian Standards (BIS), Department of Telecom (DOT), Key considerations for AI policymaking in India: Resources, Infrastructure, Markets, and Funding, Policy environment for AI innovation, Democratize AI technologies and data, National infrastructure to support domestic development: AI Data Storage, AI Networking Infrastructure, Awareness, Education, and Reskilling: Skill sets to successfully adopt AI, Early Childhood Awareness and Education, Focus on marginalised groups, Improved access to and awareness of Internet of Things, Public Discourse, Impact of AI on different stakeholders: Employees, Customers, Business, etc., How has COVID-19 affected the AI trends?, Business Transformations with AI.

Module III:**08 lecture hours**

AI in warfare and diplomacy, AI shortfalls for military applications, Transparency in AI, Audits, Tiered Levels of Transparency, AI and economic growth: Economic characteristics of AI, Private equity investments in AI start-ups, Broader trends in development and diffusion of AI, How increases automation in the production of goods and services is impacting economic growth?, How can we reconcile the advent of AI with the observed constancy in growth rates and capital share over most of the twentieth century?, Should we expect such constancy to persist in the twenty- first century?, Can AI drive massive increases in growth rates? Under what conditions, and are these conditions plausible?, How are the links between AI and economic growth modulated by firm- level considerations, including market structure and innovation incentives?, How does AI affect the internal organization of firms, and with what implications?

Module IV:**10 lecture hours**

Documentary on Impact of AI in economy, Case study: How do you organize the skills to best facilitate innovation for your firm?, Pitch deck for ideas or new application of AI, Pitch deck for ideas or new application of AI, Coordination and collaboration across stakeholders, Development of Contextually Nuanced and Appropriate AI Solutions Continuing, deepening, and expanding partnerships for innovation, Develop contextual standard benchmarks to assess quality of algorithms, Frameworks for Regulation: National legislation, Data Protection Law, Discrimination Law, Frameworks for Regulation: Competition Law, Consumer Protection Law, Sectoral Regulation, AI Policy Challenges: Intellectual Property Regime and AI issues, Catastrophic and Existential Risk, AI Policy Challenges: Security and Cyber security, Re-thinking Intellectual Property Regimes.

Text Books :

1. West, Darrell M., and John R. Allen. *Turning point: Policymaking in the era of artificial intelligence*. 1st ed. Brookings Institution Press, 2020.

Reference Books :

1. Sarangi, Saswat, and Pankaj Sharma. *Artificial intelligence: evolution, ethics and public policy*. 1st ed. Taylor & Francis, 2019.

Name of Program	Bachelor of Technology				
CSET389	Linux Apache MySQL PHP (LAMP)	L	T	P	C
Owning School/Department	Computer Science Engineering	3	0	0	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1 : To articulate an overall understanding of PHP programming through various server.

CO2 : To build modular web applications with different frameworks.

CO3 : To design and develop webpages and web sites for the needs of an organization.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	2	2	2	2							2	2	2	2
CO2	2	2	3	2	2							2	2	3	2
CO3	2	2	2	2	3							2	2	2	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

LAMP technology, Linux Operating System, Knowledge of various editors, PHP introduction, Operators in PHP, Logical and conditional operators use cases, Function Definition and Function Call, Function with arguments, Function with return value, call by value and call by references.

Module II:

11 lecture hours

Understanding variable scope, Global Variables and Static Variables, Include and Require, Built-in functions in PHP, Introduction to Array, Array in PHP, Creating an Array, Accessing Elements of an Array, Modifying Elements of an Array, Array and string related operations.

Module III:

11 lecture hours

OOPs Concepts, Defining Class in PHP, Object in PHP, Constructor, Constructor with Parameters, Introduction to Exception, Exception Handling mechanisms, Creating Custom Exceptions, Multiple Catch Blocks, Exception Propagation, Error Handling in PHP, Web designing principles.

Module IV:

10 lecture hours

HTML, CSS, Java script, Supporting tools and CMS, Introduction to MySQL, Learning the MySQL Data Types, Frequently used String functions in MySQL, Regular expressions and their uses in PHP, Cookies, Session variable, its session ID management, File handling in PHP.

Text Books :

1. McPeak, Jeremy. *Beginning JavaScript*. 5th ed. Wrox Publication, 2015.

Reference Books :

1. Welling, Luke, and Laura Thomson. *PHP and MySQL Web development*. 5th ed. Pearson, 2017.

Name of Program	Bachelor of Technology				
CSET390	Data Structures and Algorithms	L	T	P	C
Owning School/Department	Computer Science Engineering	3	0	0	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain basic data structures for storage and retrieval of ordered or unordered data. Data structures include arrays, linked lists, binary, trees, heaps, and hash tables.

CO2: To implement algorithms for the creation, insertion, deletion, searching, and sorting of each data structure.

CO3: To examine and compare algorithms for efficiency using Big-O notation.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	1	2	3									2	3		
CO2	1	2	3									2	3		
CO3	1	2	3									2	3		

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

14 lecture hours

Asymptotic notation, Recursion, Masters theorem, Array storage, Memory allocation in 2D array, Static vs dynamic memory allocation, Binary search, Array search, traverse, insertion, and deletions, Linked list, Linked list modifications (circular, doubly), Traverse, insertion, and deletions in linked list.

Module II:

12 lecture hours

Stack representation, Application of stacks, Traverse, insertion, deletions in stack, Queue representation, Application of queue, Modifications in queue (circular, priority), Traverse, insertion, deletions in queue, Tree representation, Binary tree, Heap representation, Extract min, search, insertion operations in heap.

Module III:

16 lecture hours

Graph representation, BFS, DFS algorithms, Divide and conquer algorithm, Sorting techniques, Greedy algorithm, Coin exchange problem, Frog jump problem (proof of correctness), Dynamic algorithm, MCM, 0-1 and fractional knapsack.

Text Books :

1. Karumanchi, Narasimha. *Data Structures and Algorithmic Thinking with Python*. 1st ed. CareerMonk Publications, 2016.

2. Cormen, Thomas H., Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein. *Introduction to algorithms*. 4th ed. MIT press, 2022.

Reference Books :

1. Dedov, Florian. *A Complex Subject Simply Explained (Runtime Complexity, Big O Notation, Programming)*. 1st ed. Amazon Digital Services LLC - KDP Print US, 2020.
2. Wengrow, Jay. *A Common-Sense Guide to Data Structures and Algorithms* 2nd ed. O'Reilly, 2020.

Name of Program	Bachelor of Technology				
CSET481	Software Testing	L	T	P	C
Owning School/Department	Computer Science Engineering	3	0	0	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1 : To articulate the various software testing methods.

CO2 : To make use of the various test cases for different types and level of testing.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2	2	3	1					1	1	2	1	3		
CO2	2		3						1	1	2	1	3		

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

Software testing, testing objectives. Principles of Software Testing, Testing and debugging, Test metrics and measurements.

Module II:

11 lecture hours

Verification, Validation and Testing, Software Quality and Reliability, Software defect, Manual and Automation Testing, Software Testing Life Cycle, Phases of STLC, Test Case Preparation.

Module III:

10 lecture hours

Testing Techniques: White Box Testing, Black Box Testing, Unit Testing, Integration Testing, User Acceptance Testing, Alpha and Beta Testing, Smoke Testing, Sanity Testing, Regression Testing.

Module IV:

11 lecture hours

Formal Testing, Informal Testing, Monkey Testing, Re-Testing, Load/Stress Testing, Ad hoc Testing: Pair testing, Exploratory testing, Iterative testing, Defect seeding.

Text Books :

1. Limaye, Vaishali S. *Software Testing - Principles, Techniques and Tools*. 1st ed. McGraw Hill Education, 2017

Reference Books :

1. Pressman, Roger S. *Software engineering: a practitioner's approach*. 7th ed. McGraw Hill, 2019.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET482	Computing Start-ups	L	T	P	C
Owning School/Department	Computer Science Engineering	3	0	0	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1 : Understanding of computing start-ups.

CO2 : How technology can change or upswing the scenario.

CO3 : Proposed an idea for start-up and applicability of technology (Idea to Start-up).

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2		3									1		3	
CO2	2		3									1		3	3
C03	2	1	2	3			1	1			2	1		3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

20 lecture hours

What Does It Mean to Be a Startup Entrepreneur? Engaging Others with Actionable Next Steps, Benefits vs. Features, Simple Strategies to Get Unstuck, The Financial Model, The Legal Setup of Your Startup, Meetings and Communication Skills.

Module II:

22 lecture hours

Startup Grants: Can Government Programs Stimulate Entrepreneurship? Venture Capital and Angel Investors Incubators and Accelerators Incubators and Accelerators, Moving Past the Startup Stage, How Universities Can Support Their Startups Today.

Text Books :

1. Brahim, Bachir . *The Colony Of Innovative Startups*. 1st ed. Bachir BRAHIM, 2020

Reference Books :

1. Wright, Mike, and Philippe Mustar. *Student start-ups: The new landscape of academic entrepreneurship*. 1st ed. World Scientific, 2019.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET483	Career Skills for IT Companies	L	T	P	C
Owning School/Department	Computer Science Engineering	3	0	0	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To impart aptitude, reasoning, technical and soft skills that would aid in the employability of the students.

CO2: To develop both personal and professional skills effectiveness.

CO3: To help students to gain a confidence for the upcoming placements.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	3			2	2							2		
CO2	2	2	2		3	2		3	3	3		2	2		
CO3									3	3		2			

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

11 lecture hours

Introduction to Problem Solving; Prime Numbers; GCD; LCH; HCF; Theorems of Divisibility; Divisibility Rules; Remainder Theorem; Counting; Arithmetic Progression; Geometric Progression; Harmonic; Progression; Averages; Allegations; Ratio; Proportion; Variation; Interest; Time and Work; Negative Work; Product Constancy; Equivalence Method; Efficiency; Pipes and Cisterns; Boats; Relative Motion; Opposite Motion; Straight Lines; Polygons; Triangles; Quadrilaterals; Circles; Tangent; Eclipse; Star; Functions: Tabular, Graphical, Even Odd, Inverse; Inequalities; Logarithmic; Graphical View of Logarithmic; Properties of Inequalities: Notion of Ranges, Linear Inequality, Quadratic Inequality; Permutation; Circular Permutation; Combination; Probability; Real Life Estimation of Probability: Conjunction AND, Conjunction OR; Concept for Odds for and Odd Against; Set Theory; Operations; Algebra; Coordinate Theory; Cartesian Coordinate System; Rectangular Coordinate Axes.

Module II:

11 lecture hours

Data Interpretation: Representation of Data, Nature of Data, Capturing of Data, Organization of Data, Tables, Bar Charts, X-Y Charts, Pie Charts, Cases; Logical Reasoning; Reasoning Analogies; Artificial Language; Calendar; Cause and Effect; Clocks; Critical Path; Directions; Data Interpretation; Deduction Reasoning; Statement Analysis; Data Sufficiency; Puzzles; Pattern Series; Embedded Images; Figure Matrix; Picture Series; Pattern Series; Shape Construction; Statement and Assumptions; Statement and Conclusions; Syllogism; Verbal Reasoning.

Module III:

13 lecture hours

Communication Skills; Listening Skills; Team Building; Assertiveness; Group Discussions; Personal Interview; HR Communications; Specific Applications; Essay; Memo; CV; Business Letter; Press Release; Policy and Project Proposal; Report; Minutes; Email; Social Media for Professional Purposes; Editing Documents; Crisis Communication; Getting your Ideas Across.

Text Books :

1. Sharma, Arun. *How to prepare for Logical Reasoning for CAT*. 6th ed. McGraw Hill Education, 2021.
2. Sharma, Arun. *Quantitative Aptitude*. 8th ed. McGraw Hill Education, 2021.
3. Gutbrod, Hans. *Handbook for Professional Communication: How to get your ideas across, every single time*. 1 ed. Verlag GD Publishing Limited and Co KG, 2020.

Reference Books :

1. Agarwal, R.S. *Quantitative Aptitude for Competitive Examinations*. 1st ed. S Chand Publishing, 2017.
2. Agarwal, R.S. *A Modern Approach To Verbal & Non-Verbal Reasoning*, 2nd ed. S Chand Publishing, 2017

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET484	Cyber security: Impact on Govts, Policies and Economics	L	T	P	C
Owning School/Department	Computer Science Engineering	3	0	0	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain the broader picture of cybersecurity at the world stage. To articulate the arguments for Cybersecurity related policy frameworks.

CO2: To examine the cybersecurity related challenges facing the world.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	1	2	3									1	3		
CO2	1	2	3									1	3		

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

Definition of Cybersecurity; Key Concepts in Cybersecurity; Threats to Cybersecurity; Compromises to IP, Deviations in QoS, & Espionage or Trespass; Forces of Nature, Human Error or Failure & Information Extortion; Sabotage or Vandalism, Software Attacks & Technical Hardware Failures; Technical Software Failure, Technological Obsolescence, and Theft. Security Technologies; Access Control; Firewalls; Intrusion Detection and Prevention Systems; Virtual Private Networks, Vulnerabilities in Information Assets, Understanding Vulnerabilities and Vulnerability Assessment, Vulnerability Assessment: Tools and Techniques, Vulnerability Remediation Strategies, Configuration and Change Management in Vulnerability Assessment.

Module II:

10 lecture hours

Governance, Risk Management, and Compliance (GRC) approach to Managing Cybersecurity; Management of Cybersecurity; Cybersecurity Personnel: Roles and Responsibilities; Cybersecurity & Personnel Issues; Cybersecurity Governance and Planning; Cybersecurity Strategic Planning; Cybersecurity Planning for Contingencies; Cybersecurity Risk Management; Risk Management: Models and Methodologies; Preparation for Risk Management; Risk Assessment; Risk Treatment; Cybersecurity Policy; Enterprise Cybersecurity Policy; Issue Specific Cybersecurity Policies; System Specific Security Policies; Developing and Implementing Effective Cybersecurity Policy; Enterprise Cybersecurity Policy; Performance Measures in Cybersecurity; Specifying Cybersecurity Measurements; Law and Regulation in Cybersecurity; Key Security Laws; Privacy Laws.

Module III:**10 lecture hours**

The Role of Intelligence and Information Sharing; Design and Operation of the Internet; Internet Naming and Routing Protocols; Cyber Exploits; Major Cyber Attacks; Secure Communications and Authorization; Cyber Conflict; Cyber Economics; Contingency Planning; Contingency Planning Methodology; Business Impact Analysis; CP Strategies - Data Backup and Recovery; Incident Response; Incident Response Planning; Incident Response: Detection, Containment and Recovery, Disaster Recovery; Disaster Recovery Planning; Disaster Classifications; Planning for Disasters; Disaster Preparation.

Module IV:**12 lecture hours**

History and Role of United nations in setting the norms and guidelines for cybersecurity challenges across the borders; Application in International laws to Cyberspace; Case studies of few countries exploiting the internet space to their advantage; Cybersecurity issues with the increase in Global data centres; Cloud Computing and exponential increase in data storage systems; Cyber control systems; Cyber command systems; Cyber warfare; Cyber Soldiers. and battalions; Cyber resilience; Use of Cyberspace as part of defence strategy; State surveillance; Cyber freedom; trade-off between privacy and security; Case studies to new dimensions of cybersecurity for public, Govts and Economies.

Text Books :

1. Burkart, Patrick, and Tom McCourt. *Why Hackers Win: Power and Disruption in the Network Society*. 1st ed. University of California Press, 2019.
2. Caravelli, Jack, and Nigel Jones. *Cyber security: Threats and responses for government and business*. 1st ed. Praeger Publishers Inc, 2019.

Reference Books :

1. Fields, Ziska. *Handbook of research on information and cyber security in the fourth industrial revolution*. 1st ed. IGI Global, 2018.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET485	AI and Society	L	T	P	C
Owning School/Department	Computer Science Engineering	3	0	0	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate the ethical issues related to the development and adoption of AI.

CO2: To examine the ways in which AI is impacting culture and communities.

CO3: To understand the impact of AI to make it more useful.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	2		3									1		3	
CO2	2		3									1		3	3
CO3	2	1	2	3			1	1			2	1		3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

Terminology of AI, realistic view of AI, AI Timeline, AI system illustrations, AI Failures / Challenges, Case study /Critical discussions on current applications of AI across a range of domains and sectors: Finance: Conventional risk models, Environmental, Social, and Governance (ESG) Criteria, Chatbots and Virtual Assistants, Relationship Manager Augmentation , Health care: Health data management, Treatment pathway design, Surgical robots, Supporting pharma: drug creation and clinical trials, Agriculture: Intelligent spraying of chemicals, AI-based robots for farm harvesting , National security: Emerging Threats in the AI Era, Autonomous weapons systems, Risks associated with AI-Enabled Warfare, Art and culture: Virtual visits, Language Preservation, Interpreting expressions, Painting by numbers.

Module II:

16 lecture hours

AI Ethics: Human-centred values and fairness, Ethical governance, Ethical auditing, Ethical Issues, Data access problems Healthcare Sector - Ethical Considerations, Explainable AI, Model interpretability: Global Interpretation and Local Interpretation, The Trade-off Between Accuracy and Interpretability, Interpretation Techniques: Feature Importance, Partial Dependence Plots, Local Interpretable Model-agnostic Explanations, Transparency in AI: Model Cards, Datasheets for Datasets, Fact Sheet, Fairness: AI Fairness 360, AI Explainability 360 Decision Tree, Bias in AI, Types of Bias: Observer bias, Prejudice bias, cognitive bias, Exclusion bias, Racial bias in healthcare risk algorithm, AI learning unhealthy stereotypes, Combating Bias in AI: Use less bias or more inclusive data, Diverse workforce, Legal liability: Manufacturer (Product) Liability, Product Liability Defenses Applied to AVs, Insurance Liability, Insurer's Key Challenges, Autonomous Vehicle Levels. Product Liability Defenses Applied to autonomous vehicle, Liability Due to Hacker Attacks, Role of the Regulatory Bodies.

Module III:**09 lecture hours**

Attacks on AI: Adversarial attacks, Physical attacks, Membership inference attack, The Need for Responsible AI Understanding the AI system's functioning for safe and reliable deployment, Post-deployment—can the relevant stakeholders of the AI system understand why a specific decision was made, Consistency across stakeholders, Incorrect decisions leading to exclusion from access to services or benefits, accountability of AI decisions, Dimensions of AI Accountability: Assess governance structures, Understand the data, Define performance goals and metrics, Review monitoring plans, AI and Data Privacy: AI policy options for privacy protection, Policy documents and working drafts on Artificial Intelligence, Societal Considerations in AI: Technology Based Approach for Managing AI Systems, Principles for Responsible Management of AI Systems, Societal Considerations in AI: Impact of AI on job displacement.

Module IV:**07 lecture hours**

Case study: Future in the balance? How countries are pursuing an AI advantage, Review of Global Regulatory Landscape, Guidelines for Trustworthy AI, Preparing for job transformation and building skills, Role of AI in the ecosystem for persons with Disability: Personalisation and Customisation, Neurosymbolic AI, Recommendations for Integrating Gender Equality into AI Principles, The Effects of AI On Child Psychology, Principles for AI in society, Guidelines for AI developed by stakeholders, How do countries seek to develop competitive advantage in AI?, Faster conditions, Demand conditions, Related and supporting industries .

Text Books :

1. Kumar, Puneet, Vinod Kumar Jain, and Dharminder Kumar, eds. *Artificial Intelligence and Global Society: Impact and Practices*. 1st ed. CRC Press, 2021.

Reference Books :

1. OECD. *Artificial Intelligence in Society*. 1st ed. OECD-Paris, 2019.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET486	Special Topics in Computer Engineering	L	T	P	C
Owning School/Department	Computer Science Engineering	3	0	0	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To familiarize and learn about the latest trends and research in the field.

CO2: To equip themselves with the conceptual and practical experience of few latest methods, tools, technologies or algorithms in Computer Engineering.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	1	2	3										1		2
CO2	1	2	3										1		2

1=weakly related 2= moderately related 3=strongly related

Course Contents:

Module I:

42 lecture hours

This course covers the cutting-edge topics in Computer Engineering, and these modules will be chosen by the instructor based on the requirements and relevance at that point of time. These modules need to be relevant to the industry and start-ups will also include related case studies, use cases and implementations scenarios. Students will be working on lab work and projects to get real hands-on experience of these topics and modules.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET487	Digital Marketing	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	0	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Understand, visualize, and analyse online applications based on recent digital marketing trends.

CO2: Develop an in depth understanding of digital marketing and its applications.

CO3: To create a digital marketing plan, identifying digital channels, their advantages, and limitations, to perceiving ways of their integration taking into consideration the available budget.

CO – PO /PSO Mapping:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PSO 1	PSO 2	PSO 3
CO 1	2	2	1	2	2	2	2	2	2		2	2	1
CO 2	2	2	3	2	2	3		2	2	2	2	2	1
CO 3	2	2	2	2	3	3	2	2	2	2	2	2	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I

10 lecture hours

Branding and Communications, Branding, Brand Engagement Strategy, Branding for International Markets, Marketing Communications, Marketing Data Identification and Collection, Marketing Data Integration and Exploratory Data Analysis, Predictive Analytics and Decision Making, Social Media and Content Marketing, Content Marketing Specialization, Paid Advertising and social media, Community Management, Marketing Communications, Campaign Planning, Developing creativecommunications, Public Relations.

Module II

11 lecture hours

Search Engine Optimization (SEO), Search Engine Marketing (SEM), Keywords, On-Site SEO: Optimize UX & Design, Off-Site SEO: Link-building, Programmatic & Display Advertising, Search Engine Marketing with Google Ads (SEM), Keyword Selection, Create Text Ads, CPC Bidding, Navigate Google Ads, SEM Metrics & Optimization, Jobs in SEM, Display Advertising, Display Ads& Targeting, Sales Models, Display Ads in Google Ads, Video Advertising, Jobs in Display Advertising. Email Marketing, Email List Generation, Create an Effective Email Campaigns, Create an Email Plan, Measure Results, Measure & Optimize with Google Analytics Measurability, Understand Your Audience, Evaluate Acquisition, Understand Behavior, Evaluate Conversions, Optimize Campaign Budgets.

Module III

11 lecture hours

Web Analytics, Robust Digital Marketing Strategy, designing a Web Presence, Social Media Marketing, Landscape, Channels, Content, Content Marketing, Implement & Monitor Campaigns, Measure Impact,

Jobs in Social Media Marketing, Social Media Advertising, Platforms for Social Ads, Facebook, Facebook — Create Ad Sets, Facebook — Create and Manage Ads, Jobs in Social Media Advertising.

Module IV

10 lecture hours

Digital Marketing, Digital Marketing Framework, Digital Marketing Metrics and Channels, Customer Centricity, understanding your Business, your Customer, Marketing Channels, Marketing Objectives & KPIs, Content Strategy, Content planning, Content creation, Distribute & Promote Content, Optimize Website UX & Landing Pages, Measure Impact.

Text Books :

1. McGruer, D, Dynamic Digital Marketing: Master the World of Online and Social Media Marketing to Grow Your Business (1st ed.), Wiley, 2020. ISBN 978 1119635888.

Reference Books :

1. Kingsnorth, S, Digital Marketing Strategy: An Integrated Approach to Online Marketing (2nd ed.), Kogan Page, 2019. ISBN 978-0749484224.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET488	Growth Hacking	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	0	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Integrate the data-driven and adaptive culture of Growth Hacking to improve digital marketing results.

CO2: Tell better stories and write marketing content to improve conversion rates.

CO3: Understand the economics of customer interactions and may build business tools that automate repetitive tasks in order to gain access to new data.

CO – PO /PSO Mapping:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PSO 1	PSO 2	PO S 3
CO 1	2	2	1	2	2	2	2	2	2		2	2	1
CO 2	2	2	3	2	2	3		2	2	2	2	2	1
CO 3	2	2	3	2	3	3	2	2	2	2	2	2	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I

11 lecture hours

Search engine, Search engine optimization (SEO), need of SEO, Googlebot (Google Crawler), Types Of SEO technique, Google's SEO Algorithm updates, planning and strategies for SEO, SEO tools, Technical SEO, Technical SEO Ranking factors, Type of meta tags, effect on SEO, Website architecture Optimization, Breadcrumbs, Permalinks optimization, Canonicalization, Setup of CDN, SSL, Improve Website Performance, Speed using Plugin, perform the Page Speed Test, XML and HTML sitemap creation and optimization, Add and verify a website in Google Console, Google Analytics setup and monitoring.

Module II

12 lecture hours

Keyword, importance of Keyword Research, types of keywords, Analysis of keywords using Tools, easy to rank keywords, Analysis of Keyword, find Ranking Keyword of competitor, selecting right keywords, Top ranking Keywords of your sites, Find hidden ranking keywords of your website, LSI Keywords: Easy Strategies To find LSI Keywords, Content Research, Content Structure, Content Planning with Keywords, Internal links, Outbound link, Schema Markup.

Module III**11 lecture hours**

Negative SEO, avoid duplicate content on your website, Black hat SEO, Mistake by website owners, Bad practice on the website, Common Negative SEO practices, Defense against negative SEO, local SEO, Local SEO factors, Google My Business (GMB), optimize GMB Listing, Element's optimization on GMB listing, Citations, citations in Local SEO.

Module IV**8 lecture hours**

YouTube SEO Ranking Factors, Video Optimization, Title Optimization, Description Optimization, Thumbnail Optimization, Increase YouTube Subscribers Organically, SEO Strategy from Google Search Console & Google Analytics, maintain the position of existing ranking keywords, keep Eyes On Competitor Ranking Keywords.

Text Books :

1. Das, S., Search Engine Optimization and Marketing: A Recipe for Success in Digital Marketing (1st ed.), CRC Press, 2021. ISBN 978-0367278786

Reference Books :

1. Clarke, A., SEO 2022: Learn Search Engine Optimization with Smart Internet Marketing Strategies (1st ed.), Simple Effectiveness LLC, 2022. ISBN 9780578333380, 0578333384

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET493	International Acquaintance and Externship	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

CO1. Acquire an appreciation and familiarity of the international computer science engineering professional network, as well as the current trends and developments in the industry.

CO2. Develop an understanding of the international technology landscape and how it impacts the field of computer science engineering.

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PSO 1	PSO2	PSO3
CO1	3	2			3				2		3		
CO2	2	3		3						3	2		
CO3	3	3	3	2	3				2				2

1=weakly related 2= moderately related 3=strongly related

Module 1 (Contact hours: 84)

This course is designed to provide students with an introduction to the international aspects of computer science. Students will be exposed to a variety of topics related to computer science, including computer networking, software engineering, artificial intelligence, data mining, and web development. The course will also cover the fundamentals of international relations, including international law, economic and social policy, and global trade. Additionally, students will have the opportunity to participate in an international externship, where they will gain hands-on experience in a computer science-related field. The course will also provide guidance and mentoring to help students develop the skills necessary to handle international assignments.

.STUDIO WORK / LABORATORY EXPERIMENTS:

Students will gain practical experience by using tools and technologies related International Acquaintance and Externship

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET489	Search EngineOptimization	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	3	0	0	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: Understand main search engine optimization techniques for business websites.

CO2: Analyze keyword research, writing optimized content, getting web pages indexed by search engines and tracking the outcomes.

CO3: Implement gain access to new online tools and resources to help implement successful SEO campaigns.

CO – PO /PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO1	2				2	2	2		2	2	2	2	1
CO2	1				2		2			3	2	2	3
CO3	2				2	2	2		2	2	2	2	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I

11 lecture hours

Search engine, Search engine optimization (SEO), need of SEO, Googlebot (Google Crawler), Types Of SEO technique, Google's SEO Algorithm updates, planning and strategies for SEO, SEO tools, Technical SEO, Technical SEO Ranking factors, Type of meta tags, effect on SEO, Website architecture Optimization, Breadcrumbs, Permalinks optimization, Canonicalization, Setup of CDN, SSL, Improve Website Performance, Speed using Plugin, perform the Page Speed Test, XML and HTML sitemap creation and optimization, Add and verify a website in Google Console, Google Analytics setup and monitoring.

Module II

12 lecture hours

Keyword, importance of Keyword Research, types of keywords, Analysis of keywords using Tools, easy to rank keywords, Analysis of Keyword, find Ranking Keyword of competitor, selecting right keywords, Top ranking Keywords of your sites, Find hidden ranking keywords of your website, LSI Keywords: Easy Strategies To find LSI Keywords, Content Research, Content Structure, Content Planning with Keywords, Internal links, Outbound link, Schema Markup.

Module III

8 lecture hours

YouTube SEO Ranking Factors, Video Optimization, Title Optimization, Description Optimization, Thumbnail Optimization, Increase YouTube Subscribers Organically, SEO Strategy from Google Search Console & Google Analytics, maintain the position of existing ranking keywords, keep Eyes On Competitor Ranking Keywords.

Text Books :

1. Das, S., Search Engine Optimization and Marketing: A Recipe for Success in Digital Marketing (1st ed.), CRC Press, 2021. ISBN 978-0367278786.

Reference Books :

1. Clarke, A., SEO 2022: Learn Search Engine Optimization with Smart Internet Marketing on line 416 LLC 2022. ISBN 9780578333380, 0578333384

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET491	Advanced Industry Certification	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	6	0	0	6	
Pre-requisites/Exposure	-					

Course-Outcomes (COs):

On completion of this course, the students will be able to:

CO1: Familiarize and learn about the latest trends, tools, technologies, and research in the field of computer science.

CO2: Develop the conceptual and practical understanding of the industry relevance of latest technologies and research trends.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	2	3		2	1		1					3	1	3
CO2			3	2								1	3	1	2

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I

lecture hours: 84

This course covers the cutting-edge topics from different domains of computer science through relevant industry certifications which will prepare the students for different job roles trending in the market and help them gather the required skills that will provide the students an edge over other candidates for the available job roles. Students will be working on lab work and projects to get real hands-on experience of the latest tools and technologies. This course would be equivalent to two open elective courses.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET492	Advanced Skill Enhancement	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	3	0	0	3	
Pre-requisites/Exposure	-					

Course-Outcomes (COs):

On completion of this course, the students will be able to:

CO1: Understand the latest trends, tools, technologies, and research in the field of computer science.

CO2: Implement the conceptual and practical understanding of the latest technologies and research trends related to the industry.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	2	2	2	2				1				3	1	3
CO2			3							1	1	2	1	1	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I

42 lecture hours

This course will help the students to explore their area of interest in depth and develop as well as nourish the skills required for the desirable job roles in the industry. The cutting-edge topics from different domains of computer science would be covered which will prepare the students for different job roles trending in the market and provide the students an edge over other candidates for the available job roles. Students will be performing lab work and projects to get real hands-on experience of the latest tools and technologies.

Name of Program	Bachelor of Applications				
CSET418	Global Experience and Practicum	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	0	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1. To provide students with the opportunity to gain an understanding of the global engineering landscape and develop practical skills through hands-on projects and industry-focused seminars.

CO2. To develop students' global competencies, including cross-cultural communication, problem-solving, and collaboration skills, to prepare them for success in a global engineering career.

CO – PO /PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO 2	PSO 3
CO1	1	1	1	1	1	1	1	1	1	1	1	1	3
CO2	1	1	1			2	2	3	3	3	1	1	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I 84 lecture hours

This course will provide students with the opportunity to gain a global experience in the field of engineering. It will include an introduction to engineering principles, the design process, and the engineering process through a series of lectures, hands-on activities, and field trips. Students will apply their knowledge of engineering principles to work together to develop and design a project that will solve a global engineering challenge. The course will focus on using the scientific method to solve problems, discussing engineering ethics and safety regulations, and presenting their projects to their peers and instructors. They will also explore the impact of engineering on society and the environment and develop

the skills necessary to successfully collaborate and innovate with their peers from different cultures. Upon completion of the course, students will have a well-rounded understanding of global engineering and the ability to apply their engineering knowledge in a global context.

HSS Electives

Name of Program	Bachelor of Technology (Computer Science Engineering)						
CSET307	Financial Accounting and Management			L	T	P	C
Owning School/Department	Computer Science Engineering			3	0	0	3
Pre-requisites/Exposure	-						

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate foundations of finance with an emphasis on applications vital for corporate managers.

CO2: To explain financial decisions made by corporate managers both within the firm and in their interactions with investors.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	1	2			1			2	1	1	3	1	1		
CO2	1	2			1			2	3	3	3	2	2		

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

12 lecture hours

Finance overview, Present Value Concepts, NPV Rule, Separation Theorem, Simple vs. Compound Interest, Annuities and Perpetuities, Growing Delayed Annuities and Perpetuities, Compounding Within the Year and the Effective Annual Interest Rate, Fixed Income Valuation, Valuation of Pure Discount Bonds, Yield to Maturity vs. Holding Period Return, Prices and Returns on Coupon Bonds, Semi-Annual Bonds, The Yield Curve, Holding Period Return and Yield to Maturity for Zero-Coupon Bonds, Calculating the Holding Period Return on a Coupon Bond, Equity Valuation Interest, Applying Infinite Horizon Formulas, Determining Dividend Growth, Internal Rate of Return (IRR), Comparing NPV and IRR: Accept or Reject Decision, Comparing NPV and IRR, Capital Budgeting in Practice, Expected Returns and Risk, Portfolio Analysis, Capital Asset Pricing Model (CAPM), Market Efficiency, Capital Structure, Valuation and Capital Budgeting with Leverage, Option Definitions and Strategies, Option Valuation.

Module II:

10 lecture hours

Intuition and Discounting, Compounding, Inflation, APR and EAR, Term Structure, Discounted Cash Flow: Decision Making, Free Cash Flow, Forecast Drivers, Forecasting Free Cash Flow, Return on Investment, Decision Criteria, Sensitivity Analysis, Return on Investment, Financial Reporting, Balance Sheet Equation, Assets, Liabilities, and Stockholders' Equity, Debit and Credit Bookkeeping, Relic Spotter Case, Revenues and Expenses, Adjusting Entries, Financial Statements and Closing Entries, Operating, Investing, and Financing Cash Flows, Statement of Cash Flows, Ratio Analysis, Plainview Technology.).

Module III:

09 lecture hours

Working Capital Assets, Accounts Receivable, Estimating Uncollectible Accounts, Inventory, LIFO vs. FIFO, Inventory Disclosure, Long-lived Assets and Marketable Securities, Tangible Assets, Intangible Assets and Goodwill, Long-lived Assets Disclosure, Marketable Securities, Liabilities and Long-term Debt, Time Value of Money: Future Value, Time Value of Money: Present Value, Time Value of Money: Annuities, Long-Term Debt and Bonds.

Module IV:

11 lecture hours

Deferred Taxes, Income Taxes, Deferred Tax Liabilities, Deferred Tax Assets and Tax Rate Changes, Valuation Allowances and NOLs, Income Tax Disclosure, Taxes, SCF, and Marketable Securities. Shareholders' Equity, Share Issuances and Repurchases, Dividends, Splits, and AOCI, Stock-based Compensation, Earnings Per Share, Contributed Capital Disclosure, Stock-based Compensation Disclosure, Financial Statement Analysis, Business, Strategy, and Competitive Environment, Growth and Earnings Performance, Cash Flow and Ratios, Investing and Financing Activities, Off-Balance Sheet Activities.

Text Books :

1. Irons, Robert. *The fundamental principles of finance*. 1st ed. Taylor & Francis, 2019.

Reference Books :

1. Gupta, R. K., and HIMANSHU GUPTA. *Credit appraisal & analysis of financial statements: a hand book for bankers and finance managers*. 1st ed. Notion Press, 2019.

Name of Program	Bachelor of Technology (Computer Science Engineering)						
CSET308	Communication Skills for Global Professionals			L	T	P	C
Owning School/Department	Computer Science Engineering			3	0	0	3
Pre-requisites/Exposure	-						

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To apply the dynamics of communication skills in business communication strategies and principles to prepare effective communication for business situation.

CO2: Manage information retrieval systems by performing indexing, compression, information categorization sentiment analysis, network management, search engine optimization, records compliance and risk management.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	1								2	2	1	2	2		
CO2	2								3	3	2	3	2		

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

11 lecture hours

Vocabulary Building, Concept of Word Formation, Root words from foreign languages and their use in English, Acquaintance with prefixes and suffixes from foreign languages in English to form derivatives, Synonyms, Antonyms, Standard abbreviations, Effective Communication, Purpose of Communication, Process of Communication, Importance of Communication in Business, Differences between Technical and General Communication, Barriers to Communication, Measures to Overcome the Barriers to Communication, Types of Communication, Verbal Communication, Importance of verbal communication, Significance of Non-verbal Communication, Communication Network, Scope and Types of Communication Network, Formal and Informal Communication Network, Upward Communication, Downward Communication, Horizontal Communication, Diagonal Communication.

Module II:

09 lecture hours

Listening Skills, Listening Process, Classification of Listening, Purpose of Listening, Common Barriers to the Listening Process, Measures to Improve Listening, Listening Comprehension, Pronunciation, Intonation, Stress and Rhythm, Listening as an Important Skill in Workplace, Note taking tips, Group Discussions.

Module III:

14 lecture hours

Written Communication, Principles of effective writing, Improving Writing Skills, Essentials of good style, Expressions and words to be avoided, Grammar and Usage, Steps of writing, Difference between creative

academic and professional writing, Identifying Common Errors in Writing, Rhetoric, Institutional communication, Intercultural communication, Subject-verb agreement, Noun-pronoun agreement, Misplaced modifiers, Articles, Prepositions, Redundancies, Cliches, Writing introduction and conclusion, Comprehension, Precis Writing, Essay Writing, Idea Expansion, Letter writing, E-mail, writing, Writing using latest collaborative tools like Slack, Microsoft team, issues of plagiarism, Memos and Circular Writing, LinkedIn Profile and Digital Presence.

Module IV:

08 lecture hours

Employment Communication, Job Interview, Importance and Factors Involving Job Interview, Characteristics of Job Interview, Job Interview Process, Job Interview Techniques, Manners and etiquettes to be maintained during an interview, Sample Questions Commonly asked During Interview, Mock Interviews, Expert Talk, Resume: Contents of Good Resume, Guidelines for Writing Resume, Different Types of Resumes, Reason for a Cover Letter to Apply for a Job-Format of Cover Letter, Different Types of Cover Letters, Written and verbal feedback giving and receiving.

Text Books :

1. Mishra, Richa and Ratna Rao, *A Textbook of English and Communication Skills*. 1st ed. Macmillan Publishers India Private Ltd, 2020.

Reference Books :

1. Konar, Nira. *Communication skills for professionals*. 2nd ed. PHI Learning Pvt. Ltd., 2021.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET309	Organizational Behaviour	L	T	P	C
Owning School/Department	Computer Science Engineering	3	0	0	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate concepts of organizational change, power and conflict.

CO2: To acquaint the students with the appropriate concepts, theories, models and other tools to make better understanding of behavioral dynamics.

CO3: To analyze the role of team dynamics and composition in the team performance.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	1	1				1		1	2	2	1	2	1		
CO2	1	1			3	1		1	2	2	1	2	3		
CO3	1	3	2	2				1	3	3	1	3	2		

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

11 lecture hours

Addressing Misconduct Management, Conflict Resolution, Conflict Dynamics, Varied Perspectives on Conflict, Employee Dissatisfaction's Influence, Emotional and Mood Considerations, Stress Management, Counseling Techniques, Disciplines Contributing to Organizational Behavior, Exploring Attribution Theory. Rational Model, Bounded Rationality and Intuition, Hierarchy Of Needs Theory, McClelland's Theory Of Needs, Self Determination Theory.

Module II:

19 lecture hours

Groups and teams, Group Behavior, Group Dynamics, Defining and Classifying Groups, The Five-Stage Model Of Group Development, Group Decision Making, Team process, Conformity and cohesiveness, Organizational Culture, Cultural competence, Leadership, Leading across cultures, Expatriation, Decision making and global staffing, Power and Influence, Bases of Power, Power Tactics, How People Respond To Organizational Politics, Impression Management, HR policies and practices, Theories of work motivation, Organizational Structure, Change management, Organizational stress, Nature of management.

Module III:

12 lecture hours

Understanding Organizational Behavior, Personality Traits, Values, and Attitudes in the Workplace, Perception Mechanisms, The Five Personality Model, Attitudes' Influence on Learning and Job Satisfaction, Relationship Between Behavior and Attitudes, Perception's Role in Decision-Making Processes, Cognitive Biases and Errors In Decision Making, Diversity, The challenge of diversity, Diversity management, What

motivates your people?, How to motivate your people?, Job design and employee reaction, Communication, Evaluation, Feedback, Rewards.

Text Books :

1. Buchanan, David A., and Andrzej Huczynski. *Organizational behaviour*. 1st ed. Pearson, 2019.

Reference Books :

1. Smith, Paul E., Wendy Yellowley, and Christopher J. McLachlan. *Organizational behaviour: Managing people in dynamic organizations*. 1st ed. Taylor & Francis, 2020.

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET310	Marketing Strategies and Planning	L	T	P	C	
Owning School/Department	Computer Science Engineering	3	0	0	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To identify the importance of the marketing for marketing success.

CO2: To manage customer relationships across all digital channels and build better customer relationships.

CO3: To manage customer relationships across all digital channels and build better customer relationships.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1		2	1						1	1	2	1	2		
CO2		2	1	1					2	2	2	2	1		
CO3		3	2	3					3	3	2	3	1		

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

13 lecture hours

Marketing Strategy Process, Scanning the Environment, Creating Value, Sustaining Value, Importance of Market Research, Conducting Market Research, Strategic Marketing, Segmentation Targeting Positioning, Levels of Market Segmentation, Profitable Segmentation, Targeting, Segmentation and Targeting in Practice, Branding, Marketing Strategy and Brand Positioning, Building Strong Brands, Brand Positioning, Experiential Branding, Product-Centric to CustomerCentric Management, Product-Centric Approach, Data-Driven Business Models, Direct Marketing, Customer Centricity, Communications campaign designing, Media Planning.

Module II:

12 lecture hours

Pricing, Price Sensitivity, Measuring Price Elasticity, Psychological Aspects of Pricing, Analytic Foundations for Pricing, Economic Value to the Customer, Creating Customer Access, Disruption in Information Delivery, Influencing Consumer Behavior, New Product Adoption, Managing Channel Conflict, Online Retailing and E-Commerce, Branding, Brand Elements: Choosing a Brand Name, Color & Taglines, Packaging, Persuasion, Repositioning a Brand, Building Your Professional Brand, Content Marketing, Structure for Effective Content.

Module III:

17 lecture hours

Applied Marketing, Strategic Context, 7A Framework, Agility & Authenticity, Stealing Headlines for Attention, Capturing Audience Attention, Call-to-Action & Risk Reversal, Expanding Your Network, Crafting a Content Marketing Strategy, Empathy Maps, Experience Map, Creating Irresistible Content,

Strategic Types of Content, Attraction Content, Affinity Content, Action Content, Making Action Content Actionable, Authority Content, Managing Your Content, Ethics and Professionalism, Plagiarism and Copyright Law, Multimedia Content, Visuals, Audio and Video, Digital Marketing, Digital Marketing Framework, Marketing Channels, Social Media Marketing, Email Marketing.

Text Books :

1. West, Douglas C., John Battice Ford, and Essam Ibrahim. *Strategic marketing: creating competitive advantage*. 1st ed. LID Publishing Limited, 2020. ISBN 978-1912555765.

Reference Books :

1. Ferrell, Odies C., Michael Hartline, and Bryan W. Hochstein. *Marketing strategy*. 1st ed. Cengage Learning, 2021. ISBN 978-0357516362.

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET311	Micro and Macro Economics	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	0	0	3
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To examine the understanding importance of the economy for engineering professionals.

CO2: To articulate the factors required in taking appropriate economic decisions which will help in minimizing investment, operating and maintenance expenditures.

CO-PO /PSO Mapping:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	POS 3
CO1	3	3		2		3	1				2	2	2		
CO2	3	3		2		3	1				3	3		2	

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I

8 lecture hours

Economics as a Science, Finance, Growth, and Volatility, Micro-Founded Macro Models, Markets & Competition, Supply and Demand, Consumer Theory, Production and Costs, Welfare Economics, Monopoly, Other Market Structures, International Trade, Uncertainty, Capital Supply and Capital Markets, Equity and Efficiency, Taxation and Redistribution, Social Insurance.

Module II

13 lecture hours

Microeconomics, Demand Curve, Elasticity of demand, Consumer surplus, Preferences and utility, Budget constraints, Income elasticity of demand, Deriving demand curve, Applying consumer theory, Cross elasticity of demand, Elasticity of supply, Elasticity interpreted, Surpluses & Min/Max price, Maximum rent, Minimum wage, Exchange rates, Demand for import/export, Producer Theory, Productivity and cost, Welfare economics Monopoly, Oligopoly, Factor Market, International trade, Uncertainty, Capital supply and market, Equity and efficiency, Healthcare economics.

Module III

12 lecture hours

Macroeconomics, What is GDP?, GDP Calculation, Nominal and Real GDP, Expenditure approach, Income approach, Fiscal balance and trade balance, Aggregated demand, Aggregated supply, Shifts in Aggregate Demand, Shifts in Aggregate Supply, Macroeconomic Equilibrium, Sources, Measurement, and Sustainability of Economic Growth, Business cycle, Resource Use, Housing Market, External Trade, Types

of Unemployment, Measures of Unemployment, Inflation, Indexes Used to Measure Inflation, Uses and Limitations of Inflation Measures, Cost-push and Demand-pull Inflation.

Module IV

9 lecture hours

Types of Economic indicators, Monetary and Fiscal Policy, Functions of Money, Definitions of Money, Money Creation Process, Quantity Theory of Money, Theories of Money Demand and Supply, The Fisher Effect, Effective Tax Policy, Modeling the Impact of Taxes and Government Spending. International trade, Gross Domestic Product vs. Gross National Product, Cost and benefits of international trade. Comparative Advantage vs. Absolute Advantage, Trade Restrictions, Capital Restrictions, The Balance of Payments.

Text Books:

1. Zahid A. Khan , Arshad N. Siddiquee , Brajesh Kumar and Mustafa H. Abidi, Principles of Engineering Economics with Applications (1st ed.), Cambridge University Press, 2018. ISBN 978-1108458856..

Reference Books:

1. George Abbot and Niall Kishtainy, The Economics Book (1st ed.), DK Publishing, 2014. ISBN 978-1465438409.